



SWEET Call X-2020: SWEET EDGE

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Authors The authors bear the entire responsibility for the content of this report and for the conclusions drawn therefrom.	Constance E. Allet Larissa M. Brunold Rolf Wüstenhagen, IWÖ-HSG, rolf.wuestenhagen@unisg.ch
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Summary

In Task 3.3 of SWEET EDGE, we have looked into how renewable energy installations in the Swiss Alps interact with tourism infrastructure. In particular, we were interested in understanding whether alpine wind and solar projects are seen by tourist resorts as a way to increase the attraction of the destination or whether the opposite perception prevails, and which business models can be identified to deploy renewables in alpine tourist destinations. Two separate studies were conducted, one each on solar PV and wind, combining qualitative and quantitative research methods. The results show that an existing wind project in Andermatt correlates positively with the resort's attraction to tourists, and that acceptance of alpine PV is supported by distributional justice, transparent communication, involvement of local stakeholders and spillover effects from previous renewable energy projects in the same region.

Study 1

Harnessing Swiss Alpine Solar Power - A Qualitative Study of Factors Affecting the Development of Commercial and Utility-Scale Photovoltaics Projects

Author: Constance Emmanuelle Allet

If Switzerland wants to be climate neutral by 2050 and achieve the Energy Strategy 2050 objectives (SFOE, 2020), it needs to increase its share of RE, especially in winter when there is an electricity deficit (SFOE, 2023b). Solar PV installed at a high altitude is a promising technology to cover the Swiss winter electricity deficit (SFOE, 2023a), but in 2022, only 5.7% of Swiss energy was from solar (IEA 50, n.d.). The solar energy share in winter can notably increase by private adoption of rooftop solar PV in Alpine regions, as about half of Swiss rooftop potential is not exploited (Walch et al., 2020, p.1), and through the construction of utility-scale PV projects in Alpine regions (SFOE, 2023a). Therefore, this research aims to gain insights into the factors affecting the development of commercial and utility-scale PV projects in the Alpine regions of Switzerland. To achieve this objective, fifteen interviews were conducted and supplemented by a literature review, using the social customer journey from Hamilton et al. (2021) as a flexible framework for research. Factors predominantly hindering commercial PV adoption are plot regulations and the uncertainties linked to the effective production of innovative PV models. Peer companies pioneering high-altitude PV are used to overcome these uncertainties. This peer effect extends to utility-scale PV project stakeholders seeking advice from technology pioneers. A facilitating factor for commercial PV adoption is the possibility of advertising them and having a good image. Then, the first factor hindering utility-scale project development is social acceptance from the local population. Factors hindering social acceptance are the installation's impact on nature, the visual appearance of the landscape, and the local economy. Factors favouring social acceptance are distributional justice, transparent communication, inclusion of local stakeholders and previous conduct of RE projects in the commune. Negative peer effects were observed between communal leaders and social influences between the populations of neighbouring utility-scale projects. A negative evaluation of a neighbouring project negatively affects local acceptance. It may be reinforced by the number of projects that have started in the same area over a similar period. In addition to social acceptance issues, there are technical difficulties and risks linked to high altitudes and lengthy administrative processes with time and electricity generation constraints. Finally, the main differences between utility-scale and commercial PV projects are the location of the project, the number of stakeholders with decision-making power, advertising and self-consumption possibilities, technical uncertainties and administrative processes. Based on those observations, it is recommended to choose a utility-scale project site in an area that does not have too much impact on the visual appearance of the landscape, nature and economy, in an area already impacted by humans, with existing infrastructures.



Study 2

Attraction or deterrent? Perceived effects of wind turbines on tourism in the Swiss Alps

Author: Larissa Marina Brunold

In order to implement the Swiss Energy Strategy 2050 and largely reduce greenhouse gas emissions, efforts are being made to further expand renewable energies in Switzerland. This includes wind energy, which has received less attention in Switzerland compared to other European countries, but is now also being discussed for expansion in Alpine regions. Tourist areas are being considered for the construction of wind turbines close to existing infrastructure. In this context, this master's thesis analyses the perceived impact of wind turbines on tourism from the perspective of tourists and the potential added value that wind energy can generate for tourism. This is analysed using two case studies from the regions of Andermatt and Laax in order to illustrate concrete examples of Alpine tourist regions in Switzerland. The results of the empirical study have shown that there is interest in the technology and that the proximity to existing infrastructure is considered sensible in terms of the impact on the landscape. It was also shown that the existence of showcase examples can have an effect on the perception of tourists and that it is therefore important to implement further successful wind power projects to promote the technology. Tourists do not consider their activities to be disturbed by a wind turbine, and many feel indifferent. They also see a wind farm as a sign of sustainability and progressiveness, which can have a positive effect on the reputation of a tourist destination. In addition, the presence of the Gütsch wind farm in Andermatt is a successful example of tourism development and the potential for additional regional added value generated by the wind farm.

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UNIVERSITY OF ST. GALLEN

School of Management, Economics,
Law, Social Sciences, International Affairs and Computer Science

Bachelor Thesis

Harnessing Swiss Alpine Solar Power

-

A Qualitative Study of Factors Affecting the Development of Commercial and Utility-Scale Photovoltaics Projects

Submitted by:

Constance Emmanuelle Allet

21-618-624

Bachelor candidate in Business Administration

constanceemmanuelle.allet@student.unisg.ch

Approved on Application by:

Prof. Dr. Rolf Wüstenhagen

Chair Management of Renewable Energies

Institute for Economy and the Environment (IWÖ-HSG)

University of St. Gallen

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Abstract

If Switzerland wants to be climate neutral by 2050 and achieve the Energy Strategy 2050 objectives (SFOE, 2020), it needs to increase its share of RE, especially in winter when there is an electricity deficit (SFOE, 2023b). Solar PV installed at a high altitude is a promising technology to cover the Swiss winter electricity deficit (SFOE, 2023a), but in 2022, only 5.7% of Swiss energy was from solar (IEA 50, n.d.). The solar energy share in winter can notably increase by private adoption of rooftop solar PV in Alpine regions, as about half of Swiss rooftop potential is not exploited (Walch et al., 2020, p.1), and through the construction of utility-scale PV projects in Alpine regions (SFOE, 2023a). Therefore, this research aims to gain insights into the factors affecting the development of commercial and utility-scale PV projects in the Alpine regions of Switzerland. To achieve this objective, the grounded theory method from Corbin & Strauss (2015) was followed. Fifteen interviews were conducted and supplemented by a literature review, using the social customer journey from Hamilton et al. (2021) as a flexible framework for research. Factors predominantly hindering commercial PV adoption are plot regulations and the uncertainties linked to the effective production of innovative PV models. Peer companies pioneering high-altitude PV are used to overcome these uncertainties. This peer effect extends to utility-scale PV project stakeholders seeking advice from technology pioneers. A facilitating factor for commercial PV adoption is the possibility of advertising them and having a good image. Then, the first factor hindering utility-scale project development is social acceptance from the local population. Factors hindering social acceptance are the installation's impact on nature, the visual appearance of the landscape, and the local economy. Factors favouring social acceptance are distributional justice, transparent communication, inclusion of local stakeholders and previous conduct of RE projects in the commune. Negative peer effects were observed between communal leaders and social influences between the populations of neighbouring utility-scale projects. A negative evaluation of a neighbouring project negatively affects local acceptance. It may be reinforced by the number of projects that have started in the same area over a similar period. In addition to social acceptance issues, there are technical difficulties and risks linked to high altitudes and lengthy administrative processes with time and electricity generation constraints. Finally, the main differences between utility-scale and commercial PV projects are the location of the project, the number of stakeholders with decision-making power, advertising and self-consumption possibilities, technical uncertainties and administrative processes. Based on those observations, it is recommended to choose a utility-scale project site in an area that does not have too much impact on the visual appearance of the landscape, nature and economy, in an area already impacted by humans, with existing infrastructures.

Keywords: *Photovoltaics; Renewable Energy; Customer Journey; Social Acceptance; Switzerland*

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List of Abbreviations

Art.: Article

B2B: Business-to-business

NGO: Non-Governmental Organisation

NIMBY: Not In My Backyard

PV: Photovoltaic

RE: Renewable Energies

SFOE: Swiss Federal Office of Energy

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1 Introduction

1.1 Problem Definition

Switzerland has an energy deficit in winter, running in 2022-2023, from October/December until April (SFOE, 2023b). Therefore, importing electricity from neighbouring countries in winter is necessary, causing energy safety issues (IEA 50, n.d.). Additionally, the Swiss population has low acceptance of importing energy from abroad and prefers self-sufficiency (Trutnevyte et al., 2024, p.17). Therefore, national electricity generation must be increased. If Switzerland wants to stay committed to the Energy Strategy 2050, aiming to increase energy efficiency, RE share and phase out of nuclear, and be climate neutral by 2050 (SFOE, 2020), a RE source must cover the electricity deficit. The most promising RE source is solar at a high altitude, as it generates electricity during winter. Additionally, the yield is higher since there is no fog, a thinner atmosphere, and snow reflecting the sun (Kahl et al., 2019, p.1165). Other RE sources, such as hydro and solar PV in the lowlands, are not ideal as they generate more energy in the summer (Mellot et al., 2024, p.9).

However, despite an increase in PV adoption, solar energy represented only 5.7% of the total Swiss energy in 2022 (IEA 50, n.d.). The solar energy share in Switzerland in winter can be increased in various ways: by exploiting the commercial rooftop potential in Alpine regions, as Swiss rooftop potential is underexploited (Walch et al., 2020, p.1), or by constructing Alpine utility-scale projects¹ (SFOE, 2023a). The rooftop potential focuses on commercial PV as the electricity generation capacity is higher than for household PV (SolarPower Europe, 2023). Some commercial PV projects were undertaken in constructed Alpine areas, notably by companies, to harness those advantages (Switzerland, 2024). However, about half of exploitable rooftops are not exploited in Switzerland (Walch et al., 2020, p.1). Additionally, until recently, there were few utility-scale Alpine solar systems (SFOE, 2023a). Therefore, the Swiss confederation introduced Art. 71a of the Energy Act² in September 2022 to encourage the development of such projects (SFOE, 2023a; SFOE, 2023c). As a result, many Alpine utility-scale PV systems emerged (SFOE, n.d.-b), and the SFOE (2023a) sees the most potential

¹ The Alps cover 60% of Switzerland, and the territory is divided into zones depending on the altitude, notably montane, sub-Alpine and Alpine zones, classified from lower to higher altitude. The ones considered for this research are predominantly in sub-Alpine and Alpine zones (Burga & Perret, 2001, as cited in UZH, 2024). These zones will be referred to as Alpine area for simplicity throughout the thesis.

Additionally, the distinction between commercial and utility-scale project is the following: commercial PV typically generate less than 250 kW and utility-scale PV more than 1000 kW (SolarPower Europe, 2023).

² Through this new regulation, the confederation subsidises the project's unhedged costs, at most 60% of the project costs, and facilitate the adoption process if three criteria are fulfilled (SFOE, 2023a). The park must “(1) generate at least 10 GWh per year, (2) including at least 500 kWh/kW during the winter period, and (3) the building permit application must be submitted to public inquiry before 31.12.2025” (SFOE, 2023a, para.4). Additionally, at least 10 GWh must be generated before 2025. These special conditions apply until the cumulative annual production of the utility-scale PV system reaches 2 TWh (SFOE, 2023a; SFOE, 2023c).

in such systems to increase energy security in winter. They would use only a small portion of the available land (Mellot et al., 2024, p.8). However, the implementation is complicated and faces local rejections in various cases (Rausis, 2023; “Alpine Photovoltaik-Anlage”, 2024).

1.2 Research Objectives and Research Method

Therefore, this research aims to gain insights into the factors affecting the development of commercial and utility-scale PV projects in the Alpine regions of Switzerland. Comparisons between these two types of projects will also be explored to broaden the understanding of the complications related to the conduct of Alpine utility-scale PV projects, as opposed to commercial ones. A qualitative method, the grounded theory from Corbin & Strauss (2015), will be used to explore the field of research. Following this method, semi-structured interviews (Corbin & Strauss, 2015, p.39) and existing literature will be used to derive insights (Corbin & Strauss, 2015, p.49). Additionally, a framework is flexibly used to analyse the research area, the social customer journey, from Hamilton et al. (2021).

1.3 Research Gap

This research will broaden the knowledge about factors affecting commercial PV adoption, which has not been subject to research as extensive as household PV adoption (Bach et al., 2020; Karakaya & Sriwannawit, 2015; Karneyeva & Wüstenhagen, 2017; Shakeel et al., 2023; Thormeyer et al., 2020). It will also generate insights into peer effects of PV adoption between companies, also not widely studied as opposed to peer effects in a household context (Bach et al., 2020; Baranzini et al., 2017; Cardwell, 2012; Dharshing, 2017; Graziano & Gillingham, 2014; Shakeel et al., 2023; Thormeyer et al., 2020). The research will also contribute to the literature on the social acceptance of RE projects (Olson-Hazboun et al., 2016; Tabi & Wüstenhagen, 2017; Tsoutsos et al., 2009; Vuichard et al., 2019; Walker & Baxter, 2017; Wolsink, 2012) and expand the literature on the social acceptance utility-scale PV projects, which is scarce (Cousse, 2021, p.9). Finally, it will add to the literature by evaluating the reasons for the poor social acceptance of utility-scale PV projects in contrast to smaller ones (Cousse, 2021).

1.4 Research Structure

As no particular structure needs to be followed for presenting qualitative research, the research was structured in a way that made the most logical sense (Corbin & Strauss, 2015, p.311). The methodological approach on which the research is based is presented, followed by a literature review presenting the state of the research. It includes the literature on the factors influencing the development of small-scale PV projects in companies and households, followed by the literature on the social acceptance of RE projects. Then, the social customer journey from Hamilton et al. (2021) used as a framework for the research is presented. Afterwards, the findings are presented along the customer

journey stages and then discussed with the literature. The conclusion, limitations and further research conclude the research.

2 Methodological Approach

As mentioned, the research process is based on the grounded theory method from Corbin & Strauss (2015), a method part of qualitative research. Qualitative research and the grounded theory method are presented, followed by a breakdown of the methodology used for the research process, i.e. literature review, theoretical sampling, choice of the research framework, interview guidelines and data analysis.

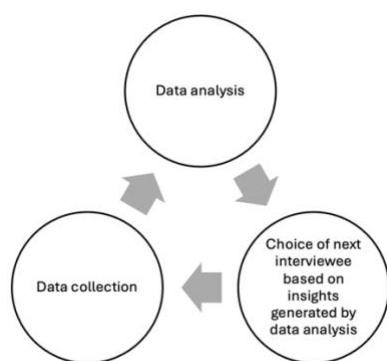
2.1 Introduction to Qualitative Research and the Grounded Theory Method

Qualitative research is an iterative and dynamic method (Corbin & Strauss, 2015, p.1), where the researcher is part of the study, collecting and interpreting data and putting oneself as a whole in the research process. Unlike quantitative research, it is not meant to be a rigorous research method (Corbin & Strauss, 2015, pp.4-5). Therefore, it makes sense not to use a method that is too rigid to analyse data, as it could prevent insights from being uncovered (Corbin & Strauss, 2015, p.1).

The grounded theory method is part of qualitative research and, despite not being flawless, has been used for 40 years with success to gain insights into various fields (Corbin & Strauss, 2015, p.4). Through data collection, theories are created. Therefore, data collection and theory creation are closely linked since the researcher is deriving insights and new data-collecting possibilities (i.e. theoretical sampling) based on the collected data, as shown in Figure 1 (Corbin & Strauss, 2015, p.7).

Figure 1

Grounded theory process



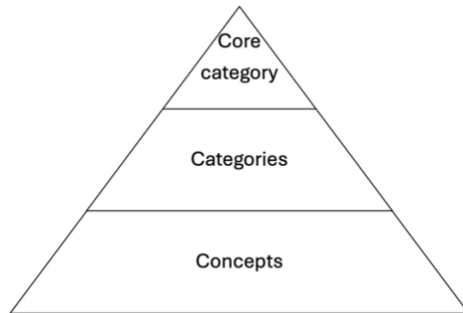
Note. Adapted from Corbin & Strauss (2015, p.8).

There are several ways to collect data, such as interviews and readily available material, such as recorded material or newspapers. Once the data is collected, it is analysed using various methods. The analysis involves grouping *concepts* sharing similarities under a common heading to form *categories*.

The researcher aims to determine a *core category* by grouping *categories*, as shown in Figure 2 (Corbin & Strauss, 2015, pp.7-8).

Figure 2

The three levels of abstraction in grounded theory



Note. Adapted from Corbin & Strauss (2015, p.77)).

The qualitative method, and more precisely the grounded method, was chosen to research the topic, as it enables exploring flexibly and holistically a field of research, connect with the study participants, and obtain in-depth insights on their point of view (Corbin & Strauss, 2015, p.5). This method allowed the guidelines and research focus to be adapted throughout the research as new themes emerged (Corbin & Strauss, 2015, p.39). This freedom made it possible to study areas that had not been envisaged at the start of the research.

2.1.1 Literature Review

As the grounded theory method is experimental, it is hard to know what literature will be necessary before beginning the research. Additionally, reading too much about a subject can constrain the findings. Therefore, a literature review was performed only to the extent required to identify issues and define points of interest (Corbin & Strauss, 2015, p.49). Literature was then used throughout the entire research process when assumptions emerged from the data analysis to build on the knowledge acquired by other researchers (Corbin & Strauss, 2015, p.49).

2.1.2 Theoretical Sampling

The sample was selected based on its relevance to the study and not the objective of having a representative sample, as it would not be possible with few interviewees (Corbin & Strauss, 2015, p.146). The target population was, therefore, initially determined as people involved in the decision-making for the adoption of commercial PV in Alpine regions. Then, based on the analysis of each interview, assumptions were formulated and tested with the next interviewee, who was, if possible, specifically chosen as a relevant candidate to test emergent assumptions (i.e. theoretical sampling) (Corbin & Strauss, 2015, p.69).

At first, the focus was on commercial PV adoption in Alpine regions. Therefore, eight interviews were conducted with decision-makers in Swiss Alpine companies who had adopted or planned to adopt solar PVs, as shown in Table 1. Some questions about the adoption of other RE were included. Then, through the analysis, a positive feedback loop for RE adoption was observed, where the successful completion of a RE project led companies to participate in other RE projects. Some interviewees notably expressed their participation in Alpine utility-scale PV projects. However, the adoption process linked to such projects was more complicated and uncertain than commercial PV projects. Therefore, the field was explored to gain insights into this discrepancy. To understand the project difficulties from various perspectives, seven stakeholders occupying different roles in various Swiss Alpine utility-scale PV projects were interviewed, as shown in Table 1. Accepted and rejected Alpine utility-scale PV projects were researched.

Table 1

List of Interviews

Participants	Interview Date	Type of Company or Commune	Role in the Company or Commune	PV Project Realised	Type of PV Project Realised or Planned	Type of Interview
Participant 1	2.04.24	Swiss Alpine Company	Member of the management	Yes	Solar PV on a restaurant rooftop	Qualitative, structured, written (email)
Participant 2	2.04.24	Swiss Alpine Company	Member of the management	No	Possible solar PV installation on their infrastructures	Qualitative, structured, written (email)
Participant 3	3.04.2024	Swiss Alpine Company	Member of the management	Yes	Planned solar PV construction on a new restaurant and solar façade on a hotel	Qualitative, structured, written (email)
Participant 4	5.04.24	Swiss Alpine Company	Member of the management	Yes	Solar PV facades on restaurant	Qualitative, semi-structured, online (zoom)
Participant 5	8.04.24	Swiss Alpine Company and Swiss Commune	Member of the management and municipal administration	Yes	Solar PV projects realised, and planned Alpine utility-scale PV Project	Qualitative, semi-structured, online (zoom)
Participant 6	23.04.24	Swiss Alpine Company	Member of the management	No	Solar PV on a renovated restaurant in the pipeline	Qualitative, semi-structured, online (phone)
Participant 7	24.04.24	Zermatt Bergbahnen AG	Member of the management	Yes	Solar PV on the “Kleine Matterhorn” arriving station and other PV projects	Qualitative, semi-structured, online (zoom)
Participant 8	29.04.24	Swiss Alpine Company	Member of the management	Yes	Solar PV on a departing station, possible collaborations for a utility-scale PV project	Qualitative, semi-structured, face-to-face
Participant 9	30.04.24	Swiss Alpine Company	Member of the board of directors	No	Possible solar PV installation on renovated restaurant	Qualitative, semi-structured, face-to-face
Participant 10	3.06.24	Farmer Community	Member	No	Landowner of a rejected utility-scale PV project	Qualitative, semi-structured, online (phone call)

Participant 11	6.06.24	Swiss Commune	Member of the municipal administration	No	Landowner and municipal actor of a rejected solar utility-scale PV project	Qualitative, semi-structured, face-to-face
Participant 12	7.06.24	Swiss energy company	Responsible for a set of utility-scale PV projects	No	Responsible for a set of projects, including a locally accepted utility-scale PV project	Qualitative, semi-structured, online (teams)
Participant 13	3.07.24	Swiss Commune	Hérémence Voter	No	Hérémence voter for the utility-scale PV project of Präfleuri	Phone
Participant 14	4.07.24	Swiss energy company	Responsible for a set of utility-scale PV projects	No	Responsible for a set of projects, including a locally accepted utility-scale PV project and a rejected one	Qualitative, semi-structured, online (teams)
Participant 15	15.07.2024	Swiss Commune	Käserstatt Voter	No	Käserstatt voter in a rejected utility-scale PV project in	Qualitative, semi-structured, written (what's app)

Note. Based on the qualitative data collected.

Four interviews were conducted in written form, eight online, and three in person. The interview lasted between 30 to 60 minutes. The process continued until theoretical saturation was reached, i.e. when the categories defined as the social customer journey (Hamilton et al., 2021) stages were dense with information (Corbin & Strauss, 2015, p.135). Additionally, all the interviews were anonymised³, following Corbin & Strauss's (2015, pp.44-45) best practice. Indeed, making the respondents anonymous can be a good alternative to providing more honest answers (McCracken, 1988, p.27).

2.1.3 Choice of the Framework for the Research

As previously mentioned, the social customer journey from Hamilton et al. (2021) is used as a framework for the research. It consists of six parts depicting the various stages a buyer goes through, as shown in Figure 3. However, the framework was not used strictly in a way that would restrain the findings but rather adapted to the interview findings. Indeed, if a framework is used, the researcher must keep their mind open about the discrepancy that could exist and not try to make the data fit the theory (Corbin & Strauss, 2015, pp.52-53).

³ The only company name that was not anonymised is *Zermatt Bergbahnen AG*, which agreed not to be. Indeed, they acted as pioneers in high-alpine PV installation, and anonymisation would have lost a considerable amount of value from the interview. Therefore, only the person doing the interview was anonymised.

Additionally, a lower level of anonymisation was required for municipal voters, as they are not recognisable even with the mentioning of the locality they live in (Erasmus University Rotterdam, n.d.).

Figure 3

Stages of the social customer journey



Note. Adapted from Hamilton et al. (2021, p.71).

There are different models of customer journeys (Følstad & Kvale, 2018, p.196), but one common feature is that they depict the multiple touchpoints of one person or entity throughout the decision-making process for a purchase. The touchpoints can be separated into three distinctive areas: pre-decision, decision, and post-decision (Lemon & Verhoef, 2016, pp.74-76). The social customer journey from Hamilton et al. (2021) was chosen based on the following considerations.

It is essential to understand the type of business analysed to select the best customer journey model. Selling solar PVs to companies can be classified as B2B interactions, characterised by companies procuring services that will not directly serve the end consumer (Kuß & Kleinaltenkamp, 2016, p.30). Indeed, solar PVs are only an intermediate good for providing the end customer with electricity. B2B has specific characteristics, such as a small number of buyers, specific requirements tailored to their needs, high customer interaction, and a formal purchasing process. Several stakeholders with various considerations, different decision-making power, backgrounds (e.g. business, technical) and attributes such as age and gender make the decision. Therefore, the purchasing process is usually lengthy, involving negotiations inside and outside the company (Kuß & Kleinaltenkamp, 2016, pp.33-34; Tomczak et al., 2018, p.5). Due to the involvement of many stakeholders in the decision-making process, it makes sense to analyse the customer journey in terms of the different actors involved and not only from the perspective of one buyer, which has traditionally been the approach of customer journey models (Hamilton et al., 2021, p.68). It is what the social customer journey offers, which is particularly interesting for B2B sales (Hamilton et al., 2021, p.87).

The construction of utility-scale projects is more complex than commercial PV sales. However, it was still helpful to continue using the social customer journey framework from Hamilton et al. (2021) for the following reasons. First, the different stages of the customer journey are still helpful for analysing the decision-making process and considerations of various actors (Hamilton et al., 2021, p.69; Lemon & Verhoef, 2016, p.85). If discrepancies were found, the framework was modified to reflect the findings. Second, the framework considers post-purchase elements that are interesting to research but not included in other frameworks used for analysing the decision-making process for such projects (e.g. the decision-making process from Díaz & van Vliet (2018, p.3) adapted from Lasswell (1971) stops at the

implementation stage). Third, keeping the same framework enables comparisons with the commercial PV adoption data, which would have been more complicated if another framework had been used.

Finally, this framework helps analyse the research question, as understanding why the implementation does not go forward can be analysed at the decision-making level (Díaz & van Vliet, 2018, p.2). It also enables understanding critical decision-making factors (Hamilton et al., 2021, p.69; Lemon & Verhoef, 2016, p.85). Moreover, various actors significantly impact the acceptance process of utility-scale PV projects, but their influence is not widely studied (Cousse, 2021, p.9). Therefore, this framework will enable the analysis of the *development dynamics* for deploying commercial and utility-scale PV projects from the perspective of various stakeholders while accounting for *social dynamics* between them. From this process, factors hindering adoption in the decision-making process will be determined.

2.1.4 Interview Guidelines

Semi-structured interviews with a defined but flexible guideline were used. They have the disadvantage of giving less freedom to the interviewee since there is a list of pre-determined questions. However, it still allows for exploring emergent topics that come with the discussion, as guidelines do not need to be strictly followed. Additionally, the interviewee was asked whether he or she wanted to add something to the discussion at the end, leaving the possibility of adding information (Corbin & Strauss, 2015, p.39). Moreover, it provided structure throughout the research, consistency between the interviews, and covering a list of pre-determined topics (Corbin & Strauss, 2015, p.39; McCracken, 1988, p.22).

The semi-structured interview guidelines were created based on the six stages of the social customer journey framework from Hamilton et al. (2021). As previously mentioned, if a framework is used, the researcher must not try to make the data fit the theory (Corbin & Strauss, 2015, p.53). Therefore, the interview guidelines were revisited after each interview to test emergent assumptions. Moreover, before the interview, background research was done on the interviewee's company, and if questions emerged, they were added to the guideline (Corbin & Strauss, 2015, p.39).

Potential interviewees were primarily contacted via email, where a condensed version of the interview guidelines was sent, covering the six stages of the framework and an additional question about people involved in the decision-making. A more extended version of the interview was created for the researcher, including the seven main questions sent to the study participant and additional sub-questions for each main question. The sub-questions could be addressed during the interview, depending on the discussion. Throughout the research, additional sub-questions were added to the interview to test assumptions (Corbin & Strauss, 2015, p.39).

The best practice for conducting interviews and creating guidelines was taken from the in-depth interview protocol from McCracken (1988), as Corbin & Strauss (2015) did not provide detailed information on this topic. Despite the age of this book, it is still used in practice (e.g. Hill et al., 2021;

Homburg et al., 2020; Thompson et al., 2006). The interview protocol was as follows. Interviewees were first informed about the overall objective of the research, and the interview then started with questions about the interviewee's life to open the discussion. Then, questions were asked along the six stages of the customer journey. The questionnaire followed two principles: (1) it must allow the interviewee to tell his/her version of the story, and (2) the interviewer tries to be the least obtrusive. Questions were therefore formulated in an undirected, open-ended way, not to prompt a particular answer (e.g. How did you find the necessary information to adopt Alpine PV?). After the answer was given, directed questions were sometimes asked to test assumptions built from the literature and former interviews (e.g. Did you look for information amongst your peers?). Asking such questions is helpful to prompt discussion that the interviewee does not immediately think about. Finally, if unintended topics surfaced, further elaboration was asked to explore those areas without inferring something the interviewee did not say. If the slip-off went too far, it was returned to the original point (McCracken, 1988, pp.34-37). Active listening was performed (McCracken, 1988, p.21).

2.1.5 Data Analysis

At the end of each interview, it was transcribed (See Interview Transcripts in Appendix B) and translated from French or German to English. Memos about the critical points of the interviews and their interpretation were written. Memos help to remember what was striking during the interview and to analyse the data to find concepts (Corbin & Strauss, 2015, p.58). They allow the researcher to think in terms of concepts, move away from raw data, and think creatively (Corbin & Strauss, 2015, p.122). Then, microanalysis, a type of open coding, was performed along the six customer journey stages of Hamilton et al. (2021). The microanalysis was not strict, and emerging themes were also included in the coding under categories not included in the initial framework (e.g. positive feedback loop, peer effects). In this type of coding, the goal is to find concepts, create ideas about the data, and spot pieces of data that seem important but unclear (Corbin & Strauss, 2015, pp.70-72). Diagrams along the customer journey stages and additional emergent themes were then created with the coding. They helped move away from raw data, think more abstractly, and boil down the data in an easily explained diagram. They make shortcomings easier to spot (Corbin & Strauss, 2015, p.123). This process enabled assumptions to be made and tested in the following interviews. If assumptions were conclusive, they were pursued; otherwise, they were abandoned.

After the first data analysis, two analytic strategies were used predominantly to uncover the messages of the data. Questioning is the most fundamental part of the grounded theory method and was used in every part of the research. Initially, it helps start the thought process and put oneself in the interviewee's position to understand his/her perspective (Corbin & Strauss, 2015, pp.90-93). The second method was to make comparisons, looking for similarities and differences. Two types of comparisons were used. The first type is the *constant comparisons*, involving comparing data within the same data collection

and with data in another data set. Similar data from different interviews were grouped under a common diagram at the end of each interview to consolidate the findings and form categories (Corbin & Strauss, 2015, pp.93-94). The second type is *theoretical comparisons*, which are used to move away from shallow analysis. Theoretical comparisons were made by taking a concept (e.g. peer effect) and seeing how it was defined in another situation/research. It helps to think more abstractly about words and concepts and gives clues about what to look for (Corbin & Strauss, 2015, pp.93-95).

The last step was integration, enabling the definition of a core category by linking the categories composed of concepts (Corbin & Strauss, 2015, pp.188-189). A diagram summarising the findings and how categories are linked was made (See Appendix A) (Corbin & Strauss, 2015, p.195).

3 Literature Review

As mentioned, the literature review was made at the start of the research to identify issues and define points of interest, but also throughout the research process to gain knowledge about assumptions made from the data, i.e. for theoretical comparisons (Corbin & Strauss, 2015, pp.94-95). Therefore, the following literature review encompasses the factors affecting commercial PV adoption. Additionally, as the literature about household PV adoption is more extensive than that of commercial PV, it was analysed to complement it. Then, social acceptance was found to be a major hindering factor for developing Alpine utility-scale PV projects. This field of research was also reviewed. The literature review follows the explained order.

3.1 Small-Scale Solar PV Projects

This section examines factors affecting the adoption of small-scale PV projects, encompassing household and commercial PV projects.

3.1.1 Factors Affecting the Development of Commercial Solar PV Projects

A study on commercial PV found that the most important motivation for adoption is financial. Environmental elements were not so relevant. Therefore, decreasing PV prices and subsidies favour adoption (Crago & Koegler, 2018, p.481). Additionally, peer effects between companies can enhance adoption (Baranzini et al., 2017, p.24). People are more likely to evaluate a new technology based on the experience of a peer rather than on scientific evidence (Rogers, 1983, Chapter 1, p.36). Indeed, peers are considered unbiased, as they are not interested in making others adopt the technology. They are just intrinsically convinced about their purchase. Companies in the solar energy industry started leveraging this marketing tool, which proved very effective. Indeed, when deciding to purchase solar panels, people want to have extensive support and expertise. Speaking with a satisfied person who adopted solar panels can foster further adoption, as some concerns are eliminated. Additionally, even without speaking, people are more likely to adopt solar panels if their neighbours also have them (Cardwell, 2012). This effect has typically been studied in household solar PV adoption but also exists in companies. Indeed,

although the peer effect for companies' adoption of solar PV is weaker than for households, it is still present (Baranzini et al., 2017, p.23).

Peer effect can spread through word-of-mouth and replication (Baranzini et al., 2017, p.24). Word-of-mouth helps mitigate company risks, as they often undertake larger-scale installations compared to households, which have the potential to be more profitable but also bear more inherent risks (Baranzini et al., 2017, p.21). Therefore, if a firm wants to learn from an installation, they are more likely to look at and discuss with project owners who possess the same size and ownership as theirs. Moreover, firms' adoption of commercial PV influences adoptions by other firms in a given area through their visual appearance. Therefore, PV facades are likely to generate more peer effects, being more visible (Baranzini et al., 2017, pp.23-24). The reason for the peer effect may be that firms adopting PV send out signals about the profitability of such installations to peer companies, as they are more likely to do it for profitability reasons and not for environmental ones (Baranzini et al., 2017, p.25). Conversely, environmental reasons motivate private individuals to adopt solar PV (Bach et al., 2020, p.5). Households in the neighbourhood also influence solar PV adoption in firms, which could be explained by the firm wanting to please its neighbouring customers who value such actions. Solar PV could, therefore, work as an advertising tool to show their commitment to sustainability (Baranzini et al., 2017, p.25).

3.1.2 Factors Affecting the Development of Households Solar PV Projects

The most influential predictor of solar PV deployment is the exploitable potential. If an area is evaluated as not having the potential to generate enough electricity with PV, it will not be exploited (Thormeyer et al., 2020, p.372). Solar PV costs are also a determining factor for adoption, as they are expensive despite declining costs (Shakeel et al., 2023, p.6). The long payback period hinders adoption if the decision is mainly financial (Bach et al., 2020, p.6; Shakeel et al., 2023, p.6). Therefore, financial support such as subsidies may be necessary to foster adoption until the technology is mature enough to be affordable without them (Shakeel et al., 2023, p.10).

Then, adoption is notably motivated by the willingness to be self-sufficient, cost-saving and environmental concerns (Shakeel et al., 2023, p.10). Early adopters are more likely to do it for environmental reasons (Bach et al., 2020, p.5). Self-sufficiency motivation is dependent on the possibility of self-consumption and is related to the desire to free oneself from reliance on large electricity companies. Indeed, people misunderstand the pricing of electricity and the reason such companies charge high prices (Bach et al., 2020, p.6). Cost-saving also depends on the self-consumption possibility. Therefore, potential adopters are more likely to invest in solar PV without government intervention if they can self-consume. However, if self-consumption is not possible, support from the government is required to reach a profitable business model, as it guarantees a stable income necessary to predict the return on investment (Karneyeva & Wüstenhagen, 2017, p.447). Financial savings also

depend on the price of other energy sources. There are more cost savings in periods of high energy prices, as electricity self-consumed is not subject to fluctuating energy prices. Therefore, such periods enhance solar PV adoption (Karakaya & Sriwannawit, 2015, p.64; Shakeel et al., 2023, p.6). Conversely, since fossil energy is a more mature and cheaper energy source, it can deter the adoption of solar PV. However, policies can overcome this barrier (Karakaya & Sriwannawit, 2015, p.63).

Personal factors also impact the decision to adopt solar PV, such as prior knowledge about solar PV, belonging to a green political party (Shakeel et al., 2023, p.8), or having environmental concerns (Shakeel et al., 2023, p.10). Prior adoption of other RE technologies (e.g. heat pumps or solar thermal) can also enhance solar PV adoption, potentially reflecting an interest in such technologies and the financial means to obtain them (Bach et al., 2020, p.5). Conversely, the lack of knowledge due to no previous experience with RE or solar PVs hinders adoption. People lacking knowledge may perceive the technology as complex and risky, leading to non-adoption (Karakaya & Sriwannawit, 2015, p.62). Therefore, sufficient information about the technology should be given (Shakeel et al., 2023, p.8), and educating customers about solar PVs benefits and their positive environmental effects should be done, as it helps overcome technology ignorance (Karakaya & Sriwannawit, 2015, p.63). Moreover, enhancing trust between the adopter and the intermediaries can help (Bach et al., 2020, p.9). The reputation of the intermediaries fosters trust (Shakeel et al., 2023, p.9).

Societal factors can also negatively influence the adoption of new technologies, such as a population being sceptical about them. Entrepreneurial culture is, therefore, an important factor when considering the adoption of new technologies (Karakaya & Sriwannawit, 2015, p.62). Conversely, a positive attitude towards the technology from the local population is an adoption driver. Finally, solar PV aesthetic is a consideration for solar PV adoption in private households, and integrated solar PV is therefore evaluated more positively (Bach et al., 2020, pp.8-9).

There are also policy barriers. Policies can be inadequate or ineffective (e.g. high-level bureaucracy, poor coordination of policies, and subsidies taking too long to be given), or subsidies can be distributed inconsistently (e.g. Insufficiency of feed-in tariffs) (Karakaya & Sriwannawit, 2015, p.64). Additionally, building permits to construct solar PV may be more complicated in certain communes (Bach et al., 2020, p.9).

Other factors affecting solar PV adoption in households are peer effects. Once again, peer effects can diffuse through word-of-mouth or replication (Baranzini et al., 2017, p.24). Indeed, people commonly look for information about solar PV through professionals and peers (e.g., relatives, friends or neighbours) for previously explained reasons (Bach et al., 2020, p.5; Shakeel et al., 2023, p.7). Additionally, the visibility of solar PV enhances the peer effect as it generates trust (Shakeel et al., 2023, p.7). However, peer effects can also deter adoption if the PV review is negative. For example, lack of knowledge may lead to inappropriate use and poor maintenance. Such adverse experiences can then

affect other people's decision-making, who may decide not to adopt solar PVs (Karakaya & Sriwannawit, 2015, p.62). Moreover, the peer effect fades as solar PV technology matures (Baranzini et al., 2017, p.19), as the distance between projects increases and as time passes. Indeed, studies in Switzerland show that the peer effect for solar PV adoption gets weaker as distance increases from the original project. More than 1km away, the effects are still present, and 3-9 km away, the effect starts dissipating. The peer effect is also more significant when new installations are built and disappear quickly afterwards (Baranzini et al., 2017, pp.16-17). Other studies about household solar PV adoption find similar results, such as decreasing effect with time and proximity (Graziano & Gillingham, 2014, p.815). However, spillovers can go beyond the municipality and the canton (Thormeyer et al., 2020, p.372) or spread to neighbouring counties in Germany, creating regional hotspots (Dharshing, 2017, p.123).

3.2 Renewable Energy Projects Facing Problems of Social Acceptance

As mentioned, Alpine utility-scale PV projects face social acceptance issues, more particularly local rejections (Rausis, 2023; “Alpine Photovoltaik-Anlage”, 2024). Indeed, social acceptance is crucial for constructing RE infrastructures, particularly those affecting the landscape, but is an obstacle (Batel et al., 2013, p.1; Wüstenhagen et al., 2007, p.2683), often provoking local resistance and leading to abandoning projects in some instances (Walker & Baxter, 2017, p.168). Such resistance is especially relevant in Switzerland, where the political decision-making process involves various actors, notably the local population, for the acceptance of RE projects impacting the landscape (Díaz & van Vliet, 2018, p.14). Therefore, the literature on social acceptance of Swiss large-scale PV projects is reviewed (Cousse, 2021; Späth, 2018) and complemented by the one on social acceptance of Swiss Alpine PV projects (Michel et al., 2015; Díaz & van Vliet, 2018; Vuichard et al., 2020) and by the wider literature about RE projects (e.g. wind, hydro), also facing social acceptance difficulties (e.g. (Tabi & Wüstenhagen, 2017)).

3.2.1. Factors Affecting the Social Acceptance of Renewable Energy Projects

First, a positive attitude towards solar PV technology is not a good predictor for social acceptance of a project with that technology (Cousse, 2021, p.7; Michel et al., 2015, pp.168-169). Indeed, compared to other technologies, such as heat pumps, solar energy is associated with positive emotions in Switzerland, such as curiosity, enthusiasm, and pride. In contrast, negative emotions such as fear are insignificant (Côté et al., 2022, p.10). However, people are more concerned about large-scale PV installations as opposed to rooftop ones (Cousse, 2021, p.7), and open-field PV is Swiss citizens' least favourite type of RE (Trutnevyte et al., 2024, p.18). It is, therefore, not enough to consider people's attitudes towards a type of energy to determine adoption (Cousse, 2021, p.7; Michel et al., 2015, pp.168-169). The attitude towards the size of the installation must be considered: as the size grows, people's

positive attitude decreases. Indeed, rooftop solar PV is more positively evaluated than large-scale ones (Cousse, 2021, p.8).

Second, various opinions about land use may cause adoption issues (Späth, 2018, pp.433-434). Swiss voters generally would rather build rooftop PV before constructing large-scale ones (Späth, 2018, p.435). Additionally, some believe the land should be kept untouched for aesthetics, to protect species, or to be used for cultivation rather than generating energy. Others believe that priority should be given to large-scale PV projects as they generate cheaper energy than other energy sources or that efforts should be made to increase energy efficiency rather than generate more energy (Späth, 2018, pp.433-434). Some people disagree with the nuclear phase-out and do not want to increase nationally produced RE, as they would not be willing to pay more for clean energy (Díaz & van Vliet, 2018, p.13). There are, therefore, diverging opinions about how energy should be generated, which is enhanced by the lack of clarity of the Energy Strategy 2050, giving no clear answer about how energy should be generated. Indeed, the various stakeholders perceive it as vague, leading to scepticism about PV installation (Díaz & van Vliet, 2018, p.13).

Another defining factor impacting social acceptance is the impact of RE installations on the visual appearance of the landscape (Devine-Wright, 2005, p.135; Kienast et al., 2017, p.398; Olson-Hazboun et al., 2016, p.176; Tsoutsos et al., 2009, p.552). The Swiss population is attached to it (Vuichard et al., 2020, p.7). Moreover, the visual impact on the landscape can affect local tourism and is, therefore, a more significant consideration in regions living off tourism (Michel et al., 2015, p.162). Indeed, the tourism industry is a significant economic driver in certain Swiss Alpine regions (FDFA, 2023d), and landscape is a critical element for tourist promotion (Mestanza-Ramón et al., 2022, p.880) and attraction (Kokosalakis et al., 2006, p.396). Therefore, local acceptance (embedded in social acceptance) increases if solar PVs do not significantly impact the landscape and blend in nature (Vuichard et al., 2020, p.8). To ensure it, RE projects must be constructed close to existing facilities (e.g. power lines) (Trutnevyte et al., 2024, p.18), in areas already altered by human beings (Kienast et al., 2017, p.398), which have few other uses, with no inherently significant economic value. Constructing on such sites limits conflicts and generates employment (Milbrandt et al., 2014, p.473).

Then, ecological repercussions are relevant for local acceptance of RE projects. As people value both ecological preservation and consuming RE, there is a preference for RE that does not impact nature (Tabi & Wüstenhagen, 2017, p.771). If a project does not have an environmental impact, it will drive the project (Díaz & van Vliet, 2018, p.13).

Understanding the local context is also primordial for successful RE project implementation (Díaz & van Vliet, 2018, p.13). As the Swiss population trusts local governments, which are perceived as being very little corrupt, the local leader's support of a RE project increases its probability of acceptance (Azarova et al., 2019, p.1181). Additionally, the identity of the project owner is important to

communities, who prefer a RE project to be owned by a local or cantonal entity rather than a privately owned local company or, worse, a foreign investor (Tabi & Wüstenhagen, 2017, p.771).

Then, transparent communication between the project developer and the population is important, as it establishes trust. If stakeholders perceive a lack of communication of certain information, they will not trust the project developer, enhancing the chances for a negative vote. Therefore, information about their worries and questions must be answered (Díaz & van Vliet, 2018, p.13). Moreover, communication about risks should be open; otherwise, there could be fear of a hidden agenda (Slovic, 1987, pp.284-285). Scientific information about a project should be conveyed in a legitimate and trustworthy manner to ensure credibility (McNie, 2007, p.31). However, inhabitants may not question the technical feasibility of an Alpine PV project but rather the difficulties that may have been overlooked (Díaz & van Vliet, 2018, p.10).

Procedural justice, associated with participation in the decision-making process, can positively affect RE project realisation (Tabi & Wüstenhagen, 2017, p.770; Vuichard et al., 2020, p.8; Walker & Baxter, 2017, p.168). Additionally, distributional justice, referred to as the benefits offered to the local population and active stakeholders, is also a relevant factor in increasing the level of acceptance (Tabi & Wüstenhagen, 2017, p.770; Vuichard et al., 2020, p.7). Indeed, advertising the benefits to the local population is a crucial driver for acceptance (Visschers & Siegrist, 2014, p.117). Therefore, benefits such as extra income to the local population can be attractive, especially in mountainous regions with fluctuating economies due to natural hazards (Díaz & van Vliet, 2018, p.3). Such benefits and participatory options should be presented to the relevant stakeholders at the beginning of a project, as it could prevent opposition from building early (Vuichard et al., 2019, p.9). It is particularly relevant as opposition builds early in the project if communication is not well-managed (Aitken, 2010, p.6074). Advertising the benefits could also combat the NIMBY phenomenon (Díaz & van Vliet, 2018, p.13), which has been used for many years as a way to explain the rejection of RE projects due to the proximity of projects to locals and the resulting inconvenience. However, one should be careful about using the NIMBY effect to explain a project rejection, as it simplifies reality, hiding the other variables explaining a project refusal. Accusing communities of NIMBY can be counter-productive, making the local population look selfish and breaking the trust, which is paramount to achieving social acceptance (Wolsink, 2012, pp.86-87). Additionally, advertising the benefits could foster the inverse-NIMBY effect (Díaz & van Vliet, 2018, p.13). When the inverse-NIMBY effect is present, locals positively evaluate the technology of projects started in their region (Warren et al., 2005, p.853). Such an effect was observed in Swiss Alpine PV projects (Michel et al., 2015, p.168; Díaz & van Vliet, 2018, p.13; Vuichard et al., 2020, p.8).

Finally, other aspects can influence social acceptance, such as social influence, symbolic representation of RE, the political context or previous knowledge about the topic (Devine-Wright, 2005, p.135).

Serving as a blueprint for similar PV projects can drive local support, as there is a willingness to be innovative (Díaz & van Vliet, 2018, p.13). Personal factors such as the political orientation of participants are also determinants. The centrists and left-wing are much more inclined to support a PV project (Vuichard et al., 2020, p.6). Additionally, political affiliation and participation in an environmental group influence the opinion of people (Tabi & Wüstenhagen, 2017, p.771). The framework used for the research will be explained in the following chapter.

4 The Social Customer Journey Framework

The social customer journey from Hamilton et al. (2021, p.69) considers the various decision-makers and the “social others” (e.g. friends, family) influencing their decision-making process. Influence can also come from people unknown to the decision-maker, such as internet reviewers. Social others may interact directly or indirectly with the decision-maker throughout some or all of the customer journey stages (Hamilton et al., 2021, p.69). There are special cases in which the customer journey is joint, where several actors have an influence and role in the decision-making. These journeys are complex and more challenging to manage since relation interactions must be considered. Such a case can also be analysed through the social customer journey framework (Hamilton et al., 2021, pp.72-73). The decision-making process for deploying commercial PV and utility-scale PVs is an example of such a joint journey.

Then, concerning the structure, the social customer journey consists of six stages (see Figure 3). However, the journey is not necessarily linear since decision-makers can move back and forth or drop out of it to resume later (Hamilton et al., 2021, p.71). The stages will be presented.

a. Pre-Decision Process

The first step of the journey is motivation. Buyers are often influenced by the decisions of others, which can motivate the start of a journey. A purchase can be a means for individuals and companies to show their identity and display values such as openness, innovation, ethics, or environmental friendliness. In joint journeys, the motivations of individuals may differ, entailing the need to negotiate and agree (Hamilton et al., 2021, pp.76-77). The second step is the information search. Buyers usually look for information until the cost of continuing exceeds the benefits. Nowadays, getting information on products has become more accessible through the internet, but traditional sources of information can also be used. Word-of-mouth is also influential and considered more trustworthy than the information provided by the company. More information is found in joint journeys since several actors are looking for it. However, the actors may interpret the information differently and seek arguments to persuade the other party (Hamilton et al., 2021, pp.77-79). The third part is evaluation, where persuasion is a defining factor. It is the part the most affected by other people’s opinions. The likeability and credibility of the social others determine their degree of influence. Additionally, when social norms are at stake, people may be influenced to act in a way that complies with the norms (Hamilton et al., 2021, p.79).

b. Decision

The previous stages and encounters influence the decision. For joint decisions, the more the journey has been made together, the more intuitive the decision will be. However, disagreements may arise at this stage since the decision entails commitment, and a return to previous stages is possible. Risk aversion likely arises at this stage in joint decisions, and the customer wants to stick with the safe option (Hamilton et al., 2021, p.81).

c. Post-Decision

The fifth stage is satisfaction with a purchase, influenced by the social, others and those who decide with them. Regrets may arise depending on the other's perception of the purchase. In joint journeys, "consumption" is usually done together, and dissatisfaction with the process may lead to not appreciating the final use. Active actors could experience different feelings about the experience (Hamilton et al., 2021, p.82). Finally, post-decision sharing about a purchase is inherent to human beings and can nowadays be done to a bigger pool of known and unknown people. Showing identity that may have motivated the purchase process is a motive for sharing the purchase experience. However, the motivation to start the customer journey and the ones to share are often different. Usually, people share an experience if they have a strong positive or negative opinion of it (Hamilton et al., 2021, p.84).

d. Modification to the Framework

The social customer journey from Hamilton et al. (2021) was used as an initial research framework. It was helpful to start exploring the social dynamics of the adoption process. However, the framework did not fit the data perfectly, and as mentioned in the methodology, if a framework is used, the researcher must beware of making the theory fit the data and not the opposite (Corbin & Strauss, 2015, p.53). Therefore, several modifications were made to it based on the interview data. First, not only the effect of social others was considered, but also the one from influential stakeholders throughout the customer journey (e.g. solar PV specialists, cantonal authority). Indeed, those actors were influential in the decision-making process despite not fitting the definition of social others given by Hamilton et al. (2021, p.69). Moreover, two additional stages were observed in the data and added to the framework. Pre-purchase communication was predominantly observed in the Alpine utility-scale journey⁴, and a positive feedback loop for adopting RE was also observed. Finally, adjustments were made to the term "social others" referred to as "peers" in the research, as it is a more common term used in solar PV

⁴ The decision-making process from Díaz & van Vliet (2018, p.3) adapted from Lasswell (1971) used to analyse an Alpine PV project on avalanche barriers highlights the promotion of the project (i.e. pre-purchase communication in the customer journey) as a second stage right after the intelligence phase, (i.e. the information collection in the customer journey). Prepurchase communication stage is therefore common for large-scale RE projects requiring the population vote.

adoption and the social customer journey (Hamilton et al., 2021) is referred to as the customer journey or journey for simplicity. With those considerations, the findings are presented in the following section.

5 Findings

The findings will be structured in the following way. First, the influential stakeholders in commercial and utility-scale projects are presented, as they are necessary to understand the social dynamics. Then, the rest of the findings will be presented according to their level of abstraction (See Figure 2). The *core category* of the research is the development and social dynamics for the deployment of commercial and utility-scale PV in the Alpine area, encompassing all the findings. It is the theory explaining the adoption process of commercial and utility-scale PV, from which insights are derived about factors affecting adoption. This *core category* is broken down into *categories* (i.e. the customer journey stages, such as motivation and information search), which will be developed and compared. Within each *category*, *lower-level concepts* are presented, highlighting the factors affecting PV adoption (Corbin & Strauss, 2015, pp.76-77). Additionally, relevant quotes from the interviews illustrate the findings.

5.1 Stakeholders Active in Commercial and Utility-Scale PV Projects

The influential stakeholders of the commercial and utility-scale PV projects are presented.

a. Stakeholders of Commercial PV Projects

In the journey to adopt commercial PV, five main categories of stakeholders are influential at different stages, as shown in Table 2.

Table 2

Influential stakeholder groups for Alpine commercial PV adoption

	Stakeholder	Role
1	Company (incl. Board of directors, management, financial department, and electrical engineers)	Mandates the project, owns the infrastructures
2	Electricity Company	Often provides funds for the project
3	Specialists' cabinet (e.g. architecture, solar PV)	Provide propositions for the PV project, the main source of information
4	Local administration	Accepts/rejects the project (building permit)
5	Cantonal and federal administration	Sets laws, grants subsidies

Note. Based on the qualitative data collected.

Members of the company are the main decision-makers (i.e., a joint journey in Hamilton et al. (2021, pp.72-73)). The other stakeholders and peers influence the decision-making with various

considerations, which mostly do not hinder adoption (e.g. Participant 4, p.c., April 5, 2024; Participant 6, p.c., April 23, 2024; Participant 8, p.c., April 29, 2024).

b. Stakeholders in Alpine Utility-Scale PV Project

There are various actors with decision-making and influential power in Alpine utility-scale PV projects, as shown in Table 3⁵. Unlike commercial PV projects, a steering committee notably composed of the project developer and the local authority leads the project (Participant 14, p.c., July 3, 2024). However, the decision does not only depend on them. Other stakeholders must accept and/or participate in the project to go forward (i.e. municipal voters, landowners, infrastructure owners, cantonal administration). Moreover, the political parties and nature conservation NGOs have influencing power: nature conservation NGOs can oppose the project if it has a too significant impact on nature (Participant 14, p.c., July 3, 2024), and political parties can influence the votes. The cantonal and federal administrations serve as a formal source of information for the legislation and subsidies (Participant 11, p.c., June 6, 2024). The specialists' cabinets are used for the technical aspects (Participant 12, p.c., June 7, 2024). People external to the project also influence these stakeholders (Participant 11, p.c., June 6, 2024).

⁵ Stakeholder groups 1-5 are similar to those considered by Díaz & van Vliet in their research on Alpine PV project on avalanche barriers in Switzerland (2018, p.6). Groups 6-9 are considered in this research, as they were determined to have decision-making power or influence in the process.

Table 3*Influential stakeholder groups for Alpine utility-scale PV development*

	Stakeholder	Role
1	Electricity company (incl. technical experts, program managers, communication department ...)	Project developer, investor, technical knowledge
2	Local administration	Co project lead, receives the project and co-presents it
3	Cantonal and federal administration	Legislators, grant subsidies, cantonal administration must accept the project
4	Nature conservation NGOs	Defend nature conservation
5	Municipal voters	Vote for or against the project
6	Political parties	Influence the voters
7	Landowners	Owns the land where the utility-scale PV project will be installed (can be farming communities, bourgeoisie, municipality)
8	Existing infrastructure owners	Own the infrastructures in place, such as power lines (can be ski resorts, neighbouring commune, commune)
9	Specialists' cabinets (e.g. solar PV, avalanche barriers experts)	Provides technical knowledge that the electricity company does not have in-house

Note. Based on the qualitative data collected.

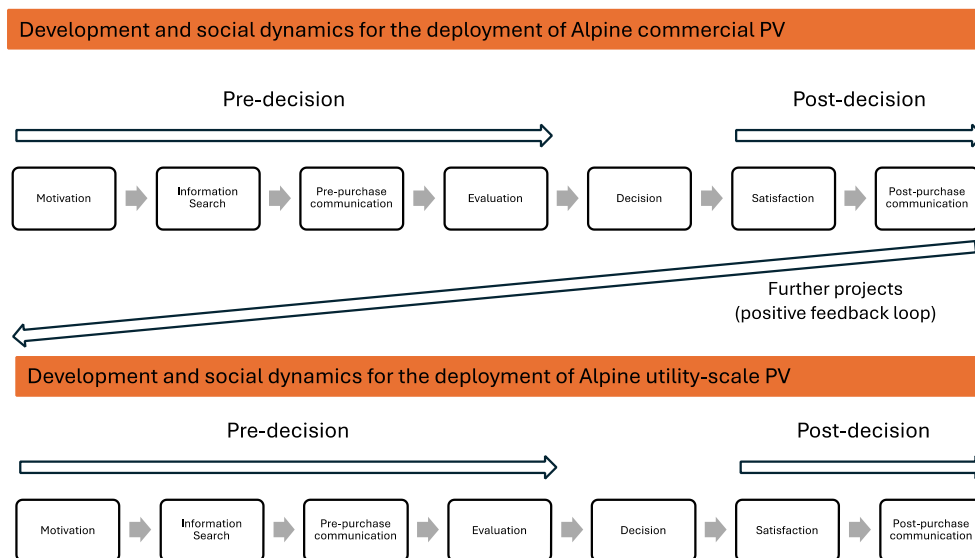
As opposed to commercial PV projects, some decision-makers, in particular the local population, were found to be hindering the project development (e.g. Participant 10, p.c., June 3, 2024; Participant 11, p.c., June 6, 2024). Indeed, as the project is subject to public communal votes, a political layer is added, complexifying the customer journey. People and political parties with various considerations can voice their opinions and potentially halt the project: *“And it [a political party] has taken the project on board and made it its hobbyhorse. So, it also made things a bit tense. [...] But it was [left local political party name] that led the way against the project. So there you have it, it's also turning into a bit of a political debate.”* (Participant 11, p.c., June 6, 2024). Moreover, the lack of consensus about energy policy in Switzerland complicates the political aspect and the adoption process, as there is no *“common political vision on energy”* about how Switzerland should phase out of nuclear, including where and how to generate electricity (Participant 11, p.c., June 6, 2024). This uncertainty can lead to scepticism about PV installations (Díaz & van Vliet, 2018, p.13).

5.2 The Customer Journey for Commercial and Utility-Scale PV Deployment in Alpine Regions

The main decision-makers in the commercial PV project are the various actors in the company (with the management usually having the final word) (e.g. Participant 1, p.c., April 2, 2024; Participant 4, p.c., April 5, 2024; Participant 6, p.c., April 23, 2024), and the customer journey is therefore analysed from their perspective. As mentioned, various actors have decision-making power in utility-scale PV projects. Describing the customer journey for each actor would not enable grasping the social dynamics between the various actors and would be repetitive. Therefore, the journey is developed from the project developer's perspective. Since it leads the project, it is active in every part of the journey (Participant 12, p.c., June 7, 2024). Additionally, it enables easier comparisons with the journey for commercial Alpine PV adoption, as both are companies. It will be supplemented by the consideration of various actors that were found influential at different parts of the journey, including peers. A graphical summary of the adoption process is depicted in Figure 4, and a more detailed version integrating all the findings can be found in Appendix A.

Figure 4

Development and social dynamics for the deployment of Alpine commercial and utility-scale PV



Note. Adapted from Hamilton et al. (2021, p.71).

5.2.1 Motivation

Motivation is the first stage of the journey (Hamilton et al., p.71). Despite nuances, the three driving motives for commercial and utility-scale projects in Alpine regions are similar: financial interest, energy security and ecology.

a. The Financial Interest to Adopt Commercial PV Requires No External Support or Reward Mechanism, as Opposed to Utility-Scale PV

In commercial PV projects, the main motivation is financial, which is only realised if there is the possibility to generate and self-consume enough electricity for the investment to pay off (Participant 8, p.c., April 29, 2024). It notably depends on solar PV sun exposition (Participant 9, p.c., April 30, 2024) but also the self-consumption possibility, as it enables the avoidance of fluctuating energy prices and, therefore, cuts costs in periods of high energy prices. If self-consumption is not possible or is only possible during a particular time of the year when the company is active, the investment may not be attractive (Participant 8, p.c., April 29, 2024). Indeed, the energy generated and not consumed in one building can often not be transferred to another building due to plot regulations, as mentioned by Participant 8: *“So, it's interesting [solar PVs], yes, if you can consume your own energy all year round. If it's only in winter and not in summer, that's no good. Because self-consumption isn't that simple, and the legal framework doesn't let us do what we want. It has to be the same plot of land, and so on (p.c., April 29, 2024).*

Project developers are financially interested in undertaking utility-scale PV projects, as they will generate and sell electricity on the market (Participant 11, p.c., June 6, 2024). However, unlike commercial projects, government subsidies are necessary to achieve financial viability and initiate Alpine utility-scale initiatives (Participant 12, p.c., June 7, 2024). The other stakeholders also have a financial motivation to join the project. The local administration usually negotiates to get a share of electricity, which they can either self-consume or sell on the market. The difference may be for the municipal household (Participant 11, p.c., June 6, 2024), and self-consumption is especially interesting if communes have high energy costs (Participant 5, p.c., April 8, 2024). Moreover, the commune usually receives royalties for each kWh produced, landowners develop their lands for more return, and infrastructure owners may negotiate a part of self-consumption (Participant 11, p.c., June 6, 2024). Local population benefits may be less direct, notably as they usually do not have the possibility of self-consumption. The lack of perception of benefits can lead to mistrust and opposition to the project, as Participant 11 states: *“That's the big problem we have in our local authorities, where people tell us ‘It's not possible, the plant is there, my house is 100 metres away, you produce at 7 cents, you sell it to me at 22. Who gets 15 in the pocket?’ “ (p.c., June 6, 2024). The perceived disparity can lead to perceptions of distributional injustice (Tabi & Wüstenhagen, 2017, p.770; Vuichard et al., 2020, p.7), which may be enhanced by the lack of understanding of electricity pricing (Bach et al., 2020, p.6; Participant 11, p.c., June 6, 2024). It is therefore critical to define and communicate the benefits for the locals to foster distributional justice, such as a reduction of the electricity price (Participant 13, p.c., July 3, 2024), or the creation of a RE transition fund for inhabitants (Participant 11, p.c., June 6, 2024). However, unlike when dams were constructed, certain communes and local inhabitants are wealthy, making it harder to sell local development opportunities (Participant 11, p.c., June 6, 2024).*

The financial viability of commercial PV projects is determined by the nature of the solar PVs, enabling self-consumption and, therefore, energy security and cost-cutting. However, utility-scale projects do not enable self-consumption for every actor involved, making the reward system more complex. Additionally, it is not profitable as such, requiring subsidies.

b. The Local Population's Inability to Self-Consume Electricity Reduces the Motivation for Energy Security

The second motivation for commercial PV adoption is to ensure energy security, especially in winter, when there is an energy deficit in Switzerland. This motivation also relied on the possibility of self-consumption (Participant 8, p.c., April 29, 2024). It was enhanced by the energy crisis caused by the Ukraine war, causing companies to fear rising energy prices (Participant 9, p.c., April 30, 2024). Despite the fear, there is the realisation that energy prices will decrease again, extending the payback period (Participant 8, p.c., April 29, 2024).

Energy security in winter is also a key motivation for the stakeholders of utility-scale projects. Indeed, project developers perceive the construction of utility-scale PV projects as an indirect confederation mandate (given through Art. 71a of the Energy Act) to increase winter production, enabling Switzerland to move away from nuclear and not to rely on foreign countries as much in winter (Participant 12, p.c., June 7, 2024; SFOE, 2023a). It is also a motivation for the local administration, who often can self-consume a part of the electricity generated. The local administration greatly values energy security and would be willing to participate in such a project even if it was not very profitable (Participant 5, p.c., April 8, 2024). As political actors, it makes sense that they have higher-level goals in mind, unlike companies that are mostly financially driven (e.g. Participant 8, p.c., April 29, 2024).

However, part of the electricity generated is to ensure national security. Therefore, the benefits of generating electricity for security are not strongly perceived by the actors who cannot self-consume, such as the local population. As a result, the construction of utility-scale PV relies on people's willingness to have an energy plant not only for themselves but for the rest of Switzerland, which can cause acceptance issues, as mentioned by Participant 11: *“Well, that was the vision [what people thought]. ‘We’re not going to denature our landscape for the Swiss plateau, which consumes too much. Then it’s us again that will have to pay by destroying our lands. In fact, just as we built dams in the past, we disfigured our valleys’”* (p.c., June 6, 2024). Although the Ukraine energy crisis strengthened the perceived need for energy security (Participant 5, p.c., April 8, 2024), the population seems not to feel the urgency of generating winter output. It might be linked to the fact that power cuts very rarely occur in Switzerland, leading to prioritising nature over energy (Participant 11, p.c., June 6, 2024).

As self-consumption is not possible for the local population in utility-scale PV projects, energy security is not guaranteed for them. Therefore, such a project partly relies on their willingness to contribute to national and not only personal security while sacrificing the landscape in their region.

c. The Ecological Aspect of Utility-Scale PV Systems is more Contested than that of Commercial PV Projects

The third motivation for commercial PV adoption is ecological, as solar PV are a clean way of generating electricity (Participant 1, p.c., April 2, 2024; Participant 6, p.c., April 23, 2024). The ecological aspect was not directly mentioned as a motivation by project developers. However, they see utility-scale systems as a way to achieve the nuclear phase-out (Participant 14, p.c., July 3, 2024). Nonetheless, the ecological aspect is important for other stakeholders in the project. Some perceive utility-scale PV systems as a way to decrease the need for silicon cells, as solar PVs generate more electricity at a high altitude compared to the plains (Participant 4, p.c., April 5, 2024). Others criticise the amount of material used to generate electricity. The re-usability of material is also an aspect to be considered (Participant 10, p.c., June 3, 2024).

Commercial PV is well-perceived for its ecological aspect and is not often contested. This is not the case for utility-scale PV systems, where there are more suspicions about their environmental friendliness. This aspect might be enhanced by the greater concern about utility-scale PV, as opposed to rooftop projects (Cousse, 2021, p.7).

5.2.2 Information Search

The second stage is the information search (Hamilton et al., p.71) and is the most extensive part of the journey. The effective production of innovative PV models is an uncertainty shared for the realisation of commercial and utility-scale PV projects. The mitigation strategies differ. Additionally, project developers of utility-scale PV systems must search for a suitable project site, accounting for factors affecting social acceptance.

a. The Need for Unbiased Information About Effective Solar Production Leads to Peer Effects and Collaboration

There is innovation in solar PV technology for Alpine areas. New models are created to generate as much energy as possible at a high altitude, regardless of the snowfalls (Participant 8, p.c., April 29, 2024). However, the effective electricity generation of innovative models is a complicated piece of information to get, and there is no certainty about which model to use for a specific situation. It is the most significant area of uncertainty for commercial PV projects, as reported by Participant 8: *“In Zermatt, at Trockener Steg, they've equipped their entire station with facade panels. So that's also productive, whatever the snow. But does it produce much? It's always complicated to get the real data. Some people communicate them easily, and others either they don't measure them, or they don't have them, or they don't give them, I don't know. There's no real exchange going on here that would be like saying, ‘Here we are, on a station of this type, we have to systematically put up dual-face panels, frontal panels’, or I don't know what”* (p.c., April 29, 2024). Therefore, companies have mitigation strategies

to overcome this area of uncertainty. First, they seek specialists' expertise (e.g. in solar PV, architecture, energy) (Participant 8, p.c., April 29, 2024). However, the latter may not have all the necessary information about the solar PV model or may be perceived as biased as being the ones selling the technology (Cardwell, 2012). Therefore, companies may refer to a person in contact with multiple Alpine commercial PV projects, which is seen as having unbiased information, as it has no stake in the project (Participant 8, p.c., April 29, 2024). Otherwise, best practice can be asked directly to peer companies with commercial PV projects, often technology pioneers, which may give the data about solar energy generation of a particular model (Participant 4, p.c., April 5, 2024; Participant 6, p.c., April 23, 2024; Participant 8, p.c., April 29, 2024). The knowledge gap is filled if the output is satisfactory, incentivising solar PV adoption. If it is unsatisfactory, there will be no adoption (Participant 8, p.c., April 29, 2024).

The utility-scale PV system's uncertainties are more considerable. There are no tried and tested solutions about which model to use, and the technology is being developed simultaneously as the project is carried out (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024). The mitigation strategies of the project developer are also to look for specialist expertise, whether in-house or in the market. Market experts can be specialised in a field linked to solar PV or other related fields that can be applied to utility-scale PV projects (e.g. avalanche experts possessing knowledge about wind and snow in a high altitude) (Participant 12, p.c., June 7, 2024). Additionally, to get the effective electricity generation of a solar PV model, competitors collaborate to set up expensive on-site tests and exchange advances (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024). Although initially competing for the project site, they have an advantage in collaborating for the technical aspect, as they are under time pressure to achieve the output (Participant 14, p.c., July 3, 2024). Finally, project developers also look for information on existing Alpine commercial PV projects (Participant 7, p.c., April 24, 2024).

There are uncertainties about the effective electricity generation of innovative solar PV models. Therefore, companies try to get unbiased information for commercial PV projects from their peers. For utility-scale projects, there is a collaboration with the competition to set up tests. Peer effects may come at a later stage when utility-scale PVs are constructed. However, best practices are asked from people leading commercial PV projects.

b. Searching and Complying with Administrative Procedures is More Strenuous and Uncertain for Utility-Scale Projects than for Commercial Projects Due to Opposition and Changing Laws

During the information search stage, the company conducting commercial projects must find the requirements for norm compliance if they choose to build according to a certain standard (e.g. Minergie, ISO) (Participant 8, p.c., April 29, 2024). Criteria must also be fulfilled to get subsidies, which are not very high and do not tilt the adoption decision balance (Participant 4, p.c., April 5, 2024; Participant 8,

p.c., April 29, 2024). Additionally, permits must be granted for commercial PV construction. However, they are usually not a problem, as they are part of a building and face little opposition (Participant 9, p.c., April 30, 2024). It makes collaboration interesting for electricity companies, who can invest in the project without going through the permit procedures to install commercial PV (Participant 8, p.c., April 29, 2024).

Utility-scale PV projects must also get a construction permit, which is more strenuous as people are more likely to oppose it, notably due to its impact on nature, slowing down the construction process (Participant 14, p.c., July 3, 2024). Opposition might be enhanced due to the concerns about large-scale solar projects (Cousse, 2021, p.7). Moreover, project developers must ensure to meet the requirements of Art. 71a of the Energy Act to get the subsidies. Criteria to get subsidies are more important to meet than for commercial projects, as financial viability depends on them (Participant 12, p.c., June 7, 2024; SFOE, 2023a). In addition to these two similar factors on which commercial and large-scale projects must inform themselves and comply, other factors complicate the search for information for utility-scale projects. First, there were periods of uncertainty concerning the regulations to follow, notably in the canton of Wallis, where several utility-scale project stakeholders were interviewed. Indeed, a decree was launched to speed up the process but then rejected (Participant 11, p.c., June 6, 2024; “Rejet du décret”, 2023). Conflicts between cantonal and federal law complicate adoption (Díaz & van Vliet, 2018, p.14). Second, the administrative burden for the communes is heavy (Participant 5, p.c., April 8, 2024).

The procedure to get construction permission for utility-scale PV projects in the Alpine area is more strenuous than commercial PV projects and faces more opposition. Getting subsidies is vital for utility-scale PV projects, putting them under more pressure.

c. Site Search as an Additional Consideration for Utility-Scale Projects: The Impossibility of Evaluating the Population Opinion Early as a Constraining Factor

Unlike commercial projects where the building defines the PV system site, the project developer must find an appropriate location for the utility-scale PV system. The project site must enable enough electricity generation (e.g. by being south-facing and not bearing too many risks) and be locally acceptable (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024). However, assessing social acceptance early in the process is complicated, as the initial project analysis is confidential until enough certainty about the project is reached. Communicating about the project and asking people's opinions too early would make it seem unthought-through, and too late could make it seem like it was made behind people's backs. It would also be a waste of time if rejected and could potentially create a “*political war*” in the commune (Participant 11, p.c., June 6, 2024). However, although local acceptance depends on the local context (Participant 12, p.c., June 7, 2024) and is hard to test early, some aspects

are systematically causing acceptance issues and should, therefore, be accounted for when choosing the project site. The following four points will focus on factors affecting local acceptance.

i. Choosing an Area Already Impacted by Nature, with Existing Infrastructures and Not Too Dependent on Tourism

The most contested aspect of Alpine utility-scale PV projects is their impact on nature and the visual appearance of the landscape (Participant 6, p.c., April 23, 2024; Participant 11, p.c., June 6, 2024; Participant 13, p.c., July 3, 2024). The impact on nature is relevant, as nature conservation actors can oppose the project, notably if species are impacted (Participant 12, p.c., June 7, 2024). People also value this aspect (Participant 13, p.c., July 3, 2024). Additionally, there are various opinions about land use (Participant 11, p.c., June 6, 2024) and an emotional attachment to the land, as stated by Participant 15 for a rejected Alpine utility-scale PV project in Käserstatt: *“The emotional attachment to the lands is by far more important than the energy source. This means that the acceptance of such a project may have been higher if it were promoted for a smaller and less central part of the commune.”* (p.c., July 15, 2024). To minimise the impact on nature, project developers should prioritise areas already impacted by humans (e.g. ski resorts, old quarries, artificial lakes) with existing infrastructures (e.g. road accessible, power transmission line). Those are believed to be the most influential aspects favouring adoption (Participant 14, p.c., July 3, 2024).

The impact on local economic activities is the second aspect deterring local acceptance of utility-scale PV systems. The project is particularly contested if it is to be built in an area depending on tourism. Indeed, there is a fear of declining tourism activities if the utility-scale PV is visible, therefore altering the natural beauty of the landscape (Participant 10, p.c., June 3, 2024; Participant 11, p.c., June 6, 2024; Participant 14, p.c., July 3, 2024; Participant 15, p.c., July 15, 2024), or if it is built on an area used for skiing (Participant 11, p.c., June 6, 2024). Project developers should, therefore, beware of choosing areas that are not too dependent on tourism and not too busy.

ii. Choosing Communes with Existing RE Projects: The Positive Feedback Loop of RE Projects in Communes

Other aspects did not systematically affect the local acceptance but were found to be influential. First, there is a positive feedback loop of RE adoption within communes, as those with existing utility-scale RE projects, which the population had to vote on (e.g. dams), were more open to utility-scale PV projects (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024). There can be several factors to explain it. First, the pioneering communes' population may perceive the benefits more clearly, as they already benefit from utility-scale RE installations (Participant 13, p.c., July 3, 2024). Moreover, they might be more open to the discussion as they had to discuss it in the past (Participant 12, p.c., June 7, 2024). Finally, there is a pride in carrying on the pioneering spirit, as stated by Participant 14: *“I think there's a certain pride in carrying out projects like this. We're in communes that are pioneers in*

hydropower and other fields. So, there is a certain pride in having carried out these projects at the time. I think there's a bit of a spirit of carrying on that pioneering spirit" (p.c., July 3, 2024). Additionally, if people are in favour of conducting a utility-scale PV project, they tend to evaluate the project in their commune more positively than the projects in other communes. It might be due to the pride in their project (Participant 13, p.c., July 3, 2024).

Even smaller RE projects, especially in the solar field, seem to favour the acceptance of utility-scale PV systems. Indeed, if little has been done in solar, the population perceives the project should be *"pushed a bit slower"*. It could be done by undertaking other smaller projects first, in which the population could, for example, participate. Undertaking such projects might help build a RE culture in the commune, sensitising the population to the subject (Participant 15, p.c., July 15, 2024).

iii. The Negative Effect of a Poor Perception of Neighbouring Utility-scale PV Projects

Another aspect that did not systematically affect local acceptance but was found influential was the number of Alpine utility-scale projects started in a region. If a utility-scale project is accepted in a neighbouring commune, the local population is not positively influenced to accept a similar project started in their commune, as opposed to what the peer effect would predict. Quite the opposite, it seems to create resistance from the local population (Participant 11, p.c., June 6, 2024; Participant 15, p.c., July 15, 2024).

A reason that could explain it is that the local population develops a negative perception of nearby Alpine utility-scale projects (e.g. due to construction noise, a project seen as a financial opportunity) (Participant 10, p.c., June 3, 2024; Participant 11, p.c., June 6, 2024; Participant 15, p.c., July 15, 2024), which leads to a negative evaluation of a similar project launched in the municipality. For example, Participant 15, who voted against an Alpine utility-scale PV project, stated that *"a similar situation [as the one in the neighbouring commune, incl. years of building and noise] could have developed at Hasliberg"* (p.c., July 15, 2024). Additionally, a project coming after many others loses the pioneering status and can be perceived as a financial opportunity, as stated by Participant 11: *"But above all, we benefited, in contrast to the general irritation about other projects, from people who said... 'Ah, but they're doing it at Anniviers, they're doing it there, they're doing it there, they're doing it there. Ah, and now again in [his/her commune name]'. So it's really getting out of hand, and everyone's jumping on it. 'It's all about money' and so on"* (p.c., June 6, 2024). Energy security can be seen as a valid reason for starting utility-scale projects, even if it affects the landscape, but financial reasons are not well perceived (Participant 13, p.c., July 3, 2024). Starting too many projects at once in a similar area seems to foster a sense of financial opportunity. Additionally, the projects are perceived to be started for financial reasons, especially if a non-local company undertakes them. Conversely, local companies are more trusted to keep local considerations at stake (Participant 13, p.c., July 3, 2024).

Locals may develop a negative attitude towards Alpine utility-scale projects if they negatively evaluate similar projects in other communes. This negative effect seems stronger for projects started in the same region, probably as people use closer projects as comparison points, as suggested by the peer effect of solar PV (Baranzini et al., 2017, p.24). Therefore, starting too many projects at once in the same region may increase the impression of projects undertaken for financial interest. Except for those factors that the project developer can account for, other factors depend on the local context and cannot be influenced by the site selection.

iv. Local Context and Other Factors That the Project Developers Cannot Influence Through Site Choice

Some people were opposed to Alpine utility-scale projects from the outset. There might be several reasons for this. There are ideological arguments about how energy should be generated, such as covering all built areas with solar panels before starting solar projects in nature (Participant 11, p.c., June 6, 2024; Participant 12, p.c., June 7, 2024; Participant 15, p.c., July 15, 2024). Some also believe that degrowth should be prioritised (Participant 11, p.c., June 6, 2024). The NIMBY effect was reported, as some people did not want a project to start close to them, as they did not want to be inconvenienced by the project (Participant 10, p.c., June 3, 2024; Participant 11, p.c., June 6, 2024). Additionally, people can negatively perceive Alpine utility-scale projects due to poor communication, which creates a bad image of such projects (Participant 11, p.c., June 6, 2024). Although technical aspects are one of the most significant uncertainties for project developers, the population was not worried about the project's technical implementation (Participant 10, p.c., June 3, 2024; Participant 13, p.c., July 3, 2024). Only some questions were raised about the recyclability of the material used and electro-smog risks (Participant 13, p.c., July 3, 2024).

Finally, a factor that project developers cannot influence is how other project stakeholders (commune, landowners, infrastructure owners, municipal voters) search for information. Although their primary source of information is the project developer (Participant 10, p.c., June 3, 2024; Participant 11, p.c., June 6, 2024), other sources that may not be influenced are used. Indeed, people do not want to rely only on the project developer, as they do not want to “*have hand and foot tied*” to them (Participant 11, p.c., June 6, 2024). Therefore, they seek information and expertise through personal, local contacts, federal guidelines, and other projects further advanced, as Participant 10 states: “*If we were unsure, we did some research on the Internet. There were already projects that were further along than ours. And on the other hand, we used the federal government's guidelines, which are also very transparent. But we also have contact with a local, regional, small electricity producer, [electricity producer company name]. ... We were able to ask them a few questions about what the whole thing would look like.*” (p.c., June 3, 2024). The information search from third parties to the project is influenced by external parties to the project, as people want neutral and external opinions, preferably from local actors. Indeed, the

project developer has a significant stake in the project, unlike external people. An interesting aspect to note is the search for *local* opinions. People tend to trust local companies more than non-local ones (Participant 13, p.c., July 3, 2024).

5.2.3 Pre-Purchase External Communication

As previously explained, this stage is not part of the social customer journey framework of Hamilton et al. (2021, p.71). It corresponds to the external communication about the Alpine utility-scale projects.

a. The Importance of Pro-Active, Transparent Communication to Foster Alpine Utility-Scale Project Acceptance

There is not much pre-purchase communication about commercial PV projects, but at most, it is a press release (Participant 6, p.c., April 23, 2024). However, information about Alpine utility-scale projects must be communicated proactively and transparently early in the process to foster local acceptance and avoid mistrust about the project at hand, as well as further projects (Participant 11, p.c., June 6, 2024; Participant 12, p.c., June 7, 2024). Indeed, poor communication may affect other Alpine utility-scale PV projects, as it contributes to creating an attitude towards this kind of innovative project. Participant 11 states: *“And the most important thing, which I think everyone paid dearly for, was the absolutely catastrophic communication about the very first project. I don't know if you remember the first images that appeared in the Upper Wallis, saying, ‘We're going to build a park of 5 square kilometres’, and then they made projections. So there were solar panels on four mountains. [...] And I think this image - you take a gun and shoot yourself in both feet - has the same effect. Because they are still paying for that image”* (p.c., June 6, 2024). Therefore, certain aspects should be stressed explicitly during the project communication to foster acceptance, such as the benefits for the local inhabitants to ensure distributional justice (Participant 13, p.c., July 3, 2024) and the urgency of generating winter electricity output (Participant 11, p.c., June 6, 2024). Having the right timing for communication is also essential but complicated, as it cannot be too early or too late, as previously mentioned (Participant 11, p.c., June 6, 2024).

5.2.4 Evaluation

Project developers of Alpine utility-scale PV systems must evaluate the local acceptance. Additionally, financial viability must be evaluated for both types of projects.

a. Evaluation of the Local Acceptance Through Communal Assembly or Comparisons

Companies do not have to seek local acceptance for commercial PV projects, as there is usually no opposition. They instead evaluate project aesthetics (Participant 1, p.c., April 2, 2024; Participant 6, p.c., April 23, 2024), financial viability (Participant 1, p.c., April 2, 2024) and safety (Participant 2, p.c., April 2, 2024; Participant 6, p.c., April 23, 2024). However, the first aspect project developers must

evaluate is local acceptance. If locals are not on board, there is no use continuing the project, as the opposition will slow down or halt the project (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024).

As previously mentioned, the population evaluation is hard to predict. Therefore, it must be addressed during a communal assembly, where the project is presented (Participant 5, p.c., April 8, 2024; Participant 10, p.c., June 3, 2024; Participant 13, p.c., July 3, 2024). The local evaluation is then mainly based on the project presentation, inner convictions (Participant 13, p.c., July 3, 2024) and the political messages of various political parties (Participant 15, p.c., July 15, 2024). As previously explained, some factors are known to be causing acceptance issues, and the population considers them when evaluating the project (e.g., impact on nature and economic activities). Other factors can influence the assessment, such as other RE projects in the commune and the number of projects that started in the same region at the time of the evaluation. Evaluation can also be based on ideological reasons (e.g. degrowth) (Participant 11, p.c., June 6, 2024). Additionally, evaluation could be influenced by how some communal assemblies' votes are done, i.e. by raising hands (Participant 13, p.c., July 3, 2024).

However, the population's opinion on an Alpine utility-sale project is sometimes not tested through a communal vote. Indeed, communal leaders want to correctly evaluate the population's attitude as early as possible, especially if they want to avoid triggering a “*political war*” in the commune and wasting energy on a project that will not be realised (Participant 11, p.c., June 6, 2024). Therefore, if an Alpine utility-scale PV project is rejected in a neighbouring commune and the local context is judged similar enough, the project may be abandoned, as Participant 11 states: “*When they [neighbouring commune] saw the result here, they said, OK, we're not going any further.*” (p.c., June 6, 2024).

The evaluation of the population is needed for an Alpine utility-scale project and not for commercial projects. Although there should be little influence as each local context is different, peer effects between communal leaders were observed.

b. Evaluation of the Financial Viability, Depending on Self-Consumption for Commercial Projects and Subsidies for Utility-Scale PV Projects

The second consideration is the project's financial viability, which depends, for commercial projects, on the possibility of generating and self-consuming enough electricity (Participant 8, p.c., April 29, 2024). However, it is complicated to know the exact payback period due to the uncertainty of the solar yield and fluctuating energy prices (Participant 1, p.c., April 2, 2024; Participant 8, p.c., April 29, 2024), but the investment has been proven to pay off (Participant 7, p.c., April 24, 2024). For Alpine utility-scale projects, the financial viability must be evaluated once the local population accepts the project. Unlike private projects, having the most profitable project is not a significant consideration, as Art. 71a of the Energy Act only subsidises the project's excessive costs. Therefore, if costs are lowered, subventions are lowered as well. As a result, the most defining factor for financial viability is the

possibility of generating enough electricity in time to achieve the output required. If the output cannot be reached in time, project developers would incur considerable losses, as they would not get subsidies and would have to dismantle the park (Participant 12, p.c., June 7, 2024; SFOE, 2023a; SFOE, 2023c). Additionally, as previously explained, considerable uncertainties are linked to Alpine utility-scale PV projects. The solar yield may vary depending on the weather and model, and opposition to the project can come late, slowing the project's realisation. For these reasons, it is not yet possible to evaluate the financial viability of such projects (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024).

c. Necessity to Ensure Installation Safety to Avoid Bad Press and Losses

The last significant aspect to evaluate for commercial and utility-scale projects in Alpine regions is the safety of the installation. Commercial PV projects notably have to account for high altitude risks (e.g. snow, freezing rain, strong wind), overload problems (Participant 6, p.c., April 23, 2024), and fire risks (Participant 2, p.c., April 2, 2024). Such issues must also be accounted for Alpine utility-scale projects. In addition, the risks of avalanches, landslides and rockfalls must be considered. Accidents would lead to a negative perception of such projects, bad press, and losses (Participant 12, p.c., June 7, 2024).

5.2.5 Decision

a. A Result of the Evaluation

The decision for companies and utility-scale project stakeholders depends on a positive or negative evaluation of previously discussed elements. However, the decision to undertake Alpine utility-scale projects cannot be taken yet, as the project's financial viability is unsure (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024).

5.2.6 Satisfaction

a. Overall Satisfaction with Commercial and Utility-scale PV Projects

There is overall satisfaction with commercial PV projects, notably linked to the relatively easy adoption process, without opposition (Participant 5, p.c., April 8, 2024; Participant 7, p.c., April 24, 2024). There is also pride in undertaking commercial solar PV projects, which may enhance satisfaction (Participant 7, p.c., April 24, 2024). However, some financial aspects inherently linked to solar energy are negatively assessed, such as the uncertainty about the payoff date due to the fluctuating energy price and solar production. The extended period required for the investment to pay off is also criticised but does not deter adoption (Participant 6, p.c., April 23, 2024; Participant 7, p.c., April 24, 2024; Participant 8, p.c., April 29, 2024). Finally, the lack of post-purchase support leading to lower efficiency is reproached, as mentioned by Participant 4: *“So in this field, there are a lot of installers, but I'm not sure there are many who have mastered the process, who go all the way. So, yes, we're happy, but we do not have the hindsight to see if we were right”* (p.c., April 5, 2024). Despite the downsides, companies are still

willing to adopt commercial PVs and start other RE projects. The positive sides of the technology, such as the guarantee of electricity, advertising possibilities and the fact that the technology pays off (Participant 7, p.c., April 24, 2024), must therefore outweigh the downsides.

It is not possible to assess the satisfaction with the finished utility-scale project, as none of the projects researched are finished. However, the project developers are happy with the stakeholder participation, which is essential for the successful completion of the project (Participant 12, p.c., June 7, 2024). Nonetheless, they would like the project to go faster, which is not possible due to the time-consuming administrative procedures and the innovative aspect of the project with many unknowns (Participant 12, p.c., June 7, 2024; Participant 14, p.c., July 3, 2024). Conversely, stakeholders involved were satisfied so far with the setup and communication process from the project developer side (Participant 10, p.c., June 3, 2024; Participant 13, p.c., July 3, 2024). Pride about the accepted project was also observed (Participant 13, p.c., July 3, 2024).

5.2.7 Post-Purchase External Communication

a. More Substantial Advertising Benefits for Commercial PV Projects than for Utility-Scale PV Projects

Post-purchase external communication for commercial Alpine PV projects is usually intensive, including press releases, sustainability reports (Participant 7, p.c., April 24, 2024), exhibitions (Participant 4, p.c., April 5, 2024), and website pages (Participant 4, p.c., April 5, 2024). There is also the possibility to earn solar prizes (Participant 1, p.c., April 2, 2024) and certifications (Participant 7, p.c., April 24, 2024). Such commercial projects are visible and well-perceived by the public (Participant 8, p.c., April 29, 2024). Although not a motivation to adopt commercial PV, their adoption can act as advertising for companies to show their commitment to the environment (Participant 6, p.c., April 23, 2024; Participant 7, p.c., April 24, 2024; Participant 8, p.c., April 29, 2024; Participant 9, p.c., April 30, 2024). It is particularly relevant as the companies interviewed depend on tourism and Swiss tourists, and the Swiss population has a positive attitude towards rooftop PVs (Cousse, 2021, p.5).

Once again, the Alpine utility-scale PV project interviewed has not been led to completion yet. However, some communes perceive an advertising possibility through such RE projects, such as being classified as a climate-neutral region (Participant 10, p.c., June 3, 2024) or an electricity-autonomous commune (Participant 5, p.c., April 8, 2024). However, such advertising benefits must be more potent than the utility-scale system's impact on nature, or they are perceived as deteriorating local tourism (Participant 11, p.c., June 6, 2024).

Completing commercial PV projects is well-perceived by the general public and, therefore, can be used as advertising. The advertising possibilities are not as straightforward for Alpine utility-scale PV projects due to their impact on nature and the landscape. Post-purchase communication is the last stage

of the social customer journey, according to the framework of Hamilton et al. (2021), but a positive feedback loop of RE adoption was observed in the data.

5.2.8 Positive Feedback Loop of RE Projects as a Result of Satisfactory Projects

Completing RE projects leads to further projects, creating a positive feedback loop for adopting RE. Indeed, before commercial PV installation, companies often already had RE projects, such as thermal energy (Participant 6, p.c., April 23, 2024) or wood pellets (Participant 8, p.c., April 29, 2024). Many were also committed to purchasing exclusively hydropower (Participant 4, p.c., April 5, 2024; Participant 6, p.c., April 23, 2024; Participant 8, p.c., April 29, 2024) and to have better energy efficiency (e.g. heating monitoring, insulation, consumption reduction) (Participant 4, p.c., April 5, 2024; Participant 5, p.c., April 8, 2024; Participant 6, p.c., April 23, 2024; Participant 8, p.c., April 29, 2024; Participant 9, p.c., April 30, 2024). Then, after a commercial PV installation, if the companies are satisfied, other solar projects or projects with various RE sources (e.g. wind, hydro, thermal energy) are started or considered (Participant 6, p.c., April 23, 2024; Participant 8, p.c., April 29, 2024). For example, the possibility of adopting commercial PV was systematically evaluated during a renovation or new construction (Participant 6, p.c., April 23, 2024; Participant 7, p.c., April 24, 2024; Participant 8, p.c., April 29, 2024).

There can be several reasons for the positive feedback loop. First, there are obligations from the State. Since September 2023, solar panels must be included if a new construction has a chargeable area of at least 300 square meters (“Parliament agrees on solar”, 2023). However, companies interviewed were not doing it for such reason (e.g. Participant 8, p.c., April 29, 2024). There are also voluntary commitments, such as having an objective agreement with the SFOE, which was more commonly found in the data (Participant 1, p.c., April 2, 2024; Participant 4, p.c., April 5, 2024; Participant 8, p.c., April 29, 2024). In such agreements, companies fix a target to achieve more energy efficiency, increase RE adoption and reduce CO₂ emissions. Solar PV can contribute to achieving this target, and subsidies can be claimed if specific criteria are fulfilled (SFOE, n.d.-a). Additionally, if companies want to construct according to specific standards, such as Minergie-A, solar PV must be included if there is exploitable potential (Minergie, 2024; Participant 3, p.c., April 3, 2024).

An alternative explanation is that because of the repeated adoption of various RE (incl. solar), sustainability and RE become part of the company's long-term vision and culture, encouraging further development. Many participants stated that ecology is part of their company's long-term vision (Participant 7, p.c., April 24, 2024; Participant 8, p.c., April 29, 2024; Participant 9, p.c., April 30, 2024). There is also pride in building commercial PV projects, especially for pioneering projects: *“I have to say, I'm a fan, of course [of the PV façade], and that's why I say a PV system, a PV façade looks really cool.”* (Participant 7, p.c., April 24, 2024). The pride can be due to the pioneering status but also to the

positive perception of the public of such installations (Cousse, 2021, p.5; Participant 9, p.c., April 30, 2024).

The positive feedback loop also extends to utility-scale PV projects. Indeed, some companies with commercial PV take part in Alpine utility-scale PV projects as infrastructure owners (Participant 4, p.c., April 5, 2024; Participant 8, p.c., April 29, 2024), and others were in favour of it or not opposed but just had not been approached by a project developer (Participant 7, p.c., April 24, 2024; Participant 9, p.c., April 30, 2024). The successful development of commercial PV projects within a company is not the reason for starting a utility-scale project, as the instigators are project developers, i.e. electricity companies. However, companies owning the infrastructures are often open to such projects, possibly because of technical know-how (Participant 4, p.c., April 5, 2024; Participant 7, p.c., April 24, 2024), and an inner conviction about the technology. They also have a financial gain in participating (Participant 11, p.c., June 6, 2024). Additionally, as previously explained, there is a positive feedback loop within communes, as those with existing RE projects were more open to utility-scale PV projects (Participant 12, p.c., June 7, 2024).

The successful completion of RE projects (incl. solar projects) creates a positive feedback loop within companies, leading to further RE projects, including the participation in Alpine utility-scale PV projects. Thinking in terms of RE becomes part of the company culture. The same process was observed in communes.

6 Discussion

Switzerland needs to cover its winter deficit (SFOE, 2023a), and solar PVs used in the Alpine area are a technology enabling electricity generation in winter (Kahl et al., 2019, p.1165), as opposed to other RE sources (Mellot et al., 2024, p.9). Recognising this challenge, the research started by analysing the adoption of commercial PV in Alpine companies, which own larger surfaces than households and, therefore, have a higher electricity generation potential (SolarPower Europe, 2023). The social customer journey by Hamilton et al. (2021) was used as a framework to start the research and was adapted based on the data collected. In parallel, the use of theoretical sampling recommended in the grounded theory method from Corbin & Strauss (2015, p.7) broadened the research to utility-scale PV system development. Findings were compared with the existing literature throughout the research process (Corbin & Strauss, 2015, p.49).

The framework and method enabled the analysis of the development and social dynamics for deploying commercial and utility-scale PV, the *core category* of this study. This *core category* explains the journey and social interactions for the adoption of commercial and utility-scale PV through various interrelated *categories* (i.e. stages of the journey) (Corbin & Strauss, 2015, pp.7-8). This method highlighted factors affecting the adoption of commercial and utility-scale PV projects, and comparing them enabled a better understanding of the difficulties linked to utility-scale PV systems. Based on the findings, several

recommendations were formulated to enhance utility-scale PV project deployment. The following findings will be discussed in order.

6.1 Development and Social Dynamics for the Deployment of Alpine Commercial PV

The adoption process of commercial PVs was found to be straightforward. Similarities were found in the literature on household PV adoption. Pride in undertaking commercial PV projects was found (Côté et al., 2022, p.10), especially in innovative projects. The motivations to adopt solar PVs, i.e. financial, self-consumption and ecological, are similar to the study of Shakeel et al. (2023, p.10), with a significant financial drive common for companies (Crago & Koegler, 2018, p.481). The decreased need for subsidies with the self-consumption possibility was observed in commercial PV adoption (Karneyeva & Wüstenhagen, 2017, p.447), and there is an enhanced willingness to adopt PV in times of high energy prices (Karakaya & Sriwannawit, 2015, p.64; Shakeel et al., 2023, p.6). Additionally, two main factors negatively affected commercial PV adoption in Swiss Alpine companies. The first one is Swiss plot regulations. Indeed, if a company generates an electricity surplus with one commercial PV installation, it cannot transfer the electricity generated to another building for self-consumption if it is in a different plot. Therefore, the electricity generated can only be partly self-consumed, and commercial PV is not profitable. Indeed, a necessary condition to ensure cost-cutting is self-consumption, as supported by the literature (Karneyeva & Wüstenhagen, 2017, p.447). Self-consumption is especially important as it was found to be a major driving motive for PV adoption in the research and the literature (Bach et al., 2020, p.6). Second, the uncertainty linked to the effective electricity generation of innovative solar PV models hinders adoption. Companies often do not want to risk adopting an innovative commercial PV system if they do not have some certainty about its profitability.

One facilitating factor is the possibility of advertising commercial PV projects. The Swiss population positively evaluates rooftop PV (Cousse, 2021, p.5). Therefore, as commercial PVs are usually visible, especially when used as a façade (Baranzini et al., 2017, pp.23-24), it gives companies a green image. Using PV as an advertising tool to show commitment to sustainability is supported by literature (Baranzini et al., 2017, p.25). Another factor facilitating commercial PV adoption is the positive feedback loop for RE adoption, sometimes leading to participation in utility-scale PV projects. Prior adoption of RE (Bach et al., 2020, p.5) and knowledge about PV (Shakeel et al., 2023, p.8) can foster further RE adoption in the household context.

Social dynamics were also analysed. The actors participating in the customer journey mostly did not negatively influence the decision-making. The only potential conflictual points were financial considerations due to the long payback period, supported by Bach et al. (2020, p.6) and Shakeel et al. (2023, p.6), as well as the aesthetic of commercial PV, also known to cause adoption issues (Bach et al., 2020, pp.8-9). Additionally, peer effects were found mainly in the information search stage to bridge the information gap for the effective electricity generation of innovative PV models. However, as

opposed to household projects, the effects were not found to be strongly bound to distance between PV projects (Graziano & Gillingham, 2014, p.815) but rather to the similarities between projects and companies, as supported by Baranzini et al. (2017, p.24).

6.2 Development and Social Dynamics for the Deployment of Alpine Utility-Scale PV

Then, the development dynamics of utility-scale PV projects were analysed by interviewing various stakeholders with decision-making power. Social acceptance was the major factor affecting the successful completion of Alpine utility-scale PV projects, as found for other large-scale RE projects (Batel et al., 2013, p.1; Wüstenhagen et al., 2007, p.2683). Factors negatively influencing social acceptance similar to other research were found, such as the impact of utility-scale PV projects on the visual impact of the landscape (Devine-Wright, 2005, p.135; Kienast et al., 2017, p.398; Olson-Hazboun et al., 2016, p.176; Tsoutsos et al., 2009, p.552), nature (i.e. ecology) (Tabi & Wüstenhagen, 2017, p.771; Díaz & van Vliet, 2018, p.13) and land use opinions (Späth, 2018, pp.433-434). Impact on the local economy was also found to be a factor affecting adoption, especially in regions living off tourism, as supported by Michel et al. (2015, p.162). The NIMBY effect was reported, but the other factors are likely more important for explaining local rejection (Wolsink, 2012, pp.86-87). Interestingly, the population did not question the technical feasibility, although being a main concern for project developers, a previously observed phenomenon (Díaz & van Vliet, 2018, p.10).

As a facilitating factor, it was found that, just like companies, communes with existing RE projects were more prone to accepting utility-scale PV projects. There is a local pride in carrying on the pioneering spirit and being innovative. The willingness of a commune to be innovative was found to be a driving motive for small-scale Alpine PV adoption (Díaz & van Vliet, 2018, p.13).

Social dynamics for the development of Alpine utility-scale projects were also analysed. There is a high number of stakeholders with decision-making power in such projects. Indeed, Alpine utility-scale projects depend on the acceptance of the project from the local community, the local administration, landowners and infrastructure owners. Political and nature conservation actors can also influence the decision-making. Additionally, several social effects were observed. Local people sought information from local companies, which were probably considered more trustworthy and less biased, as they have no stake in the project (Cardwell, 2012) and are local. The preference for local project developers to undertake the project is supported by Tabi & Wüstenhagen (2017, p.771).

Moreover, social effects were observed between communes. First, locals compared the Alpine utility-scale project in their commune with similar projects, usually in the same region. They probably take neighbouring communes as comparison points as they are seen as more relatable due to the proximity. Indeed, they were making comparisons with neighbouring utility-scale PV projects. If it is poorly evaluated (e.g., due to construction noise, visual appearance, or assumption of financial motivations only), it leads to a negative perception of utility-scale PV projects, where the local population evaluates

a similar project in their commune poorly. The poor evaluation may be enhanced by the number of projects started in the same region. If many projects are started at once, local people may feel like the projects are only financial opportunities for non-local utilities. They are not willing to sacrifice the landscape for such reasons. Their project might also lose a pioneering status. Additionally, stakeholders were researching on the internet about other Alpine utility-scale projects to find information. Nowadays, people are more likely to be influenced by information found on the internet (Hamilton et al., 2021, p.69).

Second, although the local opinion varies from one commune to another, local leaders compared neighbouring projects to try to approximate the local opinion about a utility-scale PV project. Third, information on pioneering commercial PV projects for utility-scale PV installations was requested. Therefore, innovations within companies can foster the adoption of innovative solutions for similar projects in other companies (i.e., peer effect) and the development of larger projects such as utility-scale PV projects. It supports the state-of-research study from Köhler et al. (2019, p.11), stating that companies can help transition to RE since they can develop innovative solutions.

6.3 Main Differences Between Commercial and Utility-Scale PV Projects in Alpine Regions

Finally, the customer journey for the deployment of utility-scale PV projects was compared with the customer journey for commercial PV adoption to gain insights into the development difficulties of utility-scale PV projects. The first apparent differences are the size and the location of the projects. Utility-scale projects are more significant and deployed in nature, whereas commercial projects are smaller and constructed in built areas. The size of an installation was shown to decrease the positive attitude towards PV technology (Cousse, 2021, p.8), which might explain their problems of social acceptance. Moreover, the fact that the project is constructed in nature was found to be one of the most defining factors in opposition to such projects (i.e. due to the impact on nature, landscape and local economy). Because of the negative perception of such utility-scale projects in nature, the advertising possibilities for Alpine utility-scale PV projects are only possible to a certain extent. Conversely, rooftop PV projects are positively evaluated (Cousse, 2021, p.5), and commercial PV projects benefit from a positive image.

The second difference between commercial and utility-scale PV projects is the number of stakeholders with decision-making power. With commercial PV projects, the company and its various members were the only decision-makers and usually had a shared vision for the company. Conversely, Alpine utility-scale projects depend on the project acceptance from various actors with different considerations, whom political activists can influence. Indeed, the political orientation of people can influence decision-making (Vuichard et al., 2020, p.6; Tabi & Wüstenhagen, 2017, p.771). Due to the possibility of opposition, the project cannot move as quickly as desired. The third main difference is the impossibility of self-consumption for all the actors involved in utility-scale projects. Indeed, self-consumption is

necessary to achieve two of the most important motivations in commercial PV projects: financial profitability and energy security. Not all actors can self-consume, so other benefits must be distributed to the stakeholders involved. There is a risk of distributional injustice if the benefits are not perceived as fair, which is a factor that can hinder adoption (Tabi & Wüstenhagen, 2017, p.770; Vuichard et al., 2020, p.7). However, convincing certain communes about the benefits of utility-scale projects is complicated, as they are already wealthy and may therefore favour nature over financial benefits.

The fourth difference is the technical difficulties of the project. Alpine commercial PV systems have uncertainties related to the effective electricity generation of certain models and face high-altitude issues. Utility-scale projects have even more significant solar PV model uncertainties, as the technology is developed simultaneously as the project is undertaken. High-altitude issues are also present; they face risks of avalanches, landslides, and rockfalls. Notably, because of these risks and uncertainties, utility-scale PV projects are not profitable as such and require government subsidies. Conversely, commercial PVs do not require subsidies to be profitable, probably as the technology is mature (Shakeel et al., 2023, p.10). Therefore, there is more pressure to reach a specific output in time.

6.4 Recommendations to Foster Alpine Utility-Scale PV Projects

First, utility-scale PV project development can be improved by fostering local acceptance. The points of view of various stakeholders mainly depend on the local context, as supported by Díaz & van Vliet (2018, p.13), which may hardly be influenced once the project site is chosen. Therefore, the choice of project site should account for factors systematically influencing the local opinion negatively, such as the impact of utility-scale PV projects on the landscape's visual appearance, nature and economic activities. Areas already impacted by humans, with existing infrastructures, should be chosen. Additionally, choosing communes with a culture of RE makes local acceptance easier, as the population is used to such projects and discussions and can be willing to carry on the pioneering spirit. Previous studies support this assumption in several aspects. First, previous RE adoption facilitated further RE adoption in a household context (Bach et al., 2020, p.5). Then, entrepreneurial culture fosters new technology adoption (Karakaya & Sriwannawit, 2015, p.62). It might be argued that communes pioneering utility-scale RE have such a culture.

Additionally, collaborating with local actors who already carried out RE projects in the commune can enhance trust, as the local population trusts local actors more. It might also be best to space out the projects' location, as the population may develop the impression that they are only undertaken for financial reasons. Conversely, being the first in a region can foster a pioneering feeling. As mentioned, being innovative can be a driver for locals (Díaz & van Vliet, 2018, p.13). Therefore, such benefits should be emphasised. Once the project site has been chosen, the benefits for the local population should be clearly defined and transparently communicated to make people perceive them as fair, as supported

by the literature (Díaz & van Vliet, 2018, p.13). If the communication is unclear, there might be the fear of a hidden agenda (Slovic, 1987, pp.284-285).

Then, the deployment of such a project can be improved by facilitating the administrative procedures. Indeed, because of the lengthy administrative procedure and opposition possibilities, many uncertainties persist, and project developers cannot move as quickly as they want. An attempt was made in the canton of Wallis to shorten the administrative procedures for Alpine utility-scale projects. However, it was rejected (“Rejet du décret”, 2023), even in communes that had accepted utility-scale PV projects (Savioz, 2023). Additionally, “the federal law on a secure supply of electricity from renewable sources” (Confédération Suisse, 2024, p.1), approved in June 2024, will facilitate the adoption process of utility-scale PV projects. The local population will, however, still be able to vote on it, and social acceptance is therefore still relevant (Confédération Suisse, 2024, pp.48-49).

7 Conclusion

This research aimed to gain insights into the factors affecting the development of commercial and utility-scale PV projects in the Alpine regions of Switzerland, as PV technology used at a high altitude offers a way to cover the country's winter electricity deficit (Kahl et al., 2019, p.1165; SFOE, 2023a). To explore this field of research, the grounded theory method from Corbin & Strauss (2015) was used as a methodology and enabled, with the support of the social customer journey framework from Hamilton et al. (2021), to analyse the development and social dynamics of commercial and utility-scale PV projects. Fifteen interviews with various decision-makers participating in commercial and utility-scale PV projects were conducted and supplemented with a literature review. This method revealed the significant factors affecting their development, complementing and extending the literature on the subject.

The most significant factors hindering the adoption of commercial PV are plot regulations restraining self-consumption and the uncertainties linked to the effective production of innovative Alpine solar PV. To overcome this uncertainty, peer companies pioneering in solar technology are used. This peer effect extends to utility-scale PV project stakeholders, who also seek advice from technology pioneers. Innovations within companies can, therefore, foster similar projects in other companies and larger utility-scale projects. This effect was not found to be strongly bound to distance but rather to the similarities of the projects and companies. A facilitating factor is the possibility of using commercial PVs as advertising.

As for utility-scale PV projects, social acceptance is the first factor hindering adoption. Indeed, many stakeholders with decision-making power have various, often conflicting considerations. Factors affecting social acceptance the most include the impact of the utility-scale project on the visual appearance of the landscape, nature, and local economy. Therefore, choosing areas already impacted by humans, having not too much impact on the landscape, with existing infrastructures and not relying on

tourism too much mitigate opposition. Additionally, facilitating factors include distributing fair benefits to the local population, having transparent communication, including local stakeholders and the previous conduct of RE projects in the commune, preferably large-scale ones the population had to vote on. Concerning the social dynamics, negative peer effects were observed between local leaders for the evaluation of the local acceptance of a project. Additionally, if people have a poor perception of a utility-scale PV project in a neighbouring village, they will likely develop a poor attitude towards such a project and towards a similar local project. This effect might be enhanced if many projects are started in the same area at a similar time, potentially making such projects look like financial opportunities for non-local utilities. Once social acceptance is given, other major hindering factors are the lack of tried and tested PV models for such high-altitude project, the technological risks and complexity of working at high altitude, and the time and electricity generation constraints fixed by the State.

The most significant differences explaining a more complicated development of utility-scale PV compared to commercial PV systems are the location of the project, the number of stakeholders with decision-making power, the impossibility of self-consumption for some stakeholders, technical uncertainties, and the complicated administrative process subject to oppositions.

8 Limitations and Further Research

While this research enabled insights into the factors affecting the development of commercial and utility-scale PV projects in Alpine regions, it contains limitations. First, although the social customer journey framework from Hamilton et al. (2021) was flexibly used, it may have biased the research, as stages were pre-determined. Second, as the researcher is entirely part of the research, the interpretation of the results is influenced by personal background, which can have positive and negative consequences, as mentioned in the methodology part (Corbin & Strauss, 2015, p.46). Thirdly, the sample was composed of people willing to participate in the interview and, therefore, might be more interested in the subject researched and biased to evaluate it more positively. The interviewees were also aware of the objective of the research, and it is, therefore, possible that they were influenced by a social desirability bias when expressing their viewpoint on a topic such as RE adoption or sustainability (Blome & Augustin, 2015, p.113). Then, a qualitative approach was taken to gain insights into a subject. However, such research does not give representative results due to the small number of participants (Corbin & Strauss, 2015, p.146). Therefore, new insights generated through this research should be tested in a larger sample with quantitative methods to verify their statistical significance. For example, the facilitated adoption of RE in communes with pre-existing RE projects, the positive feedback loop of RE adoption or the negative impact on social acceptance of starting several utility-scale projects simultaneously and in the same area should be tested. Finally, it would also be interesting to research if peer effects exist at the successful completion of utility-scale projects.

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Appendix A: Development and Social Dynamics for the Deployment of Alpine Commercial and Utility-Scale PV

Development and Social Dynamics for the Deployment of Alpine Commercial PV						
Company Buying Commercial PV Perspective						
Motivation	Information Search	Pre-purchase External Communication	Evaluation	Decision	Satisfaction	Post-Purchase External Communication
-Commercial PV are financially interesting given the possibility for enough energy generation and self-consumption -Commercial PV enable energy security -Generating electricity with solar PV is ecological	Energy generation and model uncertainties with Commercial PV electricity generation <u>Mitigation strategies:</u> -Search for specialist expertise -Search for best practice by existing company projects (peer effects) Legal requirements uncertainties <u>Mitigation strategies:</u> -Search for the requirements for norm compliance & subsidies	Not intensive (e.g. press release)	Evaluation of the project: - safety - financial viability - aesthetic	Result from the evaluation: - Abandon the project - Move forward with the project	-Overall satisfaction despite downsides (e.g. uncertain payback period, lack of post-purchase support), notably due to the easy adoption process -Pride resulting from the project completion	-Intensive (e.g. press release, exhibition, website) -Advertising possibilities as a side-effect (Positive attitude of the population towards solar PV in built area enables advertising possibilities)
 Further projects (Positive Feedback Loop) (e.g. utility-scale projects, wind, hydro, thermal energy)						

Development and Social Dynamics for the Construction of Utility-Scale PV Projects						
Project Developer Perspective						
Motivations	Information Search	Pre-Purchase External Communication	Evaluation	Decision	Satisfaction (cannot assess yet, but satisfaction with overall process)	Post-Purchase External Communication (cannot assess yet, but presumed)
-Utility-scale PV might be financially interesting given the possibility to get subsidies -Alpine solar PV enable Swiss energy security -Generating electricity with solar PV is enables nuclear phase-out	Search for a project site which: -is favorable for electricity generation and -is locally acceptable (incl. impact on nature, economic activities) Energy generation and model uncertainties with utility-scale PV electricity generation <u>Mitigation strategies:</u> -Search for specialist expertise -Exchange of best practice and setup on-site tests with competitors (social collaboration) -Search for best practice by existing company projects (peer effects) Legal requirements uncertainties <u>Mitigation strategies:</u> -Search for the requirements for norm compliance & subsidies	Intensive (e.g. information meeting, media release): need to foster social acceptance	Evaluation of: -the local acceptance (influenced by impact on nature /aesthetic & economic activity), -the financial viability of the project (incl. possibility to achieve output of law 71a) -the installation safety	Result from the evaluation: - Abandon the project - Freeze the project - Move forward with the project	Overall satisfaction with: -the stakeholder participation -Setup and communication process Not fully satisfied with: -the time pressure to achieve the necessary output - time-consuming permit application process	-

Third Parties Perspective (incl. population, local leaders, landowner)						
-Alpine solar PV enable Swiss (and private sometimes) energy security -Generating electricity with solar PV is ecological -there are financial benefits	General uncertainties about utility-scale projects <u>Mitigation strategies:</u> -Information search through federal & cantonal guidelines , but lack of clarity about the federal energy policy -Information search through the project lead -Information search through personal, local contacts (peer effect)	-	-Impact on visual appearance of landscape -Impact on nature -Impact on local economy -Evaluation of the benefits - ideological arguments against	Result from the evaluation: - Abandon the project - Freeze the project - Move forward with the project	Overall satisfaction with: -Setup and communication process Not fully satisfied with: -other people skepticism -lack of consensus about energy policy Pride resulting from the project acceptance	Advertising possibilities as a side-effect (e.g. climate neutral region/resort), if there is not too much impact on nature

Note. Own depiction of the findings based on the social customer journey framework of Hamilton et al. (2021, p.7).

Appendix B: Interview Transcripts

The interview structure is based on the six stages of the social customer journey from Hamilton et al. (2021). The consent for recording was requested and given for each interview. The interview transcripts were all anonymised for consistency, although most participants did not mind about not being anonymised. The anonymisation was made so that it would not be possible to find out who the interviewee was. Therefore, the level of anonymisation varied from one interviewee to another. For example, the necessary level of anonymisation for communal voters was lower, as the pool of potential people is much bigger than that of a member of a company management (Erasmus University Rotterdam, n.d.). Additionally, the company name Zermatt Bergbahnen AG was not anonymised, as the interviewee agreed to it, and the data would have lost too much value through such a process.

Participant 1
Interviewee: Member of the management of a Swiss Alpine company Interview date: 2.04.2024 Form: Qualitative, structured, written Solar Energy Adoption: Yes, solar PV on a restaurant

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Did you take the decision to adopt PV alone, or did it depend on several people?

X: [It was a] strategic decision in the Board of Directors of [his/her company name]. Thanks to the sun-exposed location of our mountain area, it was almost inevitable. Our business is energy-intensive, which is why we installed PV systems on the roof of the mountain station and the mountain restaurant.

CA: What was (were) your motivation(s) for introducing PV infrastructure in your company?

X: Doing something good for the environment; nature is our basis for action & life, utilise solar energy, reduce costs, think / be / remain future-orientated, reduce dependence on electricity suppliers.

CA: How did you find the necessary information about the introduction of PV? Did you look for subsidies from the canton/state?

X: Local electrical shop brought on board with expertise, support from the [region name] Energy Commission (affiliated with the municipality).

CA: When you had all the information and had to assess whether you wanted to introduce PV, what were the key considerations?

X: Cost/benefit and amortisation of the system (within 20 years) [as well as] aesthetics of the installation; Sunstyle the correct provider for such solutions.

CA: What were the key factors that prompted you to purchase PV?

X: Inner conviction in combination with the price.

CA: Are you satisfied with your purchase?

X: Very satisfied. Solar prize diploma won in [year].

CA: Have you communicated your investment in PV to a wider audience?

X: Media release, website, [region name] Energy Commission, Journalists on media trips in and around [region name].

---- End of Transcript ----

Participant 2
Interviewee: Member of the management of a Swiss Alpine company Interview date: 2.04.2024 Form: Qualitative, structured, written Solar energy adoption: Not yet but considering it

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

**CA: Have you ever discussed adopting renewable energy infrastructures in your establishment?
If so, did the decision depend on several people?**

X: We have undertaken initial studies into the deployment of photovoltaic panels on our infrastructure. These studies involve bringing together various experts to analyse the various issues and potential difficulties: connection, installation cost, buy-back or self-consumption, impact on building infrastructures, fire risks, etc.

CA: What were your reasons for adopting (or not adopting) renewable energy infrastructures in your establishment?

X: We are committed to energy efficiency and reducing our consumption. We want to move towards installing photovoltaic panels, but this takes time.

**CA: How did you find the information you needed to adopt renewable energy infrastructures?
Did you seek subsidies from the canton/government?**

X: We've signed an Objective Agreement with the SFOE and are supported by the company [company name]. We also have contacts with the [school of higher education name]. We are in contact with engineering firms. We'll be looking for subsidies when the time comes.

CA: When you had all the information at your disposal and had to assess whether or not to adopt renewable energy infrastructures, what were the most important factors to take into account?

X: We're not at that stage yet.

CA: What were the decisive points in your decision not to buy renewable energy infrastructure?

X: Not for us at the moment.

---- End of Transcript ----

Participant 3
Interviewee: Member of the management of a Swiss Alpine company Interview date: 3.04.24 Form: Qualitative, structured, written Solar energy adoption: Yes, on a restaurant

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Have you already invested in renewable energy infrastructures?

X: Unfortunately, we have not been producing energy ourselves so far. We are planning a solar façade for one of the hotels that is being built and that will be opening in [year], but I cannot guarantee that it will realize in the end. From what we learned recently, I assume that we will have to put solar panels on all buildings built according to Minergie standard in the future – the standards have been adjusted lately and the law of the Canton of [canton name] foresees this too. We are observing what will happen with the revised energy law. We have installed one bigger panel at the ski resort's new restaurant in [mountain village name] and the energy it produces in summer when the restaurant is closed, is fed into the local electricity net. We are just now starting to look into solar energy, for many years solar panels were thought to be unusable at our altitude with the loads of snow we get in the winter, but times are changing.

---- End of Transcript ----

Participant 4

Interviewee: Member of the management of a Swiss Alpine company

Interview date: 5.04.2024

Form: Qualitative, semi-structured, online (zoom)

Solar Energy Adoption: Yes, as a façade on a new restaurant

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Were several people involved in the decision to adopt solar panels on the [building name]? And if so, what influence did they have?

PB: Well, first of all, there's the company's entire Board of Directors. In other words, 5 of us were invited and everyone agreed [to adopt the solar panels]. In any case, we had to find a solution for heating that was as environmentally friendly as possible at altitude. You have to realise that at that altitude, you can't very well bring lorries with pellets, gas or fuel oil up to an altitude of [high altitude] metres. We'd have to find a situation [to] recover the heat from the power cabinets, then from the cable car drives. We'd have to install a heat pump to recover this energy and then, if there were problems with poor coefficients, we'd have to supplement it with electrical energy [if the solar panels hadn't been adopted]. So that's why we decided to install solar panels.

CA: And what were your motivations for adopting solar panels on the [building name]?

X: We have enough power to do without, [but it was] to have the minimum, ... try to have some self-consumption. To take advantage of the sunshine, because at an altitude of [high altitude] metres we get the maximum amount of sunshine.

CA: And have you also thought about the fact that the installation could give you a potential competitive advantage in attracting tourists to your destination?

X: Well, we have everything. Some people see that, as, well, it's clear, it's magnificent. There are other people, say yeah, but it's one of those that bothers me [the installations]. It's ugly. But logically, people really like it.

CA: And how did you find the information you needed to adopt solar panels in the mountains? Was it difficult to find this information?

X: No. A, we have an office that does studies of this kind in the valley. And then B, we went to talk to our colleagues in Zermatt who have the same type of panels at 3850 metres from the summit of the Matterhorn, and we went to see the yields and how they worked. And we worked with the same types of panels as those installed in Zermatt. They are made in Switzerland by the company [company name]. We already used panels that were known to work in the high mountains, that's all.

CA: And did you seek subsidies from the canton or the government?

X: Yes. There's a subsidy coming in for energy, but I don't know what it is, it's not huge, it's maybe CHF 20 or 30,000. On a project worth 8 million. That's not what tips the scales. That's clear.

CA: Then, in relation to the assessment, when you had all the information and you had to assess the opportunity of adopting solar panels, what considerations did you take into account?

X: We took into account that the heating had to meet the new standards as much as possible, so we thought we needed renewable energy. In the event of long periods of sub-zero temperatures, even though the average temperature rises every winter, there are still days from time to time when it can be minus 20 [degrees] or colder. So, we had to be able to justify the type of heating we have at this altitude with solar panels. If we'd only used an air-water heat pump [it wouldn't have worked], because you can't drill at altitudes like that, there's only rock. We'd also have less chance of getting the go-ahead. Maybe not, but then it's the federal or cantonal energy office that decides.

[break]

But we could see that it was impossible to bring another source of fuel to that altitude. Because the road isn't really a road, you can't get up there with a monster lorry, and it's not accessible for about 6 months of the year. Pellets don't go up anymore. Nothing works any more. The considerations were also made

to meet the new heating standards. So for us, there was an ecological part to it, and we had to respond with as little electricity consumption as possible in relation to the type of heating we had chosen.

CA: What are these construction standards?

X: The new building standards already require a thermal envelope. So the thermal envelope is the equivalent of the "JP environment". And then you're no longer allowed to install electric heating. Even if you have a solar power plant that delivers maximum output, the canton of [canton name] no longer allows you to install electric heating. So even if you had a solar array that could power an electric heater and you didn't even know what to do with the energy, you wouldn't be allowed [to install an electric heater]. You'd have to bring in a diesel-powered truck to bring in 50 or 100 tonnes of pellets. That's how it is, it's law, it's done by lawyers who don't have much knowledge of what happens at altitude.

[break]

It's restrictive, and we're obliged to have an electric part in case we have a long period of cold and then there's little traffic on the cable car, and then there's less heat dissipated. We know that a system like a heat pump on a plain, if it's minus twenty [degrees], or an air-water heat pump without drilling, there will be times when it needs to run on electricity, at home, in the boiler, to keep the water at a certain temperature. It's the same with a full tank. Except that it has more impact at high altitude because you lose about 0.6 degrees per 100 metres of altitude. At 2,000 metres, compared with the plains, you lose at least 13° in total. If it's 0 on the plains, it's minus 13 here. So there are different considerations, the standards in force, the desire to find a heating system that is as environmentally friendly as possible and that meets the new standards.

CA: And were there any differences of opinion when it came to assessing the project?

X: Let's be honest, at the same time, it was dressing up the building. It should have been dressed in a certain way anyway. So, no, there was never any disagreement. And from the start of the project, we decided that it was going to be solar, and we never went back on our calculations.

CA: And then, are you and your company satisfied with the functionality of the solar panels at this altitude?

X: We don't have enough hindsight, because the panels were put into operation at the end of December. We've barely been on them for 3 months. We are missing one or 2 tools to analyse the returns. We've gone further with batteries. Part of the surplus current recharges the batteries that are used to operate the positive and negative cold chambers at night. But the batteries interfere with the calculation of the energy returned to the grid. So you have to realise that there's a part of the solar technology that I wouldn't say has been mastered everywhere.

[break]

I have friends, ... there are companies that come and install a solar system and adjust the whole house for you. Because you can completely change the way your house is set up. They can achieve 80% autonomy. And then there are companies that come and install solar panels. And then people have 20-25% autonomy because they've left the heat pump running at night. It's useless, they haven't changed the energy consumption cycle. So in this field, there are a lot of installers, but I'm not sure there are many who have mastered the process, who go all the way. So, yes, we're happy, but we don't have the hindsight to see if we were right. It would appear that the forecasts for the power they [solar PVs] produce in relation to the amount of sunshine are right. We're slightly ahead in February, so apparently the calculations they've made are right on the mark in terms of cumulative kilowatts. What we want to see is, are we consuming everything intelligently or are we still losing energy? Or are there things we need to regulate better? But that can't be done in a week, it has to be done in a few months. The restaurant is brand new. We didn't have any load cuts before, so you have to, ... at the end of the season we'll have to analyse what happened when and then see if there are any processes we can improve to maybe get up to 85% for example. If I take, I have no idea, today, I look at, um, ... If I look at, ... If I look at the batteries, ... Today, for example, we're at 73% self-consumption.

CA: So how do you supply the other 30%?

X: Here, it's all hydroelectric. There's not a single atomic electron pointing up here. And the journey is as short as possible. So I'll be honest, if you're in St. Gallen, you'll never have a hydro-electron. Even if the school pays for what's called a certificate of origin, in fact you'll never get one, you'll just get nuclear. In fact, it's electrical energy, it's a virtual market. Electrons are used in the shortest possible way. They are first used where they are produced. After that, they diminish more and more as they go onto the grid. After that, everything gets mixed up. What's more, our company pays 0.5 centimes per kilowatt-hour for the guarantees of origin, i.e. around 30,000 francs a year. Because we know that it goes back to the municipality for a certification of origin that enables us to have sustainable development projects. We wouldn't be obliged, but we do it.

CA: And to come back to what you were saying before, you find that there's a lot of support for pre-purchase, but no necessarily for post-purchase?

X: No, but I get the impression that you're being sold a system, but I'm not always sure that everyone knows how to check that the system is working 100% and that everything is set up correctly. But I see this because, ... I see it in private projects, with friends. But you know, he says, "Look at this, it's superb" and I see that he's consuming 20 to 25% of the power for the house and then 75% goes off to the side. I say to myself that it's not right. He says no, no, we haven't changed the heat pump, it runs at night. In my opinion, it's not working. Like in all specialist offices, ... Installing solar panels is easy, it's just a matter of getting them to work and then adjusting everything behind them [which is complicated]. But no, for the moment we're happy. We'll have to analyse it at the end to optimise the processes.

CA: And do you still have contact with the company for this kind of adjustment issue?

X: Yes, we have contacts.

CA: Then, have you communicated your investment in solar panels to a wider audience?

X: There have been publications, but it's more like we're going to put an advert, ... we have a large space on the floor below the restaurant, where people come from the cable car. We're going to set up a shop there, and then there'll be some exhibition and explanatory panels. This will come this year or next to explain how the building was designed, ... and then, ... what has been done in terms of energy efficiency. But we're waiting until we've got everything in place, with the shop, before deciding how we're going to showcase it.

CA: Would you recommend adopting solar panels in the mountain, and if so, why?

X: Well, quite simply, at the altitude we're at, the panels produce twice as much as they do in [Commune name in the lowlands] and between 3 and 4 times as much as they do on the Swiss plateau. And why is that? Because, ... well, ... the solar panels that are installed, ... because we're also a test field here for Switzerland. You should know that, apart from this project here, with 380 square metres of solar panels on the facade, we have a 140,000 m² project at the top of a chairlift. This project has been completed. It has already been submitted to the canton, the municipality and the landowner. It will go to public enquiry at cantonal level at the end of June. If the park goes ahead, it will supply 2.5 times more energy than is consumed at altitude. All the cannons, all the restaurants, all the facilities. There's bound to be opposition, we'll see. It's built, it's around the chairlift, but here we have the test plant for Switzerland on different types of solar panels, ... or the alpine stuff. Clearly, those who can benefit, front, back, ... with the reflection of the snow, we need 3 to 4 times less square metres, i.e. silicon cells, compared to people who live on the plains [and install solar panels]. Silicon has to be found in the ground in other countries. And it's not a non-polluting process. So in the mountains, you need much less surface area and much less silicon than on the plains, where in winter you have mist, fog and less sunshine. You need to bear in mind that Switzerland has a deficit that runs from October 1 to March 31. From the first of April, there's a lot of sun in the plains, until 31st October. In any case, Switzerland already imports its electricity. So it doesn't know how to make its own electricity. So I think that within 2 years at most, the people who produce in the lowlands, nobody will be taking back their electricity in the summer. Because the electricity will be worthless. This is already the case in Denmark. When the weather is bad, with wind turbines, the electricity is free. Denmark is the same size as Switzerland, give or take 2000 km². It's simple: Denmark has 6,200 wind turbines, while Switzerland has 26. For the same size of country. So you need to know that it makes more sense to install in the mountains. You'll need fewer square metres to achieve a result than on the plains, where you'll need many more square metres. And above all, well, that's a lot of raw materials that you're going far away to extract in another country. You have to think about the consequences elsewhere.

[break]

On the other hand, building alpine parks in a pristine environment is just wrong. You have to build next to a dam or a chairlift, where you've already got snow cannons and chairlifts anyway. To me, that makes more sense. Particularly in the mountains, it makes sense because in terms of energy efficiency, ... in fact we had it all wrong. In itself, wind turbines make noise. I had a farm next door. You don't hear anything at all, 10 times less noise than in town. But ... what's wrong is that wind turbines should be in the open, because the air is denser. It's more efficient. Solar panels should be at altitude, where the sun is stronger. But no, we're doing it the other way round, we're putting the wind turbines in the mountains where the air is weaker and less dense, and we're putting the panels on the plains where they're less profitable. So I'm not sure we're doing the right thing.

CA: And do you think the Alpine Park has a chance of being realised?

X: Well, there aren't many projects in Switzerland. Unfortunately, you can count on one hand. Because for the moment, what has been criticised, what has caused fear, is the Grengiols-Solar project, the very large 600,000 m² project in the Upper Valais. The project has changed in size. The criticisms that have been raised concerning the environmental and the power transmission system won't be the case here because there are already existing lines. So normally, the comments and criticisms they have made about the project, which has already been put out to tender and which will be ... opposed, don't concern us. We know what we have to do for the environment because we've been doing it for a long time with our facilities. It's something new for us up there. We'll have to see. I don't know, ... in my opinion, they're going to oppose out of principle. But we've had two positions, we've had a national councillor, [councillor name] who said go ahead, you have to go, it's a project, ... If it's a valley that has to do it, it's yours. And 6 months later he comes along and says "ah, I've changed my mind a bit, you should put it on the roofs of houses in the plains". Switzerland needs electricity from October to March. So it won't work on the plains.

[break]

So you can imagine the plains, well, you can see the difference, everything in [canton name], and I know that well, ... sometimes you can't see the sky for a month. If we have a power cut in winter, we're not going to be well. We voted to dismantle the nuclear power stations, and in 11 years' time there should be no power stations in Switzerland. If we continue to oppose everything, I don't see where we're going. Today, Switzerland depends on French nuclear power.

CA: And do you have other renewable energy projects?

X: Yes, apply in projects here, professionally. Then, the only thing that really pollutes in a ski area is the snow groomers, the rest is all green electricity. At the moment there are only electric groomers for the small [slopes], we tried to get one, but it was impossible. And then, in my opinion, the big ones will

be fine the day hydrogen is mastered. It's a machine that's already been on test for 2-3 years. Personally, I see the future of large hydrogen-powered vehicles, even cars, rather than electric vehicles. Hydrogen requires a lot of electricity. It's the hydrolysis process that takes a lot of energy. Electrolysis, sorry, electrolysis.

CA: So we're going to need solar panels anyway to produce the energy, is that what you mean?

X: Take Valais. It's been 40 years that the Rhône could have been turbinéd 7 times, right? The Rhône has never been turbinéd because the Greens opposed it. So the water flows for nothing, so to speak, and you can't, ... So yes, there are fish and all, but you can do what we call ... dams and locks to bring species upstream. So you can manage that. The Rhône has never been turbinéd, and that's a huge waste. That's a telling example of what we say about not being able to change. There are lots of things we could do, but. We need to stop opposing each one of them.

CA: And then, during the whole adoption process, was there a time when there were particular difficulties?

X: No. Zero. With the cantonal energy office, I think it went through without a hitch, there was no criticism or anything like that. We also signed an agreement with the Federal Energy Office in Berne to reduce our emissions by 7% over 10 years. Which isn't easy, because there are a lot of processes involved, but ... No, we haven't had any problems at all.

CA: Have you ever heard of Swisstainable? And are you affiliated with them or not?

X: In other words, the problem isn't that we don't want to, but then what good does it [the swisstainable label] do you? We analysed how our restaurants could be as clean as possible. We've carried out tests with disposable, compostable and reusable plastic crockery. In fact, you're not allowed to compost it because it's soiled. We were fined. You're not allowed to recycle it because it's soiled. We were fined. When it comes down to it, the problem with reusable plastic is that after 3-4 years it's grey and you can't use it any more. It's so thick, it creates as much waste as the disposable plastic you buy. We've come to the conclusion that over the next 5 years, all the restaurants at altitude, especially the refreshment areas, should be equipped with a powerful washing tunnel. We'll only be using porcelain, metal services and glass. I can see what's most environmentally friendly for me. At least we've done the test all the way. And then we're down to paper napkins, um, ... and then we're trying to go through the process, where we have virtually no waste whatsoever.

CA: Are there any aspects that I didn't cover during the interview that you'd like to share?

X: No, but for you, your aim was to talk about installing solar panels?

CA: Yeah, I'm writing about the adoption of renewable energies in alpine resorts, with a focus on photovoltaics.

X: No, actually, because in fact, all the ski lifts in the whole country consume around 0.34% of Swiss energy. Cannons account for about half of that, so 0.17%. Private heating is the biggest source of pollution, accounting for 38%. At home, we've decided to lower the house temperature to 19 since it was recommended by the Federal Council. Currently, 3/4 of my colleagues have lowered the temperature from 22 to 19. One degree in a house represents 7% of your energy. So when you go into your room [laugh], in St. Gallen, if everyone was at 19, there would be enough energy. That's what's a shame, it's that... In fact, we're an industry that consumes extremely little electricity per person transported. People have trouble understanding that. They always think that it consumes a lot, and then when I talk to journalists, they say "Oh, but in fact it doesn't consume much", well we know that, "but why does everyone say..." because one of your colleagues once said that and people are convinced that lifts are big consumers. The Geneva water jet is a huge energy consumer. For decoration, that's huge. All the ski lifts in the whole country consume less energy than the hotels in Zurich. Per day, you know, I don't know how much we have. I think we have 8 million ski visitors in Switzerland, and that consumes less energy than hotels in Zurich. I've got nothing against Zurich hotels, but that gives you an idea. There's an industry working down in, ... a big aluminium industry working down in [city name], and on a big production day, it consumes a whole year's worth of electricity for the whole of the canons, restaurants and facilities.

CA: Yeah, I also read some statistics about the fact that going to the station is polluting.

X: It's travel, not necessarily to the resort. It's leisure travel that consumes energy. You go golfing, walking, ... All that, that's what consumes a lot.

[break]

People are always saying that we're going to be able to cover the houses on the plains. But I don't think the collateral damage in other countries when we put them all over the place is fair. I'm not sure it's fair to go and use 4 times as many noble materials for car batteries or for the silicon cells that make the, ... that transform light into electrical energy. I'm not sure it's right to go and do that in South America, when we could be making other systems here locally, ... or limiting the consumption of these products.

---- End of Transcript ----

Participant 5

Interviewee: Member of the management in a Swiss Alpine company and of the municipal administration in a Swiss commune

Interview date: 8.04.2024

Form: Qualitative, semi-structured, online (zoom)

PV Adoption: Yes, several solar PV projects and a solar park project in the pipeline

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Who are the people involved in the decision to install the high alpine solar plant in [commune name] and how much influence do they have?

X: Yes, so the project is in the municipality of [municipality name] and therefore the influence of it is also very high, such as where, what and when we realise. We have two partners on board for the project: [two energy company names in the region]. The influence of the municipality is very high, not in the financing, but the land belongs to the municipality and therefore the influence of the municipality is high.

CA: And do you think there's a category of people that might be problematic, like people who are concerned about the environment?

X: Yes, there certainly is. But the left, I mean, the left in the canton of [canton name], said "yes of course", if we want to build plants in pre-polluted areas, they would agree. And that's what we have in [municipality name], which is an area used for tourism, so it's very heavily polluted. But the area has

many other advantages: it is accessible by road. We have already built part of the power transmission lines in advance. And we have a large own consumption possible with the mountain railway in [municipality name]. The project would produce electricity directly where it is used. It's not quite comparable to, ... at least I think it is not comparable to a greenfield project, in an unspoilt area like in the [neighbouring valley name]. The people said no, but the areas are very different, it's an unspoilt area. The population there is much more critical. But in [municipality name], it's really already polluted with restaurants.

CA: And what are your motivations for starting the project to install a high alpine solar plant in [municipality name]?

X: The original motivations were the mountain railway. The mountain railways use about 2.3 gigawatt hours of electricity per year, which means that about 25%, 20-25% of the system could be used directly for the operation of the mountain railways. Another argument is that our aim is to organise our electricity requirements autonomously. The municipality of [municipality name], together with the mountain railways, needs around 9 gigawatt hours of electricity per year. And with this new system and the existing PV systems that we already have, we would generate around 13.5 gigawatt hours of electricity. Of course, the municipality of [municipality name] would be spread out over the year, so it's not possible [to be entirely autonomous]. We also need electricity at night. But in the summer [municipality name] would then be a place that produces more electricity than it actually consumes.

CA: And you can sell the electricity afterwards?

X: Yes, so we would envisage here, in this plant, that the mountain railway would have a preference, that we first consume the electricity that the mountain railway needs. We are now also looking at a project to build the transformer stations in such a way that we can feed the electricity directly into the municipality of [municipality name] and sell it there, possibly a little more cheaply than usual, because the line charges would be eliminated, and the surplus electricity would then be sold on the free market.

CA: And do you think that initiatives like this with the PV facilities could be a competitive advantage to bring tourists into the commune?

X: Yes, so it's certainly an advantage for the municipality, for the mountain railways also an advantage if you can work out with the argument, how you are, ... practically an electricity-autonomous municipality that produces the electricity itself. I do believe that this gives the municipalities a competitive advantage.

CA: And how did you find the information you needed to introduce solar panels in the mountains? Was it complicated to find that information?

X: Yes, we have very early [started looking for information]. We have already had the first thoughts for these PV systems in 2017. We then built a small system in 2019 with 130-140 thousand kilowatt hours, but we already had the vision in 2017 that the [region name] area would be very suitable for a PV system. There were various reasons for this. So, my son is also very involved in PV systems, and we have actually been choosing this since 2017. Of course, the, ... Ukraine, the tightening of energy availability has strengthened this idea very much and we then went through various electricity providers, be it [Swiss electricity company name], [electricity company name in the region], [electricity company name] and actually received a lot of support there, because since the war in Ukraine, the electricity demand is no longer fully secured and above all the winter output in Switzerland is lacking. And the [municipality name] PV system, which has a winter share of 47%, is actually very high for a PV system.

CA: Do you think you will manage to get government subsidies?

X: Yes, the whole, ... so the municipality can't build these on their own. It is of course an investment of 38 million, so we need an [electricity company name] and [electricity company name] to support us there, of course. But it's also the case that if you look at the PV system and the projects with this *Solarexpress*, the best-case scenario is that 60% of the subsidies will flow. That is already the case [we know we will get them]. But until these subsidies flow, it's probably 2032-33 and in that time, someone has to be able to lift the 38 million and that's why we took [electricity company name] and [electricity company name] of course, as good partners.

CA: And you said that you've done PV projects before. Was it complicated to run those?

X: No, that's, ... We've built rooftop systems, even [built] one in [area name], in the middle of the ski resort, and that was very easy. We got the one-time compensation, and we built the system and connected it to the grid, it was very easy.

CA: And it's because it's private or on existing buildings?

X: Yes, it is because it's often [already] built. The Left and the Greens are of course much more sensitive to open-air installations, and because we did them on a roof, there was no resistance at all. Because we built them in 2019, [it was] also a minor matter. Back in 2019, nobody put up any resistance or anything. It's only since 2021-22 that this has come about.

CA: When you had all the information about the solar park project, what were the considerations and were there different from the ones at the beginning of the project?

X: Say that, the basic motivation is always the same, but in the course of this time, of course, quite interesting aspects have been added. Partners, rope variants and different construction approaches. For example, there is the construction variant, which was already very exciting and new criteria or new

considerations have been added. But nothing has changed in terms of the basis, we still believe that this is the best, with the production costs. That is of course a problem [the production costs]. A year ago, they weren't great and today with electricity prices of 10 centimes per kilowatt hour it's actually difficult to achieve a return. But I don't think it's just the return, that has to be there, but it's also the electricity security that we get in winter and that can also cost Switzerland something.

CA: And was there a difference of opinion about the evaluation of the project among the different decision makers? Are there different opinions in the community?

X: So, it comes along the way. Where we are today, there are certainly differences of opinion, but of course they have been discussed out again. In the beginning, there was this rope variant from [...] that nobody actually believed in, including them. It was changed, rather sceptically. And the municipality held on to this one as much as possible and in the end an agreement was reached. There were issues and ... that had to be discussed. But it was very constructive and goal-orientated.

CA: And what are the biggest difficulties you see in this project to date?

X: Well the biggest difficulties is the canton, the licensing authority and of course the potential objectors to these projects. That is certainly the biggest challenge, because there are huge documents to draw up, you would hardly believe it. There are thousands of pages that have to be provided, which is of course the most difficult hurdle to overcome. And that's where we are at the moment, we have received the preliminary opinion from the canton, the positive preliminary opinion, and are now in the process of submitting the planning application.

CA: And do you think the project has a chance of getting through?

X: If [municipality name] isn't accepted, none will. There are various reasons for this, such as the zones that we are building, the power transmission line that is already in place, the road to the area that we already have, the restaurant at the top that can accommodate people during construction and the high level of demand. I am convinced that if [municipality name] is not built, there will be no PV system in [canton name].

CA: And do you think the project is important for the community?

X: Yes, certainly the, ...the mountain railway is 62% owned by the municipality. It is certainly important for a secure power supply and the cheapest possible power supply, but I personally also see it as a huge advertising opportunity if we can say that you make snow on our slopes with drinking water and electricity from renewable energy. I think that's a good argument for guests in the future. I think awareness of these issues is growing and I'm very confident that we'll attract a lot of guests if we can use this as a promotion afterwards.

CA: And did the people vote in the commune?

X: Yes, so the vote. Yes, we voted on this about four months ago, ... about three and a half months ago, in the castle for the primary assembly, in [municipality name], and there were about 102 people there, and it was a unanimous yes. One single vote against.

CA: So who can stop the project now?

X: So from the purely factual point of view, it can only be the building permit. If we have the building permit, we then have to conclude a building rights contract. These are the modalities for the municipality, but there are then separate details. If we have the building permit, if this hurdle is taken, then it is actually no longer to stop. The only thing would be if those with investors came to the conclusion now, that the economic viability is not given. We would have to look for solutions, but we ourselves are not against it [no economic viability], so I think that if the building permit is granted, it will be built.

CA: And is there any aspect that I haven't covered that you would like to share?

X: No.

---- End of Transcript ----

Participant 6

Interviewee: Member of the management of a Swiss Alpine company

Interview date: 23.04.2024

Form: Qualitative, semi-structured, online (phone call)

PV Adoption: No, but solar PV in the pipeline on a renovated restaurant

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Do you intend to adopt renewable energies at [company name]?

X: Yes, so we have one [project] that we're going to implement this fall, it's photovoltaic, it's solar. With the reconstruction of the [restaurant name] restaurant, we have about 600 photovoltaic panels that we're going to put on 3 sides of the building. Now, I don't know how much we're going to generate, ... I think it was around 100 to 105, ... I don't know how much we're going to generate. I can't remember. That's the first thing. And then after that, we're looking into the possibility of doing something with the wind at the cabane, the intermediate station. We've got a lot of wind there, so generating energy there [would be good]. And a slightly longer-term project would be if we put mechanical snowmaking from the [area name] at the intermediate station to the [area name], ... there's a natural lake that's going to be created on the [canton name] side, from the [glacier name] glacier, a lake of around 250,000 cubic meters, created with the glacier melting. And the idea would be to combine the snowmaking that we're going to need in 5-10 years' time, even at these altitudes, with a turbinning. The lake would be around 2700 meters, ... I think rather 2800 meters, and the first turbinning at 2500 meters and a second turbinning at

1500 meters. So in other words, we'd be using the infrastructure, ... the snowmaking pipes that are only used for a week to 20 days a year. We could use them for 6 to 7-8 months a year to create electricity.

CA: Right, and at the intermediate station, I did not quite understand, that would be wind power?

X: So this mechanical snowmaking project, using the same infrastructure to create hydraulic energy, would be a turbine at the cabin and then at the intermediate station. We're looking at other projects, but for the moment we're not looking at putting in photovoltaic panels. But then there's the [...] behind, which hides a lot of sun during the day, so it's not really ideal, and a second solution could be to use, ... There's a lot of wind there, still, so see if we could set up not large structures, but small vertical or horizontal ones. I can't remember which. I think it's vertical. As it turns, it generates energy too, but that's just an idea, we haven't gone any further yet.

CA: And what are your motivations for adopting renewable energies?

X: Well, first of all, it's to use less energy, so already save energy. And secondly to have clean energy or as clean as possible. Also, all the energy we use for [company name], it's all water energy, from Switzerland, so hydraulic energy. And we changed that 2-3 years ago. We are on the free market, so we have the Swiss hydro energy price. And then, clearly, it's also a question of price, but that's secondary.

CA: And in terms of profitability, do you think that in the long run it's still going to be a profitable investment, or is it really this aspect of being more ecological?

X: I think there are two things. One is the guarantee of energy, and stability, we know the price. And the second point, I don't think it's really going to be much cheaper than energy to buy on the markets, because then the price will drop very sharply again. And I don't think it's primarily the financial aspect, it's more about being as self-sufficient as possible with natural energies, if you want to put it that way, and always having the guarantee of having enough energy to run the cable car, et cetera, to be as self-sufficient as possible.

CA: And then, do you think that initiatives like this could be a competitive advantage, for example in attracting tourism?

X: 5 years ago, I would have said no. Now I'd say yes. Not directly I think, it's just, is it, ... the excursion point it's still pretty special I think. In a hotel, I realize with my own experience, when I have 2 or 3 hotels, ... I don't know if I go to somewhere in Zurich and I see 3 hotels that I like and then there's one that has the echo thing, a label or something that's very strong. Clearly, I'm more attracted to that one. Now the lifts, if you want to go skiing in April, and there's only the glacier open or if you absolutely want to go on [resort attraction name], will that make a difference or not? I'm not sure. But it's all positive, isn't it? And secondly, in the longer term, it's clear that we don't want temperatures to rise too quickly, so we're also prepared to make savings, to try to do our bit to ensure that global warming doesn't

happen so quickly. Then how can I put it. There's energy creation that we've just discussed and the second thing is also using less energy. I don't know if that interests you too or if it's really just energy creation?

CA: Yes, also the energy reduction.

X: Deviating from the conversation a little further, we've set up a production company at the [area name]. We've bought the restaurant at the [area name] and we've redone the kitchen. Firstly, with new tools that use much less energy. And secondly, we've installed an intelligent system that manages peaks of the machines, that manage each other, to avoid big energy peaks. And with that we manage to save quite a bit. Secondly, we have a project to change the motors on the two cable cars. They were old motors. And now, between the, ... the grey energy, which couldn't be re-injected into the system... And now we're using less energy with these new engines, these new systems, and we're not creating any grey energy. There's just one project we'd still like to implement, and that's at the cabin. To see how much it costs and when it's possible, we'd like to take the energy when the cabin is full, store it in batteries directly at the cabin and reinject it when the cabin leaves. This means cutting off the peaks with this energy that we can produce on site, just store for a few minutes or a few seconds, and then reinject it. And that's a project we're looking at right now. And then, just to give you an example, we also try to groom black runs when there's snow, [but] when it's good powder skiing, we're going to make the slopes safer, but we're not going to groom them. And with that we also gain or we can reduce quite significantly the use of grooming oil.

CA: Right. And then, when you have to make decisions like that in your company, who is involved in the decision-making process?

X: So it depends on the project. Now in the kitchen, it's clear, it's myself who was involved, it's the restaurant manager and it's the kitchen suppliers, ... the kitchen designer. So it's the specialist, let's say, still a generalist, the company, plus the specialist afterwards from our in-house company. If it affects the cable car to the building, it's often with the technical manager of [company name]. And then it depends on how much it costs, it has to be validated by the Board of Directors, either by the President or the Board of Directors. We also monitor energy use with [name] who is our specialist who's been monitoring with us for I don't know, ... 3-4 years now to try to find solutions, on how to reduce our energy consumption.

CA: And in these different people who are included in the decision, are there differences of opinion on adoption?

X: No, that's normally good. I mean, when you're doing something new, it's easy to isolate things better, to do things better together. When we did the new [restaurant name], I would have liked to have seen a [restaurant name] that was completely covered with photovoltaic. But the architect wasn't happy with

that, because for him, a facade was alive and well, and that's also a question [for discussion]. And then there were the fire safety guys, who weren't too keen on having all the facades completely covered in photovoltaic panels. And sometimes it's also a question of age, I think. Younger people are much more sensitive, and it's people who are, let's say, 65-70 years old [who disagree about such installations]. Sometimes they're a little less concerned about renewable energies. Or saving energy. We realize that too. But between us in the team, no, there's no objection, we try to find the best solution. That has to play a part, because at [high altitude] metres, you can't just do tests. We have to go for things that hold up, that work. Apart from that, it's going very well.

CA: Is it complicated find the information needed to install solar panels at such high altitudes? How do you go about it?

X: Yes, I thought it was still on the edge of feasibility. But there aren't really that many suppliers doing it. There aren't really many panels that can withstand these winds. The price is also very high, it's not comparable to a system on a, ... on a flat roof. So we're hoping it will work, but it's not an exact science yet. But I do think that the potential is there. And if we want to build, it would be completely stupid not to use facades to at least partially install photovoltaic panels.

CA: Right. And are you comparing it to other projects that have been done before to see if it's going to work, to see what project might be possible?

X: There aren't many projects over [high altitude] meters, so we looked at Zermatt. The specialist who did the follow-up on this project or who did this project, he also knew the Zermatt part well, the Kleine Matterhorn. And there, it has to be said, there are also quite a few problems in the early years. Now it works pretty well. And then there are one or even two suppliers who can deliver the panels, the solar panels. There isn't a wider choice.

CA: And then, what is your decision regarding the external communication of these projects?

X: We held a press conference and there we communicated on these panels and that the restaurant should be more or less autonomous. After that, the other projects are ongoing. We haven't communicated on the smaller projects, such as changing the engines and so on. But I can see that more and more companies are doing so, and maybe if we had a tab, an environmental tab, where we also put the little projects we've made over the years with the savings [it would be good]. To be able to communicate more on that, I think it's becoming more and more important too.

CA: And what are the biggest difficulties you see in installing these solar panels at a high altitudes?

X: Stability. You've got winds of up to 250 km an hour. Snow can get in anywhere there's a small opening. There's also the problem of the strength of the sun. That means if some of the panels are in the

shade or if there's snow sticking to them, you have to be careful not to overload. The sun is very, very intense.

CA: And what are the solutions to this snow problem?

X: We put it on the facades because we can't put it on the roof because it's under the snow. We have snow 7-8 months a year, at [high-altitude] meters. That's why we went for facades and then we worked with the engineer, the supplier, we calculated, we found solutions specific to the location. It's not standard stuff to guarantee the highest level of safety.

CA: And are you counting on government subsidies?

X: No, there are no subsidies for the [restaurant name] restaurant. But yes, it's clear that on other projects, it's often quite large amounts. And to encourage, ... it always depends on the price of electricity, ... it was around 4 cents, and then electricity prices have risen to 30-40 centimes from one year to the next. Clearly, I think differently. But it's, ... I think it's good that there's help to launch these projects, not to finance them completely, but to break this peak of financing a bit too.

CA: And in general, do you think it's important to start installing solar panels at altitude?

X: Personally, I think they should be installed on existing buildings. Down in the valley and up in the mountains. You should really do that first. After that, I'm not sure about putting panels in a completely untouched area, ... like solar parcs, ... I'm a bit mixed. It's good for creating energy but it's often, ... I think, there are two things that don't really go together: protecting nature and the landscape and environmental issues. You understand, it's like it is for us, if we want to be the most ecological, the most effective, to groom tracks, we need "highways". We have to break every stone, everything, it has to be flat, completely levelled. And then we need 20-30 cm of snow, we go over it once and it's good. We take the example of the [...] which is full of stones, we have a canyon there. We need to put 5 to 7 meters of snow in it. It's full of holes, we have to dig out stones. We need 2-3 meters of snow [to cover it up]. It's clear that we're burning fuel. It's all about having pretty, ... how can I put it, ... using as little machinery as possible, while breaking up nature. And on the other side, we have to keep the beauty of nature, the stones, the flowers and all. But it takes more energy to clear that track. And then there's the question of whether we should have a track there or not... that's a third question. It's always a bit contradictory, I think. The two things are not always, ... protection of nature is not always equal to using more renewable energy, being more efficient.

CA: So you're not planning to participate to the construction of an alpine solar park?

X: No, it's complicated here. We had 12 meters of snow this year. We have freezing rain up to [high altitude] meters, so that means we have a lot of ice. In November, December and January, we had tons of ice around the cables, and so on, because it was snowing and the temperature was rising. There was

a lot of wind. It was getting wet, and then it was really pure ice, which broke everything. And it wouldn't be so easy to set up too with the avalanches. So on the flat we'd have too much snow. We'd really have to put them at altitude and on the slopes at [...], ... for example in [...], just behind, there, that could make sense because there's the sun in the evening. We really need the energy, ... but it's clear that it's an ifp zone [most beautiful landscapes in Switzerland], and I don't think they'd be happy to put panels there. And there are avalanches.

CA: Okay. Then I saw that you've been awarded the Swisstainable level 1 label, and I wanted to know if you think this label can give you an advantage?

X: So it's a pretty easy entry into it. It's not complicated to receive it. So we've done it, but I think it's good that the first step isn't complicated, because then you've got 1 and you're interested in 2 and 3. But you [the label owner] also have to be careful that it's not just a logo you can buy and it's useless. Yeah, you really have to be careful.

CA: And you think that would be a risk?

X: I think there's quite a lot. Yeah, I think, like everywhere else, it's a question of companies wanting to be the most, how should I put it, ... the best seen on the outside, but that the logo is just something you buy and it's not much use. Is there really an overall vision behind it? I think there is in every field. Swisstainable I'm talking in general terms, I couldn't give you an example [of a company] who wouldn't do it properly, but there's a danger anyway.

CA: I'm done with all the prepared questions, but I wanted to know if there was anything I didn't cover that you'd like to share?

X: I can't think of any. No. I think you've covered enough, but one thing that's very important, I think for us, is the question of efficiency, in energy. It's about trying to be as efficient as possible with what we do. That's one of the big points, and [energy] creation really has to be done where it makes sense. I don't think we should each try to create our own energy. I think it's better to get a few of us together [to build a project] and use the energy [together]. After that, it's not so simple with the laws and so on. Solar farms or whatever, they're highly regulated.

CA: Ok, just a little clarification on what you've just said, when you say "getting a few of us together", what do you mean?

X: Even then, ... Well, I don't know because if we put in photovoltaic panels, just to take this example, ... but I can't give you the answer, ... we have photovoltaic panels at the [restaurant name] restaurant that get the sun until 9:10 in the evening. We use no more energy from 06:00 because we've stopped everything. Ideally, we could use this energy ourselves. So the network is ours up to the [area name]. But ideally, we could use this energy to run the hotel in the evening until 9-10am. But from a regulatory

point of view, I don't think that's possible, because then we're on the [network name] network, so it's really, ... it's stable if you're on your own network. But as soon as we go beyond that, the only thing we can do is resell the energy. Well, I'm not a specialist in that. I haven't really thought about it yet. As we're not an energy reseller at the moment, we're more of a buyer.

---- End of Transcript ----

Participant 7
Interviewee: Member of the management of a Zermatt Bergbahnen AG Interview date: 24.04.2024 Form: Qualitative, semi-structured, online (zoom) PV Adoption: Yes, solar PV on the “Kleine Matterhorn” arriving station in 2009

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Who was responsible for the decision to install solar panels on the Kleine Matterhorn and were there differing opinions on their acceptance?

X: Well, you are referring specifically to the Kleine Matterhorn project. Altogether, it was actually the first project with solar panels and that was in connection with a new project, a new restaurant and new infrastructure. Yes, that was a decision by the management and, of course, the electrical engineering department indicated that this could perhaps be examined, and the decision was actually made immediately that we could realise it in this way. Yes, and then we included it in the project [the solar panels]. And we installed solar panels on the entire south-facing façade and put it into operation. And the whole thing was actually also in connection, I have to say, in connection with the Minergie P, consideration. The whole restaurant building has been equipped to Minergie P standard and so of course, the kind of discussion or the idea came up, that we could also include renewable energy. And equipping the entire south façade with a photovoltaic system. That would actually be the decision or the consideration.

CA: And what were your motivations for starting the project to install solar panels on the Kleine Matterhorn?

X: Well, the motivation was certainly that we were offered the opportunity to have a south-facing façade. The restaurant could be built that way, and it would be a complete façade. You have to imagine that the façade is actually equipped with an exterior façade and an interior façade. And in between you have to imagine is like an air canal, which means that the [...] air is warmed up. The warm air is then utilised in the canal and converted into heating, hot water, etc. and equipped with photovoltaics on the front and integrated directly into our electrical system. In other words, thrown into the electrical installation. The motivation was also that at this altitude, 3800 metres, we achieve an extremely high level of efficiency with our photovoltaic system, i.e. we are talking about up to 80% more efficiency, more utilisation, more production than at sea level, for example, at 0 metres above sea level. The optimal orientation to the south and also with an inclination of 70 degrees was able to provide optimal conditions for this façade, installation. And a very big decision was also the reflection through the glacier, i.e. through the snow. This means that we have production even when there is no sun. Purely through the radiation reflection from the glacier. You don't have that anywhere else in a green field, or where this radiation is not present. And on top of that, we have a very high and long sunshine per day. So, sun early in the morning until later in the evening. So, there is a pronounced energy production, and it is far above the production, for example, at sea level. That was actually our motivation. And then I also have to say, I've looked at all the questions, perhaps I'm coming a bit too soon, but there was also one other consideration. We have to bring electricity energy up the mountain to 3000 almost 4000 metres. So that's also ecological and economical of course. We also questioned that and then we decided that if we build a photovoltaic power plant up there, we can use the energy produced directly. Short, so no transmission lines, but produce and use it directly on site, because this is purely daytime production. Solar energy is mainly used during the day, but I have also said that due to the radiation from the glacier, there is also electricity production at dusk or so. But we only use it during the day because we only have the restaurant open during the day. So, we use the production directly in our daily operations. For the first question, that's it. But that's a bit part of the whole project at the [restaurant name] with this restaurant. That's actually the motivation and also the considerations and the decision to take this step, which we did.

CA: Additionally, you realised the project in 2009. Did you already think about the competitive advantages that this facility could bring you back then?

X: So perhaps we didn't do it now because of this motivation, that the tourists, more tourists would come or something or [that we] could identify with a label. Rather, I need to add something to it, that you need to know, is that we were cooperating with an energy company at the time, with a large energy company. And that is a cooperation, a collaboration, so we actually provided this façade and [electricity

company name] actually made the solar package. It's a co-operation. We provided something, but the costs, or rather the production costs, were then borne by an energy company, of course, with the background that they can, of course, market renewable energy. Do you understand this connection? We later did other projects with our local energy suppliers and that actually had the same effect. They have a solar power plant here in the mountains. And [the solar PV were built] in our infrastructures, that means they didn't have to go to any great effort with any licences, because the licence was already there. Because the facade and the building were already there, and nothing special had to be built. In addition, we simply co-operated with energy suppliers who have shown an interest in investing in renewable energy and then marketing it. And of course we have the utilisation. Of course, we also have this, I'll say, for the public or so for our marketing, of course, we can show that.

CA: And for the other projects that you have at Zermatt Bergbahnen AG were the motivations similar?

X: Exactly, our goals were always when we were organising something new, making new projects, for instance [a new] building, operating buildings for machines, garages, but also lift stations [then we would install solar PV]. We have built new cable cars. You may know the Alpine Crossing. We now have the first section from "Trockener Steg" to "Kleine Matterhorn" and last year we made a second section with Italy to the Italian border. And these are buildings and we have also integrated this photovoltaic system into these new buildings. So, we pursue every new project that we do, or conversions or renovations, we try to somehow integrate such renewable energies, photovoltaic systems. But we've also already installed solar thermal energy at two locations and if it's ecological and economical, of course, we have to think economically, don't we? What really helps is not to install 2-3 panels and to realise a small ambience with a small lamp, no! Of course, we would like to look at the whole thing and make sure that it's economically, also commercially thought through, it also brings something. And to round off the question, we check every project, every object, to see whether there is any possibility of realising such things.

CA: And for the Kleine Matterhorn, how did you find the information for the installation? It was one of the first projects, wasn't it?

X: Yes, exactly. Then of course with the cooperation with [energy company name] we also had external advice from possible suppliers of photovoltaic systems. It was special, nobody had any experience of installations at this altitude. I'm talking about 4000 metres. We're talking about external conditions such as wind, let's say, wind speeds of up to 250 kilometres per hour. We also had to take into account the mechanical stability of the façade. And then we also had to install special PV panels. We couldn't actually install conventional panels because they weren't suitable at this height, so that was a challenge. Nobody could tell us exactly because there was no experience at this height. So we already had external advice, external, yes specialists, who stood by us.

CA: And since you were the first to do this, did you have companies that later asked questions about how you implemented the project?

X: The interest and demand for the realisation of this project in general is Minergie P, but photovoltaics has also triggered enormous demand. So, there were weekly requests for a tour, for explanations, for enquiries, for information and so on. It was a showcase project "par excellence" and there were really a lot of enquiries. How do you do it, what experiences have you had and so on? What kind of products did you use and so on?

CA: And have you also received questions from companies that want to build alpine solar parks?

X: Yes, the interests are there, yes, there are interests because, um, [happy] ... there are various projects that are now known and are being examined. And then perhaps I will come to realisation because various components or various things are not, ... just not optimal or something. But the altitude, the mountains, of course, the production, the economic efficiency, the energy that this can bring is of course enormous. Higher, so even more than at other altitudes, for example, and that's why it's actually quite in demand. Yes, the locations are being examined very, very closely, now yes.

CA: And did your original motives change after you had found the necessary information for the Kleine Matterhorn?

X: Yes, the motivation or the goals and what the consequences are, have actually always been the same. I have already mentioned that we are examining every possibility of somehow utilising such plants and such PV areas and then also implementing them. In the meantime, if you're interested, we have an annual production of around 400,000 kWh per year with all the systems we have in operation. Oh well, that's not little, so yes. You'd have to think we'll soon be able to be self-sufficient, no? But of course not, because we do have a large energy requirement, of course, with the railway and snowmaking facilities. That's clear, but we can already cover a fair bit with our production.

CA: And were there differences of opinion about the assessment of the project among the various decision-makers? And why?

X: Actually, we had, ... we actually have all the facades integrated now, ... buildings are not like detached or outside, [where it would be problematic landscape-wise] ... no that has really, that has had no, ... objections from any institution. On the contrary, it was always welcomed that you could use it, wasn't it? Instead of a just a facade. I have to say, I'm a fan of course [of the PV façade], and that's why I say a PV system, a PV façade looks really cool. With glass too, partly with glass, and then the daylight gets into the floors where the restaurant is, for example. It's a nice combination, isn't it?

CA: Yes, that's right. And are you happy with your investment and the process of buying solar panels on the Kleine Matterhorn or were there any problems?

X: No, we're actually very satisfied and happy, because there's practically nothing to worry about, the operation is, ... it all works, ... yes, you don't actually need to do anything. So let's say, let's be honest, expectations are generally high. But otherwise, the investment has paid off and we are also happy with this investment and the production also tells us that we are on the right track, because we can draw about 80% more production from this plant than we could at sea level.

CA: You also have other PV system installations. Are you satisfied with these? And was it easier to do these other projects after the Kleine Matterhorn project?

X: Exactly, then of course we had experience. Also, with all the wind loads and so on, how does this facade have to be constructed so that it can withstand wind, cold, moisture, you see? And that's where we've gained our experience, of course, and we've been able to use it in other projects and implement it successfully, yes. We didn't change much because it was actually ideal.

CA: And have you publicised your investments in PV systems on the Kleine Matterhorn to a wider audience and do you think it's important to share this?

X: Yes, of course, we also publicised and announced this to the outside world in connection with the Minergie P investments and the demand was, as I said, huge. And we have this on our website. We also publicised this with the ISO certification, of course. We have the management and environmental certificate, which is... ISO 14001. We are now doing a replacement for the sustainability certification and the topic has now come up again. So, I'm in the middle of it now [happy]. To prepare for the audit, to prepare for the final audit and at the end of the certification audit. We have also prepared a, ... so to speak, ... a sustainability report and there is a sustainability report, which will then be published. And then of course this information, all these considerations and everything are in this report, and it will then be published, it will then be available to the public on our website. That will be around summer 2024, you can look it up already. Unfortunately, I can't tell you any more about it yet because I'm in the middle of the certification process. So, there's also [the answer to] your question, we'll publicise it. We also want the public to be aware of this, that they know about it.

CA: Would you recommend the introduction of solar panels in the mountains in general, including in the other projects you have?

X: Yes, I would definitely recommend it. It has to be said that the economic efficiency is always a different question. Because the procurement and the whole thing has to be taken into account, apart from choosing the location, the realisation, the institution and when will it be profitable. And these are already far-reaching investments. So, these are investments of 10-20 years, you're already reckoning with several years, or even more, yes, ... exactly, so in any case, because the, say now, ... the use for the energy production is several times greater than in lower regions and therefore it will also, ... it will also be economically, let's say, interesting, yes? But you have to be convinced, and we are convinced and

we would certainly like to see further projects, larger projects, [happy] ... we already have it in our pipeline and yes, you will certainly hear from Zermatt, I think.

CA: And what were the biggest difficulties in installing such a solar system?

X: Yes, not really any major difficulties. Because we have always taken this [difficulties] into account right from the start and included it in projects. And so, we are now building another company building. For the management buildings etc. at the valley station in Zermatt and it was clear from the very beginning that we would equip every possible area with solar energy, with solar power plants. And we are now also doing this in the village, so not only in, on the mountain, but also in the village. And I think it only makes sense that smaller, several smaller properties should also be equipped with solar. I'm just saying that this is the opposite of the large projects, large solar plants that simply exist or are built somewhere in the area. Many small ones also give everyone a large amount of energy production. Because they already exist, let's say, ... the small buildings with facades, roofs and so on. You don't have to give up new areas, give it away, use it, look for it, exactly.

CA: So also build them privately in the village, you mean?

X: For example, exactly yes.

CA: Okay. And do you think it's important for Swiss Alpine companies to produce its own energy?

X: I think so. I think so. I have explained in detail ... that on-site production is of course very interesting - you don't have to transfer any energy. We can convert and utilise the energy that is produced directly on site. So of course, it's ecological and economic, isn't it? So just a rudimentary example, there is a very large solar energy plant somewhere down by the lake. And the energy has to be transported from there to the mountains somehow. And we can do something in the mountains, realise it and use it directly. I think that's a big, big consideration with these things, solar plant projects that are still being discussed or are in the pipeline. It's exactly the same as hydroelectric power plants, you build them in the mountains because the water is there, and you can use it directly. You can transport the surplus and take it further, that's for sure.

CA: The "Swisstainable" project, which was launched by the government, aims to make Switzerland the most sustainable holiday destination. I read that you have the Level 3 label. Do you think this is important for Swiss tourism?

X: Mhm. I think so, because it's really ecologically renewable and I also think what we have in the mountains is sun, water, so also take the hydroelectric power stations as renewable energy. There's also wind. That's an important consideration, isn't it? And where do we have that? We have it all in the mountains. And I think it's also a big plus that we communicate this to our guests and say why we

support such projects or local renewable energy and also push it. I think that's a good label that we also use for our guests, I think in any case.

CA: Okay, so do you think that the Swisstainable label can bring you advantages?

X: I just think it has advantages, but it ... we are signalling with it that we value our renewable energy and don't buy electricity from somewhere, let's say, transport it to the mountains, with transmission lines and so on, and we know ... where some of the electricity comes from, right? The, this is also a consideration of sustainability for us for sure.

CA: And so, are you also planning to build or take part in an alpine solar park, and what is your opinion on that?

X: Well, we ourselves wouldn't get, let's say, a solar park. We have our, let's say, infrastructure that we could make available, for example. That would certainly be a possibility or worth considering, yes. But that would have to be realised by Zermatt Bergbahnen AG, but we can certainly do that with [a partner] ... that we could realise something. I'm quite clear, yes, of course, that we would also support that.

CA: You produce a lot of electricity yourself. Would you consider introducing electric vehicles that can be charged with the electricity you produce? For example snow groomers?

X: Yes, yes, of course we have also looked into that. But at the moment they're still not sure whether it's suitable for our conditions. You have to imagine that we are at 3000-4000 metres. We are in a high alpine environment. The temperatures go far below 0, especially in winter. It has not yet been fully proven that even at low temperatures the performance of such electrically powered vehicles is really useful for our purposes. And we're not quite there yet, ... I think it's also naturally not fully developed for such applications. But of course, everything else, we have electric vehicles for our transport, we have battery-powered forklifts and so on. Yes, everything that is possible and also realisable for our needs, we have of course also with, ... I'll say now, electric, as we also drive iTronic cars to some extent, [happy] ... not only for business but also privately, ... yes, of course. And of course, [village name] is actually predestined for electric vehicles. Because [village name] is car-free. All taxis, all transport, all cars are electric and have been for decades. So, we are a bit infected with these electric drive options.

CA: I've asked all the questions I had prepared. But are there any aspects that I haven't covered that you would like to share?

X: No.

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Participant 8
Interviewee: Member of the management of a Swiss Alpine company Interview date: 29.04.2024 Form: Qualitative, semi-structured, face-to-face PV adoption: Yes, solar PV on departing stations and possible collaborations for a utility-scale PV project as infrastructure owner

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

X: We get them [request for interviews] nearly every week. Afterwards, yes, we'll follow up with an interview but, it's really become, as I said previously, ... the fashionable theme in the ski area. Well, that's normal. But there are still major, ... major margins for progress in [the ski] areas where we can't influence. Like snow grooming.

CA: Yes, snow grooming, there's discussion about going electric, but there are only small machines, right?

X: Yeah, there are only small machines and that's not going to work at all, there's no winch and then there's no, ... well, hydrogen is just at the beginning, just prototypes. Snow groomers are a niche market. There aren't many of them in the whole world, so ... development is going much more slowly than in the car industry, for example.

CA: And have you already considered projects with electric snow groomers or other electric vehicles?

X: Electric snow groomers, we really can't use them because we have an area that requires, ... we have 26 snow groomers. Half of them are winches, so there's already no such thing as an electric vehicle. And then 100% electric is just small machines for making a cross-country ski run. For flat terrain, that's fine, but we don't have that type of activity here. I mean, we have cross-country skiing, but we don't need a 100% electric machine. Then there are hybrid machines that exist, so they're powered by a diesel engine that drives the electric motors behind it, and the savings are really, ... that's borderline greenwashing. There are 2-3 litres of savings per hour. So yes, you'll tell me that out of 26 snow groomers working 1000 hours, that's so many litres of diesel saved. But in terms of ... the machine is heavier. In other words, the technology is not yet sufficiently advanced to make it really worthwhile. So we're not there yet. But that doesn't depend on the manufacturers.

CA: Can you give me an overview of all the renewable energy projects you have at [company name]?

X: We started in 2014 with a project that was subsidised by the SFOE [Swiss Federal Office of Energy] and then a private company in the region that had launched this project. [The aim] was to create an energy monitoring platform for all the company's heating systems, from small sheds to large buildings like this one. This platform is still running today, 10 years later. Every year we add more things. That's the heating monitoring part. In other words, even at the start of an installation, the little shed there [for example], there's a heater. It's very simple, this heating switches itself off at night, it switches itself off as soon as you open the door. It's a small saving but, ... it's the same number of heaters as on the estate, so at the end of the year it's still a good saving. We started with this, 10 years ago now. And then we moved on to more ambitious projects. We've started on the new platform that's about to arrive for buildings, but it's a huge project because more and more buildings are being monitored. But you have to realise that to monitor a building, you need a sensor every time, you need a relay, and you need an interface with the platform, which means that we have, ... we're close to having a team of electricians working for us monitoring these heaters, connecting new things every year so that it's complete in the end. We don't want to have any unmonitored receivers running all summer, because we forgot to turn it off. That should never happen again. So we're still in the deployment phase, but please understand that it's a huge area. There are a lot of buildings. There are buildings, but there's no network, so we can't necessarily get the information, so there's still a bit of room for improvement there. That's the heating part. After that, to save on heating, you have to insulate the buildings. [Company name] is [age] old next year, so it's an old building. And big buildings like this one were built using technology that is obviously not at all current today. But what are we going to do with a building like this one, which houses 3 cable cars? We're not going to knock it down and start again from scratch. So we have to carry out what are known as CECB analyses, and so on, to see where we can optimise heat loss. Interior perimeter insulation, et cetera, et cetera. So that's where we have the biggest challenges, on buildings. So we've already changed everything, ... so for a building like that, it was 80,000 litres of heating oil.

Now it's distance heating. So we're connected to the district heating system, which means we have more production, ... we don't use fossil fuels to heat this building. That's the first step. The main one.

[break]

Then there's reducing consumption. To reduce consumption, you have to monitor it. You have to go back to all the heaters that are 40 years old, ... well, you have to read again. It's a constant battle. And when you look at it [points to an old radiator], I mean, it's like a heating table, it's just bad. It's against a wall that's not well insulated, under a shelf that doesn't allow the heat to spread. I mean, you've got to imagine that it's times X building. You just can't stop [improving]. Every year we have to redo something. So yeah, we still have a few oil-fired boilers that we're going to get rid of. And then we're systematically replacing them with either pellets or a heat pump, depending a bit on where we are, the altitude, and so on. We tend to use pellets. So when we redo a boiler at over 2000 meters, et cetera.

[break]

So that's the big part of the buildings. And yeah, when you're building new, of course it's a lot easier, but it's rare for us to build completely new. We often salvage buildings, we add to them, we rebuild a ski lift in an existing building, it's a massive challenge. And then each time we have to deal with the energy side as well. So we're well supported, whether it's the Confederation or the canton. When we know where to look for a boiler, when we renew one, we get good grants.

[break]

We're also working on the new building, where we have to systematically fit solar panels every time. When we can get a surface that's well exposed, like what we're going to build this summer, we're of course going to put panels on, it's necessary. But that's when you start from scratch, it's easier.

CA: Do you envisage solar panels only when you're doing new buildings?

X: We already have areas that are rented out, they're made available to operators in the region, energy suppliers. So we typically have the buildings at [name of two departing stations] that are already in use. The roofs are already being used. Even at [name of a third departing station]. And that depends, either we install it ourselves, or we have contracts with a supplier. In the end, the important thing is to recover this energy, that's the main thing. It's about using this roof, these façades and so on. In the end, it doesn't matter if it's us. We're interested in self-consumption, so reducing our bills, of course. But the main thing is that this energy is not lost, because it's not used. It always works out one way or another. But we're still, ... I don't think we're the most advanced in terms of solar panels. Still a lot to do on that.

CA: And this is a field that you intend to continue developing?

X: Yes, of course. Yeah, especially as we're well placed here. We've got buildings at high altitude, and there's always plenty of sunshine. The radiation on the snow means that there's better production. So there's quite a lot of potential here.

CA: And for these adoptions of renewable energy, are you only going for solar power or are there other alternatives too?

X: No, we already have thermal energy, typically for buildings that face south like those above [region name]. We already heat the water with thermal energy. That's another aspect we already have.

CA: For the adoption of renewable energies, solar panels for example, what are your motivations behind adopting them?

X: There are two reasons. It's both environmental, because the more we self-produce for Switzerland, ... on a global level, we realise that we're dependent on foreign countries, so the more we produce internally, the better off we'll be. It's on the bill too. Because the self-consumption part is necessarily reduced [in price], so [also] the electricity bill behind it. I think these are the two most important aspects. But the return on investment is very long indeed. You can't fool yourself, especially when you're building at altitude [it takes long to get a return].

CA: How do you mean?

X: It takes more than 15 years to make it profitable, a solar installation, at today's energy prices. If what happens tomorrow is... energy will go down again, because the energy crisis didn't last long. There was the panic after the invasion of Ukraine, prices went up very, very high and the trend now is [going down]... we're already buying our energy very early on. And in 2027-28 we will be in the same range as before, ... the kilowatt-hour rate, just so we're clear, is virtually the same as it was before the war in Ukraine.

CA: Yes, energy prices have already decreased.

X: Yeah, so the investment in solar energy has been extended. Because the cost of energy means that, well, the higher it is, the more attractive the solar installation. Because the feed-in tariff is also higher, so all that will start to fall again. So investment in solar energy is going to, ... the payback period is going to get longer. So it's interesting, yes, if you can consume your own energy all year round. If it's only in winter and not in summer, that's no good. Because self-consumption isn't that simple and the legal framework doesn't let us do what we want. It has to be the same plot of land, and so on. So if you install solar panels on an adjacent plot, self-consumption doesn't work today. It really has to be on the same plot. So typically, if we put panels anywhere on the mountain, we can't connect them to our grid [and cannot do self-consumption].

CA: So, in fact, you won't necessarily be able to use your own energy, and it will be sold on the market. So it's not very interesting, because it's at the market price?

X: Yeah, but that's not up to us. So, yeah, solar is great, but there are a lot of “buts”, aren't there? So we've started to do it, we're doing it, and this summer we're doing it too, we're continuing to build solar power. But I wouldn't say that you have to, ... today they're putting panels on all the facades, because, ... you can see that the trend is also calming down, which, ... so the price part is one thing. Then there's the part where, as I said, self-production is important. If everyone does it, then we reduce our dependence on foreign countries. Energy supply. Especially in winter.

CA: And what is the process internally to take decision about renewable energy adoption?

X: Well, if we start with new buildings, typically the three departure stations behind you that we're going to build this summer, we have commissioned a specialist company to do the CVS, all the thermal analysis. It's a specialist office, and then they come up with this: “We need as much insulation, we need, ...”. Well, that's a mandate we give them. When we renovate, ... We've are nearly finished, we've commissioned a study to map all our buildings, so that we can plan for the next 10 years. Each year we build a particular building, and so on, on the basis of this report. So we rely heavily on specialist consultancies. We don't necessarily have the skills to do this kind of thing in-house, or to go and find the subsidy programmes that go with it, and so on. We work with offices in the [region name] region that specialise in this area.

CA: So you decide internally that you want something that's as renewable, ecological and profitable, and then you give it to the office that's going to look after it?

X: Well, they come up with a budget, then they say here's what we're going to do. Because there are always several options. There's X amount of solar panels, there's such and such an energy solution. After that, there's a budget behind it, so that's another thing we have to work with. If it's a building that's only used in winter, maybe we won't put the panels on the facade and on the roof, we'll just put them on the roof for example. It all depends on each building.

CA: And why is that?

X: Well, typically the investment is going to be much more substantial. I'm talking about a restaurant we did last year, where we had the option of putting up panels all over the place. This restaurant only operates in winter. We can't self-consume, as I explained earlier, when the restaurant is closed because it's a plot in the middle of the mountain, and there's no direct link with other buildings. Unfortunately, solar energy isn't profitable. It will never be profitable. Additionally, there is still the fact that you have to clear the snow off the roof for the panels to produce. But on the facade, it really didn't produce much. So it's a pretty detailed analysis every time. OK, it's going to cost us X thousands of francs, a hundred thousand francs more, but there's a return on investment in X time, OK, let's go. So we didn't go ahead

with it because it just wasn't cost-effective. Instead, we put in a pellet boiler. We heat with air, with renewable energy, with wood. So the share of electrical energy remains marginal.

CA: I have the impression that every time you do something new, you try to go for the most renewable energy possible. Is that part of your corporate vision?

X: Completely. What's more, as I said at the beginning, we have [certification name] certification, which covers everything to do with the environment. This means that we also have targets to achieve. And then we have a target agreement with the Federal Office of Energy. So we're obliged to make a commitment over the next 10 years to measure them and pass them on to the Swiss Federal Office of Energy, in order to receive what they call the FIT [feed-in tariff] reimbursement, which is 2.3 centimes per kilowatt-hour that they pay on each of our meters, on condition that we undertake to reduce our energy consumption. And every year we send the SFOE our data. So these are commitments that we have to keep. That's why we have so much available each year for renewable energy.

CA: And the state subsidies that come in cover roughly what percentage of an investment?

X: Or if I take one, ... on a boiler, it can be between 20 and 30%. Yeah, that's substantial, yeah.

CA: And, we started talking about it a little earlier, but do you think initiatives like that can attract tourism, or be a competitive advantage over other resorts?

X: I'm not sure that many tourists are aware of that. There are labels now. I think the label that's coming out of the woodwork a bit, because there are 25,000 of these labels today, ... but there's Swisstainable which has been taken up by a lot of tourism players. And our destination is committed to this. So it's clear that it's a nice label to have, I think. But we can't do it on our own because Swisstainable also affects mobility. And the bulk of [company name]'s carbon impact isn't us, it's the people who come to stay with us. And here we already have, without counting the train coming to [commune name], we have public transport which runs all year round except today, unfortunately for you, but which allows us to bring our customers here. So we're already pretty good when it comes to mobility. But let's not kid ourselves, the train is great if you're young, if you don't have a family or children. If you have to carry your skis, your backpack and the two children under your arms, taking the train is not necessarily easy, depending on where you live with regards to the station. So there's, ... that's the biggest challenge I'd say. And in Swisstainable, we have to be, ... in liaison with the local or tourist authorities, to have the same 10-year vision, to say, here's how we want our customers to get to [station name] in 2030. That's why we're all doing Swisstainable together here. Destination, [company name], commune. Yeah, the whole valley too. It's a shared vision.

CA: And after the label, it's for the whole region?

X: Yeah, because it makes no sense for us to do it on our own. We have our labels in relation to the agreement on the objective with the SFOE, and things like that. But they are labels, yes, ... you can find them on the site, but no one is going to be interested in that.

CA: Because with this Swisstainable, Switzerland wants to become the most sustainable destination in the world. Do you think this vision is possible? Or what do you think of this label at Swiss level?

X: I'd say it has the merit, at least, of being somewhat unanimous. Because there was a time when we were approached every month by companies proposing that we be awarded a label. And finally, too many labels kill labels.

CA: Totally, it loses credibility.

X: Of course, yeah. This one has the merit of saying that SBB is on it, that the major destinations are on it. There's probably some greenwashing involved, I don't know. But I think it's sufficiently state-run, not private. So whoever launched it has no interest in making money out of it. Which is the case for most of the others who have come to see us. So I think it has a good future. I'm not sure that it's an argument for foreign tourists to go and see if [resort name] has the Swisstainable label, which is a 100% Swiss label and not an international one.

[break]

Yeah, labels. It's not really my thing, but ... it's the same thing as I said at the beginning, you always have to be careful. But this one, in any case, it doesn't hurt, it's nice, it's not restrictive, in the sense that there's nothing behind, it's not intended to be lucrative for the person who launched it. On the other hand, in the end it's all government companies [who launch it]. So yes, I think it's a good idea to get into that. Then I think everyone should get involved. After that, for Switzerland to be the most sustainable, that's ambitious.

CA: And then, when you want to adopt solar panels, for example, how do you find the information you need? Is it just the office to which you delegate that task?

X: Yeah, it's very, very technical and specific. Then it also depends on the architecture for integration. Because you can't really do everything you want. Solar panels are investigated. They also have to be integrated into the building. Do you put them on the roof? Do you put them at an angle? Do we, do we, do we, do we what? And those are studies too [who do the plans about it]. I'd rather have a report that says, here are your panels, you put them like this, this colour, and it costs so much. Because we don't have the skills to do that in-house.

CA: And did you have any concerns about using solar panels in the mountains?

X: Oh no, on the contrary, because we can see only the good. The controversial projects are the solar farms in the Alps. Because that's where we can produce the most electricity all year round. So there's a project with the local council here, on the [site name], with a huge solar farm laid out in crosses. So it's also innovative. The panels are laid ... the support is vertical. The panels are arranged crosswise on either side, like this, so they are bifacial. So with the reflection, it produces in all conditions, well if the weather is good. And the impact on the landscape here is less, in the sense that it's already a colonised [polluted] region, already impacted by tourism. It's not in the middle of nowhere like Griengols, or some of these projects that are causing a bit of a debate because they're in areas that are still unspoilt. In the end, it's already been impacted. Yeah, so that's interesting, because we're actually a partner too, in the ski area. In the future, with this solar farm, we're talking about producing hydrogen on site. So in the future, in 10 years' time, the machines will be hydrogen-powered. So if the production of hydrogen, which today is much criticised because it requires a lot of energy, ... so if we can produce it on site, ... So there's a start to this project. That's interesting.

CA: And what is your role in this project?

X: Yeah, well, we're already a partner because we run the ski area, so they come to us. Well, it's not on our land, but they're using our existing infrastructure, in particular for everything to do with underground pipes and things like that. So it's a partnership, yes indeed. And we're interested in these hydrogen projects, because for me, it's the only source that will enable us to propel a snow groomer with the same efficiency as diesel today, so yeah.

CA: Right, and do you think there's any chance have a solar park like that accepted?

X: Yeah, I'd say that here, yes, because the landscape impact, it's less, compared to some of these farms there, yeah.

CA: And would this be for your own consumption, as you said before, or would it just be for hydrogen production?

X: No, no, it's a whole thing. It's for the region, for us, for all the local partners.

CA: And how far along is this project?

X: This winter, it was undergoing testing, so several crosses were put together to see what the actual profitability would be, and then the enquiry is scheduled for this year.

CA: Going back to private solar panels adoption, when you received the information from your specialist office, do the initial considerations change, are there any other considerations that came along the way?

X: Yeah, often. Well, if it's renovation, then yes, there are choices to be made because we're proposing things. Are we going for Minergie class, ... there are different classes of, ... it's AEEG energy labels, it's these colour rules. It's the same as for cars. When it consumes A, it consumes very little, then G it consumes a lot. And it's the same for buildings. So after that, we find ourselves in a range that we'd like to stay within, in this zone, in this range we'll say from here to here. And that's a choice that also depends on your finances, and where, ... how far you're prepared to go in terms of insulation, renewing a heating system, and so on. So yeah, it's each building that has its own information. That's why we have this report, this vast study that was launched a year ago and that we'll be receiving very soon. It's a map of the current state [of buildings], and then the measures to be taken to achieve neutrality in buildings.

CA: And are there sometimes differences of opinion at [company name] about adoption decision?

X: If there could be one, it would be about the financial aspect. Fortunately, we have enough, ... we're a healthy company, so generally we always go with the, ... because we're building, we always say to ourselves it's for the next 40 years, so we do it well. As long as we have the means to do it, we'll continue to do it. So I'd say that could be the only source of divergence. Because internally we all have the same common vision. So yeah, it's clear that you have to spend a million more on a building, and I'm giving random numbers here, to be Minergie A++, yeah, ... do you really need that for a ski lift station, which is in operation 4 months a year? It's always a question of weighing up the interests.

CA: And in relation to Minergie standards, as soon as you renovate a building, you have to comply with their standards?

X: Well, yeah, but Minergie is extreme, it's more for a dwelling, let's say, a private house. We are often not in a Minergie class. Unless you start from 0, which unfortunately is rarely the case because we have existing stations. But we're going to focus on ranges that are, ... to get a subsidy, you have to commit to a certain energy class anyway. So we're going to go for that class anyway. After that, if it's A or B, that's when we'll look at the aspect, how far you have to go to be in A, how much more it costs, and so on. It's like the "nutriscore". It's all colours. A is green, then G is red. We've still got G and we'd like to be in A and B everywhere.

[break]

Yeah, that's kind of the rule, it's just always the aspect, here's the financial balance and then [the ecology].

CA: And I suppose that as you can't know the price of energy, it's complicated too?

X: No. And the price of materials and everything. Because everything went up two years ago. It's now calming down a bit. But it had become ... it had become extremely expensive to build in 2022. The end

of the COVID plus the war was a total disaster. Raw materials skyrocketed, so it just became impossible to build at a reasonable cost. Now it's becoming a bit more reasonable, so to speak.

CA: So for all building materials, or just the ones for renewable energy?

X: Everything, wood, everything, it's incredible. Let's not even talk about steel. So we're very dependent, it's exogenous factors, but that's part of it, ... so when you want to build a 30 million plant and there's a 20% price premium for raw materials, well, ...

CA: And are you satisfied with your investments in renewable energies, the ones you've already made?

X: Well, yes. Yes, clearly. What's more, it's measurable because, well, we already have to report to the SFOE for the target agreement. So each measurement point can be compared from one year to the next. You can already see the investment when you change a ski lift engine, and you can see the result for one year directly at the bottom of the line [it is worth it]. After that, of course, it depends on the number of hours of operation. If the season is longer or shorter, it has an impact on the price, because the installation runs for more or less time. But if you make ratios, or more precise calculations in relation to, ... you'll find out very, very quickly, won't you? Changing an electric motor in an installation that runs for 1500-2000 hours a year goes very quickly. For 1% better efficiency. So it's very interesting. Monitor speed. Automate the operating speed. The employee doesn't always realise that he has to ... that he has to lower ... He doesn't always think about it, that when there are fewer people, he has to lower the speed. So that's become automated, for example. And then, yeah, at the end of the day, you can see the bill. All the little things, but put together [it makes a difference].

CA: And being part of the SFOE, that's voluntary?

X: Yes, it is. Yes, clearly [it's voluntary]. But on the other hand, it's a condition for being reimbursed from the FIT. So it's 2.3 centimes per kilowatt hour. If there's no agreement on the target, then there's no way of getting that money back.

CA: And there are a lot of stations committed to this?

X: There are quite a few. Yeah, there are more and more. However, administratively, it's quite heavy. You also have to get the service behind you [in your company] because you have to produce all the invoices and all the measurement points. It's not a total nightmare [to deal with], but you have to take care of it. After that, you have to monitor and send it to the SFOE. I'm working with an office in the [Swiss canton] to do that. He's a specialist in this field who deals a lot with ski lift companies. But there are quite a few.

CA: And what were the biggest difficulties in adopting renewable energies? If there were any.

X: Well, there's already the choice of, ... because electricity in our company is about 10 gigawatts a year. And then there's the choice of using only hydroelectric power. So already that's a choice that needs to be deliberated, it's more expensive. Because we could buy the energy, ... an energy mix that comes from German coal, for example, which is much cheaper. So we don't do that. So that means we buy energy that is, ... it's a [label name] label. So we have certification, we have the assurance that in the end, ... that our energy comes from hydro. So it's renewable. And we voluntarily pay more for it.

[break]

So that's the first thing, which is that ... it's all very well to say that everything runs on electricity, but if the electrical energy that comes to us isn't, ... produces CO2 in Germany in coal-fired power stations, the result isn't achieved. On the other hand, well, it costs us hundreds of thousands of francs more per year in electricity because we've chosen a green label, in other words, hydraulic, 100% hydraulic. So yeah, that's already the first choice. It's strategic. That's why we can say that our energy is renewable.

CA: And this is also in line with the SFOE's direction?

X: So the SFOE is sensitive to the fact, ... yes, of the origin, ... of the labelling. These things, yes. But it's not the main argument. It's not even a sine qua non condition. At least not to my knowledge, so that could mean that, ... yeah, they just look at the energy bill, the targets and that's it. But given that energy is more expensive, the rebate, well, ... since it's a calculation that's made each year between the company's turnover and the proportion of electrical energy consumed, given that we pay more for electrical energy, well the rebate is greater too. So we are not basically losing out either. That's more or less how it works.

CA: But that would have more to do with communication than with the SFOE?

X: Yeah, clearly.

CA: Right. And we've talked about that a bit before too, but in terms of your investments, you said that you communicate them but not very openly. Can you elaborate?

X: We've put this page on the site, which you've seen. There's the main list, and then there it is, and then there are no details. That's still, ... yes, we're doing something, here's what we're doing, but we're doing a lot more behind the scenes. But it was a bit like, yeah, I've got quite a few colleagues, they've done, ... who've gone really far in this. If you're interested in Villars, they've actually hired an environmental specialist, maybe full-time or 80%. Anyway, her job is to make of Villars in I don't know [how much time]... they've set themselves the target, which you can see on their website, of achieving... it's a pretty ambitious target. They've hired this lady to do just that. She communicates a lot, because her job is also about that. So after that, ... well, you have to achieve the targets set, prove them, show them, and then you're committed, ... the company is committed to its public, its customers. And if it doesn't succeed?

CA: Yeah, that's a risk too.

X: Or they fudge the figures. Well, I'd be surprised if they did, that's not the objective. So they're more likely to find themselves, ... well, ... I think they'll manage it but, ... I am just saying, communicating about it, hiring someone just to do that, that's very ambitious. I'm going to work more discreetly and do things well. But after that, everyone does as they please. Another colleague in Saint-Moritz, to name but one, were attacked because they were using a much less polluting fuel, but in the end a whole study came out showing that this fuel wasn't much better than the one they were using. And then it all came crashing down on them. So that's why I'm in favour of the less said, the better. But you shouldn't say nothing either. Because we have customers who are interested in this. The proof is that a lot of them come to ask us questions, so...

CA: And St. Moritz shows a lot about what they are doing.

X: Type in GTL, GTL fuel. That's what got them into trouble in the first place. I've got GTL too, but I don't give a toss, I haven't mentioned it anywhere.

CA: And would you recommend the adoption of renewable energy in the mountains in general?

X: Well, clearly, a resort like this, which is superbly well oriented, is interesting. But if you go and have a look around [commune name], there's still a lot of solar power being installed.

CA: So also in individual installations you say?

X: Yes, yes. And the chalets they build here are energy pits. They're not poor people. They've got the pool, they've got everything that goes with it, so they're also thinking about how they could, ... Some of them don't give a toss, but on the whole now, it's part of the complete analysis of everything that's built. Solar energy is automatically incorporated into new buildings. So no, I think things are changing in the Alpine valleys too, and very quickly. Even in the valley where I live it's, ... soon there won't be a roof left unused. It's happening fast. There are also some nice municipal subsidy programmes, so...

CA: Do companies sometimes come and ask for best practices on how to adopt solar panels in alpine regions, given that you started a few years ago?

X: I wouldn't say we're a reference in this area. I've looked for information myself. Then I saw that in Klosters, for example, they had installed vertical panels on a roof, bifacial panels. And I was interested to know exactly ... what they had produced during the winter. Because these panels are never hidden under the snow. On a roof, the problem is that even if you angle them at 45° or more or less, they're bound to get covered in snow and stop producing. So what I saw with Kloster was that they had installed these panels straight onto a roof. Then yes, best practice, it's rather to "steal" ideas from others. Or maybe we're pioneers, but when it comes to solar panels, we're not pioneers. As I said before, we're still ... we don't have much equipped [with solar panels], so it's more a case of seeing what others are doing.

In Zermatt, at Trockener Steg, they've equipped their entire station with facade panels. So that's also productive, whatever the snow. But does it produce much? It's always complicated to get the real data. Some people communicate them easily, and others, either they don't measure them, or they don't have them, or they don't give them, I don't know. There's no real exchange going on here that would be like saying, "here we are, on a station of this type, we have to systematically put up dual-face panels, frontal panels", or I don't know what.

CA: Yeah, so there's sharing, but ...

X: There's sharing, but you have to go and get the information. You have to go and find it, you have to ask for it, you have to find the referent.

CA: And I imagine that the solar panel studies are in competition and don't necessarily want to share this information?

X: Yeah, yeah, it's a business. Especially since the explosion in energy costs. Yeah, everyone's getting into it. And then there are the good ones, and then there are the not-so-good ones. It's always the same thing. You have to find the right one first [PV company], and then figure out, are the data he has accurate, reliable, and so on and so forth.

CA: Yeah, and that's in relation to the company installing the solar panels or to the study?

X: No, to the study. Because when it comes to installing the panels, it's not the technical part that's important, it's really the study upstream. Is it cost-effective, what type of panel, and so on? That's the determining factor.

CA: And you say that sometimes it's a bit difficult to know if the studies are right?

X: [You need] to have real feedback on what's happening in the mountains, reliable data, and then what we could transpose to our own company, whether or not it's profitable, and so on. So that's why I was interested in Klosters, and then I saw the data. At first glance, what they are producing there isn't all that great. But it's only been a year. So once again, before drawing any conclusions, you need 2-3 years.

CA: You need to make adjustments, I imagine.

X: Yeah, then you have to go and get the information, and it's not communicated.

CA: And as soon as you go and ask a company, are they generally open to share?

X: Yeah, if you know them. But it's a small business, ski lifts. It's clear that when we know each other, it's easier. But they're not obliged to give it to us, so it still takes time and energy.

[break]

We're not “born” with these information, we don't have easy access to it. We don't have a database. So yeah, there's a theoretical part to it that's still important. Then, do we also do it for the aspect, just, ... “Ah, there are panels on the roof”, then the customer sees the panels on the roof, then he says, “ah yes they're doing solar”. There's that marketing aspect too, isn't there? Solar panels are interesting anyway. It's just a question of finding the right for the right building, and so on and so forth. It's clear there's something to be done.

CA: And do you think that this non-transparency problem relates to solar panels in the mountains, or is it a general problem with solar panels?

X: No, but it's still sensitive data. Energy consumption, things like that, so they're not necessarily easily accessible. So to get real data, you really need to know the people you're dealing with, and make sure they're prepared to give it to us. And then what do we do with it, does it do us any good or not? That's always what you have to do, ... yeah.

[break]

That's why I like to rely on offices that work with other companies, such as [company name], which is very familiar with the [canton name] region, because they have a huge amount of experience with these large stations there, because they've already, ... they monitor, he's a monitor, they take care of making declarations for the SFOE, these objective agreements. They do this and they have contracts with each station. So with him, with this representative, well, he shares information with me from the other stations, but I prefer to get it from a specialist. Then [after I ask him] what do you suggest for [commune name]? What could we do? “Ah well, you've got to put up panels here, in this place here, and so on”. Then he told me that Klosters was still pretty average for the moment.

CA: I imagine he has a more neutral point of view?

X: Oh yes, he's neutral, yes.

CA: With regard to the energy you produce yourself, do you use it to recharge your company's electric vehicles, for example?

X: No, because we don't have any. We have some at [commune name], for customers. So there's a whole series at the station in [commune name]. But the municipality makes them available. And our fleet of vehicles isn't electric. So we only have commercial vehicles. Unfortunately, the market for electric utility vehicles is ... not yet won. It's like snow groomers. There's no demand. So it's still something totally, ... marginal, with astronomical prices and which don't perform well. After that, we only use our vehicles 5 months a year, in the summer. In winter we hardly ever use them, so once again it's a question of ... consumption, of ... what is the actual gain and then on the environment [what is the gain]. So, no, we don't have any. We don't have an internal charging strip. But I think it's going to happen, I can already

see that for our employees alone, I think we're going to equip the car park here. Because we also have an internal demand. For private vehicles. I think that's the next step. After company vehicles, the day when, ... yes, it exists on the market, it's available, it's affordable, yes. It will come, today, it's not the case.

CA: And this, for example, if it were for employees, would it be possible to self-consume as well, in relation to your production?

X: Yeah, of course, it would be interesting to have the building equipped to supply the terminals. Yes, yes, it's clear that this is a project that we're going to be looking at fairly quickly, I think.

CA: And what are the next steps in the transition to renewable energy?

X: Well, we have targets every year. After that, the big steps are, ... that remains, ... our CO2 balance, we've done a balance, ... what do you call it, energy-climate. The overall consumption of [company name] is around 10 and 18 gigawatts, taking all energy sources together, then 3800 tonnes of CO2, and so on. So how can we reduce our carbon footprint? Buildings account for 30% of our carbon footprint. So we have this objective of changing the boilers, so it's not at all excessive, we're going to get there. 70% is that, that's the machines there, ... the snow groomers. So, unfortunately, as I said at the very beginning, we don't have the leverage. There really is no solution today. After that, yes, the snow groomers are extremely low-polluting, but they're still diesel. So today, there's no alternative. Or very few. And it's not going to happen in 5 minutes. So it will remain our, ... our weak point in terms of greenhouse gas emissions.

CA: Right. And in general, do you think it's important for Swiss alpine tourism to move towards more renewable energies?

X: Yes, so there are 2 aspects, there's the aspect, ... already that the ski area has already been under attack for, ... a few years. I'd say since 2020, since the COVID in any case. For various reasons, but above all for its impact on the climate because, ... the images of snow channels, of a grooming machine, ... it is perceived by the general public, like one, it's useless, it's for leisure. And two, it pollutes. So yes, I think that communicating that yes, OK, so we consume, ... [but] we produce energy, ... for us it's 3800 tonnes of CO2, it's 4 return journeys, with an A330 [plane] Geneva-New York. Just to put the church back in the middle of the village. So where's the impact here, compared with all the people who fly every day? It's just ridiculous. The impact is from the people who come here. That's a big one. How can we improve that part, but ...

CA: In what sense are you saying that?

X: It's the car, to get to [commune name] or [commune name]. That's where the worst thing is, ... that's the worst thing, in the end, it's all those cars every Sunday evening in traffic jams, because they've all

come to ski in [canton name]. There's a huge lever for action here, but for that you need ultra-efficient public transport. I mean, you need local facilities where you can leave your equipment, showers, I don't know what, but you really need to be able to leave home and get your equipment, and then go skiing with your family, without any major constraints. That's a huge challenge. That's where we'll be able to, ... I think the day we have, ... reduced mobility. And in Switzerland, we have a great public transport network. But it won't be enough. Today. To accommodate all those customers who would come by car. So there's a huge lever here. And what were we saying? What was the question? I got lost.

CA: It was about the adoption of renewable energies in Alpine tourism.

X: Well, yes, there was the aspect of impact for the customer, saying that the ski area is doing a lot for the environment. And then there's the aspect of the energy bill at the end of the month. It's as simple as that. For me, it's always these 2 levers. There's the aspect where we have to show customers that we're doing everything we can to reduce energy consumption, water consumption for the snow cannons, fuel consumption for the machines, and so on. And then, well, at the same time, it's also good for us because we're spending less. We spend less on heating, electricity, and so on.

CA: Well, I've gone over all the questions, but are there any additional aspects that you would like to approach?

X: No, I think you'll find lots of interesting studies on the web, but if you ever do, well, ... I've given you 2-3 contacts. But there are some interesting things to look for, ... I'd say there's, ... It's a very trendy theme, that's all. And the ski areas are very much in demand, but often in Switzerland we're very critical, even our media. RTS is fundamentally anti-ski. You can feel it in most of their publications and articles. It's never very positive in favour of skiing, even though it's a huge source of income, because a ski resort like [resort name] indirectly generates thousands of jobs in the surrounding area. Not just the ski area, but everyone who works in the area.

[break]

So it's in Switzerland, it's something extremely important. And unfortunately it's often misperceived, even though the impact, when you measure it and compare it with other sectors of activity, is really small. Especially now that we're all committed, most of the big companies, to, ... what we do, they all do in one way or another. So for these two aspects. Communication and then internal consumption. So I wouldn't say that's very much in the spotlight these days. The, ... yes, the media don't necessarily pick it up. They prefer to put up a photo, ... of a white strip in the green grass, of snow produced. Something that will say "Oh come on, they're making snow, but in fact it's all green all around, there's no snow". Nothing positive. It's rarely positive and that's something that's... I'd say, for us, it's a bit of a pain. So unlike France, which always has a bit of praise for a ski resort, pretty pictures, something that's much more positive for the activity, whereas skiing in France represents a much smaller proportion of GDP

than in Switzerland. But they're much better at communicating positively than we are. So it takes its toll on us in the long run because we're, ... during COVID, we were the only ones who were open, who could work, but we were under attack all the time. Because there was a lot of jealousy too. The fact that, yes, we could go skiing, but we couldn't go to restaurants. All those things fell back on us a bit. And then now it's energy and then tomorrow I think it will be, but, ... that's the heaviest burden for us I'd say.

CA: And do you think that's why communication is so negative in Switzerland and not in France?

X: I don't know. The French media are very biased too. They're ... it's a lot, ... we don't have many deported journalists who do, ... the big cities, in the end, RTS is, it's Geneva, it's, ... Rather left-wing oriented. I'm not saying this in a negative way. But there you have it, it's a [political] party. And skiing is considered a leisure activity, so it's useless. I'm simplifying, but in the end it only generates ... problems and pollution. So it's rarely to our advantage. I see when we're doing reports, either there's been a death, so that's interesting, or, ... there's no snow. So we take pictures of people walking around the resort because there's no snow. Except that everything's open, but they're going to film them in a place where it's all green. It's always a bit like that, and that's what really weighs on us. So there you go, and that's that. And we've just had a record winter, so the skiing isn't going to stop any time soon. Despite the strange conditions, lots of rain, we've never had so many customers.

CA: So after the COVID it's back on track?

X: Ah yes, yes. We're much higher than 2018-19, which was our best year. Yeah, since COVID came out.

CA: Why do you think that is?

X: There are several aspects, ... There's, ... generally speaking, there's been a resurgence in activity. There's an explanation for that, ... not quite, but the fact, ... there's also the, ... the [pass name] subscription that came out and did a lot of good. We're not into it but it's not, ... we're not into it. [Pass name] is great because it allows everyone, with a reduced-price pass, to go skiing in 50 resorts, and then, ... the proximity of these resorts to the towns, well there you go, it's proven. And for us, it means new skiers. Children will learn to ski at [ski resort name], but not in [their ski resort name]. What's important is to have these local resorts. [Pass name] has democratised that. Which, for us, is a good thing. It's put people back on skis. And sooner or later, these people still come to [their ski resort name] one day or another. So these are customers that we get back one way or another.

[break]

And then the aspect, well yeah, we've had some nice winters. Despite the fact that we only remember the rain. Well, we've had winters with good snow cover, which makes people want to go skiing again. We've also had a lot of international visitors coming back since the end of Covid. Also the fact that we

don't depend solely on the national clientele. But the majority of the clientele remains Swiss. Then more than 50% are national customers, or even more than 55%. But we have a lot of foreign visitors. There are all kinds of factors. In any case, we're not going into the wall at all. So, it will be more complicated, we have an extremely complicated winter, a lot of precipitation at very high altitude, so rain, wind, in short appalling conditions. But I must admit that when the weather's fine, it's packed with people. The future lies in 4 seasons, of course, and summer development, we have to be there. Today, summer doesn't even account for 10% of our revenue. We can't operate without winter. It's winter that finances summer, to be clear. In the meantime, we work very hard. We work from June to the end of October all summer, we work, ... A lot of facilities too. We're offering bike packages, and we're developing more and more in the summer, but we make the turnover for 5 months of summer in less than a week at Carnival. So skiing is still our bread and butter. We still have a long way to go to break even. We do that a lot, yes. But I don't think we'll ever have as many people as we do in winter.

---- End of Transcript ----

Participant 9
Interviewee: Member of the board of directors of a Swiss Alpine company Interview date: 30.04.2024 Form: Qualitative, semi-structured, face-to-face PV adoption: No, but planned solar PV installation on a renovated restaurant

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: [start recording during the company presentation] **What renewable energy projects do you have?**

X: Right now we've replaced the ski lift with a chairlift, we replaced a ski lift last year, with a new ski lift, then the next objective is to replace the chairlift with the renovation of the restaurant. It's a very big project, I think we'll be looking at around 10 million. We're in the process of talking about it, ... the big "secret" is the financing. So for the moment, what we're going to do is renovate this facility, this year and next year, so that we can continue operating until 2038. After that, we'll be doing a lot of renovation work, renovating the restaurant with a new energy source, but I'll also explain 2-3 of the problems we're having with this and the chairlift. To give you some information about energy, the new ski lift we've installed means that as soon as the lift notices that there's no-one on the line, it turns very slowly. As soon as there's someone on the line, there's a "beep, beep, beep" and the thing starts up again as normal. It saves energy. So that's not bad. I think it's the new chairlifts that will be arriving, it's more or less the same principle. I think I've said it all.

CA: **And do you have projects with renewable energy?**

X: Yes, well, the project we've already been working on, the restaurant, is all set to use solar panels. But, there's a problem, you've already been to [commune name], up to [high altitude] meters, it's all in the shade. That's the problem. So those who are studying this, because we've got an architect who's made himself available to draw up a project for us, he's studying all this, what he can do. Because at the beginning of the season there's no sun. So we're hoping that in a few years' time, well, the solar panel, ... already, I think it's improved, but the aim is clearly that it will be, ... the other thing is that we currently have electric heating everywhere. The building is [age] years old. The insulation is not the same as we build nowadays [...], it's very thin. We are clearly heating the [valley name] Valley. We're wasting a lot of energy, that's clear. A huge amount of energy, and now there's also a huge amount of money [being lost].

CA: So this project is renovation?

X: Yeah, the restaurant was, ... The project is really just keeping the foundations. The foundations, then the load-bearing walls, everything else would be destroyed.

CA: And now the architect is looking at where you might be able to fit solar panels. Is that right?

X: Exactly. If we take this example, it's fine, but it's very short, there's a setting sun [he shows a map he's drawn showing that there isn't much sun]. So it's not long, but it's true that it's, ... logically it's in the process of, ... we're going to go there, it's clear [towards solar PVs].

CA: And for the decision-making, who is involved?

X: The Board of Directors, we incorporate it, because these are big decisions, they involve large sums of money. We work on the project, to put the finishing touches to it. And that's it, we work on it, we move forward. We always have to move forward anyway. Because there are authorisations, there are things that take a lot of time, so we're always moving forward.

CA: And are there different considerations from different people on the board of directors?

X: No, I think we're all convinced that we need to make savings and then we need to renovate. In any case, we need to conserve energy and find new sources of energy, according to what we can do. Well, not what we can't do. At the moment, we're saving energy because we don't have to run the cannons very much. I think we've done everything we can to save energy. The only thing left to do would be to say, well, we're closing the lift from such and such a time to such and such a time. Yeah, some people have asked us "But why? To save money". But I say, do you realise that it takes a lot more energy to get the lift up and running again? It's much more expensive than letting it run without stopping. It's enormous to start something up. So that's it. I think we've done all we can.

CA: And what are your motivations behind this adoption of solar panels that you've started talking about?

X: Motivation, it's very clear, it's a restaurant, a working tool that's [age] years old, that was made for 20 to 30,000 skiers. Today, we have 60,000 skiers. It's still the same tool. That's the main reason. Because it's better suited to, ... we can work miracles with what's there. But the tool has to be better adapted to the current situation. And it's ageing. And as I say, we're heating up the [valley name] valley.

CA: And do you think that in the future such initiatives could be a selling point for tourists who come?

X: Well, it's certainly in line with our philosophy of nature, natural snow, and what's more, we did it for ourselves. Yeah, that's clear. We've got to keep digging in there because we stand out from all the others. And people also come to us a lot because it's natural snow, so it's also easier to ski. But we have to keep going in this direction, and that's something we can fight for. The rest ... we can't expand the area. That's what we're banking on, clearly.

CA: And so this is part of a vision of the company that you want to be as natural as possible?

X: That's clear. That's clear, we don't have, ... that's the only thing we can, ... that we can do, because we stand out from the others. And the aim is really to stand out from the other stations.

CA: And after this project, do you plan to do other renovation projects like this in the future?

X: Well, the big project is a chairlift and something like that. 10 million, you have to imagine that, ... I think for the next 20 years, there's nothing left at all. Clearly. This project has to be finished before [year]. And why is that? Because in [year], the concessions expire. We have concessions, authorisations to operate until [year]. To renew that, we need to be prepared for the long term. But we also have to wait, and this is a bit crucial now, as you know, there are new elections. The [commune name] local council is the majority shareholder, owning 52%. We also need to know what their vision is. As their vision is more on the green side, I think they'll go along with us. But it's also a big investment. And we also have a special parameter, which is that you know there's a merger being discussed with [commune name]. And there too, if it goes ahead one day, there would be capital coming in to modernise the whole thing.

CA: Right, because where are you with the project at the moment? Are you waiting for authorisations?

X: No, not at all. For the moment, we're just studying the projects.

CA: And just how do you go about finding the information you need for this implementation?

X: Well, we've commissioned an architecture network to look after that, and they've already shown us a project. The manufacturer who has already done the other installations is working on it. But we were talking about, ... clearly, we were talking about having the whole bottom of the installation [of the

chairlift] with solar panels. He said, but I can't offer you that, there isn't a ray of sunshine all year round. It's clearly not possible over there. They'll never be able to help out, the investment will never pay off. And that's an important point. It's an advantage for us because we have great conditions, and we need a lot of snow, it's a fridge. But it's not possible to install solar panels for the installation itself. And if you take the chairlift, if you have point A for the start and point B for the finish, then you have the restaurant. That's the [valley name] Valley, isn't it [shows a map drawn by himself]. The [river name] descends like this. Up to here it's pretty much in the shade. We say why don't we put all the engines and everything up there [where there's sun]? But the engines have to be at the start to intervene, they can't put all that stuff up there. So it can't go anywhere else. And there's no solution there. So it will be electric. Then there's the restaurant, as I said. The sunny area we have is about, ... at the beginning, it comes here [shows on map], at the beginning of winter it comes here, then it comes here, then it makes the terrace, then it comes about, then it goes like this, sunset to here. But at the beginning of winter, that's all we get. It comes in very nicely but, ... 12:30, ... yeah 12:00 to 12:30, at first there's very little sun. So the idea was maybe to have a system of solar panels maybe like that [shows arched solar panels on the roof to track the sun during periods].

CA: And is it difficult to know exactly what yields solar panels have?

X: We haven't gone that far yet. So it's clear that if we commission a company, we'll have several studies, with different styles and types of solar panels. I think ... I imagine there are lots of them. So for the moment we're working on the project, but not yet in depth.

CA: And do you expect any state subsidies for projects like that?

X: Yes. We expect it. Now, the ski lifts, it's not really, ... the subsidy thing. But we're hoping to get subsidies from the state clearly, ... I don't know if it's the environment department, but we're hoping to get them in any case, ... because it's still a renovation thing. Then there's also something we've got for the mountain region. Maybe we can also get something there.

CA: And have you yet received feedback from the architect on the exact plan?

X: No, we've just done, ... you have to imagine that we've done the, ... they came to do everything that needed to be done in relation to the load-bearing wall, to see what already existed, to see what we could keep. What we got was more of an, ... simply an exterior project. Not to mention the interior.

CA: And if you decide to invest in them, or once you've installed these solar panels, do you plan to communicate it to a wider audience?

X: Definitely. As I said, it's part of our policy, so we're not going to shy away from showing that we've done this as well.

CA: And you're not reticent about the fact that there might be some criticism coming back?

X: No, because it's precisely these criticisms that we've transformed. We use it as a positive side. So, on the contrary, we're reaching some customers who want that. Especially these days. So, no, on the contrary. There won't be any negative criticism, there'll only be positives, that's for sure. I can't think of anyone who, ... anyone who'd be stupid enough to say "what on earth did you put that there for"? On the other hand, it's clear that if we had a chairlift with solar panels at the bottom, then yes, they'd say "you're completely stupid, there's no sun". But not there.

CA: Have you ever considered other kinds of renewable energy, given that there's less sun there?

X: Clearly not. Clearly not, I don't even know what other energy would be, because going into the ground up there, ... I don't really see how much would have to be dug up to get there. So I don't see what we could do.

CA: And once you've potentially produced your own energy, would you be interested, for example, in adopting electric vehicles that you could recharge with your own energy?

X: So there are 2 things, the region, "[region name] région tourisme" has an electric car that it lends out to everyone. After that, it's clear that these cars aren't used much in winter because they're small cars, to be honest, ... I'm not sure that they can get up to the slopes when there's snow.

[break]

The groomers, now that's an interesting point. Today I visited *Prinoth*, a manufacturer of snow groomers for MIT, the COVID year. It's the only place we could go, so we went to see it. They showed us what they were developing electrically. So you have to imagine, a snow groomer, ... I'm schematising [draws a snow groomer]. Like this, like this, this is the front. There are no wheels here, but there are tracks. And then we've got something on top, like this, all wound up, called a winch. So to have top quality tracks, you need snow groomers with a winch, with a long cable, the famous cable that you hear talking about from hikers, 1200 meters. They're going to hook it up there. And then they'll have to go back up and push the snow to get an even carpet. Today, the snow groomers have to use electricity, ... they never manage. We'd have to replace the whole surface here [shows the top of the snow groomer on the drawing] with panels, but that wouldn't be technically possible because there's, ... you have to imagine that the pressures are enormous, especially for snow. Now there's hydrogen. Hydrogen is on its way, and here too we have seen the machine. Electricity, on the other hand, for snow groomers, which you may be interested in, for cross-country ski runs, is slowly coming on stream. If they have the sun, it's coming along nicely. Well, for cross-country ski trails. It doesn't concern us. After that, the winch, for hydrogen, you'd have to imagine a hydrogen tank on top, ... there would be more room to put a winch. So they're also looking into that now. Then there's the investment for the hydrogen, the power station, it costs 1.5 million, just to build a hydrogen power station. So I think that oil has a great future for ski lifts. It's not going to disappear any time soon, that's clear. There have been 2-3 resorts that have taken

on test groomers. I think it's fine if there's no winch, if it's not too steep, if it's cross-country skiing. I don't think there's any problem with cross-country skiing. So the groomers are getting cleaner and cleaner. That's for sure. But we're not yet into renewable energy. You need so much power that it's not possible. And they work at night [with the snow groomers]. They often work at night. So often these machines are in garages during the day, and in the evening they go out to work. So as far as electricity is concerned, we're not there yet, ... not with recharging, not with solar panels, not with hydrogen, ... we're a long, long way off.

[break]

You can go and have a look at *Prinoth* Switzerland on the website. And then you can see, they all put forward ecology, and they had made in green the electric machine. And you'll certainly see things that are being done, that are being prepared.

CA: And do you feel pressure from outside to move towards renewable energy?

X: No, because we're already doing a lot. We're not going to go and build snow cannons. After maybe for survival one day, we might not have the choice of building a ski slope, ... having more cannons. But for the moment we're not there. But you can feel the pressure to get a runway at all costs, so to say, you don't give a damn about energy, you have to get a runway. But the pressure doesn't come from us, it comes from outside, when there's the frustration of not being open for Christmas-New Year. That's where it comes from. "But then why don't you make an artificially snow-covered slope to ensure that? [people say]. So we're a long, long way away because there's no water in the region. It comes nicely, but we're very, very far away. It's very expensive to use snow cannons, the energy, the snow grooming, the staff who have to work nights too to get things up and running. And we once did a study to create a water basin. The water basin cost 1.5 million euros. 10 mobile cannons, that was also 1 million, that made 2.5 million, ... we were at 3 million. We had snow in November, the slope was hit by a foehn wind, so there was more water in the basin. It wasn't, ... that's also why we said we weren't going to incur unnecessary costs.

CA: And do you also have partnerships to create alpine parks in your area or do you have any offers in this respect?

X: That, nothing. So afterwards, I analysed what was being constructed around here, in the [neighbouring valley name] and all that, ... So it's clear, for the landscape, it's not very beautiful [to have solar panels]. BUT, it's still a solution. Then I also see, I notice, I often go by, well ... if I go to Aoste, the dam where they put the solar panels in the summer, I think it's a great idea. But we've never been approached. I don't think there's any risk. Then there's the [area name], which is an important natural area for us, where it will apparently never be possible to do anything. So I don't see where they could install that.

CA: Well, I've gone over all the questions, but are there any additional aspects that you would like to approach?

X: Not about renewable energy, we talked about it. What were the motivations for adopting energy infrastructures, well, we talked about that. The information needed for energy is in the office. Well, it's clear that the day when we finally ... we're going to be looking for subsidies, that's for sure. Because in the world of ski lifts, the word subsidy is part of our basic vocabulary. Subsidies, fund-raising, grants, ...

CA: Because you often receive them?

X: When we have projects, we have to apply for interest-free loans. It's not "given" money, it's never "given" money. As for the solar panel, they give you something the day you get it. They also give you an amount for a house [if you set them up]. What we have here are projects with 0% interest rates. But there's no "given" money. So when it comes to the decisive point and making decisions to buy infrastructure, ... well, there's no decisive point because that's what we want to do. We have to, ... the day we change, it's to go there. Clearly. For me, the big obstacle is the financial side. We pay a lot of electricity. To give you an idea, until 31 December we had a locked-in contract at 6.8 centimes per kilowatt. To give you an idea, we pay an average of 55,000 francs for electricity, for installations, ski lifts, heating, everything. At 6.8 centimes. The new price on 1 January is 17 centimes. So you do times three. So 55,000 times three makes 165,000. The energy costs department is enormous. If we don't have a good season, a normal season, we'll never be able to pay that, it's not possible. We have to move forward with renewable energy. That's clear, but as I said, we're in a bad way up here [because we have no sun].

CA: And it would be for self-consumption?

X: Yes, yes, clearly. Clearly, well, it takes so much energy [ski resorts] that it's self-consumption. We're never going to say, "we're going to sell to X-Y".

CA: Are there any aspects that you think are important that I haven't covered?

X: With regard to energy, no. For me, there's one thing in relation to energy, maybe not renewable energy, but energy in general, and that's the number of people who come by car and the number of people in the car. So I understand. Try to play the game, you say I'm in, ... [village name], I'm going skiing in [station name]. You use public transport. Imagine that there are three buses going up to the facilities during the day. It's really, ... and I understand people who say "I want to ski for a while in the morning", ... with the [pass name] especially, it's changed a lot, people come to ski then they leave at 12:00, or that, they have to wait I think at 2:00 to get a bus. They prefer to say I'm going skiing by car, I take my car, I ski, I leave. That's also the [pass name] effect, which means that people ski in a completely different way. But public transport here is a disaster. On the other hand, I admire Le Châble:

Verbier-Genève, Le Châble station, directly to the top. I often go, ... well I'd say once a year to Pila, I don't know if you know it's in Aosta. There's the town of Aoste, and there's a ski resort up there. You start right from the town of Aoste. The town of Aoste, a cabin, you leave and ski back down. It's a means of transport to go skiing up there.

[break]

You may have heard about the [two village names] mountain link project. This is a project to get people who live in the area, or who don't, to commute up. Instead of taking a coach, there would be a cabin at [village name] that arrives at [village name] village. Then it's 3.5 km to the ski lift, and again it's a shuttle bus. Here you have to imagine, every 7 minutes there's a cabin, the time it takes to go up and down, it's not, ... yeah. But we have a big problem, more in terms of people. It's about mobility. But we can't blame them because the area is so poorly served that it's impossible. We've been trying something for several years now, and it's working better and better. It's a free shuttle. Once in the morning, once in the evening. People go on the shuttle, but the person in [village name] says, I'm going skiing in the morning, then I'll come back afterwards. But he can't. He'll take his car. He's often alone.

CA: And I imagine that huge subsidies would be needed for a small area like that to be able to implement better public transport?

X: Well, it's public. First of all, it's not about the domain, it's public. This is public. The postal line you see with the post buses, that's public. It has nothing to do with the ski lifts. We, with the commune of [commune name], have done more to take the shuttle. In this case, we have taken on the responsibility with the commune of [commune name], the commune of [other commune name]. We're paying for this to try to, ... save parking spaces. But the rest is public transport. We had a post bus that arrived first thing in the morning at 10:00. We held a meeting with the post bus to change it, so that it would arrive at the opening at 9.05am. Oh, they said, there's no problem, it costs 7,000 francs a year, ... it's theft. Now, I also understand that some people are looking for other alternatives to the post bus. BUT once again, the link in the middle of the mountains is great for commuters, but then there's the monster job of lateral transport. To get people from the resort to the villages. So there's, ... it's working on it, it's the commune that's working on it.

CA: And you, do you also work in collaboration with the commune?

X: Very much so. Enormously because we currently have people on the local council, including the president who has been there for [X] years, who is a former [company name] employee. So he knows the issues, he knows everything. And he's always available. And we've got a local council that's behind us. That's also what allows us to move forward. But it's seven people who have been there for many

years, and of those seven we have four people who are leaving at the end of this year. That's a big change. Will they still be behind us? We don't know, of course.

CA: Behind you in which sense?

X: Well, in terms of supporting all that, of course. Everything, the ski lifts, ... do they want to keep the ski lifts? Ah well, that's also a consideration. Because here we had a local council that was completely in favour. Ski lifts are important, they bring people in, they build land, they create value or chalets, they increase value. Will they still support the new local council? That's another important point. Apparently, what we're hearing is that all the people who are already close to the ski lifts, ... well in the village, are skiers, people who like it, so in my opinion they're behind it, but it's a big, ... it's a big change too.

[break]

We'll see what happens in October. But politics plays a role, because it's the majority shareholders. You have to imagine that in our case it's the commune of [commune name], the commune of [other commune name], the commune of [other commune name] and the commune of [other commune name]. Those are the big municipal shareholders. Then there are lots of private shareholders. There are about [x] shareholders. But it's true that if we have a majority like the commune of [commune name], which has a majority share, if they're with us, well, they raise their finger, that's fine. After that, they've always been with us. They've always supported our projects. That's also why there's always a member of the Board of Directors from each municipality. And the link is really important.

CA: Do you have to take it to the General Meeting, which will then be voted on?

X: I'll tell you what, first we take it to the commune president. That's what we did recently for the renovation of this chairlift. We called all the commune presidents to [town name], to get their support. Or their support is given with a financial contribution. A loan, not a donation, but a loan. To see whether they are in favour or not. Everyone is OK. Then we can move forward with that. Then go and see the Board of Directors where we met together. But the General Meeting of shareholders takes place in September. We can't wait until September to move forward with projects. We need to get going.

CA: But the General Meeting could say no, right?

X: Yeah, obviously. It's never happened, because in general, we bring stuff in, we've already got the support of all the communes. I mean, it's not, ... But it could lead to comments, for example. But we've always had 100% support. We've never had, ... in any case since I've been here, a rejection or a restricted vote. Of course, there are spectators at the general meeting, but they've never had to do their job and count. So things go very well.

CA: And when you launch new projects like this, do you also go and see what's been done elsewhere?

X: Oh yes, clearly, yeah. When we did the new chairlift, we went to Chamonix because they had the manufacturers, and he said "we've installed this project here, come and see". And then you have to imagine here, we have a chairlift where the operator has a control, here, around his neck, and then he manages everything from there. It was fantastic to see how it works. And clearly. Of course. We were looking at changing this chairlift. We said to ourselves, yeah, we'd need a detachable chairlift, but people would have to be able to go down without snow, with skis on their feet, and also go up with skis on their feet. So we went out and tested it out. We took the chairlift, went up, took a turn, then came back down, there was no snow to get out with our skis on. If we want to be able to use this chairlift in 30 years' time, ... skiing might be 2000 metres higher. So this installation must be used to get people up there. Snow or no snow. And above all to bring people back down. Snow or no snow. And that's a problem when there's no snow at the bottom, you come once, you queue for 1 hour to get back down the chairlift, you don't come any more.

CA: And in relation to solar panels too, have you been to see other installations?

X: Not at all. You have to imagine that in terms of ski lifts, it's slowly starting to happen now. So when it comes to lifts, I don't see ... what we could have. And then everything that's restaurant and, ... there we see the new projects but we've never been to visit another project.

----- End of Transcript -----

Participant 10
Interviewee: Member of a farmer community, landowner of rejected utility-scale PV project Interview date: 3.06.2024 Form: Qualitative, semi-structured, online (phone call) Local acceptance of utility-scale project: No

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Can you explain what role and influence you as well as the farming community had in the decision to build the solar farm?

X: Well, part of it was certainly because we, from the corporation of farmers, i.e. the farming community, we were approached, ... by a colleague from Bern, who is an area canvasser [responsible for finding lands for solar parcs] for photovoltaic areas in the region and is also a member of the cantonal council of Bern. And he asked us about this [the solar parc] for the [electricity company name/project developer], who was the project developer. And it was clear to us that if we could lend a hand, we would give it a try, yes.

CA: So it was for the land?

X: Yes, exactly, it was for the land.

CA: And what other actors were responsible for the decision about the solar farm?

X: Yes, that was just [electricity company name/project developer] as the project developer. And the farming community as the landowner, of course. There was a resolution passed at a meeting [of the

farming community], which was approved with a large, positive vote. Then the old co-operative, i.e. the exploiters of the alp, were involved in the search for a location.

[break]

The mountain railways, as tourism drivers, I must perhaps say, were sceptical at the beginning, but not against it. In the end, they were found to be the driving force against the project.

CA: And why were they against it?

X: That's still difficult to say. Presumably it was a certain loss of monopoly for the mountain railways, certainly at one point. Or the fear that tourism could perhaps be damaged, that was certainly also the motivation for them [to refuse the project]. And then there were other players, certainly the local council, with whom they got together. That also needed the approval of the local community. There was also an assembly resolution, which unfortunately, as you know, was certainly rejected. Then other players were also needed, the technical report, which was prepared by [photovoltaic company] in Bern. There was also an environmental impact report, which was done by [spatial development company] in Bern. These were actually the main players involved in the whole decision.

CA: And what were the motivations from the farming community to participate in the solar farm project?

X: Well, certainly on the one hand a future-orientated project with climate-friendly energy generation, which is becoming increasingly important and which, in my view, should also be pursued for the future and for the sake of the climate and future generations. On the other hand, a certain independence and autonomy both locally, meaning financials for the municipality, and nationally, a certain independence from energy sourcing from abroad certainly also plays a role.

CA: Why do you think such initiatives are important for Switzerland?

X: Well, I think it's forward-looking. Although it's probably still too early for such Alpine regions. I think the impact on the landscape is still significant, perhaps too significant. There will be a rethink over time, I hope.

CA: And do you think the park would have been profitable?

X: I think so. I think it would have been profitable. We also got a lot of information from the Internet or via [electricity company/project developer], which gave us very good, transparent advice. Sustainability is always the emphasises, and for us, the procurement of materials was always a big issue. Iron, I mean iron used for construction, can be recycled. And regarding solar panels, we enquired and found that 30-year-old panels were taken down from the roof, ... and they still had a capacity of up to 85%. So we think that's very sustainable. Surprisingly, yes.

CA: And did you have to seek out information to participate in the solar farm project? If so, where did you find the information?

X: Basically, we didn't have to search for information. If we were unsure, we did some research on the Internet. There were already projects that were further along than ours. And on the other hand, we used the federal government's guidelines, which are also very transparent. But we also have contact with a local, regional, small electricity producer, [electricity producer company name]. They produce [drinking] water from hydropower in [commune name]. Water springs here on the [region name] and we are actually in contact with them. We were able to ask them a few questions about what the whole thing would look like.

[break]

But what was also very important was that we received very transparent and very good advice and support from [electricity company name/project developer]. So when we had questions, they answered to us in writing and also gave us the source of the answers. So where the whole thing came from, it was very transparent, yes.

CA: And what were the questions about the project?

X: So for us, it was certainly not just about what happens within the perimeter of the solar installations. The transformer station and how things would proceed were also important. For us, it was also very important whether there is harmful electrosmog [electromagnetic pollution], that could harm animals or people, or how the power discharge is managed and whether the power company can handle it. That was all very important as well, yes.

CA: I don't think you did this, but could you get subsidies from the canton or state for this project?

X: So there is, yes. I don't know, 40 or 60 percent, so about half would have been subsidised. But the subsidies would only have gone to the developer, not the landowner. The landowner would only have been compensated by the developer on the basis of the area made available.

CA: So you didn't do the administrative part? You participated to the project and then finding all the information like government subsidies was the role of the electrical company?

X: Yes, exactly right, that's [electricity company name/project developer] who had acted as the project owner. And the forest community, where the land would have been made available.

CA: And to find information, you said before that there were other projects that were further along in the project development. Did you seek information from those projects?

X: Yes, that was actually very easy to research on the internet. There were two or three solar parks in the Surselva or Bündnerland that were already much further along than we were, ... where we actually found a lot of information about it.

CA: And was that also farmer community that had the land and were there the same concerns?

X: Some of the same concerns were certainly there, yes. And it was also partly co-operations that provided the land, but also municipalities that had the land.

CA: When you had all the information and had to assess whether you wanted to join the solar farm project, what considerations were important? Always the same as you told me before, or did it change?

X: No, so in part for sure, as mentioned before with the motivation, it was the climate-friendly energy generation for sure and a certain independence. But also the fact that everyone has to take the first step to make a difference. An important guiding principle for us was that everyone needs electricity, everyone wants electricity, but please not on my or our doorstep. We thought that was selfish. It should be different for us and we were of the opinion that everyone should contribute their share to a future-orientated electricity supply. These were the considerations, which of course also motivated us to receive an enquiry. We've already said to inhabitants that we're not just saying no, it's worth supporting.

CA: And were there differences of opinion about the assessment of the project among the various decision-makers and why?

X: Yes, as perhaps already mentioned at the beginning, there were certainly those in the municipality, i.e. the municipal council, the commission and alps people actually, there was already agreement that it would be a good thing. But of course everyone said, yes, in the end, we don't know if we like it, we don't know what it will look like in the end. That's why there was a lot of scepticism. People say they have photomontages of what it might look like, but it's difficult to say whether it will look like that in the end. And I think there was a lot of scepticism.

CA: And what were the deciding factors that made you want to take part in the solar farm project?

X: Yes, the three stakeholders, the farming community, the local community and the alpine co-operative were actually in agreement from the outset that we would utilise it as a tourist advantage in the region and that both in summer and winter, many regions advertise with a certain climate friendliness. And with this project, i.e. this region, but also the [ski resort name] ski resort, could be rated as climate-neutral with such a system. It was quite a large area, where the industry plus households were supplied on the [region name]. We thought that would have been a great addition.

CA: Are you satisfied with the set-up process?

X: Yes, with the set-up process clearly yes. When you consider that the whole project was worked out down to the smallest and most precise detail in the preliminary phase. Yes, which makes it all the more difficult that there was so much scepticism from the local population. And that the municipal assembly rejected the project by a very large majority. And this is, after all, a political decision if it is to be accepted. And as was said earlier, perhaps the time is simply not yet right and the people are not yet ready for a certain cut in the landscape.

CA: Do you think that because it was rejected, it will perhaps be more difficult in the future projects?

X: That's what I'm afraid of. So the fear is great. We've been saying that from the beginning, the farming community, ... from the beginning. It will be difficult to find someone to invest so much energy in a project again if the uncertainty is so great that it is not accepted at all. So no investor will probably dare to do it again.

CA: What were the biggest difficulties in the adoption process, do you think?

X: That it wasn't received. I think what was also noticeable was the effort involved, or rather the transport over the [region name] to the site, because there are so many individual parts, so many lorry journeys and all the construction noise and everything, which probably also led to it being rejected.

CA: Did you communicate your project to a wider audience? For example, the press, and do you think it's important to do that?

X: Well, we communicated it very openly. The press was very active, but only when the whole thing became more concrete. In the beginning it was only superficial, but when the whole thing became concrete, we organised three public events. So if anyone wanted to find out more, they could do so very well and very transparently.

CA: And why do you think people still didn't want it?

X: Yes, I think many have already formed an opinion from the outset without questioning it, that's the problem. So as I mentioned earlier, we also asked ourselves, or questioned, what are the material costs. The fact that you can recycle a solar panel today, and that the metal that is there remains, it doesn't break, it can be recycled. And many people were probably sceptical about the fact that a lot of resources are used for very little electricity, which is not really the case.

CA: Would you recommend starting another Alpine solar park project, and why?

X: Yes, I'm still interested in things like that. I read another report today, there are between three and nine projects in the Oberland. They would have been the same size, or slightly smaller, which are still in progress at the moment, and some of them are already very far along without the landscape

conservationists or anything like that objecting. So I am still interested in such plants and I am convinced that things are moving in the right direction.

CA: And do you think that such initiative would have been a competitive advantage to attract tourists to [region name]?

X: Yes, I would have thought so. That's what I said earlier, many regions advertise themselves as being climate friendly. And this project could definitely have achieved a lot. Also in terms of tourism, I think it could have been noticed, as it goes underneath a hiking trail, and if you had a screen there with the initial PV system, where you could describe how much is produced during the time you walk through, that would certainly have been a marketing gimmick.

CA: I've asked all the questions that I had prepared, but is there any aspect that I didn't cover that you'd like to share?

X: No.

---- End of Transcript ----

Participant 11
Interviewee: Landowner and member of the municipal administration of a rejected Alpine utility-scale PV project Interview date: 6.06.2024 Form: Qualitative, semi-structured, face-to-face Local acceptance of utility-scale project: No

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Can you introduce the solar park project?

X: The commune and the bourgeoisie were involved in the project. And that's it, in terms of this project, that's all we represent, because we weren't the initiators of the project, they [electricity company name/project developer] came to present it to us. And then we took it on ourselves as the owner of the land, the bourgeoisie, and then the commune, because we were interested in the project.

CA: And you were approached by an electrical company for the project?

X: It's [electricity company name/project developer] who approached us, as part of discussions we were having, which were quite different, it was on another matter. They had started to investigate the Alps as part of the Solar Express project. And then, in the commune of [his/her commune name], they analysed six potentially interesting locations. And then, using a grid of criteria, risks, opportunities, environmental problems and others, they settled on the [solar parc site name] site, which had the most advantages and the fewest disadvantages. They came to us proposing this solar farm. At first, we were

talking about 200,000 meters square. After the initial studies, we were looking at 120,000 meters square. It's a very interesting location, close to the [ski area name] ski area. Because we're on the border with the commune of [adjacent commune name], in fact. So there were already amenities to get the energy out so to say. There was no need to build a new power line. That was important. [And it was] In an area that wasn't in a nature protection zone, which was used for grazing in the summer. But it wasn't the most interesting plot from an agricultural point of view. With an access road that wasn't too far away, so it wasn't too complicated to build. And with limited avalanche danger. So, really, there were quite a few points, at least orange, not to say completely green, about the project. And we've made a decision in principle [agreement on the initial project]. So when I say we, I mean the local council, which also represents the interests of the bourgeoisie. We made a decision in principle, as I remember, but I can tell you just in case, on [date of the decision], we said OK, we'll study the project. We'll give the go-ahead to study the project, to see whether it's feasible or not. That doesn't mean that at the end of the day we're going to sign off and say OK, we'll do it. We have launched this study. Rapidly, within the Municipal Council, we did not have unanimity. Some people were completely opposed for reasons that were more ideological than factual. "Before you put solar panels in the wild, you put them on roofs and façades" [people were saying]. These were the arguments we also heard during the cantonal referendum campaign, ... you may have followed it, ... on facilitating authorisation procedures, in fact, for solar farms. It did not prevent us from starting the study nevertheless. The project was not originally a municipal one, but the municipality was involved from the outset.

CA: As you said, there are several parties involved in this decision. What were the considerations of the different parties?

X: Well, there was one more private player who was approached. It was [ski resort company name], which owns the ski area right next door, which could have possibly made it possible, ...[for context] they were building a new chairlift, which was planned in any case. There could have been synergy between the two projects, just to clarify. So there was the [ski resort company name], and then there was the commune of [adjacent commune name], which could have evacuated the energy through its own network. So there were quite a few partners. Then, if I take you in order of the various partners, the [his/her commune name] bourgeoisie was interested because it was developing its land. So they would have received an annual fee for the use of the land for a minimum of 30 years, renewable for a further 30 years. The local authority was interested because it received a share of every kilowatt-hour produced. So it was money that it could have made available to its constituents, for example to set up a renewable energy fund, that sort of thing. For example, the more shares it would have bought in the company [project of solar parc], the more energy it could have sold directly and used as it saw fit. So we had both aspects.

CA: So it's also self-consumption?

X: Yes, self-consumption. Or it could be sold on the open market via a trader. You could do whatever you wanted with it. We would have had a share of energy as well as money directly. [Neighbouring ski resort company name] could have supplied the [ski area part name] part of the ski area directly, in fact. If you know the [ski area name] ski area, you have [ski area name], then [ski area part name]. Here, we're really adjacent to [name of the ski area part]. So they had a direct interest in using this energy. And they would also have had an image, I don't mean doing greenwashing but, ... they would still have a renewable image. And that was interesting for them. The municipality of [name of an adjacent commune] also would have recovered some of the energy on its network. I mean, they could have put it to good use. [Electricity company name/project developer], or even other potential partners, ... that's something we would have had to discuss after, we didn't get that far, ... would have recovered the energy left, and then could do what they wanted with it. So there were quite a few stakeholders with concurrent interests.

CA: And with hindsight, do you think there was one group that was more of a problem than another for the realisation of the project?

X: Well, the group that caused problems was the commune of [his/her commune name]. Not in the sense of us, the municipal administration, but the population, in fact. Because when we made progress on the project, we kept it confidential for a few months, just long enough for us to carry out the studies and to be clear about the potential, the risks, the financial and environmental considerations and so on. Then we called an information meeting, where we presented the project with [electricity company name/project developer]. And then, very quickly, we felt that there was a problem, ... because we sent out a mail to all households 20 days before the meeting, something like that, and I got quite a lot of direct or indirect feedback from people who were vehemently opposed. Even before they knew what we were talking about. Alpine photovoltaic solar panels, no way. Really, it was a no-go. So we said to ourselves, we'll do the information session anyway, and then we'll see. And at the information session, we had about 250 people, which is quite a turnout. I'd say 150 of them were sitting with their arms crossed, and the legs crossed. When you see the body language, you think, this is going to be complicated. We presented for an hour. [Electricity company name/project developer] was there with us and had a bit of experience, because they had already presented 2-3 other projects in other municipalities. [The two other municipalities name], where they were involved. And the presentation went well, so to speak. But when we open the floor to questions, at the end, I give the floor to the audience, the first question, a hand went up, and then someone said, "could we still ski in the [solar parc site name] ?" But nobody skis there. It's more a route used by some guides and a few people in the valley who know it. Then I saw a hundred or so people starting to shake their heads and say, "that's the question". From then on, we knew it was going to be very complicated. And above all, [his/her commune name] is a large commune. There are a lot of people in the room who live on the hillside, which is right under the solar parc site. They wouldn't have seen the panels at all, because it's physically

impossible, it is hidden away at the top. But we felt that we were going to touch their piece of land, their nature, and that it wasn't possible. So we thought, wow, ... this is going to be complicated. So we went back to the local council and said, what are we going to do with this? Because we already felt that there were some around the table who weren't convinced, who were reinforced in their vision of saying, no, we don't want that. And then, just afterwards, there was the famous cantonal vote, the decree on the solar express, let's put it that way. So we said, well, we're going to let the people of [his/her commune name] vote on the cantonal decree anyway, and then we'll analyse the result. Even if it's not directly a vote on the project of [solar parc site name], we'll still understand the political message behind it. And then [name of his/her commune] voted a two-thirds no in the cantonal referendum. So we thought, let's pull the plug. We would have had to do, ... there was a very high turnout [people in the commune participating in the vote] compared with the rest of the canton. So 750 people said no, more or less. This meant that if we held a primary meeting to vote on the project, we could reasonably expect there to be 500 to 600 opponents in the room. So we needed to find 500 to 600 people in favour. So we would have ended up with an insane assembly to manage. And then we said to ourselves, it's not worth it. I just summed it up for you. So we took the formal decision to abandon the project. It was announced in the media. We were, I think, one of the first local authorities to actually say, well, we're stopping. [Another commune name] said, we're freezing, seeing what's going to happen over the next few months. We said no more.

CA: And the arguments they were in relation to the landscape you mentioned, right from the start, they were fixed on their opinion?

X: Right from the start, that was it. It was partly an alliance of circumstances, I think, between people who were fiercely opposed for ideological reasons. As I said at the beginning, one should put up solar panels on the buildings first. And then out into the wild. It doesn't matter whether it's coherent or efficient. Well, that was the vision. "We're not going to denature our landscape for the Swiss plateau, which consumes too much. Then it's us again that will have to pay by destroying our lands. In fact, just as we built dams in the past, we disfigured our valleys "... ignoring the fact that this allowed us to develop our valleys economically, but that's another debate that affects people much less today. So that's it. That was the main argument. It was to say no, we don't want it, it's disgusting, we're damaging the landscape. Then there were the local people who said, I go ski touring up there, I'm skiing there. A lot of guides, ... we felt that there were quite a few guides living in the area. In fact, it's a route that they use, when they come down from the ski area, they go down this way [through the solar parc site]. And then they go for in the forest, skiing. And that's it. Those are the arguments.

[break]

Beyond that, I think there's also a local political movement that's rather left-wing. It's called [left local political party name]. Officially, it's not the Socialist Party, it's not the Greens, but it's more a left-wing

tendency, which is represented on the Local Council, by the way. And it has taken the project on board and made it its hobbyhorse. So, it also made things a bit tense. Within the centre, there were also people who were against it. And within the LRP [liberal-radical Swiss party] too. But it was [left local political party name] that led the way against the project. So there you have it, it's also turning into a bit of a political debate. You can't make people happy without their agreement. They don't think it's appropriate. We've decided to stop. It should also be said that on the territory of the commune of [his/her commune name], we are in the middle of negotiations for the return of the [his/her commune name] power plant concession, [power plant company name]. This means that we already have quite a lot of clean hydroelectric power. So why start a war across the municipality over this [solar parc] project, when in a few weeks' time we're going to be voting on the return of the [power plant company name] concession. Politically, we didn't want to start a war across the municipality either, at the risk of undermining the major issue of the return of the concession. It happens every eighty years in the communes. It would have been a real disaster for us. It's also a question of political interest.

CA: I see. And do you think it's perhaps too early to undertake such projects? Do you think that perhaps with future generations, it will be easier?

X: So looking at the representation in the room when we had the info session, I think the problem is more the reverse. Back home, the older generation, if you want to put it that way, has been pretty much won over to the idea of saying, ... in any case, they didn't come out loudly to say we don't want this. "Well, after all, why not?". On the other hand, the younger generation has quite a lot of people who have, ... we heard the word degrowth a lot before we put up panels, we consume less. Fair enough, but how do you do it from one day to the next? I'm not passing political judgement, it's just to say, we've had this argument in the new generation of saying "no, but it's not possible. We don't want panels. All we have to do is manage to put them up everywhere [else than mountains]". It's a bit an idealistic vision to say that "you can put panels on every roof, they'll have the same output, and then, if it doesn't have the same yield, we put twice as many". Knowing that the solar farm we had planned would have produced around 25 million kilowatt hours. That's a quarter of the output of the [his/her commune name] plant, which is quite a lot. And to produce that much photovoltaic energy, I think you'd have to cover practically all the roofs in the commune of [his/her commune name]. Without thinking about how much sun they get. If you put panels on all the roofs, and then you calculate how much they'll yield, you'll no doubt realise that you don't get the same yield from these south-facing panels at an altitude of 2,500 metres. But the younger generation was still saying, let's not waste our nature, let's find other solutions. I felt more like that was how it played was.

[break]

There's a form of local conservatism, but it's perhaps linked to degrowth, in other words a slightly idealised vision of nature, saying, "OK, we've polluted enough, you've polluted us, now we don't want

that". That's surprising. But afterwards, it represented a movement that took the... The opponents always shout louder than the people in favour. So maybe on the whole, the younger generation would have been.... I don't realise. In any case, we've heard a lot of young people saying we don't want that.

CA: What were your basic motivations for participating to this Alpine solar park project?

X: We saw it as an opportunity, we the communes. After, I will talk about the bourgeoisie again. For the commune, we saw a double opportunity. Firstly, to have additional energy of our own. Self-consumption or sale on the market. Then, with the difference [surplus], the money would have been used to supply the municipal household. That's for the energy part. For the immediate return part, as I was saying, we had an amount that we had negotiated where for each kilowatt hour produced, we received a direct amount that would have gone into an environmental transition fund. Things like that. If someone wants to put panels on their house, we can subsidise them. If someone wants to convert their electric heating to solar heating, that sort of thing. That's what we saw. Then, if we want to be a bit more global, there's also an image of the municipality giving itself a green, renewable image. But we already have a huge stake in hydroelectricity. We don't have much to prove on that front, but all the same. We're a player in the transition.

[break]

Then the bourgeoisie, there was an immediate return linked to the development of this land which today earns us three times nothing. It's leased to a consortium for grazing, but that brings in a few hundred thousand francs a year. If the land had been allocated to a solar farm, it would have brought in several tens of thousands of francs a year. For a bourgeoisie like ours, it wasn't the goose that laid the golden eggs, but let's just say that it would have been a very interesting source of income compared to the usual operation of our bourgeois finances.

CA: Would this project have been considered if there had been no state subsidy?

X: Then I think you'd have to ask [electricity company name/project developer]. They were really the ones who came up with the project. We had a range of subsidies for the project. We didn't know the exact amount. We knew it was going to be between 32% and 38%. Finally, there were the ratios to be determined. It's clear that if the municipality had to replace the federal authorities to provide the money, we wouldn't have had the means. So these would have been considerations that we would have had to put on the table at the outset, in order to find other partners. Would the [local energy group name], a major energy group or a bank? I don't know who else, also would have put money in. But one way or another, at national level, if we go up a level in relation to these projects, it's clear that I can understand the criticism that these projects are massively subsidised via federal funds. On the other hand, we have subsidised nuclear power, hydroelectric power and all forms of energy. And we realise that if we want to have a credible policy in terms of, not complete energy independence, [but] if the country wants to

have its own energy, without subsidies, it's complicated. So I think the project would still have gone ahead. And then, if we really had been told it was zero [subsidies given], we might have had to find partners. I make the analogy with... I don't know if you saw a small article in today's [local journal name] explaining that we've just invested 13 million to rebuild the [commune name] power plant, in which the commune of [his/her commune name] has a 51% stake. We launched this 13 million project to increase production at the plant, even though there was no federal subsidy at the outset. And then along the way, we learned that we were still eligible. So that was the icing on the cake, a big icing on a big cake. But to say that it's still possible to develop energy projects, despite the lack of subsidies. And then, if they come, all the better.

CA: And do you think that the solar park would have been profitable?

X: The idea was to find a balanced economic model. But then, it would have been profitable based on federal subsidies or through our partner assuming the risk. After that, it depends on how you value the energy. Because you have two models. Either you take the energy yourself and then distribute it to your subscribers. Which we could have done, for example, directly to our subscribers. Or you sell it on the market. You sell the energy to a trader, who makes a profit on it. So that depends on your business model. For example, we produce all our own energy and sell it to a company called [local trader company name], which is based in [adjacent commune name], who is a trader, and does only that [energy trading], and of which we are a shareholder. This allows us to mitigate the risks. Because if the municipality, directly, has to go and sell energy, beyond its subscribers, ... I say really, go to the market and then say "OK, hello, we've got 5 gigawatts, who knows who wants them?". It's a real job. So there you go, you have to put your trust in professionals somewhere.

CA: How did you find the information you needed to adopt this solar park in the mountains?

X: [Electricity company name/project developer] has know-how, knowledge, a network, lobbying in Bern. I think they were involved in creating the law at the federal level too. In any case, they were following it very closely. So we had [electricity company name/project developer] on one side. And then we also had regular contact with the cantonal energy department, which also gave us input on the legal bases and possibilities, because we did not want to have hand and foot tied to [electricity company name/project developer] either. There were also meetings with the State, to which we were invited, because we were a municipality that had announced that it had a project. In particular, we had a meeting with two State Councillors. One in charge of energy, the other in charge of the territory. Their respective heads of department. And then we had live explanations of the procedures and so on. So there you have it, we had both sides. And then there was personal interest too. When I say personal interest, I mean the elected representatives, the members of the administration who also have contacts, who were able to get information.

CA: Is the procedure complicated?

X: So, in terms of administrative and legislative procedure, I know how things work. Obviously, I don't know everything inside out. But all the same, I have the entry doors to the cantonal administration level. After that, we were in a bit of a blur because there is a federal basis, and then there was an attempt to create a cantonal basis. The Great Council adopted it, there was a referendum, then the population had to vote. So we were in between. There was the federal basis. Then there was perhaps a cantonal basis that specified certain elements. So we worked on two scenarios. In the first [scenario], there was only the federal basis. Then in the second [scenario], there's the federal basis plus the cantonal one, in case it's accepted. And then it was rejected. So we said we'd stop everything because we understood the political signal the public had given us. That was it, in fact.

CA: And were you apprehensive about the actual production of energy in solar parks in the mountains?

X: In that case, we relied completely on the technical expertise of the [electricity company name/project developer]. Of course, we still had questions. But afterwards, we said to ourselves that if a company like the [electricity company name/project developer] is interested, it must be worthwhile in some way. But obviously even they were a bit... How should I put it? Technically, there were several solutions being considered. Normally, we would have had to install test installations, a few panels for a few months to see how they reacted in terms of production. And here, we were relying on them, obviously.

CA: Because it's often complicated too, I imagine, with the snow, how does it work, what gradient? I know there are questions about that.

X: Or what system? We even found out over there that they had two systems. In fact, for skiing, it would have been funny. Because there was a system where you put up masts and then you put your panels on the masts. You put up, I don't know how many, 24,000 masts or whatever, with 5 panels each. And that's it, you can't ski anymore because there are masts every three metres. Or they had developed a second system with clotheslines. You put two masts maybe 50 metres apart, then you pull a cable between the two, you put the panels on the cable at a height of 3-4 metres, in which case, it would still have been possible to ski a little. That was the big argument for skiing. They explained this when the question came up that there might still be a way to ski, but we can't commit ourselves at this stage [that it will be possible to ski]. There are various technologies that are still being studied. I don't think it's finished yet. From what I understand, they still have possible variants.

CA: It seems like with each project, there are new tests to see what is the most efficient.

X: If you make the analogy with dams, it's exactly the same thing as saying... At the time, we built heavy dams, the Grande Dixence, for example, is a gravity dam. It's a huge mass of concrete, 200 metres at the base and 20 metres at the top, something like that. You have arch dams. Emosson, for example, is a large arch dam. Then you have double-arched dams. If you go to Toules, not only is it vaulted on

the... It's not a weight that's put down, it's an arch. But what's more, it goes up... There are several technologies for dams. There are several technologies for solar panels. I think it also depends on the configuration of the site. The slope, the sunshine, the wind, the snow and so on.

CA: Do you think that the fact that more and more projects are being stopped or rejected by the public could be a brake on future solar farms?

X: Yes, that's the great paradox in Switzerland. Because we're one of the richest countries in the world. We have the means to invest in trying to develop our own technologies. And then, as I was reading in [local journal name], a week ago there was a graph showing how far behind the rest of Europe and the rest of the world we actually are in wind and solar power. It's staggering. There are 40 wind turbines in Switzerland, or something like that, or 50. There are 1,500 in Austria, right next door, which has more or less the same topography as ours. So we really do have the impression that there is a gap between the understanding, or rather the popular will, which says "We want to consume less, we want to consume cleanly, etc. But not here." In English, they talk about the syndrome "Ok, I'll do it, but not in my backyard." I'll go and put it in the neighbour's. And then we really get the impression that more and more people are thinking like that, saying, "we live well, we don't need this, so you're very nice, but go put it in the neighbour's garden". I've heard people say to us, ... they've asked the question, "Why don't you put it in the next municipality?" Really, the commune next door, 500 metres away. [The reason we put them here is] because it's more worthwhile technically and technologically in terms of sunlight to put it there. That's funny. We'll see what happens this weekend, in this case the vote on 9 June. The right-wing populists say, if you have seen the posters "No to fattening up the electricity barons" - well, I have to say that some are lining their pockets. And the extreme environmentalists, since even the WWF and pro-natura is in favour of the law, said yes. On the other hand, it's really the extreme groups who are saying that we're sacrificing the landscape for the sake of energy. If the law is passed, I think it will open up the situation a bit, because then the cantons will have to define the sectors in which they agree to develop this [solar parks]. We'll have a little less power, the communes, so much the better perhaps on the one hand. Perhaps it'll be a little more coordinated, and in the cantons, we'll have to say, ok, solar farms, we're putting them in, I'm schematising, but in this sector. Or I don't know, at the bottom of the [valley name], and then if there's a commune that comes up with a project that's completely outside the sector that the canton has defined, well that's a no go. Perhaps this will help rationalise development a little. It remains to be seen. But it's true that the population is a bit schizophrenic, we want to consume less, consume cleanly, but not at home.

CA: Do you think this has something to do with the fact that electricity is a commodity that we only think about it when we no longer have it? A "boring" consumption good?

X: Well, typically, or when it becomes very expensive, when there is a crisis. Because we lived through the crisis. Two years ago, Guy Parmelin said that you had to cook by putting the lid on the saucepan,

etc., that there were going to be shortages. What I mean is that, “unfortunately”, there hasn't been a shortage. Unfortunately, in the sense, not that I want there to be one so it's a mess, but the population needs to realize how quickly, in the event of an energy problem, it can turn into chaos, you know? And there, we have rich people problems, you know. In fact, we don't realise what it's like to have no energy, no electricity. For an hour OK, that's a blackout. Three hours, that's a major blackout. Half a day, it gets complicated. From one day onwards, I think, it's a disaster. I read this not long ago, the secret services, I think it's the British, in their risk analysis linked to the functioning of the country, estimate that England is three hot meals away from anarchy. This means that if, for a day and a half, there are really massive problems, such as a lack of energy or something else, people's behaviour can become so irrational that you can end up with people shooting at each other in the street because they don't have any food or they're afraid of [not having enough food]. We experienced this kind of thing, all things considered, at the beginning of Covid, when people were rushing in shops to buy toilet paper. You see, these are deep-psychological phenomena, which we don't realise, but it can quickly get out of hand. I mean, I'm not at all conspiratorial, but when you put it like that, that's the kind of thing that hangs over your head. And so you don't realise it at all, because everything's fine.

CA: Maybe there is not the awareness yet of the possibility of blackouts.

X: Maybe it will come. It'll probably come in countries where it happens a lot. I have two or three friends who have lived in countries where the blackout, or rather the shortage, happens two or three times a day. You know you have no energy between 8pm and 10pm. You organise yourself. But that means that between 8pm and 10pm, depending on the people, you don't go out into the street because there's no light and so on. There's security [in Switzerland]. We have a different way of acting [as a result].

CA: When it came to building the Alpine solar park, what were the key factors to be taken into account? And have they changed since the project began?

X: Well, first of all, we needed to know whether it was technically feasible. [Electricity company name/project developer] told us yes. What was the municipality's financial commitment? There were ratios, criteria that said, OK, at the end of the day we take X percent, etc. Then, once we'd decided to go ahead with the project, we had to make a decision. Then, once we had all the technical and financial elements, did the population agree? And as there was this cantonal vote on the cantonal decree, we didn't hold a communal vote on the project itself, since the timing was a bit off. It all happened at the same time. We said that we were going to interpret the cantonal vote as a plebiscite on the municipal project. Because we had held the information session before the cantonal vote. So people in [his/her commune name] knew about the project when they voted at cantonal level. It's clear that it's a bit far-fetched in some respects. But it was clear that people weren't going to vote for or against the decree. They were going to vote for or against the photovoltaic solar park projects. And in [his/her commune

name], in particular, [they were voting] against this project. And we had... I think the cantonal average turnout for the vote was 33%, 34%. We had a more than 50% turnout. So people really voted. It's not like people just don't care and don't vote. With two thirds of people saying no, the result was really clear. And from then on, when the public told us all the good things they thought of the project, the bad things in this case, we said OK, we're not going any further.

CA: With regard to the process of adopting the solar park, are you satisfied with it as such?

X: Yes, in the sense that we were able to put the project together and then indirectly question the population via this cantonal vote. Otherwise, we would have had to hold a communal vote, a primary assembly. So it might have been a bit more complicated to manage. It might have been more emotional. A perhaps complicated meeting with I don't know how many hundreds of people in a room shouting at each other. And then a vote. And then at the end, the results are announced. And then one side explodes with joy. We've seen that happen with some of our local authority projects. It's not the most pleasant moments to deal with, but it's manageable. But in the end, the process itself was... I don't think I'd change a thing if I had to do it all over again. Because we were able to... There's a question of timing, perhaps. But in this case, it's beyond our control. We started the project a bit later than other projects. So, in a way, we could have benefited from the experience of others. But above all we benefited, in contrast to the general irritation about other projects, from people who said... "Ah, but they're doing it at Anniviers, they're doing it there, they're doing it there, they're doing it there. Ah, and now again in [his/her commune name]. So it's really getting out of hand, and everyone's jumping on it. "It's all about money" and so on. That's the argument we've heard. And every time a project was added, it was, in a way, a question of purely financial opportunity. They were thinking that, "there's a way of making money, so let's get on with it" [which is negative]. And the most important thing, which I think everyone paid dearly for, was the absolutely catastrophic communication about the very first project. I don't know if you remember the first images that appeared in the Upper Wallis, saying, "We're going to build a park of 5 square kilometres", and then they made projections. So there were solar panels on 4 mountains. But that was produced by the promoters of the project, even! And I think this image - you take a gun and shoot yourself in both feet - has the same effect. Because they are still paying for that image. So there really was a communication blunder on the part of the people who had the very first file in hand, which was a bit complicated to manage for everyone involved.

CA: And during this process, did you go and see how things were going in other projects too?

X: Mostly telephone contact with local representatives, elsewhere I know fairly well. There was a briefing session with the canton, where all the communes that had a project were on the table. So yes, there is information. Contact with the federal authorities, including the federal representatives who had voted for the law in Bern. That's mainly how we got our information. And after that, we also had quite a lot of contact with the [his/her canton name] power companies, because we're in the process of

negotiating the return of the concession for the [his/her commune name] plant. I'm in close contact with this sector at the moment.

CA: And when you get together with the local authorities involved in this kind of project, do you think there's influence between one another?

X: Well, after that, every project is different, with different local context and so on. But we still try to have a... The ideal is to have a common vision or a common message. Because some people say, I'm developing a project at home. I'll say whatever, but to... To fight nuclear power. Then there's the other one who says he supports nuclear power. People are going to say, wait a minute, what's going on here? So we try to explain things in the same way. It's really linked to the environmental transition, to local opportunities, in fact, to gain energy and money for local populations. Then, in [his/her canton name], there's the sacrosanct autonomy of the communes, which is virtually untouchable. We really are a canton where the communes have enormous power in terms of their willingness to develop projects within their boundaries. So that's a good thing when it's used properly. And then, in this case, it allows each commune to say to itself, I'd like to be able to develop a project or not. So we had a project. [Name of another commune] did not have one. [Name of another commune], when they saw the result here, they said, OK, we're not going any further. In [name of another commune], there's one, I think there's still one under study, maybe even two. So there you have it, each municipality does what it wants, but in the right way, to defend the interests of its citizens.

CA: Looking back, what do you think were the biggest difficulties in this solar park project?

X: The general and popular context linked to the lack of a common political vision on energy, in the sense that we still haven't found a consensus at national or even cantonal level on what kind of energy policy we want. Between those who say we need to build a new nuclear power station, those who say we need to put panels on all our roofs and facades, and those who say we need to develop hydroelectricity even further, we are not clear. We have an energy strategy for 2050 in Switzerland, but it's pretty vague for the everyone. And that's a bit of a drawback of our direct democracy, where in fact, by calling a referendum with 100,000 signatures, you can completely overturn the political reality. In 2013, there's Fukushima, we vote on the nuclear phase-out. In 2022, there's a war in Ukraine, ten years later, nine years later, which completely overturns the paradigm in terms of energy supply, and prices in particular, ... How do we deal with that? We decided ten years ago to phase out nuclear power. Ten years on, we're in trouble because we're going to run out of energy at the current price. As direct democracy allows us to vote four times a year, we could still have more surprises. We're voting this weekend.

[break]

It's an incredible richness, we really need to keep it that way in Switzerland. But it does mean that sometimes we can make 90-degree, not to say 180-degree, turns in the space of a few years. Unlike democratic countries, where you take a line and say we're building 50 nuclear power stations, and that's that. Or you dismantle them all, put 10,000 wind turbines in the North Sea, and install hundreds of square kilometres of solar panels, as Germany has done. We can't do that here.

CA: Do you think that, with this political system, it will ever be possible to reach a consensus - it's hard to say - on this kind of project?

X: Sunday's vote will already give an idea about it. On the preponderance of energy over nature. Because in a way that's what it's all about, it's the balance of interests between whether we favour nature or energy. If this is accepted, the balance will tip a little more towards energy. That doesn't mean we're going to put up panels anywhere and everywhere. But we'll already be sending out an initial political message. After that, I think it's important to give local people a voice. I think that's important. Because if you end up with local people who don't see the point of having a facility of any kind in their area, it's complicated to manage. I'm not even talking about closing down a valley, building a new dam and relocating 5,000 people. This isn't China, where they moved the entire population of Switzerland to build a dam. We're not there. But really, the local population has a say. After that, the debate has to be rational, if possible. And that it's not just a case of 'not in my back yard'. Saying 'no, I don't want it because I don't want it'. It's also the job of local authorities to be able to pass on projects and explain them properly.

CA: Do you think there's a better way of explaining these projects so that people understand the benefits?

X: If there was a magic formula, there would already be 1,000 wind turbines in Switzerland and a few more solar panels. I think it's complicated. What might be interesting is really to make people understand the direct influence that this could have on their electricity bill. Even though it's very difficult to do so directly. It's not like water. You turn on the tap and then you pay directly for what you use. Electricity has been turned into a commodity that is quoted on the stock market. The electricity that you consume now, you will pay, ... may already have been sold on the European energy market, and so on. It has become a value, ... I mean there is no longer a direct link between what you consume and what you pay for. From that point on, it's not possible to have the direct value on the bill. That's the big problem we have in our local authorities, where people tell us "it's not possible, the plant is there, my house is 100 metres away, you produce at 7 cents, you sell it to me at 22. Who gets 15 in the pocket?" That's got nothing to do with it. It's not like a water pipe where you pay directly for what you use. Getting people to understand that is really complicated. That's the big challenge. Even for us, it's complicated. We're in the thick of it. As a local authority, we sometimes find it hard to determine, when we touch the 'electricity production' lever, what will the end result be for the consumer's final bill?

Because we have a stake in production, distribution, electricity networks and energy recovery, and we sell energy. How can we get the right levers moving to have a direct influence on the bill? But after a while, it almost becomes a populist argument to say that there should be a direct correlation between the factory and the home. But it doesn't work like that because the whole network is European, it's balanced at European level, so you have no choice but to work together on pricing.

CA: I understand, it's a challenge. Solar Park get media coverage. Do you think it's important to share this kind of project with the press, and why?

X: Yes, because we live in a democratic society of transparency where if you don't communicate actively, people are going to come down on you anyway and say that if you don't communicate, you must have something to hide. So in any case, you have to communicate. The question after that is to know at what point you communicate in terms of the project's degree of maturity. If you just say hello, we've got an idea, we'd like to put a million solar panels here, that's not enough. In the sense that you have to say why, who's paying for it, what's in it for them, and so on. So you need to have a minimum amount of information to give. And that's a bit of a challenge, too: how you inform people, at what point, and how you build a public opinion that has the impression that it's being informed and that we're not simply imposing a project on it [the people] that it has nothing to say about. Then you'll always have people who are against it on principle, who will in any case use every possible argument to say that you're trying to pull the wool over their eyes, and so on. That's part of the game. You need a majority, that's it. So communicate, yes, actively, but at the right time. That's it.

CA: We touched on this a bit at the beginning, but do you think that an initiative like this from alpine solar parc could have been an advantage for [his/her commune name] in terms of tourism?

X: Tourism? Not necessarily, it wouldn't have revolutionised local tourism, because we already have very strong tourism. We're an exception in [his/her canton name], we have more people in summer than in winter. It might have brought a few people, but it's at the top of the [ski area name] ski area, so people might have gone for a walk to see. But other than that, I don't think it would have added much. A little, but not a lot. I don't think so.

CA: Would you recommend adopting an alpine solar park? Why or why not?

X: Let's just say that if, in a given local context, we realise that by producing via a solar farm, you have enough energy either for your local subscribers or to generate revenue for the municipality, and then you manage to convince the population that it's worthwhile, then yes, we should look into it. It's like, what's the point? It's like any form of revenue for a municipality, anything that brings in money for a municipality, why not. You see, in the commune of [commune name], are currently studying a floating photovoltaic solar park on the lake. You have the [dam name] dam, then on the lake, there are currently 6,000 meters square of panels, there is a small installation in the east, they are currently studying, for

example, making a much larger one. Typically, I can't see people shouting that you're denaturing nature, because you're on a lake that's already artificial, so why not? So there may be potential to develop this kind of thing. And then, there are populations that have accepted. In Hérémence, for example, in the old quarries where they extracted the material to build the power station, which is in any case a site of no natural landscape or environmental interest. Apparently, they're going to make a park there. So why not? As long as it's profitable, it's a question of taking risks. The municipality has to go with it alone or not. I think it's complicated for a municipality to go it alone. From a financial point of view, and also from an industrial risk point of view, you could say. Because hiring local authority employees to manage a park, you know what I mean [it's complicated]? After that, you can hire librarians, public works people, secretaries, but hiring park management specialists for a local authority seems completely implausible to me. So, after that, you need a partnership with a specialist company.

CA: Would you do anything differently if you had to start another Alpine solar park project?

X: I'd say cut the main power cable to all the houses, then I'd cut off the electricity for half a day [laugh]. It comes back to the notion of scarcity and understanding. I'm not saying that people haven't understood, I'm not making that claim. I'm simply saying that if people had been more aware of the risks of shortages, they might have thought differently. If we draw an analogy with the development of dams at the time, what tipped the balance was the fact that the democratic debate was much less advanced at the time. There was no website to explain why it was a good idea to build a dam. People would arrive and say "Hello, we're going to build a dam." There were votes, but it wasn't as tense as it is now, I think. Different in any case. Because we also had hopes of economic development. At the time, the commune of [commune name], what started to make [commune name] rich was the construction of the dam of [dam name], but not so much the dam itself, by saying that there was energy produced, etc. It was the 3000 workers who came to work on the dam. It was the 3,000 workers who arrived in the municipality who started paying workers' tax. And from one day to the next, the local authority saw its tax revenues skyrocket. That gave them the means. Then there were all the local farmers who were still making a bit of a living, who gave you cheese, who gave you meat, who found themselves employed in the dams. These were really tough jobs, but they were starting to earn some money. Maybe it wasn't the first time they had seen money, but all the same, there was financial income. It developed the regions economically. Nowadays, when everyone has an iPhone 14 worth 1,000 francs and drives a Tesla or almost a Tesla and then flies off on holiday to other parts of the world, it's a bit complicated to tell people that if you build a photovoltaic solar park in the Alps, you're going to develop the local economy. It's also an element that has changed a lot in the vision in 50-60 years.

[break]

"You'll see, thanks to this, we're going to earn 1 million for the commune." Let's say there are 3,000 of us, we divide that by 3,000. What do you do with that million? The smallest project we have costs

millions. You have to say something. At the time, to come and say that with the energy that [dam name] is going to produce, it's going to give us back free of charge to the commune the equivalent of the commune's annual consumption, that was a thing that shook up the [his/her commune name] commune, for example. When it was done 50 years ago, over a period of time.

CA: I've asked you all the questions I'd prepared, but if there are any aspects I haven't covered that you'd like to add?

X: No, I think we've covered everything. Then there's the NGOs. Because they're the ones who want to get a handle on promoting renewables, getting away from fossil fuels, and at the same time absolutely rejecting this kind of project, saying that it's a complete distortion of nature, and then trying to convince people that putting panels on roofs and facades will solve all the problems, when in other countries it's still being done [solar parcs]. Wind turbines too, they're often fiercely opposed to them, ... whereas in Austria there are a thousand of them, for example, and we're a long way behind... I'd be interested in that, ... I've already discussed it with 2-3, so after that it's, ... they drape themselves in the defence of the higher interests of nature, that man is just a creature anyway, ... after that it's a philosophical vision of man's place in nature, but after that I'm not going to go into that too much. No, I think we've gone around the topic.

---- End of Transcript ----

Participant 12

Interviewee: Responsible for a set of Alpine utility-scale PV projects for an energy company, including an accepted one

Interview date: 7.06.2024

Form: Qualitative, semi-structured, online (teams)

Local acceptance of utility-scale project: Yes

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: What motivated you to launch the solar park project in [project site name]?

X: Well, on the one hand, [project site name] is the mandate that the Confederation has given to the major Swiss electricity companies. In that sense, it's not an official mandate, but it's a very clear expectation on the part of politicians that the big electricity producers should participate in these efforts to improve and increase winter production. So we saw this as an opportunity to meet that expectation. The second [motivation] is [to create] a joint development with the commune of [commune name] and other local partners. So it's not just a project that we, [electricity company name/project developer], decided to do alone, but one that we decided to do together with the municipality president, the municipality and the other stakeholders, who are [electricity company name] and [electricity company name]. It was a joint, group decision. The third [motivation is that] this decision actually allows us to have good relations, to build something together with the same partners as [dam project name].

CA: Would you have considered the project if there had been no state subsidies?

X: No, no Alpine project would be considered if there weren't the 60% subsidies under Law 71a. It's too expensive to do and the cost is too high to do it without subsidies.

CA: How did you find the information you needed to adopt these solar parks in the mountains?

X: Firstly, we brought together the experts in each field. We do a lot of hydroelectric projects in the mountains, so we brought in the civil engineers [for the solar park project]. We have a solar projects group, so we put the solar engineer [on the solar park project]. We put the structural engineer who works on hydroelectric structures [on the solar farm project]. We put all the technical experts together, who each took on their own little piece of the challenge, and then we went looking for the best available information. And then in the end [the objective] is to get the right experts together. If the in-house engineer doesn't have the expertise, [the objective] is to find the necessary expertise on the market. For example, the wind and snow studies [for the solar park project] are the same wind and snow studies that the experts do for the avalanche barriers. It's the same areas, the same conditions, the same altitudes. And there's a lot of expertise in Switzerland for mountain projects.

CA: And is there still some uncertainty around these solar panel projects at such a high altitude, and is it difficult to navigate this uncertainty?

X: Yes, it's difficult because, on the one hand, we have to move quickly given the deadlines we've been given and, on the other, there is no tried and tested solution. So we're following three technologies at the moment at [electricity company name/project developer]. The first [technology] is fixed tables, as we've seen on other projects. The second [technology] is the [test area name] fir trees or the cross-shaped formations at [test area name]. And the third [technology] is a cable-suspended system. Depending on the situation, the wind conditions, the snow conditions, one or other of the solutions is better suited. But for each technology we had to carry out a fairly expensive on-site test to see the strengths and weaknesses of each solution, and there are always unknown factors, which is a risk we're willing to take, but which we need to reduce. The main unknown is the snow drifts created by mountain winds.

CA: There are several electricity companies doing this kind of project. Is there exchange of best practice from one company to another?

X: Well, everyone has to find their own balance of risk analysis. But in fact there are a lot of exchanges between the major companies developing these solar farms. We have, let's say, regular technical calls to exchange information with the other major electricity companies, like [2 electricity company name], or with other companies. And then we have, for example, the test at [area name], which is this cable-suspended system, which we're doing jointly with others, with [2 electricity name]. We decided to invest in a system together and test it.

CA: And was the fact that other projects had already been rejected a hindrance to the [project site name] project?

X: No, not at all. We think the [project site name] project is a good site. The framework conditions are favourable. I don't know if you're familiar with the parameters of Law 71a, but perhaps you looked into it a little before doing the interview. You need to have a certain winter output, 500 kilowatt-hours per kilowatt-peak installed. You need to have an overall production total of 10 gigawatt-hours per year. But above all, NGOs and politicians want these sites to be sites that have already been impacted by man and with the infrastructure in place so that we can achieve this 10% by 2025. So [project site name] is perfect for this, it's a quarry, it's a high mountain desert that has already been heavily impacted by man. Nature has reclaimed some of its rights in certain areas that we are protecting and not going to impact. Otherwise, it's a field of rocks.

CA: And for the other projects, not the [project site name] one, do you find it a hindrance that there are rejects?

X: Well, we're currently going through the [other project site name] permit application, because we submitted the permit application quite a few weeks ago and we're now going through the analysis of the permit application. Each project is analysed independently, and each project is either rejected more or less strongly or accepted by local people. It's important to understand that there are two things. The first is public rejection. There are municipal assemblies that have not accepted projects, but that's purely local. [Project site name] was accepted by the community of [commune name], and what people think of it in [other commune name], what NGOs think of it, has very little impact on a bourgeois or communal assembly. So acceptance by the owner and the municipality is very local. In fact, what had the greatest influence on municipal decisions were those municipalities that already had energy assets, in the sense that [they already had] hydroelectric dams, hydroelectric plants. So the population was used to making decisions about energy assets. For these communes, the solar park was another energy producer and they could discuss it quite openly, they were quite in favour. The other communes, which don't have this type of installation on their territory and have never had these joint discussions on the subject of energy, were more, I would say, political-theoretical discussions and not practical ones. This influenced the municipal decision. After the decision to oppose these projects, the only opposition I know of was from national NGOs. The regional public interest groups don't oppose the solar farms, the big national NGOs do. And there are two kinds of NGO. There are those who are fundamentally opposed and have decided that it's no good, that's it. And then they look for every argument to oppose it. I'd say the fundamentals are a bit procedural. Is the law applicable? Because the application and authorisation procedure is very short, is the short circuit that the law gives us legal? Is it acceptable? Doesn't it contradict other parts of Swiss law? And then there's the second type of NGO, which is really about the detail of the impact study. Are a certain class of animals affected? *Bartavelles* [for example] are small

birds that nest on the ground. Has the work been done properly to protect these animals, this mountain fauna and flora?

CA: When you had all the information, and you had to assess whether it was appropriate to build the Alpine solar park, at [project site name] for example, what were the important factors to take into account when assessing the project?

X: The risk study, that's important, because we've seen with accidents [like] the wind whirlwind in the Jura, and then the fire in Vétroz, that when damage occurs in these projects, the impact in the press, the visual impact is very important. So [electricity company name/project developer] wants to make sure that all the risk reduction work is considered and undertaken. That's one thing, the risk analysis. The second [element to be taken into account] is the capacity, or the possibility of achieving the objectives of the law, i.e. the 500 kilowatt hours per kilowatt-peak. This winter output is not as easy to achieve as people think, because once the park has been built, the Confederation will simply measure the amount of energy produced over three years, take the average, and then say to us, you haven't achieved the 500 kilowatt-hours, we won't give you a subsidy. So in fact it's even worse, we have to dismantle the park if we don't reach the 500, because we've lost the permit. The permit is granted subject to the 500 [kWh] being reached. So it's a critical value to reach, and one that we have to achieve with a high degree of certainty. If one year we're at the limit of 500, it's no big deal, but the other two years we have to be above it if we want to be on the right side. And in the mountains, some winters are bad, and then there can be three winters in a row where there's less sun. So that's one of the parameters for achieving the objectives set out in the law, and that's important. And can the projects be designed to have as little environmental impact as possible on flora and fauna? That's the third aspect, the environment. These days, you can't just go ahead with projects and say, "I'm going to destroy a piece of nature and then come what may". It's not done, ... it's not done any more. And the bigger the company, the more sensitive it is to these kinds of issues. The economic side is less critical, because in fact the subsidy system brings everyone back to the same economic parameters. To explain, I don't know if you're familiar with the principle of the subsidy under the law, which is that the subsidy is up to 60%. But no more than is necessary for the project to achieve a zero return on investment. So even if we have a great project that doesn't cost much, and has a great economic return, it doesn't make any difference, because it's the subsidy that will be reduced. So we'll all end up with more or less the same economic return.

CA: And what were the decisive points, in the end, that made you decide to go ahead with the construction of the solar farm?

X: [Project site name] is the surface area available, the infrastructure that's already in place, in terms of access by road, access by tunnel, the connection to the electricity grid, the fact that it's an area that's already been heavily impacted by man, and then the excellent relations with [dam company name] and the local council, which are enabling us to make rapid progress on this project.

CA: And are you satisfied so far with the process of adoption of the solar park at [project site name]?

X: I'd say that all the players are positively involved in facilitating the process. And we wouldn't have done it if everyone wasn't involved. In the sense that we have on the [project site name] project management team: a representative of the municipality, a representative of [dam company name]. We have a representative of all the stakeholders. So in fact, all the partners in the project are co-investors in the project. We meet regularly for project management meetings and we manage the progress of the project. So the project implementation process is very efficient. And then the approval process, [in other words] the permit application process is a little more complicated, but it's not very complicated.

CA: If there were any, what were the main difficulties in this project?

X: The main difficulties are [firstly] the altitude at which we're working, in the sense that we're at [high altitude] metres, with snow and wind conditions that are even more difficult than most Alpine projects, which are at around 2200 metres. [Then there's] the remote location, in the sense that you can only get there by helicopter or on foot at the moment, because the road isn't open. And then the complexity of the site. It's actually a quarry, which shouldn't be very complicated. But technically, it's not an easy site, because it's a site that has been very much modified by man, but which has remained untouched for years. So there have been rockfalls, there have been landslides, there are a whole host of technical components to which we have to pay attention. And so the choice of technology, the choice of solar solution, is the most difficult.

CA: Solar park projects get media coverage. Do you find that communicating about these projects is complicated?

X: Well, I wouldn't call it complicated, but you have to manage it proactively. It's not enough to react to news and calls from journalists, you have to be proactive. I have presentations every two months at different events to manage this communication.

CA: What are the reasons why you think the project has been so well received so far?

X: Because it's a site that's conducive to this kind of project. It's not an untouched nature where we're going to impact something that has already been impacted and turned upside down. And the infrastructure is available, so it's a place that makes sense.

CA: I asked all the questions prepared. Do you have another aspect that you want to speak about?

X: No.

---- End of interview ----

Participant 13

Interviewee: Municipal voter of the accepted solar park project in Prafleuri

Interview date: 3.07.2024

Form: Qualitative, semi-structured, online (phone)

Local acceptance of utility-scale project: Yes

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: Why did you choose to vote for or against the Hérémence solar park project?

X: But because there's a glaring energy shortage that needs to be filled. In this case, it's going to supply 25 kWh a year. So it's not just a little bit. It's quite substantial. On the other hand, up there is the combe of Prafleuri, that's where they extracted the materials to build the Grande-Dixence dam. It was lunar soil. So there's no deterioration of the soil. Now it's lunar. So you can't see it. So it won't bother hikers. You have to be on a plane to see this solar alpine park. It will face to the south. It will provide a lot of electricity, even in winter. It's a really appropriate situation for the occasion, which lends itself beautifully, without deteriorating the nature or anything. It's really ideal. That also helped me to vote yes, because I'm also one of nature's protectors. Every mountain dweller is one. But in this case, it doesn't prejudice anything. It's an opportunity to have more energy. And that's that.

CA: For clarification, at primary assemblies, for example in Hérémence, are votes anonymous?

X: No. It was an open vote.

CA: And at this primary assembly, were there any differences of opinion? And if so, what were they?

X: No, there were no differences of opinion. The project was well put together. It was well presented by all parties, and concluded with [a presentation of] our commune president. No, it was unanimous. There were no opposing views. There were questions, of course, of a technical nature. That was all. The transport of energy, all that. But no, otherwise, really no opposing views. Everyone was in for it.

CA: And on a technical level, did you have any concerns about how it would work?

X: Not at all, not at all. I trust people in the business who know what they're doing. That's one of my fundamental principles. Trust and confidence are essential to me. These are people who know the project, which has been presented in a magnificent way. They're specialists. No, no, apprehension. Total confidence.

CA: And the stakeholders leading the project, were they people in whom you and the local population trusted?

X: Ah, but they're also partners for the commune. It's Grande-Dixence, in this case, if I need to cite only one. They're people we've been working with for a long time, ever since the first Dixence. So they're people we can trust. And it will bring something to them too, of course. But everyone will benefit. We're going to pay less for electricity, filters and so on. I think there will be benefits in that sense too. But the people behind the project will also recoup their costs, I mean, afterwards. It's a big investment at the outset. But afterwards, it will pay off. And it's going to produce a lot of... Not just energy, but money in the till too, on their side. But it's a partnership. There are... I don't remember who, what or how, but there are still 2 or 3 institutions involved in the construction.

CA: Do you think that if it had been other players with whom you hadn't had the Grande Dixence project, for example, it might have caused problems in terms of the trusting the people running the project?

X: I think so, yes. That's true. It's just there... No, but the people there knew the place well, the situation and everything. They've been part of Grande-Dixence since the beginning. So all in all, yes, it's a vote of confidence. But if it had been another company that had done this for financial reasons and all that, no, no, I don't think so.

CA: That it wouldn't have gone down as well, in that sense?

X: Yes, yes, certainly. Very definitely. No, no, because it's a really serious company. The people who presented it are really serious.

CA: And what are the benefits for the people of Hérémence of accepting an alpine solar park like this?

X: Certainly a reduction in the price of kWh. Most certainly. The municipality of Hérémence already does a lot for its taxpayers when it comes to electricity. Every year, we receive a bonus bill that helps us to pay. But here, in this instance, it's going to be stepped up.

CA: So, at the presentation, during the primary assembly, did they promise you a certain cost reduction?

X: Yes, yes, yes. They did hint at the possibility of reducing the price per kWh. Yes, yes, that was part of the discussion.

CA: I see. So I think you've been following the discussion about the solar farms in the Alps. Some have been rejected...

X: Yes, yes. I know, I know. Ours was so well located and so well presented. And you know that it will be operational by the end of 2025. After that, you must know that it passed on the Swiss level [recent vote about Swiss alpine solar parks]. Well, not in Grenchols, way up there, but down here, there's no problem. It will happen.

CA: And why do you think the other projects are overwhelmingly rejected?

X: There were too many... They disfigured the landscape. It was too intrusive, too substantial, unlike ours, which nobody sees.

CA: And if I'm not mistaken, you were involved in one of the first Alpine solar parks in Wallis?

X: Well, yes. It's been thought through. It was a guy who worked at Grande Dixence, who was on the local council. I spoke with him. He thought about it and everything. And then it took off. He was in a good position to talk about it, as he was working in Grande-Dixence, you see, in the office, in Sion. And then they launched the idea. He was on the local council. They spoke about it to the council. It grew. Gently, it took shape. They went to see interested, higher authorities, ... well, the developers, but... Those who could carry out the project and so on. And then it happened. They worked on it constantly. It progressed. It took its course. It was finalised. It was accepted.

CA: When the project for the Alpine solar park at Prafleuri was launched, did you go and see, for example, other solar park projects that had been carried out elsewhere? Did you talk to people in other places?

X: I read a lot. I'm very curious by nature. I read a lot. I read about all these projects in the newspaper. OK, now you're going to tell me... It depends on who's writing what. You have to read between the lines. No. I read a lot. I got interested. And then, in my opinion, they didn't hold up, frankly, the others [solar

park projects]. It was made to “look good” in the landscape. They all started doing it [solar parks], but it wasn't... No, it wasn't well put together.

CA: Right. And in that case, the rejections from the other solar farms didn't influence your decision at all with regard to the Prafleuri project?

X: Not at all, no. Quite the opposite. It gave me the boost to go ahead with it.

CA: And how did you go about finding the information you needed to make up your mind, to vote for or against the project?

X: Ah, but it was presented to us from A to Z. But I must say, it was a long meeting, with lots of partners, but there was no veil. It was presented from A to Z in a really concise, precise way. Everything was really developed. Nothing was hidden. Everything was presented to us. So much so that, in the end, it was almost impossible to refuse. Ask technical questions, yes. That, yes, the material of the panels, the dimensions and so on. But everything was presented to us. Really, really well. It was really well done.

CA: How many people were at this meeting?

X: 70 people.

CA: And during this primary assembly, were you able to discuss with the other people before making the decision? And did the discussions influence the result?

X: No, no, I'm repeating myself, it was so well presented. The project is so well done that there were no questions asked. There are no [questions], if not technical, then none at all. Everyone was won over to the cause. Afterwards, we had another chat over a drink. There was a drink at the end. But everyone was really enthusiastic and on board, without reservation.

CA: You've already partly answered this question, but did you feel that the project was well communicated?

X: Oh yes, the project was really well presented. From every angle, without hiding anything, it was really perfect in that respect. And with all the diagrams, sketches, everything you could want, the plans... No, it was really well done. They didn't do it lightly. It was really... Yes, super, really well done.

CA: When you had all the information at your disposal and you had to evaluate whether you wanted to accept or not, were there any additional aspects that came into play?

X: No, everything was said. Nothing was hidden. We really took the decision with full knowledge of the facts.

CA: And did you vote on the cantonal decree to speed up the construction of alpine solar farms in Valais?

X: Yes. I rarely miss a vote. And it's a duty for me.

CA: So in relation to this cantonal decree, when you voted, did you associate this vote for alpine solar parks with the Prafleuri project or were these two things totally separate for you?

X: [break] No, I mean, it was part of the package. It wasn't dissociated. We voted on a global plan, I mean. But as far as we were concerned, it had every chance of being accepted, of passing the test. And what's more, it wasn't called into question, unlike others.

CA: Yes, absolutely. I've asked you all of the questions I prepared, but I don't know if there's anything I haven't covered that you'd like to share about this subject?

X: Well listen, I told you it was a wonderful situation. There's a power line nearby, power transmission and everything, it's really only partially built. It might need consolidating, I don't know. There's a tunnel, if you had to go up to Prafleuri, there is a tunnel, you can go up there with a Jeep. It's all the elements that worked in favour of this project too.

CA: Absolutely, the infrastructure in place.

X: Exactly, yes. And then, I said, I think it should be finished by the end of next year, by the end of 2025. So, they've already started everything, but it's really moving forward. I've seen the situation, so this photomontage and everything, we've seen it all on site and everything, they've presented it, there's really nothing that's been hidden, I repeat. It was perfect, the presentation, the value of the file and everything, it really holds up, it's solid, strong stuff.

CA: So total confidence in the project developers.

X: Exactly. All the partners were there at the primary assembly, so it was all the more interesting.

CA: Is there are any aspects I haven't covered that you'd like to talk about?

X: No.

---- End of Transcript ----

Participant 14
Interviewee: Responsible for a set of Alpine utility-scale PV projects for an energy company Interview date: 4.07.2024 Form: Qualitative, semi-structured, online (teams) Local acceptance of utility-scale project: One accepted, one rejected

Introduction:

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:

CA: We can also talk a bit about the [commune name] project, I would be interested to know why you think it didn't succeed as hoped.

X: Well, it was the vote on... I can't remember the date, but there was a vote in [canton name] on the decree [about the acceleration of alpine solar park projects], which finally killed the project, and we thought we'd follow the vote in the commune of [commune name]. And then, the negative votes, it had won, not much, but it had won. So we thought, well, it's like a signal. So this project, we, the steering committee, decided to put a stop to it. So it was mainly the municipality that was involved. But I think it was... So I wouldn't know the whys and wherefores. I think it was political, in the end. I think it had a big impact on the ski slopes and the mountain guides. And I think it was, ... because the project wasn't very visible, so it wasn't really a question of visibility, it was relatively hidden, ... so it was more in terms of operations and the impact on existing activities. That's what made the project a bit less attractive.

CA: Concerning this topic, in [commune name], it was decided to abandon the project because of the negative vote on the decree. But in [other commune name], for example, people rejected the

solar panel decree, but in the end, they accepted the project in [project site name]. So do you think it was right to draw conclusions in that sense?

X: The sequence was different. In [first commune name], the population had not yet expressed its opinion. So that was one of the future stages, in fact. After the decree, if the decree had been positive, there would have been a vote by the Primary Assembly. What happened at [second commune name] was the opposite: the Primary Assembly had already expressed its opinion. The bourgeoisie assembly also did I believe. And then, indeed, the decree was passed. Subsequently, the decree was very close to a yes and no vote. So given that the population had already voted in favour, the commune president and the steering committee wanted to continue [in the second commune name]. Because the population had already expressed its opinion, unlike [first commune name].

CA: And which people were or are involved in the decision to adopt the large-scale alpine solar park?

X: Well, there's a steering committee made up of... It's mainly the local authority that is concerned. And then we work in partnership. There's [electricity company name/project developer], there's the municipality, there's the ski lifts, there's [other energy company] and there's the mountain pasture [owners], who owns the land and is also involved in the project. So I'd say that it's these five stakeholders who will decide.

CA: And do you find it complicated to work with many different parties who may or may not have the same interests?

X: No. So, listen, things are going rather well. The interests tend to converge, because in the end everyone wants to move forward. The ski lifts also have experience in this type of alpine project. So that's a plus for us. The company [energy company name], which is involved in the project, is also responsible for all the connections and electricity involved. So it's a plus for us to have them on board, I'd say, in the decision-making too. We are in the mountain pastures, so it's important to have them [pasture owners] around the table as well. And the commune is motivated by the project. So there's a good dynamic. A good dynamic going in the right direction.

CA: And what motivated you to launch this alpine solar park project?

X: Well, I'd say that at the [electricity company name/project developer] level, we're looking to develop the energy mix. So the energy mix involves all types of energy, whether it's wind power, as I said at the beginning, solar farms, hydroelectric power or even nuclear power. Today, nuclear power is part of the Swiss energy equation. We want to get out of it. So we need to find new solutions. We also realise that we need a lot of energy in winter. In summer, we can have a lot. If we develop solar panels in the plain or elsewhere, we'll have plenty of energy, even a surplus of energy in summer. But in the end, many countries have a surplus of energy in summer. And it's in winter that we're often running out of energy.

In Switzerland, we really need to be able to fill this gap, which is already there and which is only going to get bigger as society becomes increasingly electrified. So the demand for electricity is only going to increase. And it's true that in winter, the energy sources that can meet winter demand are mainly wind power. So we're also pushing for wind power. And above all, the large solar farms in the Alps, which have a much higher output than lowland photovoltaics in winter. So it's really these objectives that we're pursuing with this type of project. It's electricity in winter.

CA: And how do you go about finding the information you need to adopt alpine solar farms in the mountains?

X: So, I would say it depends on the type of information. If it's administrative, we'll approach the canton. We work, I'm not going to say in partnership with them, but we go to them for all administrative procedures. For example, the major plans project has gone through a stage called the request for information. We submit a preliminary file to the canton, which they analyse. They'll tell us whether it corresponds to what's required for an enquiry or whether additional information is needed. So, we're at this stage of the [project name] project. In other words, we're going to get this information from the canton.

[break]

After that, when it comes to technical matters, we either have in-house technical specialists or we rely on partners. If we're talking about all things electrical, for example, we'll be relying on [partner name], who will supply us with the electrical part, the panels. The panels are not conventional panels. They're what we call bifacial panels. These are panels that will produce electricity using both sides. It's just that they produce electricity on both sides. So we're going to rely on technical specialists. So either we have people in-house who know about what we need to know. Then we go to suppliers who will provide us with the panels. Or then we can rely on partners to advise us. After that, I'd say that a big part of the equation for the [project name] project, for example, is the fairly innovative system for supporting the panels, since we're using cable systems. I don't know if you've seen some of the photo montages on the site. A company called [company name] is developing this technology. It's brand new. At the same time as we're developing the project, we're also developing the technology. Some things are a bit *rock'n'roll*. But in the end, they are a manufacturer of cable cars and alpine lifts. They're used to working in these environments. They adapt their technology to solar alpine. Large cables are pulled across the site using special masts. We install solar wings with 12 panels. We install wings like this. This makes large rows of solar wings that are aligned. So we're relying entirely on [company name] to develop this. We're going to find the skills where they are. What we have in-house is limited to electricity and solar panels. But as soon as we get into things that are a little more innovative, we'll call on the specialists. The same goes for everything to do with excavations. We have to dig for all the cabling and so on. So we really rely on the specialists who are going to provide us with the services.

[break]

Another major aspect of these projects is the environment and environmental protection. And I think this is one of the project's biggest reports. So we also rely on specialist environmental consultancies to produce all these impact reports, to make sure that we can demonstrate that these projects do not have too much impact on flora and fauna and can be acceptable from that point of view too.

CA: And when you develop a solar farm project, like [project name] for example, do you also draw on the knowledge you've gained from other projects or do you talk to other companies working on other alpine solar farm projects?

X: Yes, absolutely. At [electricity company name/project developer], we have a pioneering project, the [project name 1] project, which is an [electricity company name/project developer] project with a different technology. They don't use the famous cables, as I was saying. They use something else. But of course it's a project that's ahead of ours. So we're going to use all the feedback we can get, whether from other companies, if we have any particular contacts with them, or from our other projects. For example, on the [project name 2] project, we're slightly ahead of schedule. Typically, things that happen at [project name 3] will be used at [project name 2]. So yes, we're trying to make the most of it. There are so many uncertainties and unknowns surrounding these projects today. What's more, the time schedule is extremely tight. So we have to use everything we can find. So yes, that's for sure.

CA: And you've started to talk about that, but in terms of inter-company relations, is there also communication?

X: Yes, there is communication, I would say. I think it's the same as for any development, [or for] a bit of a new development, there's the identification of sites, which may be a bit competitive between companies, because that's where you're going to put it. And then, once the sites have been identified and the players are known, we're more likely to enter a collaboration phase. So, even if it's not [electricity company name/project developer], we're still going to talk to them. Several of us are using this famous [system name] system. So we have a test installation at [project site name] that will be installed in partnership. So we have other companies who are going to install this system who are also with us in this test project. So yes, there is good cooperation between the various players in the sector.

CA: Do you think that the fact that [project site name] was one of the first, has had a positive impact on local acceptance?

X: I'm not sure. I'm not sure. But there was the vote on the Electricity Act on 9 June. It was a fairly favourable vote. So I'm not sure. I think it has a lot to do with the local context. There has to be a [commune] president who is in favour. There may have to be a bourgeoisie or, ultimately, local players who are perhaps concerned by the terrain and that sort of thing [that agree with the project]. I think there really is a local context that also favours this acceptance. I don't think it's just the global context.

I think it's really a very local context with local particularities. I suppose that can change. Would we have a positive vote today? I can't say for sure. But I do think that there is a... I remember that when the vote was held, the president of the commune said something quite important. I think there's a certain pride in carrying out projects like this. We're in communes that are pioneered in hydropower and other fields. So there is a certain pride in having carried out these projects at the time. I think there's a bit of a spirit of carrying on that pioneering spirit. I think you can see it a little bit in the population.

CA: I was wondering about that because [two commune names] have dams, and I did not know whether that might also encourage discussion about renewable energy projects.

X: Yes, I think so. Yes, absolutely. Especially as [project site name] is on a former quarry, a [dam name] materials depot. So there's a real link. In the end, it's a great conversion, so to speak.

CA: And for all these projects, in general, is there a benefit that is also offered to the local population? And in your opinion, does giving benefits to the local population encourage acceptance?

X: Yes, that's for sure. Of course. The benefits, ... there are royalties which go to the local authority and the landowners. So it goes straight back into the pockets, so to speak, of the citizens. And then there's a guarantee. The idea is also to be able to inject the electricity into the villages concerned. So there's also an electricity supply guarantee. We know that the electricity will come from the park as a priority. So I think that this is also, in a way, an interesting element for the people. And of course, the fact that it has a positive outcome for them will help, that's for sure.

CA: And, in general, do you think that the rejection of certain Alpine solar park projects is an obstacle to the development of additional solar farms?

X: It's difficult to answer. As I said at the beginning, I think the context is fairly local. So, in the end, they won't all happen, that's for sure. So, if one stops, it could be an opportunity for another to become a player. As if, for example, at [electricity company name/project developer], we lost one park, perhaps we'd put more effort into another. So, yes, I think that if one stops, it could be an opportunity for another.

CA: When you had all the information at your disposal and had to assess whether or not to build an Alpine solar park, what were the most important factors to take into account for assessing the project?

X: So, the first elements. The problem is that we still don't have all the information we need to make a decision. I'd say that we'll make an investment decision the day we really have a finished project with a real budget and the exact costs of everything, which is not the case today. Today, we're still working on estimates. I would say that for this project, for these projects, they are still subject to Solar Express, so the law, this specific law. So, in the end, the main thing was first to know that the people of the

municipality and the people concerned are OK [with the project]. Once we have these approvals, we have to make sure that the project falls within the scope of the law. This means that it has to produce enough, that it has to have a sufficient winter production, and all these criteria. Because if you have one criterion, if you have one killer criterion and we don't meet one of the criteria in the law, it's doomed anyway. There's no point in continuing. So I'd say these are the first things to check.

CA: And what do you think you're going to need before you can assess the financial profitability, if it's going to be possible to match the costs?

X: We need to continue developing the project. At the moment, we don't have studies that are sufficiently advanced. So we need to be able to continue the development with [system name], for example, so that at the end they can tell us OK, such and such will cost you so much. Then we will have a firm offer, so to speak, from them. So that [partner energy company name] can tell us, the connection of all the panels, the connections, it will cost so much. And with that we can, in fact, have all the final bids. With the final bids, we know, we can calculate how much subsidy we can get, and at the end of the day, that allows us to know whether it's worth it or not. But today, we're too far away from all these bids. Our project is still too... It's still too much of a "project" to have a real idea of the final costs. So we're really at the stage of estimating based on the experience we have, based on what we can imagine, but we don't yet have costs that are sufficiently detailed to be able to say OK, let's add it up, let's draw the line, then OK, it's plus or it's minus [the project value].

CA: And are you satisfied with the progress of the large-scale solar park project so far?

X: Yes, yes. Things are going well at the moment. We've got some very good partners, so I think there's also a good development dynamic. So it's not going as fast as we'd like, because we really have a time criterion that's the most difficult. And then, as I say, we're developing this project at the same time as we're developing the technology that would go into it. Sometimes there are blocking factors, factors that we don't know about. So we have to wait for certain studies and analyses before we can continue. So, overall, I'd say we're happy. After that, in terms of planning, I think we'd perhaps like to go faster. But if we want to develop properly, technically we can't.

CA: What are the main difficulties you are encountering?

X: That's it, of course, the deadline and then a certain number of things that we don't control. It's true that when the file, ... we put the file in the hands of the canton, and we're waiting for it to come back now. Their feedback, their analysis, their critical thinking of the support we'll also get from the state, so to say. How accommodating they'll be, less, more, I don't know. We'll see. And then there's also the opposition, the opponents, the associations. All the things we need to work on with them. In any case, try to involve them as much as possible.

CA: Because at the moment, even if the municipality has accepted the project, the environmental associations could still be stopping the project?

X: Yes absolutely. We're currently in the development phase. Once we're ready, once we've got the go-ahead from the canton, we'll apply for a permit. This permit application is an official procedure that anyone can oppose. Associations, citizens... After that, some people may object because they are affected by the project. Typically, neighbouring citizens or others. Not everyone can object. You have to have the character of... You have to be affected by the project to be able to object. But associations can object. Indeed, we could have associations for the protection of the landscape, birds or others, as we generally have in major infrastructure projects, such as the projects in the background behind me [windmills]. It's usual for these oppositions to oppose [such projects].

CA: And do they have to get a petition signed?

X: No. When you apply for a permit, you put it out to public inquiry. For 30 days, the file is accessible and can be consulted by anyone. People can object to the documents they have access to, saying "I think there's too much impact here and there". They are opposed. After that, the opposition has to be processed. That's what takes time. First, it's the commune. Then it's up to the State to deal with the opposition. If people don't agree with the lifting of the objections, they can lodge an appeal. That's when we usually go to court. That's where it takes time, because there are two court stages in Switzerland. It takes a long time.

CA: That's where it would also be a big risk, I imagine.

X: Yes, exactly.

CA: There is media coverage about such projects. Do you find communication on these projects complicated?

X: No. It's true that at [electricity company name/project developer], we have a communications department that takes care of that sort of thing. I'm less involved. I'm more involved in providing technical content. But no, I don't think it's complicated. It was not extremely publicised. There are subjects that get more coverage. There was a small article on our projects yesterday, I think, or the day before, because they came in for an inquiry. But there were relatively little reaction, I found.

CA: I'd like to come back to what we said before. It had to do with the fact that communes with dams are more open to questions about RE. In [commune name], they had a dam. In your opinion, why did they not want the solar park project?

X: So [commune name] is close to the [area name] ski area. And I think there are interactions, and I think there's a big influence from mountain clubs, guide clubs, that sort of thing. And I think they were the ones who tipped the balance in the direction of the no. So it was the impact on the ski area. In

[commune name], it's true that it was an off-piste slope that was, I think, relatively used in winter. And I think that's what held it back. And then, having [ski area name] next door, with its major infrastructures and so on, I think that was a hindrance. After that, I don't know for sure, but it's what I imagine. And then they have a small infrastructure. It's the [commune name] power plant. It's not the Grande Dixence [big dam in Wallis], it's still a small infrastructure. But yes, there's also the pioneering character of hydraulic power there, of course.

CA: Some projects are accepted, others are rejected. In your opinion, what are the most influential factors in these acceptances or rejections?

X: I think it's the integration of projects into existing infrastructures. If we take [project site name], we're talking about an "equipment depot", so it's not an area, ... we're not in the middle of a pasture or in the forest or in a place that would be perceived as unspoiled nature where I think it's more difficult to put a project there. If we take the [project name] project, we're on the ski slopes, we're close to an infrastructure, the area is already anthropized, there are already ski lifts, there are already electrical connections, so in the end, all the necessary infrastructure is already there. So yes, we're adding a certain number of things that may affect the landscape or other elements, but we're not putting in a project [in a pristine area], ... So I'd say the whole context is important for acceptance, and ultimately, people say, well, we're adding energy production elements, but we're not putting them right in the middle of a virgin pasture where there's nothing. So I think that's a good thing for the acceptance of a project.

CA: And if you were to start another Alpine Solar Park project, is there anything you would do differently?

X: No. At the moment, I don't think we've made any particular mistakes.

CA: I've asked you all the questions I had prepared, but is there are any aspects I haven't covered that you'd like to talk about?

X: No, I think we talked a bit about the main aspects, the development of the project, technical, communication, partners, that sort of thing. No, I think we've covered a little bit of everything. As I was saying, this interest in winter solar production is the main driver for us as energy players.

---- End of transcript ----

Participant 15**Interviewee:** Municipal voter of the rejected solar park project in Käserstatt**Interview date:** 15.07.2024**Form:** Qualitative, semi-structured, written**Local acceptance of utility-scale project:** No**Introduction:**

Self-presentation	My name is Constance Allet. I am studying business administration at the university of St. Gallen and am writing my bachelor's thesis with the Institute for Economy and the Environment under the supervision of Prof. Dr. Wüstenhagen.
Research objective	This research aims to gain insights into the decision-making process for the adoption of RE with a focus on solar PV in Alpine resorts.

Interview transcript:**CA: Why did I vote for or against the Solar installation in Käserstatt?**

X: I did not fact check the whole project and who would have been involved directly. I know the Bürtnergemeinde pushed the project, which means there must've been an advantage for the landowner, as well as possibly the local construction firm(s) taking over the project. This interest is rumored to have won over a project on the other side of the valley in a different commune. The project there will involve years of building streets and further logistic systems to access the space which will be covered in solar panels, as well as the installation of the panels themselves and their long-term entertainment. A similar situation could have developed at Hasliberg. Personally, I voted against, not because of purely fact-based reasons. First of all, I think that every free spot on already existing building should be utilized before ruining diverse and purely natural grass, forest or rock landscapes. As this is not given yet in the commune of Hasliberg, nor on the buildings of the Bergbahnen or Restaurants in the resort on the same altitude or higher than the planned panel field, I do not see why destroy nature for complex technological installations. The second point, which was quite interesting, as it was one of the first political actions of this scale in Hasliberg since I can remember, focuses more on the visuals of such a project. In the

middle of the most passed village (first village of Hasliberg), right next to the road in the field, an imitation of a solar panel made of wood was displayed. On there was a poster with some phrase and a big NEIN across the whole panel. This was impressive, as it gave an idea of the scale of one single panel and would let imagination take over the rest of the field of panels possibly installed.

CA: If another solar farm project were to be undertaken in the region, what measures do you think could be taken to improve community acceptance?

X: As already mentioned, a similar project was accepted in the valley. I am convinced that the majority of the people living in the valley do not necessarily care about renewable energy sources and are neutral towards it. The accepted project will be in a less frequented region of the valley and is not the focus of tourism. In addition, it is a huge and longer term win for the local construction and electricity firms, who can divide the project among themselves. I think the people are in general more conservative and value the lands, the biodiversity and the peace of the commune. In addition, the topic is not as pressing or interesting for the average habitant of Hasliberg, as it might be in cities. The emotional attachment to the lands is by far more important than the energy source. This means that the acceptance of such a project may have been higher if it were promoted for a smaller and less central part of the commune. It might have also been more interesting, if the topic were pushed a bit slower, and solar panels would have been integrated one by one by the Bergbahnen and their public houses (restaurants and hotels), maybe subsidized to the inhabitants etc.

CA: What do you think would have been the advantages of voting yes to a solar park project like the one in Käserstatt?

X: Advantages would have been mostly the economic value of the renewable energy field, meaning that local firms could have benefited from such a project. I doubt it would attract more people, as winter sports do not attract or repel tourism based on political issues like the energy used. It would have, however, help normalize investing in renewable energy and show the importance and long-term thinking behind it, maybe saving a few years of skiing in the alps.

CA: Is there are any aspects I haven't covered that you'd like to talk about?

X: No.

---- End of transcript ----

Directory of Aids

Aid	Use	Parts
DeepL	Used to identify synonyms, translate unknown words and cross-check sentence structure	Entire thesis
	Used to translate the interview transcripts	Appendix B (Interview Transcripts)
Grammarly Pro	Spelling, grammar and style of writing check	Entire thesis
TurboScribe	Creation of transcript	Appendix B (Interview Transcripts)
MyBib	Compilation of the references and creation of in-text citations	In-text citations, References
ChatGPT	Used as a search engine to get unknown word/concepts definitions to enhance comprehension. Information were then cross-checked with real sources and not used as such. GPT was not used for generating new content, text or restructuring text.	No specific area
MS office Features	Creation of tables	Tables
Private, non-paid readers	Spelling and comprehension check	Entire thesis
Powerpoint	Creation of figures and Appendix A	Figures and Appendix A

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19.08.24, Constance Emmanuelle Allet



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Hochschule für Wirtschafts-,
Rechts- und Sozialwissenschaften,
Internationale Beziehungen und Informatik

Masterarbeit

Attraktion oder Abschreckung?

Wahrgenommene Auswirkungen von Windkraftanlagen auf den
Tourismus in den Schweizer Alpen

eingereicht am 6. November 2023 von:

Larissa Marina Brunold

16-924-458

Referent:

Prof. Dr. Rolf Wüstenhagen

Korreferent:

Prof. Dr. Moritz Loock

Abstract

Um die schweizerische Energiestrategie 2050 umzusetzen und die Treibhausgasemissionen weitgehend zu senken, sind die Bestrebungen hoch, die erneuerbaren Energien in der Schweiz weiter auszubauen. Dazu gehört auch die Windenergie, der in der Schweiz im Vergleich zu anderen europäischen Ländern bisher weniger Beachtung geschenkt wurde, jedoch nun auch darüber diskutiert wird, diese in alpinen Gebieten auszubauen. Um Windkraftanlagen in der Nähe von bestehender Infrastruktur zu bauen, werden Tourismusgebiete in Betracht gezogen. Die vorliegende Masterarbeit untersucht in diesem Rahmen die wahrgenommenen Auswirkungen von Windkraftanlagen auf den Tourismus aus der Sicht von TouristInnen und welche potenzielle Wertschöpfung durch die Windenergie für den Tourismus entstehen kann. Dies wird anhand von zwei Fallbeispielen der Regionen Andermatt und Laax untersucht, um konkrete Beispiele für alpine Tourismusgebiete in der Schweiz abzubilden. Die Resultate der empirischen Studie haben gezeigt, dass das Interesse an der Technologie vorhanden ist und in Bezug auf die Auswirkungen auf das Landschaftsbild die Nähe zu bestehender Infrastruktur als sinnvoll betrachtet wird. Zudem wurde gezeigt, dass das Vorhandensein von Vorzeigebeispielen einen Effekt auf die Wahrnehmung von TouristInnen haben kann und es daher wichtig ist, weitere erfolgreiche Windkraftprojekte zur Förderung der Technologie umzusetzen. TouristInnen sehen sich in ihren Aktivitäten nicht durch eine Windkraftanlage gestört, viele empfinden dabei Indifferenz. Sie empfinden einen Windpark zudem als Zeichen der Nachhaltigkeit und Fortschrittlichkeit, was sich positiv auf die Reputation einer Tourismusdestination auswirken kann. Zudem zeigt Andermatt mit der Anwesenheit des Windparks Gütsch ein erfolgreiches Beispiel einer Tourismusentwicklung und das Potential für zusätzlich durch den Windpark generierte, regionale Wertschöpfung.

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1 Einleitung

Im Rahmen der Energiestrategie 2050 des Bundes setzt sich die Schweiz zum Ziel, einen bedeutenden Anteil der eigenen Energieproduktion aus erneuerbaren Quellen zu gewinnen (Bundesamt für Energie, 2023a). Dies unter anderem auch, um die Unabhängigkeit vom Ausland zu stärken. Diverse internationale Geschehnisse, wie beispielsweise der anhaltende Konflikt zwischen Russland und der Ukraine, haben besonderen Anlass dazu gegeben, die Versorgungssicherheit der Schweizer Bevölkerung ganzjährig zu verbessern und das Bewusstsein der Bevölkerung um den Energiekonsum zu schärfen. Dies zeigte sich beispielsweise, als der Bund im vergangenen Winter eine Kampagne zu Energiesparmassnahmen lanciert hat (Eidg. Departement für Umwelt, Verkehr, Energie und Kommunikation UVEK, 2023). Weitergehend wurde im Juni 2023 durch die Schweizer Bevölkerung über das Klima- und Innovationsgesetz abgestimmt und angenommen. Dabei wurde festgehalten, dass direkte und indirekte Treibhausgasemissionen durch die Energieproduktion oder deren Einkauf vermindert werden sollen (Bundesrat, o.D.).

Die einheimische Energieproduktion stützt sich in der Schweiz weitgehend auf die Wasserkraft. Um die Energieversorgungssicherheit anhand „neuer“ erneuerbare Energien in der Schweiz auszubauen, untersucht beispielsweise die Eidg. Forschungsanstalt für Wald, Schnee und Landschaft WSL die Potentiale, welche in den Schweizer Alpen sowohl für den weiteren Ausbau der Wasserkraft, aber auch der Wind- und Sonnennutzung vorhanden sind (Energiepotenzial Schweizer Alpen - WSL, o. D.). Während in benachbarten europäischen Ländern spezifisch die Windkraft in ihrem Potential erhöht genutzt wird, scheinen sich in der Schweiz bisher gewisse Hindernisse zu deren Entwicklung zu zeigen. Dazu gehören unter anderem ein bis zu fünfundzwanzig Jahren andauerndes Planungs- und Bewilligungsverfahren, wobei primär lokale Vereine, Privatpersonen, Landschaftsschutzorganisationen, Pflanzen- und Tierschutzorganisationen Einsprache erheben und damit für Verzögerungen oder Verhinderungen von Projekten sorgen (Suisse Éole, 2020). Die schweizerische Bevölkerung sieht trotz Zustimmung zum Ausbau von erneuerbaren Energiestrukturen ein Tabu darin, Anlagen erneuerbarer Energien wie Solar- oder Windkraftanlagen in unberührten alpinen Gebieten zu platzieren. Jedoch sind diese vorstellbar in Regionen, in welchen gewisse Infrastrukturen bereits vorhanden sind. Damit sind vorwiegend touristische Infrastrukturen gemeint, wo beispielsweise bereits Zufahrtsstrassen und Netzwerkanschlüsse gegeben sind (Kittl, 2022). Auch der Tourismussektor leistet seinen Beitrag an der Schweizer Wirtschaft und damit auch am Energiekonsum. Besonders der Wintersport wird zukünftig von den Auswirkungen des Klimawandels zunehmend betroffen sein.

Nun möchte zukünftig auch die Windkraft mehr genutzt werden, wozu unterschiedliche Projekte zur Untersuchung von Potentialen sowohl auf nationaler, aber auch auf kantonaler Ebene im Gange sind. Darunter befinden sich mittlerweile verschiedene Potentialgebiete im Kanton Graubünden, welche Windparkprojekte in alpinen Tourismusgebieten, wie beispielsweise

Laax, vorsehen (Kanton Graubünden, 2023b). Ein bereits bekanntes Beispiel für einen alpinen Windpark besteht heute auf dem Gütsch oberhalb der Gemeinde Andermatt. Die Tourismusregion um Andermatt wurde in den Jahren 2010 und 2012 durch den Windpark Gütsch um vier Windkraftanlagen ergänzt und ein weiterer Ausbau des Windparks steht bereits bevor.

Bisher haben sich besonders Studien im Ausland, beispielsweise in Deutschland, Tschechien, den USA oder Island, damit befasst, welche Aspekte von Relevanz sind, dass Windkraftanlagen zu einer Tourismusattraktion etabliert werden können oder eine Windkraftanlage erfolgreich in ein bereits bestehendes Tourismusgebiet integriert werden kann. Nur selten wurden dabei alpine Tourismusgebiete untersucht. Besonders die Erscheinung im Landschaftsbild und die Sinnhaftigkeit einer Anlage am jeweiligen Standort wurden in Untersuchungen als entscheidende Faktoren eruiert. Des Weiteren wurde das Potential untersucht, welche zusätzliche touristische und regionale Wertschöpfung durch eine Windkraftanlage entstehen kann.

Anhand der beiden Referenzgebiete Laax und Andermatt untersucht die vorliegende Arbeit wahrgenommene Auswirkungen von Windkraftanlagen auf alpine Tourismusgebiete in der Schweiz. Um sowohl die unterschiedlichen Aspekte der Wahrnehmung von TouristInnen und ihren individuellen Betrachtungshintergründen als auch konkrete Wertschöpfung und zusätzlicher Nutzen und Potentiale herauszuarbeiten, wurden für die vorliegende Studie sowohl eine Umfrage mit TouristInnen in Andermatt und Laax durchgeführt als auch verfügbare Statistiken zur regionalen Wertschöpfung ausgewertet. Als Ergänzung wurden Interviews mit den beiden Tourismusdirektoren aus Andermatt, Thomas Christen, und aus Laax, André Gisler, geführt.

Angesichts der ambitionierten Ziele der Energiestrategie 2050 und des Bedürfnisses, die Energieversorgungssicherheit der Schweiz anhand des Ausbaus erneuerbarer Energien zu erhöhen, dies jedoch anhand von schnellgreifenden Verfahren und an einem für alle Parteien geeigneten Standort zu erreichen, möchte die vorliegende Studie die Perspektive darum erweitern, welchen Beitrag Windparkprojekte in schweizerischen, alpinen Tourismusgebieten leisten können.

Die vorliegende Studie geht einleitend auf den Tourismus in der Schweiz und dessen Energiekonsum ein. Es wird erläutert, welche Ziele die Energiestrategie Schweiz verfolgt und welche erneuerbaren Energien dabei von grösster Bedeutung sind. Anschliessend wird spezifisch auf die Windenergie eingegangen, wobei die aktuelle Situation in Europa und in der Schweiz beschrieben wird sowie auf das zukünftige Potential aufmerksam gemacht wird. In diesem Zusammenhang wird bereits dazu eingeführt, wie der Richtplan des Kantons Graubünden die Windenergie einzubinden plant. Abschliessend für dieses Kapitel wird erläutert, welche unterschiedlichen Gründe vorliegen, weshalb Windkraftanlagen in der Schweiz in ihren Bau gehindert sind.

Das anschliessende Unterkapitel befasst sich tiefer mit der bisherigen Forschung rund um Windkraftanlagen im Zusammenspiel mit dem Tourismus. Zuerst wird darauf eingegangen, welche Aspekte für die Akzeptanz von Infrastrukturen erneuerbarer Energien in Tourismusgebieten von Relevanz sind. Anschliessend wird erläutert, wie Windkraftanlagen und andere Anlagen erneuerbarer Energien als Teil von Tourismusattraktionen in bisheriger Forschung untersucht wurden. Im Schlussteil des Kapitels wird spezifisch auf untersuchte Tourismusgebiete eingegangen, die durch Infrastrukturen erneuerbarer Energien ergänzt wurden.

Im nachfolgenden Kapitel werden die beiden Untersuchungsgebiete Andermatt und Laax vorgestellt. Dabei wird auf das Angebot und die Tourismusentwicklung der beiden Regionen eingegangen. Bei der Tourismusregion Andermatt wird zudem der Windpark Gütsch und seine Geschichte beschrieben.

Das Kapitel über den theoretischen Rahmen der Arbeit gibt zuerst eine reflektierte Gesamtübersicht über die Literaturrecherche und geht anschliessend auf die relevantesten Aspekte für die vorliegende Studie ein, von denen die Hypothesen zur empirischen Untersuchung abgeleitet und formuliert werden. Im Fokus stehen dabei die beiden Aspekte Standortwahl und Windtourismus. Anschliessend wird im darauffolgenden Kapitel das Forschungsdesign beschrieben, wobei das Vorgehen und die Datensammlung erläutert werden. Zudem werden die untersuchten Variablen beschrieben und die genaue Analyse und Messung derjenigen erklärt. Die Resultate der Umfrage, der Interviews und der betrachteten Statistiken werden im nachfolgenden Kapitel ausgeführt. Anschliessend werden diese gesamtübergreifend diskutiert und daraus konkrete Implikationen abgeleitet und ausformuliert. Abschliessend werden die Limitationen der vorliegenden Studie beschrieben.

2 Literaturrecherche

2.1 Tourismus in der Schweiz

2.1.1 Beitrag an Wirtschaft und Umwelt

Der Tourismus in der Schweiz trägt mit einem Anteil von rund 4% der landesweiten Arbeitsplätze und 2.9% der Bruttowertschöpfung einen bedeutenden Teil an der Wirtschaft bei (Schweizer Tourismus-Verband STV, 2022). Entsprechend ist auch der Beitrag dieses Sektors an den Klimawandel nicht vernachlässigbar. Weltweit entspricht dieser ca. 8% aller Treibhausgasemissionen, die zur globalen Erderwärmung beitragen (Burch, 2020). Obwohl besonders der (Flug-)Verkehr bedeutend ist, steuern auch der Verpflegungsbereich in Gaststätten und die Hotellerie ihren Teil dazu bei. (Bundesamt für Statistik, 2023a)

Die Nachfrage in der Hotellerie in der Schweiz steigt und sinkt einhergehend mit globalen politischen und wirtschaftlichen Schwankungen. Beispielsweise führte die Aufhebung des „fixen Wechselkurses“ des Euro der Schweizerischen Nationalbank (SNB) im Jahr 2015 dazu, dass leichte Abnahmen in der Nachfrage zu spüren waren, wobei die Logiernächte von 2017 bis 2019 wieder einen Anstieg erlebten und im Jahr 2019 sogar einen neuen Rekord verzeichneten. Ab dem Jahr 2020 wurde der Tourismussektor wiederum durch die Covid-19 Pandemie stark negativ beeinflusst. Die Nachfrage steigt seither und insgesamt jedoch kontinuierlich an. (Bundesamt für Statistik, 2021b)

2.1.2 Zusammenhang mit dem Klima in der Schweiz

Besonders bekannt ist die Schweiz für ihren Wintertourismus. Die Spuren der globalen Erwärmung sind aber auch am Schwinden der Gletscher und somit auch dem schwindenden Schnee, der für Wintersportarten wie Ski- und Snowboardfahren genutzt wird, immer stärker zu erkennen. Die Notwendigkeit, Schnee dafür künstlich zu erzeugen, steigt und entsprechend erhöht sich der dafür benötigte Energieverbrauch stetig. In den vergangenen Jahren konnte ein deutlicher Trend hin zu rückgehenden Tagen mit schneebedecktem Boden festgestellt werden. Gemäss Prognosen rechnet das WSL-Institut für Schnee- und Lawinenforschung SLF (2023) mit einem Rückgang der natürlichen Schneedecke von 70% bis Ende des Jahrhunderts, sollten die klimaschädlichen Emissionen nicht reduziert werden können. Dies bedeutet, dass die Skisaison einen halben bis einen Monat später beginnen würde als heute und nur noch oberhalb 2500 m ü. M. genügend Naturschnee für einen rentablen Betrieb eines Skigebiets vorhanden ist. Obwohl es in der Schweiz einige Schneesportgebiete gibt, die bis über 2500 m ü. M. reichen, betrifft dies jeweils nur einen gewissen Teil des jeweiligen Gebiets (Schweiz Tourismus, 2023).

2.1.3 Energiekonsum

Gemäss einer Pilotstudie des Bundesamts für Statistik (2023) ist der Anteil an erneuerbaren Energien im Tourismussektor nach wie vor klein (siehe Abbildung 1) und die Energiestrategie 2050 sieht vor, diesen Anteil erheblich zu erhöhen. Es ist zwar zu verzeichnen, dass die Verwendung von erneuerbaren Energien im Tourismus stärker anwächst als im gesamtschweizerischen Durchschnitt, jedoch trifft dies in der genannten Studie im betrachteten Zeitraum bis 2019 auch für die nicht erneuerbaren Energieträger zu. In Betrachtung des Vergleichsjahrs 2014 zeigen die Resultate aber erste Hinweise darauf, dass „für einen Teil der touristischen Aktivität eine Substitution der Energieträger möglich war und stattgefunden haben könnte“. (Bundesamt für Statistik, 2023a)

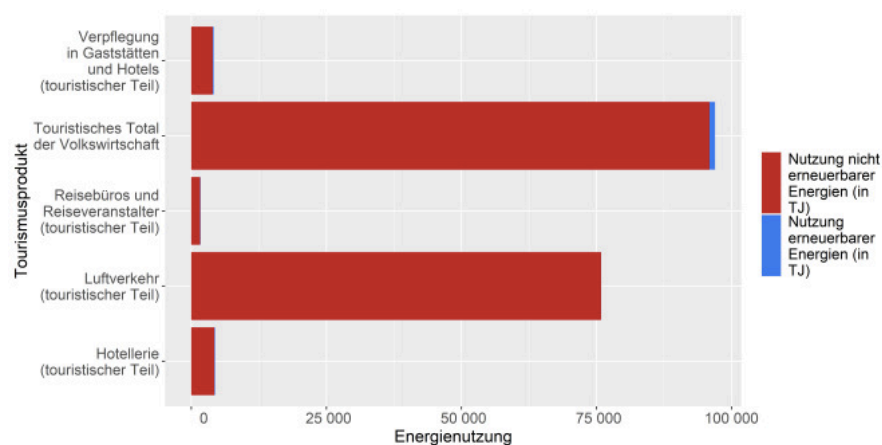


Abbildung 1: Verteilung der Energienutzung nach Tourismusprodukt und Art des Energieträgers (Bundesamt für Statistik, 2023a)

2.2 Energiestrategie 2050 und bisherige Entwicklungen

In dem im Jahr 2021 veröffentlichten *Energie: Panorama* Bericht des Bundesamts für Statistik (2021a) wird festgehalten, dass die Schweiz 24,1% des Endenergieverbrauchs durch erneuerbare Energien abdecken kann. Im Sommerhalbjahr verfügt sie dank der heimischen Stromproduktion über Stromüberschüsse, welche exportiert werden können. Im Winterhalbjahr besteht jedoch ein Defizit, was erfordert, dass Strom aus dem Ausland importiert werden muss. Dies entspricht den letzten veröffentlichten Zahlen nach 70% des gesamtjährlichen Endenergieverbrauchs (Bundesamt für Statistik, 2022). Besonders während des noch anhaltenden Konflikts zwischen Russland und der Ukraine hat sich im letzten Winter 2022-2023 gezeigt, dass die Schweiz die eigene Versorgungssicherheit unter anderem durch den Ausbau von einheimischen erneuerbaren Energien zukünftig nachhaltig stärken möchte (Eidg. Departement für Umwelt, Verkehr, Energie und Kommunikation UVEK, 2023).

Die Schweiz hat sich gemäss dem Pariser Klimaabkommen dazu verpflichtet, ihren Treibhausgasausstoss gegenüber dem Stand von 1990 bis 2030 zu halbieren. Im Jahre 2019 hat sich

der Bundesrat dazu entschieden, das Netto-Null Emissionsziel bis zum Jahre 2050 zu erreichen. Derzeit laufen unterschiedliche Massnahmen, diese Strategie zu verfolgen, wie beispielsweise die Steigerung der Energieeffizienz. Dazu gehört auch der Ausbau der erneuerbaren Energien. (Bundesamt für Energie, 2023a)

In erster Linie werden die Betreiber von Stromproduktionsanlagen aus Sonnen- oder Windenergie sowie aus Geothermie (Erdwärme) und Biomasse anhand von Investitionsbeiträgen vom Bund unterstützt (Bundesamt für Energie, 2023b). Aber auch die Projektdauer, die in ihrer Länge stark durch das Bewilligungsverfahren geprägt ist, sollte verkürzt werden (Beschleunigungserlass), wie sich dies beispielsweise am „Windexpress“ dank dessen Annahme des Bundesparlaments im Juni 2023 gezeigt hat (Suisse Éole, 2023a).

Die Wasserkraft wird in der Schweiz bereits seit geraumer Zeit genutzt, weshalb neben deren Ausbau laut Bundesamt für Energie (2023c) vor allem die „neuen“ erneuerbaren Energiequellen – also Windenergie, Photovoltaik, Holz, Biomasse und Biogas – entsprechend gefördert werden möchten. Im Jahr 2021 betrug der Anteil dieser Energiequellen 8% der Netto-Stromproduktion in der Schweiz, welcher als „relativ klein“ bewertet wird. Abbildung 2 zeigt die Entwicklung der genannten „neuen“ erneuerbaren Energiequellen für die Stromproduktion von 1990 bis 2021, wobei besonders der Anteil an Sonnenenergie einen sich abzeichnenden Auf-

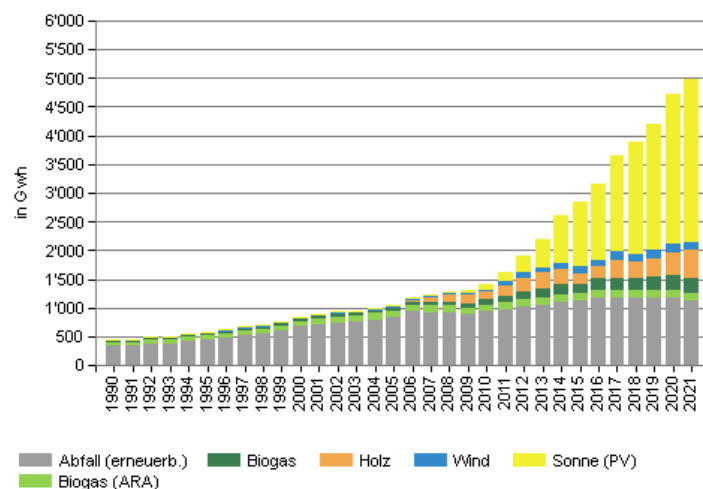


Abbildung 2: Erneuerbare Stromproduktion nach Energiequellen (ohne Wasserkraft)
(Bundesamt für Energie, 2023c)

schwung erlebte (Bundesamt für Energie, 2023c). Dies hat einerseits mit der Förderung dieser Energiequelle anhand von Subventionen sowie mit der schliesslich damit einhergehenden starken negativen Preisentwicklung von Photovoltaikanlagen zu tun. Ersichtlich ist in dieser Graphik auch, dass die Windenergie seit 2010 keinen deutlich erkennbaren Zuwachs in ihrer Nutzung aufweist.

2.3 Die Windenergie

2.3.1 Aktuelle Situation

2.3.1.1 Europa

Die Windenergie hat in Europa in den vergangenen Jahren einen Aufschwung erlebt. Allein im vergangenen Jahr 2022 wurden in Europa 19 GW Windkraftleistung zur Erreichung der Klimaziele gebaut. Dies bedeutet einen Anstieg von 46% in der EU im Vergleich zum Vorjahr 2021, was jedoch immer noch nicht der erforderlichen jährlichen Kapazität entspricht, um die gesetzten Klimaziele bis 2030 zu erreichen. Dazu müsste der Ausbau verdoppelt werden. Die installierte Windenergiekapazität in Europa entspricht heute 255 GW. (WindEurope, 2023)

2.3.1.2 Schweiz und ihre Nachbarn

In der Schweiz entspricht der Anteil an Windenergie an der Energieproduktion bisher weniger als 1% (EnergieSchweiz, 2023), obwohl das Potential als sehr hoch eingeschätzt wird. Es bestehen 41 Anlagen, welche insgesamt 87 MW produzieren (Stand 2022; Bundesamt für Energie, 2023d). Vergleichsweise hat Deutschland einen Anteil an Windenergie bei der Stromproduktion von 24.1% (Stand 2022), im Jahr 2021 waren es bereits 21.4% (Statistisches Bundesamt, 2023), und zählt aktuell somit als das „ausbaustärkste“ europäische Land betreffend Windkraftanlagen (WindEurope, 2023). Auch beim Landesnachbar Österreich stellt die Windenergie 11% der heimischen Stromerzeugung dar, was im Jahr 2022 einer Gesamtleistung von 3573 MW und somit der Versorgung von mehr als der Hälfte der österreichischen Haushalte entspricht (IG Windkraft, 2023).

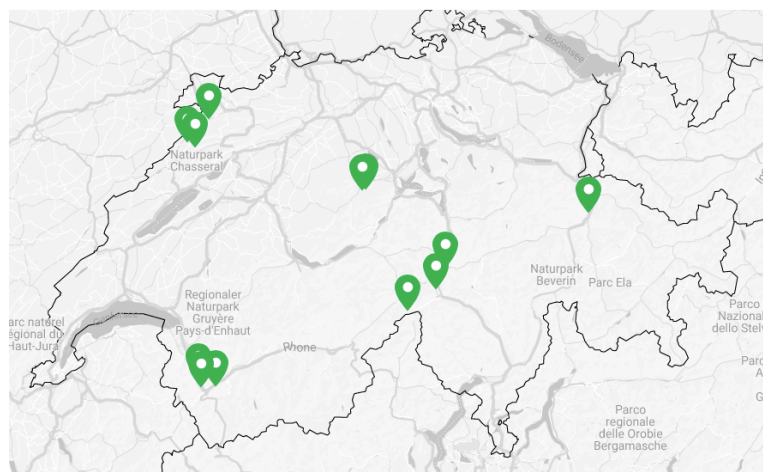


Abbildung 3: Betriebene Windparks und -anlagen in der Schweiz (SuisseEole, 2023b)

Die erste Windenergieanlage in der Schweiz wurde 1986 in Langenbruck, Baselland mit einer Leistung von 28kW errichtet. Der grösste Windpark in der Schweiz befindet sich im Berner

Jura auf dem Mont Crosin mit einer Leistung von 37.2 MW von insgesamt 16 Windturbinen. (Bundesamt für Energie, 2023d)

Grundsätzlich sind die Kantone dafür verantwortlich, geeignete Gebiete für die Nutzung der Windenergie in ihren Richtplänen festzulegen. Dies ist durch das Energiegesetz definiert. Weist ein Windpark eine Leistung von 20 GWh pro Jahr auf, ist dieser von nationalem Interesse (Bundesamt für Energie, 2023d). Da der Ausbau der Windkraft zur Energiestrategie gehört und beschleunigend gefördert werden möchte, sind mehrere grössere, geplante Projekte entsprechend von nationalem Interesse.

2.3.2 Potential

Ein signifikanter Grund, weshalb die Windenergie in Europa gefördert wird und werden möchte, wird vorwiegend durch ihr Potential begründet, einen bedeutenden Teil des Winterstrombedarfs zu decken (Bundesamt für Energie, 2023d). Im Vergleich zur Solarenergieleistung und Wasserkraftleistung, die erhöht während den Sommermonaten generiert werden, produziert die Windkraft in der Schweiz rund zwei Drittel ihres Stroms im Winterhalbjahr. Dies wird in Abbildung 3 anhand des Vergleichs von Stromproduktion durch Wasser-, Wind- und Solarkraft der Jahre 2017 und 2018 verdeutlicht (Bundesamt für Energie, 2023d).

Im Jura, im Mittelland, in den Voralpen und auf den grossen Alpenpässen liegt der Anteil der Jahresproduktion, der in den Monaten Oktober bis März produziert wird, bei 55-70% (Meyer et al., 2022).

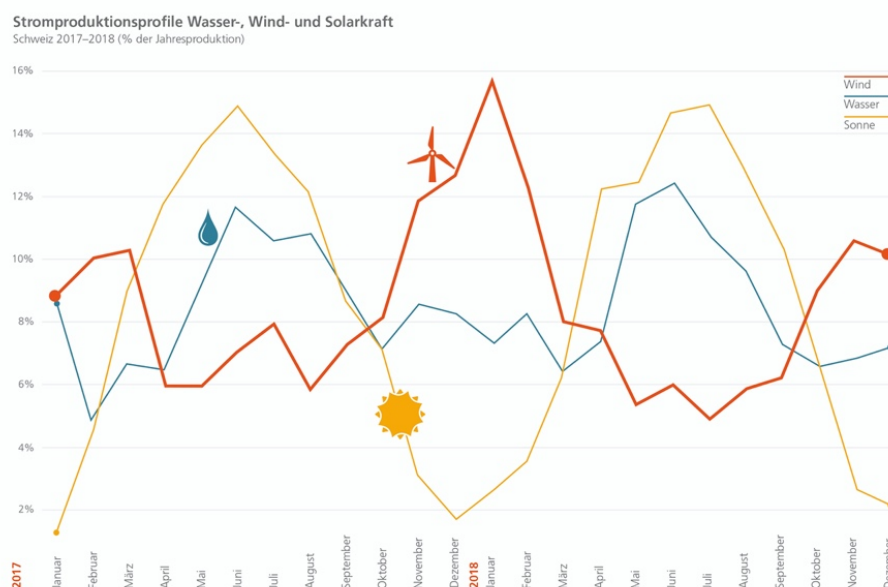


Abbildung 4: Stromproduktionsprofile Wasser-, Wind- und Solarkraft zwischen 2017-2018 in der Schweiz (Bundesamt für Energie, 2023d)

In einer Studie von Suisse Éole (2020) in Zusammenarbeit mit dem Bundesamt für Energie BFE und EnergieSchweiz zeigt sich ein weiterer Vorteil der Windenergie gegenüber anderen Energiequellen in der Ökobilanz von Windanlagen. Mit 15 g CO₂-eq/kWh weist sie die beste

Ökobilanz unter den „neuen“ erneuerbaren Energien auf und spart im Vergleich zum importierten Energiemix 412 g CO₂-eq/kWh ein. Des Weiteren sind im Vergleich auch die Gesamtauswirkungen auf die Umwelt nahezu am geringsten.

Nicht zuletzt werden auch die Windverhältnisse in der Schweiz als „hervorragend“ beschrieben, oder zumindest „ebenso gut“ wie in den Nachbarsländern, was schweizerische und europäische Windatlas zeigen (Suisse Éole, 2020). Mittels neuer Technologien und Methoden wurde demnach im Forschungsprojekt „Maximum Wind Energy Production (MaxWEP): Maximizing Winter Production of Energy by Exploiting Terrain Potential“ von Kristianti et al. (2022) der ETH Lausanne untersucht, wie man den schweizerischen Windatlas¹ optimieren kann, um bisher unerkennbare, aber geeignete Windpotentialgebiete neu zu entdecken und für den Ausbau der Windenergie zu nutzen. Die Erkenntnisse zeigen, dass weitere Gebiete durchaus in Betracht gezogen werden könnten, die bisher nicht berücksichtigt wurden.

Das Windpotential beläuft sich gemäss einem Bericht des Bundesamts für Energie basierend auf dem Windatlas 2019 auf 29.5 TWh/a (Meyer et al., 2022). Dies sei jedoch eine konservative Schätzung und Suisse Éole (2020) schreibt bereits in ihrem Bericht aus dem Jahr 2020, dass das Potential auf 30 TWh/a nach oben angepasst werden dürfte.

2.3.3 Die Zukunft der Windenergie in der Schweiz

Angesichts der hochgesteckten Ziele der Energiestrategie 2050 plant der Bundesrat nun, die Verfahren für die Planung und den Bau grosser Kraftwerke für erneuerbare Energien, wie im vorherigen Kapitel erwähnt, zu beschleunigen. Im Jahr 2021 veröffentlichten Dujardin et al. (2021) der ETH Lausanne und des WSL-Instituts für Schnee- und Lawinenforschung SLF Ergebnisse ihrer Untersuchung, wo sich welche Art von erneuerbarer Energie besonders eignet, um eine maximale CO₂-Neutralität und Energieunabhängigkeit sicherzustellen. Es wurde gezeigt, dass sich vor allem das Jura für die Windenergie anbietet, aber auch die Voralpen und Alpen. Spezifisch betreffend die Windenergie hat sich zudem eine kürzlich veröffentlichte Studie von Spielhofer et al. (2023) der ETH Zürich unter anderem damit beschäftigt, welche spezifischen Gebiete von nationalem Interesse für den Bau von Windenergieanlagen in Betracht gezogen werden können, und inkludieren hierbei erstmals auch Flächen, auf denen bisher eigentlich keine Windkraftanlagen gebaut werden dürfen. Während die Studie beschreibt, dass sich ein Ausbau von grösseren Anlagen vor allem im Mittelland und den Voralpen lohnen würde, wurden 36 Standorte identifiziert, die sich besonders gut für die Produktion von Windenergie eignen würden – dazu gehören auch Standorte im Kanton Graubünden, wo die Nähe zu den Alpen gegeben ist.

¹ Der Windatlas Schweiz modelliert Windgeschwindigkeiten und Windrichtungen und erschliesst somit Windpotentialgebiete, welche für die Planung von Windenergieanlagen berücksichtigt werden können. Siehe dazu: https://www.uvek-gis.admin.ch/BFE/storymaps/EE_Windatlas/?lang=de

2.3.3.1 „Spezialfall“ Graubünden

Der Kanton Graubünden hat im April 2023, gestützt auf die aktuellen Entwicklungen im Rahmen der Energiestrategie 2050, einen neu angepassten „Richtplan Energie“ entwickelt, für welchen bis im September 2023 noch schriftliche Einwände und Vorschläge entgegengenommen werden. In diesem wurden Potentiale zum Ausbau der Wasserkraft sowie der Windkraft thematisiert und aktuell geltende Bestimmungen für die Solarkraft gezeigt. (Kanton Graubünden, 2023a)

Für die Windpotentiale wurden „[u]nter Berücksichtigung von Nachhaltigkeits-, Landschafts- und Umweltschutzkriterien“ 25 Eignungsgebiete festgehalten, die alle nationales Interesse vorweisen und auf einer interaktiven Karte aufgerufen werden können². Abbildung 5 zeigt die Ge-

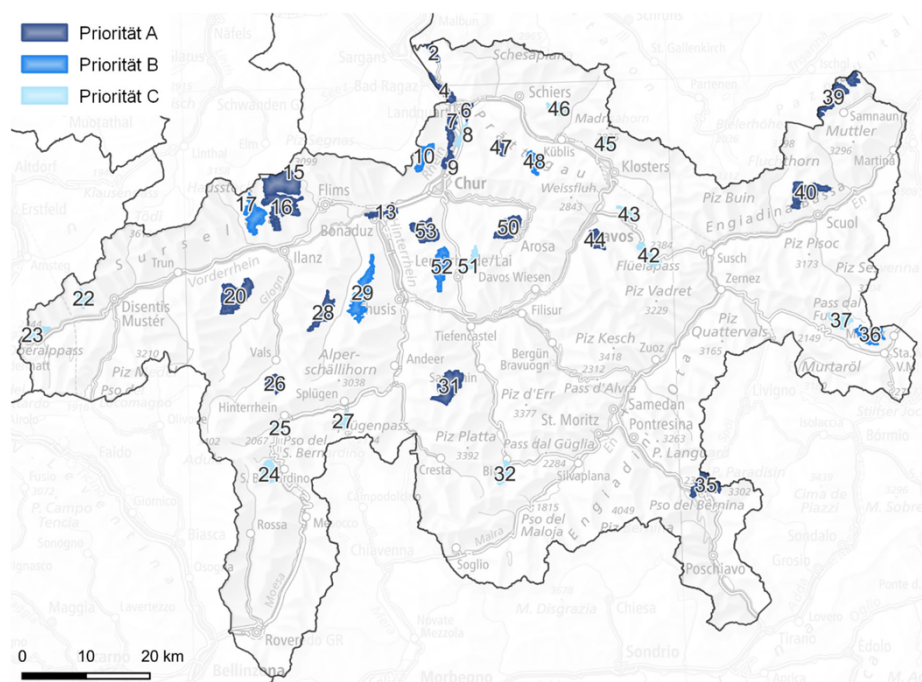


Abbildung 5: Überblick der Eignungsgebiete für Windkraftanlagen in Graubünden (Kanton Graubünden, 2023b)

gebiete mit Priorisierungskennzeichnung im Kanton Graubünden auf. Im Bericht werden nachfolgend die Gebiete gemäss ihrer Priorisierung aufgelistet, wobei die bei dem in der vorliegenden Studie untersuchten Tourismusgebiet Flims-Laan gelegenen Eignungsgebiete Nummer 15 „Crap Ner“ den dritten Rang und das Eignungsgebiet Nummer 16 „Crap Sogn Gion“ den elften

² Siehe dazu: https://kt-gr.maps.arcgis.com/apps/Embed/index.html?web-map=09375bccb1894f7e837bb8e4f35aa8d9&extent=8.3516%2c46.0829%2c10.848%2c47.1142&zoom=true&previewImage=false&scale=true&legend=true&disable_scroll=false&theme=light
Aufgerufen am 23. Juli 2023

Platz belegen. Laut aktueller Planung werden beide Gebiete in den kantonalen Richtplan aufgenommen.

2.3.4 Gründe für die Verzögerung oder Verhinderung von Windparkprojekten

Suisse Éole (2020) hat sich in ihrer publizierten Windenergiestrategie damit befasst und in einer Studie untersucht, welche Gründe für die Verzögerungen von Windparkbauten vorliegen oder weshalb diese bereits in einer frühen Phase blockiert werden: Es wurde gezeigt, dass die meisten Projekte durch systematische Einsprache verzögert oder gar verhindert werden. In 65% der von Suisse Éole untersuchten Fälle wurde dabei von Privatpersonen und lokalen Vereinen Einsprache erhoben. In 40-50% der Fälle handelte es sich um Einsprachen von Landschaftsschutzorganisationen, in 30-35% der Fälle von Pflanzen- und Tierschutzorganisationen, in 18% der Fälle von Gemeinden und schliesslich in 6% der Fälle von anderen NGOs oder vom Bundesamt für Umwelt. Besonders Lärmemissionen, Schattenwurf, Auswirkungen auf die Landschaft und Tierwelt (hauptsächlich Vögel und Fledermäuse) waren Hauptgründe für die Einsprachen. Aber auch Gründe wie Energieproduktion und -effizienz, Gefahren durch Raureif und Eiswurf wurden genannt sowie Umweltbelastung (Wasser, Pflanzenwelt), die Landnutzung, Auswirkungen auf die Immobilienpreise und finanzielle Aspekte der Projekte. Spannenderweise wurden Auswirkungen auf Tourismus- und Freizeitaktivitäten nur selten erwähnt. In der breiten, lokalen Bevölkerung wurden durch Suisse Éole nur geringe Akzeptanzprobleme erkannt. Die Windenergie geniesst in ihrer Studie Unterstützung der Bevölkerung, wenn diese in die Projektierung einbezogen werden.

Unter dem Projekt „Windexpress“ wurde darüber diskutiert, das Verfahren zur Realisierung eines Windparks zu beschleunigen. Heute dauert das Verfahren rund 25 Jahre. Es ist hauptsächlich dadurch geprägt, dass verschiedene politische Stufen durchlaufen werden müssen und die Erstellung des Nutzungsplans während des Bewilligungsverfahrens, bei welchem bei Gemeindebehörden, Kantons- und Bundesgericht Rekurse eingelegt werden können, ungefähr 10 Jahre dauert. Durch eine Anpassung des Energiegesetzes soll nun die letzte Phase der Baubewilligung um 2 Jahre verkürzt werden, sodass ein Rekurs vor dem Kantonsgericht nach wie vor möglich ist, jedoch der Gang vor Bundesgericht eingeschränkt ist. Aktuell betrifft dies vor allem Windparkprojekte, die teilweise bereits vor fast 20 Jahren mit ihrer Planung begonnen haben. (Suisse Éole, 2023a; Fedlex, 2023)

Dass Windparkprojekte nicht rasch umgesetzt werden können, liegt daran, dass die Dauer des Planungs- und Bewilligungsverfahrens eine beträchtliche Zeitspanne in Anspruch nimmt. Auch der Bundesrat erkennt, dass die zahlreichen politischen Einsprachemöglichkeiten während diesen Planungs- und Bewilligungsverfahren zudem die Gefahr für weitere Verzögerungen bergen.

2.4 Die Beziehung von Infrastrukturen erneuerbarer Energien (insbesondere Windenergie) und Tourismus

2.4.1 Akzeptanz von Infrastrukturen erneuerbarer Energien und insbesondere Windkraftwerken als Tourismusgebiete

Weltweit untersuchten bisher unterschiedliche Studien die Akzeptanz und den Einfluss von Energiekraftanlagen aus erneuerbaren Quellen und deren Infrastrukturen auf die Tourismusentwicklung an einem Tourismusort, die TouristInnen selbst, die lokale Bevölkerung und entsprechend auch auf (lokale) wirtschaftliche Veränderungen. Schliesslich sind deren Resultate beeinflusst durch unterschiedliche kontextuelle Gegebenheiten, wie beispielsweise die Art der erneuerbaren Energiequelle, Landschaftselemente, Lokalkultur, demografische Merkmale von befragten Personen, politische Bedingungen und Einstellungen, erwünschte oder erwartete Erlebnisangebote, individuelle Präferenzen oder die räumliche Planung. Während gewisse Untersuchungen einen schwachen, aber nachweisbaren negativen Effekt auf den Tourismus feststellten (u.a. Gardt et al., 2018; Broekel & Alfken, 2015), gelang es mehreren Studien, keinen negativen oder sogar einen positiven Einfluss darzulegen (u.a. Bidwell, 2023; Frantál & Kunc, 2011; Gardt & Litmeyer, 2017; Weiss, 2015).

2.4.1.1 Landschaftsbild

Häufig ist der Hauptgrund der Einwände von Gegnern von Windkraftanlagen, dass das Landschaftsbild durch das Errichten einer Anlage „zerstört“ wird und somit ein Verlust der wahrgenommenen Landschaftsqualität vorliegt. Dies (oder deren Annahme) ist häufig auch die grösste Besorgnis von TouristInnen, wenn sie nach ihrer Meinung zur Möglichkeit der Errichtung einer erneuerbaren Energiekraftanlage an ihrem Reiseziel gefragt werden (Aschenbrand & Grebe, 2018; Bidwell, 2023). In der Studie von Hunziker et al. (2014) der Eidg. Forschungsanstalt für Wald, Schnee und Landschaft WSL wurde die Akzeptanz von Wasser- und Solarkraftwerken in alpinen Gebieten in der Schweiz untersucht, wo sich gewisse TouristInnen kritisch zeigten, weil sie die „sichtbare Modernisierung“ in einem Erholungsgebiet nicht als „authentische Landschaft“ betrachten. Werden TouristInnen und Personen aus der Bevölkerung jedoch zu bereits bestehenden Projekten befragt, fällt ihre Haltung meist positiver und als „nicht störend“ aus (Bidwell, 2023; Gardt & Litmeyer, 2017; Tverijonaite et al., 2022; Frantál & Kunc, 2011; Spiess et al., 2015). Dabei spielt die Visibilität (demnach die Grösse und die Distanz zu der erneuerbaren Energiekraftanlage) und Art der Energieanlage eine Rolle. Je weniger sichtbar oder bemerkbar die Anlage ist, desto eher wird sie von befragten Personen unterstützt, unter anderem auch weil Sicherheitsbedenken und Lärmemissionen reduziert werden (Tverijonaite et al., 2022; Spiess et al., 2015)

2.4.1.2 Standortwahl

Der NIMBY-Effekt („not-in-my-backyard“) ist ein Phänomen, das besagt, dass Personen grundsätzlich eine positive Haltung gegenüber etwas haben (wie z.B. Windenergie), dieses jedoch aus persönlichen, egoistischen Gründen ablehnen, sobald sie direkt damit konfrontiert werden (O’Hare, 1977). Dieser Effekt wird in Zusammenhang mit der Akzeptanz für erneuerbare Energien in der bestehenden Literatur insofern herangezogen, als dass die soziale Akzeptanz gegenüber erneuerbaren Energien grundsätzlich hoch ist, Kraftwerke erneuerbarer Energien in Nähe zum Wohn- oder Erholungsgebiet jedoch abgelehnt werden. Frantál & Kunc (2011) haben diese Theorie um die Dimension „Tourismusakzeptanz“ (engl. tourism acceptance) als eine räumliche Zwischenstufe von globaler und lokaler Akzeptanz von Windkraftanlagen ergänzt, in dem sie gezeigt haben, dass sich die Wahrnehmung des Landschaftsbilds von der Lokalbevölkerung und TouristInnen unterscheidet. Hunziker et al. (2014) haben zudem gezeigt, dass Stammgäste die zur Debatte gestellte Erhöhung einer Staumauer stärker ablehnen als andere TouristInnen.

Was sowohl für die lokale Bevölkerung als auch für TouristInnen bei der Wahl eines Standorts für eine Windkraftanlage relevant ist, ist die Nähe zu bereits bestehenden Infrastrukturen (Hunziker et al., 2014). Die unberührte Landschaft möchte nur ungern verbaut werden, da nebst dem Anschluss ans Stromnetz auch Strassen für den Zugang für Bau und Unterhalt der Anlage gewährleistet werden müssen. Wenn die Aussicht jedoch bereits durch technische Anlagen oder andere Bauwerke geprägt ist, ist die Akzeptanz für die Errichtung einer Windkraftanlage höher (Weiss, 2015) und sie werden teilweise sogar als „Teil eines umfassenderen Naturverständnisses“ gesehen und nicht nur als Infrastruktur (Spiess et al., 2015). Aschenbrand & Grebe (2018) gehen weiter darauf ein, dass es eine Frage der Darstellung von Tourismusangeboten in Werbematerialien ist, da Windräder aktuell noch weniger abgebildet werden. Jedoch wurde aber auch am Beispiel der Windmühlen in den Niederlanden eine „Romantisierung“ dieser möglich gemacht.

2.4.1.3 Individuelle Präferenzen

Bei der Akzeptanz von Anlagen erneuerbarer Energien sind auch persönliche Präferenzen, demografische Merkmale oder die politische Haltung weisende Faktoren. Altruistische Personen befürworten ein Windprojekt eher als traditionell eingestellte Personen (Bidwell, 2013; Bidwell, 2023), so wie auch jüngere TouristInnen stärkere BefürworterInnen sind als ältere (Gardt & Litmeyer, 2017). Diese Faktoren beeinflussen schliesslich auch die individuell wahrgenommene Sinnhaftigkeit eines Projekts, wovon die Unterstützung oder Akzeptanz eines Windkraftprojekts entsprechend abhängt (Bidwell, 2013). Sinnhaftigkeit wird beispielsweise darin gesehen, dass wirtschaftlicher Nutzen für die Gemeinschaft entsteht oder dass die Errichtung eines neuen Kraftwerks als notwendig wahrgenommen wird (Bidwell, 2013; Gardt & Litmeyer,

2017; Tverijonaite et al., 2022). Je sinnvoller eine Windkraftanlage bewertet wird, desto weniger wird das Landschaftsbild als „durch die Anlagen gestört“ empfunden (Gardt & Litmeyer, 2017).

2.4.1.4 Wahrnehmung von bestehenden und sich vorzustellenden Energiekraftanlagen

Bisherige Studien haben gezeigt, dass die Akzeptanz für bereits bestehende Energiekraftanlagen höher ist als für solche, die sich befragte Personen imaginär vorstellen sollten. Dies gilt besonders für Windkraftanlagen wegen ihrer hohen Sichtbarkeit. Die Studie von Bidwell (2023) untersuchte die Errichtung eines Windparks in Küstennähe einer Feriendestination in den U.S.A. von der Planungsphase über die Bauphase bis zur Betriebsphase. Obwohl die an der Studie teilnehmenden TouristInnen anfänglich die Erwartung hatten, dass der Windpark einen negativen Einfluss auf Natureigenschaften und Naturressourcen haben wird, zeigten sie eine positive Haltung gegenüber dem Projekt und die Akzeptanz zugenommen hat von der Bau- zur Betriebsphase.

Beispiele aus der Schweiz bestätigen dieses Phänomen, wobei die Akzeptanz der Lokalbevölkerung vor und nach der Errichtung einer Windenergieanlage untersucht wurde. Seitz (2011) hat in ihrer Studie untersucht, wie sich die Anwesenheit und Sichtbarkeit der Windkraftanlage in der UNESCO-Biosphäre Entlebuch in Luzern auf die Akzeptanz der Anwohnenden auswirkt. Die befragten Personen haben angegeben, dass sie sich als „nicht sehr innovativ und eher konservativ“ beschreiben, jedoch aber durch die Anwesenheit der Anlage eine hohe Akzeptanz gegenüber dieser zeigen. Denn Personen, die die Windkraftanlage sehen, besuchen diese öfter, als solche, die sie nicht sehen, bewerten die Windenergie generell auch positiver und interessieren sich entsprechend auch mehr für deren Technik. Die befragten Personen scheinen sich bewusst, dass „wenn man Strom braucht, dieser irgendwo erzeugt werden muss“.

2.4.1.5 Weitere Aspekte für die Akzeptanz

Besonders für die Lokalbevölkerung ist es relevant, dass der (lokale) wirtschaftliche Nutzen einer Windkraftanlage gegeben ist und bereits in der Planungsphase transparent ist, um ihre Unterstützung zur Errichtung einer neuen Anlage zu gewinnen (Bidwell, 2013; Spiess et al., 2014; Spiess et al., 2015). Dazu gehören die Schaffung von Arbeitsplätzen, tiefere resultierende Energiepreise, Gewinnbeteiligung, Steuervorteile oder die Erweiterung des Tourismusangebots. Auch für TouristInnen ist dies nicht unwichtig, auch wenn sie die direkten Vorteile nicht unmittelbar sehen können (Bidwell, 2023). Wenn die Bevölkerung wirtschaftlich nicht beteiligt sein kann, ist Transparenz betreffend der Erträge (monetär und Energie [kWh]) wünschenswert, wodurch Vertrauen geschaffen werden kann und eine gute Grundlage für die Errichtung von

weiteren Windkraftwerken gebildet werden kann (Spiess et al., 2014). Eine transparente Kommunikation zu sämtlichen Stakeholdern in frühen Planungsphasen erweist sich als äusserst sinnvoll und zielführend für den Erfolg eines Windkraftprojekts, da Bedenken aller Beteiligten bereits früh erkannt und gemildert werden können, die Akzeptanz erhöht werden kann und Einsparungen somit minimiert werden können (Spiess et al., 2014; Spiess et al., 2015).

Nebst einem wirtschaftlichen Nutzen sieht die Lokalbevölkerung in hier betrachteten Studien aus der Schweiz einen Vorteil darin, das „Ortsimage“ aufzuwerten (Hunziker et al., 2014; Spiess et al., 2014). Die Errichtung eines Windparks, einer Wasserkraft- oder einer Solarkraftanlage bedeutet ein Zeichen der Umweltfreundlichkeit und somit auch der Fortschrittlichkeit.

2.4.2 Energietourismus

Die Aufwertung des „Ortimages“ durch die Anwesenheit einer Energiekraftanlage ist für eine Gemeinde von noch höherer Bedeutung, wenn dabei die Tourismusattraktivität gestärkt werden kann. Dies gelingt unter anderem durch die Erweiterung des Tourismusangebots. Frantál & Urbánková (2017) definieren das Zusammenspiel von Energiekraftwerken und Tourismus als „Energietourismus“, eine Nische des Industrietourismus, anhand von drei Beispielen: so genannten „Kohlesafaris“ (Führungen durch Kohleminen), einem Informationszentrum bei einem Kernkraftwerk und einem „Dragon Kite Festival“ unter Windrädern. Die grösste Motivation der BesucherInnen dieser Attraktionen waren das Interesse an den jeweiligen Technologien oder an der Energieproduktion generell, aber auch das Bedürfnis, einen „aussergewöhnlichen“ Tourismusort zu besuchen, war von Interesse. Etwa zwei Drittel der in dieser Studie befragten Personen haben eine positive Veränderung ihrer Haltung gegenüber der jeweiligen Energiequelle gezeigt, nachdem sie diese besichtigt haben. Dies zeigt, dass solche Unternehmungen als effektive Marketingwerkzeuge verwendet werden können, um negative oder problematische Ansichten gegenüber Energiequellen zu verändern und die soziale Akzeptanz dazu erhöht werden kann.

Beispiele aus Deutschland, Dänemark oder Island zeigen auf, wie Bildung über erneuerbare Energien und Spass auf Wander- oder Radwegen verknüpft werden können, beispielsweise anhand von interaktiven Tafeln am Wegrand. Informative Besucherzentren, um erneuerbare Energiequellen zu bewerben, sind teilweise kostspieliger, aber unterstützen die Initiative, wie Beispiele aus Dänemark, dem Vereinigten Königreich und den U.S.A. zeigen. (Beer et al., 2018) Deutschland hat im Jahr 2020 einen Reiseführer „Energiewende Deutschland“ veröffentlicht, um aufzuzeigen, dass die Energiewende auch attraktive Ausflugsziele mit sich bringt (Bundesministerium für Wirtschaft und Klimaschutz, 2020). Dabei wird mit Attraktionen wie „Tour de Wind“ geworben, wo man entlang eines „Windpfads“ mehrere Windkraftwerke besuchen kann.

Ein Beispiel aus der Schweiz zeigt einer der ersten Windparks, den Mont-Crosin, der erst mit seiner Erbauung für Tourismusaufkommen sorgte. Im Ort wird nun eine Standseilbahn von St-

Imier auf den Mont-Soleil betrieben, im Hôtel Restaurant Chalet Mont-Crosin werden Unterkunft und lokale Gerichte angeboten und auch in der Auberge L'Assesseur kann man einkehren. Zudem werden Pferdekutschenfahrten durch den Solar- und Windpark angeboten, oder man reitet selbst auf einem Pferd durch die Landschaft. In der Sternwarte auf dem Mont-Soleil kann man zu Tageszeiten die Sonne beobachten und während Nachtzeiten die Sterne. Besonders lokale Geschäfte haben durch die Tourismusentwicklung der Region profitieren können. Das Beispiel zeigt auch, wie die Akzeptanz der Bevölkerung für Windenergie durch die Verwirklichung des Windparks erhöht werden konnte. (*Sonnen- und Windkraftwerk, Mont-Soleil | Grand Chasseral Tourismus (CH) | Führung, o. D.*)

Dem Potential der Ergänzungsnutzung von Windkraftanlagen in einem Tourismusgebiet scheinen keine Grenzen gesetzt zu sein. Während im bereits erwähnten Beispiel von Frantál & Urbánková (2017) des „Dragon Kite Festivals“ einmal jährlich Familien mit Kindern dem Kitefliegerwettbewerb zusehen und dazu eine Führung der Windkraftanlagen möglich ist, fliegt Hannes Arch mit einem Flugzeug Slalom durch einen Windpark in Österreich (Halloran, 2022). Wo in manchen Fällen eine Aussichtsplattform an einer Windkraftanlage angebracht wird, um die Höhe für eine gute und weite Sicht zu nutzen, wird auf einer solchen Plattform in Zusammenarbeit mit IG Windkraft eine Musiksession zweier DJs Camo & Krooked live aus den Alpen in Österreich für UKF On Air (2021) übertragen. Ein Beispiel aus der Schweiz ist die Verzasca-Staumauer im Tessin, an der Pierce Brosnan als James Bond im Film „GoldenEye“ Bungee Jumping macht und die seither zahlreiche BesucherInnen anlockt („*GoldenEye*“ *Bungee Jumping*, Vogorno, o. D.).

Im Rahmen der Energieforschungsgespräche Disentis der Fachhochschule Graubünden, der ETH Lausanne und EWZ wurden das Thema „Energie und Tourismus“ und die Potentiale der beiden Bereiche im Jahr 2020 bereits einmal diskutiert (Fachhochschule Graubünden, 2020). Wobei in der Schweiz vor allem vom Zusammenspiel von Wasserkraftwerken und Skigebieten die Rede ist, wird die Windkraft bisher noch ziemlich aussen vor gelassen. Ein Beispiel für eine kreative Kombination von Tourismus und Energiekraftwerk ist die Nutzung von Staumauerflächen für künstlerische Installationen wie Lichtshows oder Open-Air-Filmvorführungen.

Die Mehrfachnutzung von (existierenden) Infrastrukturen und die Aufwertung des Images von Energieinfrastrukturen sind Ziele, die aktiv verfolgt werden möchten. Das Potential, die Zusammenarbeit zwischen Energiewirtschaft und Tourismus zu stärken, wird vor allem auch in Kommunikationsmitteln gesehen. Anhand von Vermarktungsplattformen, auf denen touristische Angebote, wie beispielsweise Wintertourismus und Energietourismus in einem Skigebiet gebündelt vermarktet werden könnten, würde die Verschmelzung der Interessen der Akteure aus den beiden Bereichen gestärkt. Edutainment-Beiträge auf Medienkanälen wie Social Media werden als weitere Mittel vorgeschlagen, um dem Effekt der Akzeptanzerhöhung Beihilfe zu

leisten. Höchst relevant in der Zusammenarbeit der beiden Bereiche ist eine gute Kommunikation und Koordination aller Stakeholder bereits in frühen Planungsphasen, auch um die jeweiligen Verantwortlichkeiten und die gegenseitigen Unterstützungsmöglichkeiten zu klären und weitere Synergien zu nutzen.

Tverijonaite et al. (2022) heben in ihrer Studie hervor, dass befragte TouristInnen in Island erneuerbare Energiekraftwerke als positiv ansehen, wenn diese als Beiträge zu einer nachhaltigen Entwicklung des Landes und zur Abmilderung des Klimawandels durch TourführerInnen dargestellt werden. Frantál & Kunc (2011) sprechen sich in ihrer Studie aus Tschechien ebenso für die Wichtigkeit von unterstützenden Marketing- und Kommunikationsmitteln zur Förderung der Akzeptanzerhöhung aus.

2.4.3 Um Infrastrukturen erneuerbarer Energien ergänzter Tourismus

Betrachtet man Tourismusgebiete, die nicht spezifisch durch ein Energiekraftwerk zum Tourismusort wurden, sondern deren Infrastrukturen durch ein solches ergänzt wird, stellen die höhere Zugänglichkeit durch die Erweiterung des Strassensystems oder von Wander- und Radwegen weitere Vorteile für den Tourismus dar, was erlaubt, neue Tourismusregionen zu erschliessen. Tverijonaite et al. (2022) haben gezeigt, dass sich durch die bessere Erschliessung gewisser Gebiete bessere Möglichkeiten für den Wintertourismus ergeben oder Touren diversifiziert werden konnten. Sie verglichen dazu Beispiele aus Alberta, Kanada und Burgenland, Österreich. Beer et al. (2018) zeigten, dass erneuerbare Energiequellen die Besucherzahlen einer Tourismusregion in gewissen Fällen durch ihr modernes Design, ihre Proportionen, das „Öko-Image“ und manchmal durch ihre Einzigartigkeit erhöhen und zudem höhere Gewinne von Verkäufen an Tankstellen oder Restaurants verzeichnet werden. Nachteile des infrastrukturellen Ausbaus sind mehr Stau oder illegales Parkieren an Strassenrändern und ein höherer Druck auf die natürliche Umwelt.

In der Studie von Frantál & Kunc (2011) zu Tourismusgebieten in Tschechien spricht sich eine deutliche Mehrheit von Befragten dazu aus, dass sie sich durch die Anwesenheit von Windkraftanlagen nicht gestört fühlen und diese ihre Wahl des Tourismusorts nicht beeinflussen würden. Bei dieser Betrachtung wird in bisherigen Studien häufig unterschieden, welches Erlebnis verfolgt wird. WandererInnen fühlen sich eher durch die Anwesenheit von Infrastrukturen erneuerbarer Energien gestört, da das Landschaftsbild eher gestört wird, wenn sie nach „unberührter Natur“ suchen und sie befürchten, dass zukünftige Wanderwege solchen Infrastrukturen zum Opfer fallen könnten (Weiss, 2015; Tverijonaite et al., 2022). Dies ist jedoch abhängig von den Routen, der Reisedestination, des Reisemodus und des Erlebnisses, das gesucht wird. Nicht alle WandererInnen suchen nach der „unberührten Landschaft“ (Spiess et al., 2015). Hingegen zeigten in der Untersuchung von Weiss (2015) WintersportlerInnen hohe Akzeptanz gegenüber Windkraftanlagen und nur eine geringe Meidungsabsicht der Tourismusregion der

Windkraftanlagen wegen. Zudem wird hervorgehoben, dass die Windkraftanlagen im untersuchten Gebiet in Salzburg, Österreich ohnehin bereits an einem Ort gebaut wurden, der von Infrastruktur, technischen Anlagen und anderen Bauwerken geprägt ist.

2.5 Fallbeispiele für Windparkprojekte in alpinen Tourismusregionen in der Schweiz

2.5.1 Tourismusstandort Andermatt und Windpark Gütsch

2.5.1.1 Tourismusangebot in Andermatt

Die Andermatt-Urserntal Tourismus GmbH (*Ferienregion Andermatt*, o.D.) wirbt für die Ferienregion Andermatt in erster Linie mit ihrer zentralen Lage, den umliegenden acht Alpenpässen und der hohen Vielfalt der Tourenmöglichkeiten und wird somit als „idealer Ausgangspunkt für erholungssuchende Naturliebhaber“ beschrieben. Besonders auch der historische Hintergrund der Region gibt Anreiz, die Gegend in allen möglichen Jahreszeiten zu besuchen. Zur Ferienregion gehören nebst der grössten Gemeinde Andermatt auch die umliegenden Orte Göschenen, Gurnellen, Hospental, Realp und Wassen, die jeweils die benachbarten Kantone mit der Region verbinden. Die Alpenpässe werden als Attraktion mit dem Auto, Motorrad oder Postauto erkundet.

Im Sommer bietet sich die Landschaft der Ferienregion Andermatt vor allem zum Wandern, Velofahren oder Mountainbiken und Klettern an. Die lokale Tourismusorganisation bietet zu diesen Aktivitäten jeweils geführte Touren an. Weitere Angebote bieten vier Golfplätze oder eine Sportanlage zum Padelspielen. Für Wanderer und Wandererinnen wurde ein „Berghüttenpass“ entwickelt, mit welchem man Stempel in 23 Berghütten sammeln kann und schliesslich etwas gewinnen kann. Zudem wirbt die Organisation mit dem „Hashtag“ #urnerhüttenpass auf den Sozialen Medien für einen Besuch der Hütten. Auch für die Kinder gibt es einen entsprechenden „Murmeli-Pass“, womit auf Spiel- und Grillplätzen fünf Kisten für einen Stempel zu finden sind. Weitere Attraktionen für Kinder bieten beispielsweise Bogenschiessen oder spezifische Themen- und Erlebniswege. Dazu gehören einige „Audio-Touren“, mit denen es die Region auf spielerische Weise zu entdecken gilt. In jeder der zugehörigen Gemeinden werden auch zu fast jeder Jahreszeit Dorftouren oder Museumsführungen angeboten. Für Schulklassen, Vereine, Familien und andere Gruppen gibt es auch eine Schnitzeljagd, die einen digital mit dem Smartphone durch Andermatt führt. (*Ferienregion Andermatt*, o.D.)

In den Wintermonaten ist das Tourismusangebot den Schneeaktivitäten Skifahren, Schlittschuhfahren auf Eisfeldern, Langlaufen, Schlitteln, Schneeschuhlaufen, Skitouren und Winterwandern gewidmet. Auch im Winter können zu den meisten der genannten Aktivitäten geführte Touren gebucht werden. An Tagen, an denen schlechteres Wetter herrscht, bieten verschiedene Spas ein Wellnessangebot an. (*Ferienregion Andermatt*, o.D.)

Sowohl im Sommer als auch im Winter bietet die Andermatt-Urserntal Tourismus GmbH eine Führung „Energie und Umwelt“ an, mit welcher man durch den Windpark Gütsch sowie das Heizwerk Göschenen geführt wird. Auf der Webseite der Andermatt-Urserntal Tourismus

GmbH liegen aus Sicht einer interessierten Person jedoch keine konkreten Informationen zum Inhalt der Führung vor. (*Ferienregion Andermatt*, o.D.)

Wenn man sich im Tal und somit im Dorf Andermatt befindet, sieht man den Windpark Gütsch nahezu nicht. Sobald man sich mit der Bergbahn von Andermatt auf die Bergstation Gütsch begibt, erscheint dieser aus Fahrtrichtung links der Bergstation. Eine übliche Wanderoute leitet vom Schneehüenderstock am Lutersee vorbei bis zur Bergstation Gütsch, wobei der Windpark Gütsch aber eigentlich in seiner gesamten Erscheinung erst nach dieser zu sehen ist. So ist es möglich, die Bergstation Gütsch zu passieren und die Wanderung durch einen Rundweg im Windpark zu verlängern oder hinterher eine andere Wanderoute einzuschlagen.

Bei Winteraktivitäten, wie beim Ski- oder Snowboardfahren ist der Windpark entsprechend dann ersichtlich, wenn man die Bergbahnen und Lifte um die Bergstation Gütsch benützt. Aus dieser Perspektive sind jeweils nur ein bis zwei der vier Turbinen zu sehen.

2.5.1.2 Tourismusentwicklung in Andermatt

In den Jahren 2009 bis 2011, als der Windpark Gütsch zu gewissen Teilen bereits bestand, wurde vom ägyptisch-montenegrischen Unternehmer und Investor Samih Sawiris im Namen seines Unternehmens Andermatt Swiss Alps AG die Errichtung eines Tourismusresorts geplant, welches in den Jahren 2012 bis 2014 in die Bauphase übergang und schliesslich zwischen 2016 bis 2020 Schritt für Schritt in Betrieb genommen wurde (Ecoplan, 2017). Im Jahr 2013 wurde das erste „5-Sterne-Deluxe Hotel The Chedi Andermatt“ eröffnet, im Jahr 2015 folgte die Eröffnung eines Golfplatzes, wonach im Jahr 2016 das Skigebiet „SkiArena“ mit dem Baustart der Gondelbahn von Andermatt auf den Nätschen und weiter auf den Gütsch erweitert wurde. Im Jahr 2018 wurde das zweite Hotel, das „Radisson Blu Hotel Reussen“ und die „Gotthard Residences“ eröffnet. In diesem Jahr wurde auch der „Schneehüenerstock-Express“ eröffnet, eine zusätzliche Erweiterung der Bergbahnen. Im Jahr 2019 wurden zehn Apartmenthäuser fertiggestellt. (*Geschichte Andermatt Swiss Alps*, o.D.)

Das Projekt wurde in vier Teilen durch die Hochschule Luzern mit einer Studie zu den soziokulturellen und sozioökonomischen Auswirkungen begleitet. Insgesamt wurden bis Ende 2020 über CHF 1.2 Milliarden in die Ferienregion Andermatt investiert, was den Ort sowohl im Aufkommen von neuen Bauten und Anlagen, aber auch wirtschaftlich durch „das Aufkommen eines neuen Gästesegments“, neugeschaffenen Arbeitsplätzen und Erweiterung des touristischen Angebots verändert hat. Sowohl die Logiernächte, die Anzahl verfügbarer Betten, die Ankünfte als auch die Zimmernächte sind seit 2013 erkennbar angestiegen. Weitere Auswirkungen der Investitionen zeigen sich in den steigenden Preisen für Eigentums- und Mietobjekte. (Durrer Eggerschwiler, 2021) Entsprechend werden auch der Bahnhof samt umliegendem Areal modernisiert, das Skigebiet ausgebaut und die Gemeinde Andermatt hat dabei auch in anderen

infrastrukturellen Belangen in einen Ausbau investiert, so zum Beispiel um den Wasserverbrauch sicherzustellen oder eine Erschliessungsstrasse zum Golf-Clubhaus zu bauen (Ecoplan, 2017).

2.5.1.3 Windpark Gütsch

Der Windpark Gütsch ist ein Energiekraftwerk des Elektrizitätswerks (EW) Ursern im Schweizer Kanton Uri und liegt auf rund 2300 m ü. M. auf dem Gütsch oberhalb der Gemeinde Andermatt. Er stellt eine Ergänzung zur Wasserkraft, die die Region hauptsächlich im Sommer mit Strom versorgt, dar. Der Windpark erzeugt mit einer installierten Leistung von 3.3 MW jährlich rund 5.5 GWh Strom, was ca. 1'250 mittlere Haushalte der Region versorgt. (Kanton Uri, 2023a; Elektrizitätswerk Ursern, 2023)



Abbildung 6: Windpark Gütsch im Sommer 2023 (eigene Fotoaufnahme)

Die Idee zum Windpark Gütsch entstand zum hundertjährigen Jubiläum des Elektrizitätswerks Ursern, das als Pionier gelten wollte, und sollte im Jahr 2002 errichtet werden. Der erste Anlauf war jedoch ein Misserfolg, da das beteiligte Bauunternehmen im Verlauf der Planungsphase Konkurs ging. Im Jahr 2004 hat die Errichtung des Windparks jedoch seine erste Etappe erreicht, da ein neuer Partner gefunden werden konnte und man nach wie vor vom ausgesuchten Standort überzeugt war. In den Jahren 2010 und 2012 wurde der Windpark in seinen zweiten und dritten Etappen fertiggestellt. Gründe für den passenden Standort waren nebst den optimalen Windverhältnissen, dass Zugangsstrassen und der Anschluss zum Stromnetz an diesem Standort bereits gegeben waren und auch ein Windstärkemesser von MeteoSchweiz vor Ort installiert war. Der Dialog zu Umweltverbänden wurde schon in frühen Phasen des Projekts gesucht, mit denen eine schnelle Einigung gefunden werden konnte, da die Umwelt am Standort

durch Strassen und Skianlagen „vorbelastet“ war, so dass lediglich noch Abklärungen mit der Vogelwarte gemacht werden mussten. (Kanton Uri, 2023a; Russi, 2019)

Der Kredit für die Finanzierung des Projekts musste vom Volk bewilligt werden und wurde gemäss Markus Russi, damaliger Geschäftsführer des EW Ursern, „mit Applaus“ einstimmig angenommen, da der Glaube an die Windenergie gross war. Weil das Elektrizitätswerk Ursern finanziell unabhängig bleiben wollte und dies zu dessen Leitbild gehört, wurde eine Beteiligung anderer ausgeschlossen und so konnte arrangiert werden, dass das EW nicht auf finanzielle Unterstützung angewiesen war. (Russi, 2019)

Laut dem kantonalen Richtplan des Kantons Uri (2023) ist vorgesehen, dass der Windpark Gütsch durch drei zusätzliche Windkraftanlagen ausgebaut wird und drei bestehende Anlagen erneuert werden. Daraus ergibt sich mit einer installierten Leistung von 14.7 MW eine erwartete mittlere Jahresproduktion von 23 GWh, was eine beträchtliche Erhöhung gegenüber den heutigen 5.5 GWh darstellt. Des Weiteren wird bereits über das Potential diskutiert, den Windpark in ein zusätzliches Gebiet mit sieben Anlagen zu erweitern.

2.5.2 Tourismusregion Laax und das Greenstyle Konzept

2.5.2.1 Tourismusangebot Laax

Die Tourismusregion Flims Laax Falera (Weisse Arena Gruppe, o.D.) befindet sich im Kanton Graubünden und präsentiert sich als dynamisches Ferienziel, das sich dafür begeistert, Outdoor-Aktivitäten zu bewerben. Das Weltkulturerbe Sardona, die Rheinschlucht und der Caumasee sind wohl die bekanntesten Landschaftsattraktionen, die die Region zu bieten hat. Dazu zählen des Weiteren auch das Martinsloch, das Hochtal Bargis und die Gletschermühlen Alp Mora.

Die Region bietet ein Ski- und Snowboardgebiet mit fünf Snowparks an, in welchen verschiedene Wettbewerbe durchgeführt werden, die nebst den Teilnehmenden auch viele Zuschauende anziehen. Sogar der Tiefschnee wird als ein sicheres Winterabenteuer beworben. Eine „Free-style Academy“ steht den AnfängerInnen zu Verfügung, um Tricks für die Snowparks und den Tiefschnee zu lernen. An einigen Tagen im Jahr ist es möglich, in der Nacht Ski oder Snowboard zu fahren und es besteht auch die Möglichkeit, eine geführte Vollmondabfahrt zu machen. Im Winter bietet sich die Region aber auch für Schneeschuh- und Winterwanderungen, Schlitten und Ski- und Snowboardtouren an. (Weisse Arena Gruppe, o.D.)

Das Sommerangebot fokussiert sich vorwiegend auf die Aktivitäten Wandern und Biken. Die Region bietet unterschiedliche Arten von Wanderungen, ob etwas gemütlicher, familiengerechter oder abenteuerlichere Berg- und Talwanderungen. Zudem gibt es zehn Themenwanderungen, auf denen es viel zu lernen gibt – beispielsweise zur Alpenbildung, zu Pflanzen, über die rätoromanische Sprache oder regionale Spezialitäten auf dem Kulinariktrail. Diese können auf Wunsch auch durch eine Führung erlebt werden, beispielsweise mit einem Wildhüter, einer

Orchideenkennerin, einem Geoguide oder einer Kräuterexpertin. Es ist auch möglich, Wanderwegweiser zu fotografieren und anhand einer App verschiedene Daten, wie Höhenmeter, Wanderzeit oder Routenkarte online abzurufen. Zwei Bergsportschulen bieten zudem geführte Bergtouren an. Zum Biken bieten sich zahlreiche Biketrails an und das Unterstützungsangebot für AnfängerInnen sowie die Verfügbarkeit von Mietstationen sind breit. Zur Entspannung gibt es vor allem im Sommer auch ein Yoga-, Meditations- und Wellnessangebot. (Weisse Arena Gruppe, o.D.)

2.5.2.2 Tourismusentwicklung und Nachhaltigkeitskonzept Greenstyle LAAX

Seit den 2000er-Jahren setzt sich die Marke LAAX als Kennzeichen für den Winter- und insbesondere Snowboardtourismus in der Ferienregion Flims Laax Falera durch, da seither vermehrt in diese Wintersportart investiert wurde. Bereits im Jahr 2001 fanden die ersten internationalen Snowboardwettbewerbe in Laax statt. In diesem Jahr wurde auch das erste Design-Hotel in den Alpen eröffnet, das Riders Palace, welches auch durch seine Event-Location ein „junges, freestyle-begeistertes Publikum“ anzieht. Mit dem Bau des rockresort an der Talstation in Laax ab 2007 beginnt die Umgebung zu wachsen und Laax in ihrer zielgruppenfokussierten Vision aufzublühen. Seit 2016 findet auch das LAAX OPEN, das grösste europäische Snowboardevent, dort statt. In der Wintersaison 2014-2015 wurde auf dem Crap Sogn Gion die grösste und längste Halfpipe der Welt errichtet. (Weisse Arena Gruppe, 2023)

Für Sommeraktivitäten wurde für einige Jahre mit der Marke „Flims“ geworben, jedoch werden seit 2019 alle Angebote und Produkte ganzjährig unter beiden Marken („Flims Laax“) vermarktet. Besonders der Caumasee, aber auch der Crestasee, wurde dank seiner türkisblauen Farbe aufgrund eines „Instagram-Hypes“ vor allem im Sommer regelrecht überrannt, sodass die Gemeinde Flims die Eintrittszahl limitieren musste. 2019 wurde der GALAAXY Park auf dem Crap Sogn Gion eröffnet, welcher auch im Sommer Freestyle-Aktivitäten anbietet, darunter beispielsweise einen Skatepark. (Weisse Arena Gruppe, 2023)

Im Jahr 2016 wurde die Greenstyle Foundation gegründet, deren Namen bereits durch das im Jahr 2010 gestartete Konzept aufkam, mit dem Ziel, die Natur, die Tiere und das Klima kontrolliert zu schützen. Sie ist eine non-profit Organisation, die mit dem „Greenstyle“ Konzept einen Sechs-Punkte-Plan aufgestellt hat, um eine „nachhaltige Tourismusregion“ zu etablieren. Dieser beinhaltet vor allem die Steigerung der Energieeffizienz, die Dekarbonisierung von Gebäuden und die Elektrifizierung der Mobilität. Nebst Grosskraftwerken der Wasser-, Solar- und Windkraft sollen auch Solaranlagen auf Dächern und Fassaden installiert und schliesslich anhand von Speicherseen, Wasserstoff und Batterien die Energiespeicherung gesichert werden. Das Konzept sollte der Region erlauben, den gesamten Energiebedarf zu hundert Prozent mit klimafreundlichen und regionalen Quellen zu decken und dabei gänzlich auf fossile Energien zu verzichten. (Weisse Arena Gruppe, 2022)

2.5.2.3 Alpines Windparkprojekt

Vor rund zehn Jahren wurde die Idee ins Leben gerufen, auf dem Vorab bei Laax einen Windpark zu errichten und dafür bereits die ersten Windmessungen vorzunehmen. Auf dem Crap Sogn Gion konnte nicht genügend Windvorkommen gemessen werden und laut Herrn Reto Fry, Umweltbeauftragter der Weissen Arena Gruppe, haben sich auch Umweltschutz- und Landschaftsschutzorganisationen dafür ausgesprochen, einen Windpark eher auf dem Vorab zu bauen, da dieser vom Tal aus nicht ersichtlich wäre. Die Projektleitung wurde damals von einem schweizerischen Energiegeneralunternehmen übernommen und von der Greenstyle Foundation im Sinne ihrer Vision, sich als Feriendestination „vom Konsumenten zum Produzenten“ zu entwickeln, unterstützt. Dazu haben sie auch Befragungen der BesucherInnen in der Ferienregion Laax unternommen, wobei sich etwa 80% der Befragten positiv zu einem möglichen Windpark auf dem Vorab und lediglich 20% negativ geäußert oder enthalten hatten. Positive Rückmeldungen haben laut Herrn Fry unter anderem beinhaltet, dass die BesucherInnen bewusst sehen, dass die konsumierte, notwendige Energie vor Ort produziert wird, was insofern auch „ihrem Gewissen guttut“. Aufgrund eines Führungswechsels beim Energiedienstleistungsunternehmen zog sich dieses aus einer Vielzahl von Windparkprojekten zurück und begründete dies damit, dass die finanzielle Rentabilität nicht gegeben sei. Dazu gehörte unter anderem auch, dass der zurückzulegende Weg für die Installation des Windparks stark hätte ausgebaut werden müssen und dass dieser Weg bei einem Defekt an der Anlage im Winter zurückzulegen „chancenlos“ gewesen wäre. Damit verbundene Einbussen in der Stromproduktion während des Defekts hatten in dieser Rechnung ebenso Gewicht. Die Herausforderungen auf dem Vorab betreffend Kosten und Nutzen, extremen Wetterbedingungen, Vereisungen und Bedenken um die Sicherheit der SchneesportlerInnen unterhalb der Windturbinen haben diesen Entscheid weitergehend geleitet. (R. Fry, persönliche Kommunikation, 26. September 2023)

Das Greenstyle Konzept unter Herrn Reto Fry schlägt nun vor, unter anderem auch wegen des Eingriffs ins Landschaftsbild, Kleinwindkraftanlagen auf dem Vorab zu errichten, für deren Transport und Unterhalt der Aufwand nicht im selben Ausmass anfallen würde und sich diese auch durch ihre Erscheinung etwas besser ins Landschaftsbild einfügen würden. Aktuell ist auf dem Vorab ein alpiner Solarpark geplant, der in seiner Projektierung stetig voranschreitet. Laut Herrn Reto Fry würde sich der Standort sowohl für den Solarpark als auch für Kleinwindkraftanlagen gut eignen, da die Umgebung auf dem Vorab durch Hochspannungsleitungen visuell und strukturell „vorbelastet“ ist, die Biodiversität nicht besonders hoch ist und bereits heute gut erschlossen ist. Sollte der geplante alpine Solarpark umgesetzt werden, würde es das aktuell installierte Stromnetz nicht erlauben, zusätzlich einen Windpark in der Dimension, die vor zehn Jahren geplant war, zu errichten. Es wäre ein beträchtlicher Ausbau des Stromnetzes erforderlich, was besonderer finanzieller Ressourcen bedürfte. (R. Fry, persönliche Kommunikation, 26. September 2023)

3 Theoretischer Rahmen

Die aktuelle Situation um die Windenergie in der Schweiz und Europa zeigt sich fortschrittlich, da der Bedarf am Ausbau erneuerbarer Energien bedingt durch den Klimawandel und gemäss der Energiestrategie 2050 in der Schweiz, sowie aufgrund ihres Potenzials als Ergänzung in der Winterstromproduktion, steigt. Das Ausbaupotential möchte in der Schweiz, unter anderem auch in den Alpen und darin gelegenen Tourismusregionen, genutzt werden. Durch das Zusammenspiel von Tourismus und Windkraftanlagen ergeben sich potenzielle Interessenskonflikte aber auch unterschiedliche Synergiepotentiale.

Die Ferienregion Andermatt bildet mit dem Windpark Gütsch für die Schweiz ein einzigartiges Beispiel für einen in einem alpinen Tourismusgebiet gelegenen Windpark, wobei bereits vor dessen Errichtung Tourismusbetrieb in dieser Region stattgefunden hat. Mit ihrem breiten Tourismusangebot und der Entwicklung in den vergangenen Jahren ist die Region aus Tourismussicht eine erfolgreiche Reisedestination.

Die Nachfrage nach der Errichtung von Windkraftanlagen im Kanton Graubünden und spezifisch in Tourismusgebieten zeigt sich in den Richtplanverfahren des Kantons. Die Destination Laax ist mit ihrem breiten Tourismusangebot sowie dem Greenstyle Laax Konzept ein attraktives und fortschrittliches Tourismusgebiet und erweist sich als ein passendes Untersuchungsgebiet für die vorliegende Studie.

Verschiedene Studien erwähnen (u.a. Aschenbrand & Grebe, 2018; Bidwell, 2023, Spiess et al., 2015), dass der Einfluss von Windkraftanlagen auf das Landschaftsbild ein besonders wichtiger Faktor für deren Akzeptanz von TouristInnen darstellt. Es wird teilweise von einem „Verlust der Landschaftsqualität“, „sichtbarer Modernisierung“ oder „nicht authentischer Landschaft“ gesprochen. Dies bezieht sich vorwiegend auf eine gewisse Nähe der Anlage zu einem Wohn- oder Erholungsgebiet, obwohl die grundsätzliche Befürwortung von erneuerbaren Energien hoch sein mag (NIMBY-Effekt). Manchmal ist aber auch von einem „Teil eines umfassenderen Naturverständnisses“ die Rede. Befindet sich eine Windkraftanlage in der Nähe von bestehender Infrastruktur, mildert sich die Abneigung gegenüber der Anlage (Hunziker et al., 2014) und dies ist vor allem auch durch die Sinnhaftigkeit eines Standorts beeinflusst (Bidwell, 2013). Daraus lässt sich für die vorliegende Studie folgende erste Teilhypothese ableiten:

1a. Die Nähe eines Windparks zu bestehender, touristischer Infrastruktur wird von TouristInnen (beider Orte) als sinnvoll empfunden.

Eine Unterscheidung der Wahrnehmung von Energiekraftanlagen zeigt sich unter anderem auch in der Betrachtung einer bestehenden im Vergleich zu einer sich vorzustellenden Anlage. Wird eine bestehende Anlage betrachtet, wird diese als „weniger störend“ betrachtet als die

reine Vorstellung einer Anlage, die neu erbaut werden soll (Bidwell, 2023; Seitz, 2011). Folglich leitet sich daraus folgende zweite Teilhypothese ab:

1b. Jeweilige TouristInnen der Tourismusregionen Andermatt und Laax empfinden das Landschaftsbild durch den bereits existierenden Windpark Gütsch weniger gestört als durch einen potentiellen Windpark in Laax.

Spezifisch TouristInnen in einem alpinen Tourismusort sehen sich bei Sommeraktivitäten, wie Wandern oder Biken, eher durch eine Windkraftanlage gestört als bei Winteraktivitäten, wie Ski- und Snowboardfahren oder Schlitteln (Frantál & Kunc, 2011; Tverijonaite et al., 2022; Weiss, 2015). Dementsprechend geht eine weitere, dritte Teilhypothese wie folgt hervor:

1c. SommertouristInnen fühlen sich bei ihrer Tourismusaktivität eher durch den Windpark Gütsch respektive einen potentiellen Windpark in Laax gestört als WintertouristInnen.

Des Weiteren steigt die Akzeptanz von Energiekraftanlagen, wenn die Sinnhaftigkeit derjenigen für eine Person hoch ist (Bidwell, 2013; Gardt & Litmeyer, 2017). Durch die Aufwertung des „Ortsimages“ durch eine Windkraftanlage im Zeichen von Nachhaltigkeit und Fortschrittlichkeit kann die Sinnhaftigkeit erhöht werden (Hunziker et al., 2014; Spiess et al., 2014). Daraus ergibt sich für die vorliegende Studie folgende vierte Teilhypothese:

1d. TouristInnen der beiden gewählten Tourismusorte Andermatt und Laax, denen Nachhaltigkeit und Fortschrittlichkeit wichtig ist, empfinden das Landschaftsbild durch eine Windkraftanlage weniger gestört als solche, denen diese Aspekte von geringerer Wichtigkeit erscheinen.

Auch demografische Merkmale, die Regelmässigkeit von Besuchen – damit sind spezifisch Stammgäste im Vergleich mit gelegentlichen BesucherInnen gemeint – und die grundsätzliche Haltung gegenüber erneuerbaren Energien stellen Faktoren dar, die die Akzeptanz einer Windkraftanlage beeinflussen können.

Beispiele aus Tschechien, Deutschland oder Island haben gezeigt, dass durch sogenannten „Windtourismus“ das bestehende Tourismusangebot ergänzt werden kann. Dies kann beispielsweise anhand einer Führung durch das Energiekraftwerk oder eines damit verbundenen Events geschehen, wodurch schliesslich auch die Akzeptanz solcher Anlagen im Allgemeinen erhöht werden kann (Frantál & Urbánková, 2017; (Beer et al., 2018). Dies unterstützt die zuvor erwähnte Erkenntnis, dass Personen eine positivere Haltung gegenüber solchen erneuerbaren Energieanlagen aufweisen, die bereits bestehen. Folglich lässt sich für die vorliegende Studie eine weitere Teilhypothese formulieren:

2a. *TouristInnen, die den Windpark Gütsch (oder einen anderen Windpark) im Rahmen einer Führung kennengelernt haben, fühlen sich in ihrer Tourismusaktivität weniger durch den Windpark Gütsch oder einen potentiellen Windpark in Laax gestört als solche, die diesem nicht im Rahmen einer spezifischen Veranstaltung begegnet sind.*

Durch „Windtourismus“ wird auf direktem und indirektem Weg wirtschaftliche Wertschöpfung generiert. Eine Windkraftanlage kann zwar auch dafür verantwortlich sein, dass die Besucherzahlen eines Tourismusorts sinken können (Gardt et al., 2018; Broekel & Alfken, 2015) und die Ausgestaltung eines zugehörigen Tourismusangebots (z.B. Führungen, Informationszentren) nimmt finanzielle und personelle Ressourcen in Anspruch (Beer et al., 2018). In gemeinsamer Betrachtung mit der bisherigen Entwicklung der Tourismusregion Andermatt lässt sich für die vorliegende Studie folgende, letzte Teilhypothese ableiten:

2b. *Andermatt erzeugt mit der Anwesenheit des Windparks Gütsch und dem damit verbundenen „Windtourismus“ (zusätzliche) positive regionale Wertschöpfung für die Tourismusregion.*

Abbildung 6 stellt nachfolgend die Relationen bildlich dar, welche den formulierten Hypothesen zugrunde liegen.

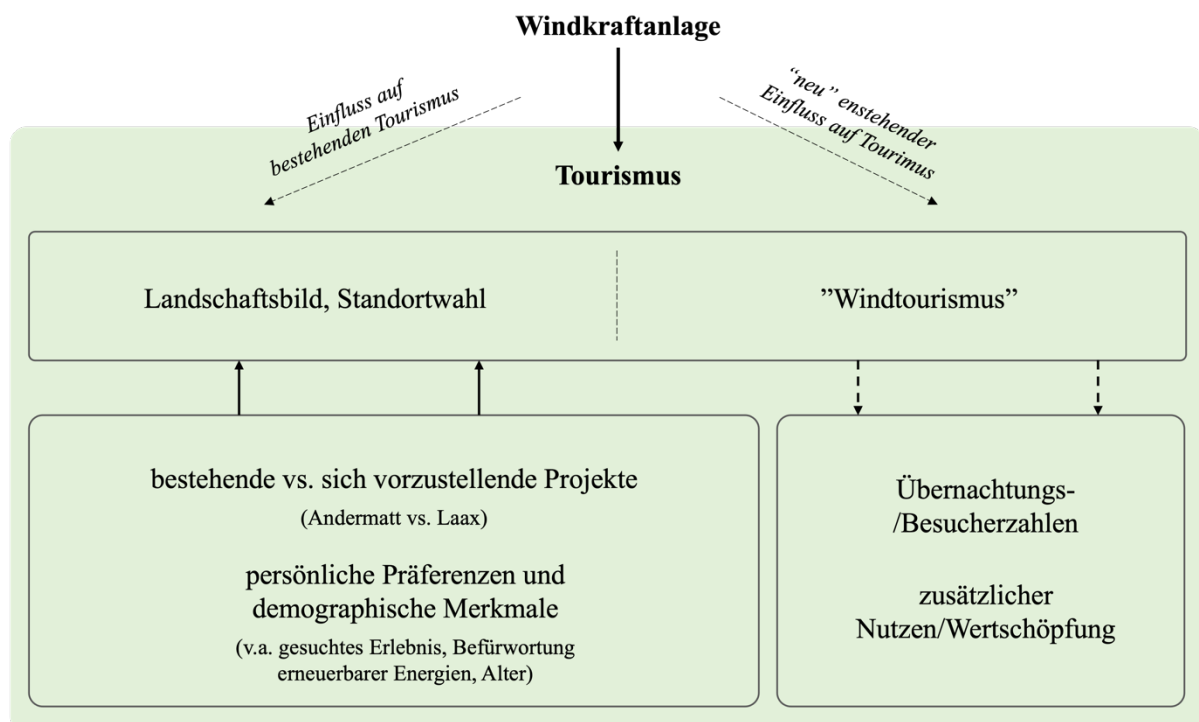


Abbildung 7: Theoretische Grundlagen und Beziehungen (eigene Darstellung)

4 Methodik und Daten

4.1 Forschungsdesign

4.1.1 Vorgehen und Gestaltung

Bisher wurden unterschiedliche Ansätze verwendet, um den Einfluss eines Windparks oder anderer Infrastrukturen erneuerbarer Energien auf den Tourismus zu messen. Einige Studien arbeiten mit quantitativen Mitteln, wozu sie Umfragen mit betroffenen Personen, wie TouristInnen oder TourismusbetreiberInnen, durchführten (z.B. Bidwell, 2023; Bidwell, 2013; Frantál & Kunc, 2011; Gardt & Litmeyer, 2017; Hunziker et al., 2014). Eine andere Möglichkeit stellt die Auswertung von statistischen Kennzahlen, wie beispielsweise die Übernachtungszahlen eines Tourismusorts oder die Besucherzahl eines Kraftwerks (z.B. Broekel & Alfken, 2015; Gardt et al., 2018), dar. Weitere Mittel sind qualitativer Natur, wozu unterschiedliche Stakeholder in 1:1-Interviews oder in Fokusgruppen für eine meist offenere Diskussion befragt werden (z.B. Spiess et al., 2014; Tverijonaite et al., 2022).

Für die vorliegende Studie wurden zur Untersuchung der breitgefächerten Hypothesen unter Kapitel 3 alle drei erwähnten Methoden zur Datensammlung in der einen oder anderen Ausführung beigezogen.

4.1.1.1 Quantitative Umfrage

Um die direkte Wahrnehmung der BesucherInnen eines Tourismusorts zu untersuchen, wurden während der Monate September und Oktober 2023 zwei unterschiedliche Online-Umfragebögen für die jeweiligen Tourismusgebiete Laax und Andermatt verbreitet. Sie wurden einerseits an TouristInnen vor Ort anhand von QR-Codes abgegeben, während sie beispielsweise am Bahnhof, bei einer Bergbahn oder an einer Bushaltestelle warteten. Andererseits wurden sie beim Wandern oder im Dorf direkt angesprochen. Die QR-Codes wurden auf einem kleinen „Flugpapier“ ausgehändigt, das zudem den Titel der Umfrage sowie die Kontaktdaten der Autorin beinhalteten. Zudem wurden einzelne Personen aufgefordert, die Umfrage an einem iPad direkt auszufüllen. Des Weiteren wurde die Umfrage im direkten und indirekten Umfeld der Autorin gestreut, da die Rücklaufquote der vor Ort angesprochenen Personen bedauerlicherweise eher tief war. Die Links zu den beiden Umfragen wurden in mehreren WhatsApp-Gruppen verbreitet sowie auf LinkedIn und Instagram unter den privaten Profilen der Autorin veröffentlicht. Alle ausgewerteten Antworten der Umfrage wurden auf freiwilliger Basis von den Teilnehmenden erfasst und bestätigt.

Eingangs des Fragebogens wurden die Befragten nach ihrem Einverständnis zur Teilnahme gefragt. Bei beiden Fragetypen wurde anschliessend die Regelmässigkeit der Besuche abgefragt, wobei angegeben werden konnte, ob die Person bisher zum ersten Mal oder in einer gewissen Regelmässigkeit in Andermatt oder Laax war. Daraufhin wurde auch der Grund bzw. die Gründe für den Besuch am jeweiligen Ort erfragt.

In einem nächsten Schritt (Block A: Einstiegsfragen) wurde die Person anhand von drei Fragen zu ihrer Haltung zur Energiewende und zum Klimaschutz befragt. Diese Fragen wurden teilweise an solche angelehnt, die jährlich durch die Studie der Universität St. Gallen „Kundenbarometer erneuerbare Energien“ (Côté et al., 2022) abgefragt werden, sodass die Antworten in ihrer Repräsentativität mit den Ergebnissen der Studie unterstützt werden konnten. Darauf folgten drei weitere Fragen, welche die Teilnehmenden nach der Wichtigkeit von Nachhaltigkeit und Umweltfreundlichkeit bei der Wahl einer Reisedestination, nach dem Interesse an Windkraft und ihrer Technologie sowie dazu, ob es für die Person sinnvoll ist, einen Windpark in Nähe bestehender Infrastruktur zu platzieren befragt wurden.

Im nächsten Block (Block B: Fragen zur Wahrnehmung) zur individuellen Wahrnehmung wurden den Befragten zur Reisedestination Andermatt zwei Bilder gezeigt, auf denen der Windpark Gütsch einmal im Sommer und einmal im Winter abgebildet ist. Die Bilder sollten den Befragten als Gedankenstützen dienen, um die nachfolgenden Fragen zu beantworten. Äquivalent dazu wurden auch in der Umfrage zum Tourismusort Laax zwei Bilder gezeigt, die das Gebiet sowohl im Sommer als auch im Winter zeigten. Jedoch wurden auf diesen Bildern jeweils drei Windturbinen künstlich ins Bild eingefügt, so dass sich ein potenzieller Windpark in Laax vorstellen lässt. Zur Wahrnehmung des Landschaftsbilds und des Einflusses des Windparks auf die jeweilige Tourismusaktivität wurde eine direkte Frage mit Beantwortung anhand einer Skala gestellt. Zudem wurde erfragt, ob der Windpark Gütsch bzw. ein potenzieller Windpark in Laax als Zeichen der Nachhaltigkeit und Fortschrittlichkeit angesehen wird und wie die befragte Person diesem generell gegenübersteht. Auch wurde erfragt, ob der jeweilige Standort als sinnvoll für die Erzeugung von Strom erachtet wird. Sofern die Frage eher negativ beantwortet wurde, wurde zusätzlich nach einem sinnvolleren Standort gefragt. Dieser Block wurde mit einer Frage dazu abgeschlossen, wo sich die nächstgelegene Windturbine zum Wohnort der befragten Person befindet.

Der darauffolgende Block (Block C: Fragen zum Windtourismus) beinhaltete Fragen zum „Windtourismus“. Dazu wurde bei beiden Fragetypen erfragt, ob die teilnehmende Person bereits einmal an einer Führung durch den Windpark Gütsch bzw. eines anderen Windparks teilgenommen hat. Falls die Frage mit Ja beantwortet wurde, wurde auch erfragt, in welchem Rahmen (z.B. einer beruflichen oder schulischen Veranstaltung) daran teilgenommen wurde. Anschliessend wurde eine Frage dazu gestellt, welche Attraktionen oder Angebote die befragte Person ermuntern würden, an einer Führung teilzunehmen. Dazu wurden folgende Auswahlmöglichkeiten gelistet:

-
- a. Flugshow von (Solar-)Flugzeugen oder Drohnen um die Windräder
 - b. Flugshow von Fallschirmspringern oder Wind-Drachen um die Windräder
 - c. Eine Aussichtsplattform
 - d. Ein Foxtrail von Windrad zu Windrad oder andere spielerische Ansätze, um die Windkraft näher kennenzulernen
 - e. Ein Kombi-Angebot von Transportmitteln (Bergbahn oder SBB) und einer Führung durch den Windpark
 - f. Eine künstlerische Darstellung (z.B. eine Lichtshow) auf den Windrädern
 - g. Eine spannende Führung durch die Windkraftanlagen, um dabei mehr über die Technologie zu erfahren, würde mich bereits überzeugen
 - h. Ein Musikfestival beim Windpark
 - i. Wenn ich auf eines dieser Angebote auf Social Media aufmerksam gemacht würde oder eines dieser Angebote gewinnen könnte
 - j. Ich würde den Windpark niemals besuchen.
 - k. Andere: ...

Der Fragebogen wurde abgeschlossen mit Fragen zum Alter, Geschlecht, dem höchsten Ausbildungsstand und dem Heimatland der befragten Person (Block D: Allgemeine Fragen). Wurde bei Heimatland „Schweiz“ angewählt, wurde zusätzlich der entsprechende Heimatkanton erfragt. Als abschliessende Frage wurde gefragt, ob die Umfrageteilnehmenden ein Abonnement für den öffentlichen Verkehr besitzen, was als Indiz für umweltbewusstes Verhalten beigezogen werden kann. Der vollständige Fragebogen ist im Anhang zu finden.

4.1.1.2 Qualitative Interviews

Um die Umfrage ohne Vorbehalt unter den TouristInnen in den beiden Tourismusregionen zu verbreiten, wurden in erster Linie die beiden Tourismusbetriebe Andermatt-Uri-Region Tourismus GmbH und Flims Laax Falera Management AG angesprochen. Dazu wurde direkter Kontakt zu den jeweiligen Tourismusdirektoren Thomas Christen (Andermatt) und André Gisler (Laax) aufgenommen. Beide Kontaktpartner haben sich sehr unterstützend gezeigt und sich zudem für ein schriftliches Interview bereit erklärt.

Die beiden Interviews wurden in schriftlicher Form durchgeführt, um den zeitlichen Aufwand der beiden Interviewpartner möglichst klein zu halten. Die Interviews wurden insbesondere durchgeführt, um die Umfrageantworten durch professionelle Stimmen aus dem Tourismus der beiden Orte zu ergänzen. Die Fragen wurden daher jeweils an den Fragebogen, der im vorangehenden Kapitel beschrieben wird, angelehnt formuliert.

Beide Interviewpartner wurden eingangs nach ihrer persönlichen Haltung zur Windenergie und dem Windpark Gütsch respektive einem potenziellen Windpark in Laax³ gefragt. Anschliessend wurde eine Frage zur Wahrnehmung des Windparks als Zeichen der Klimafreundlichkeit und Fortschrittlichkeit für die jeweilige Tourismusregion gestellt. Auch wurde der empfundene Einfluss des Windparks auf das Landschaftsbild und gewisse Tourismusaktivitäten (vor allem im Vergleich zu Sommer- und Winteraktivitäten) erfragt. Des Weiteren wurde nach dem generellen Einfluss des Windparks auf die Tourismusentwicklung der jeweiligen Region gefragt. Die letzte Frage hat sich schliesslich darauf bezogen, welche Attraktionen oder Aktivitäten sich die beiden Tourismusdirektoren zur Gewinnung zusätzlicher Wertschöpfung in Verbindung mit dem Windpark vorstellen könnten. Der Leitfaden für die beiden Interviews befindet sich im Anhang.

Separat wurde auch ein mündliches Interview mit Herrn Reto Fry geführt. Reto Fry ist Umweltbeauftragter der Weissen Arena Gruppe und zuständig für die Umsetzung der Massnahmen, die unter der Greenstyle Foundation im Rahmen des Greenstyle Konzepts für die Ferienregion Flims Laax Falera erstellt wurden. Die Fragen haben sich darauf bezogen, wie es um die Projektierung und Planung eines Windparks innerhalb des Greenstyle Konzepts steht und welche bisherigen Bedenken Herr Fry als Projektbeauftragter bei der Umsetzung eines Windparks in Laax sieht (z.B. gesellschaftlicher Druck, politische Hindernisse, Einfluss auf Tourismus). Die Antworten wurden besonders zur Beschreibung des Tourismusorts Laax und spezifisch unter Kapitel 2.5.2.3 beigezogen. Seine Aussagen wurden ergänzend auch in den Resultaten zu den Interviews einbezogen.

4.1.1.3 Statistische Kennzahlen

Um die Tourismusentwicklung in Andermatt mit der Anwesenheit des Windparks Gütsch statistisch zu untersuchen, wurde beim Amt für Raumentwicklung des Kantons Uri nach Zahlen und Berichten gefragt. Als Kontaktperson wurde Herr Marco Achermann, Amtsvorsteher / Kantonsplaner im Amt für Raumentwicklung der Justizdirektion Kanton Uri, dafür angefragt. Er hat freundlicherweise auf verschiedene Statistikportale hingewiesen. Konkret wurden folgende Daten und Statistiken beigezogen:

- Angebot und Nachfrage der geöffneten Betriebe nach Gemeinde (Bundesamt für Statistik, 2023b) und Hotels und Kurbetriebe: Gäste nach Inland-Ausland und nach Gemeinde (Bundesamt für Statistik, 2023c) zwischen 2005 und 2022, spezifisch die Kennzahlen Logiernächte (in Andermatt und gesamtschweizerisch)

³ nachfolgend einfachheitshalber in beiden Fällen „der Windpark“ genannt

-
- Beschäftigte und Vollzeitäquivalente nach Geschlecht und Branche im Kanton Uri zwischen 2011 bis 2022 (Kanton Uri, 2023b), spezifisch die Angaben für Beschäftigte im Tourismus (Beherbergung und Gastgewerbe, so wie diese Kennzahl auch im Controllingbericht zum kantonalen Richtplan Uri (2023) betrachtet wird)

Um die Nachfrageentwicklung einer Tourismusregion zu untersuchen und darzustellen wird in häufigen Fällen auf die Kennzahl „Logiernächte“ zurückgegriffen, da sie als Indikator für die wirtschaftliche Leistungsfähigkeit der Tourismusbranche in einer bestimmten Region gilt. Logiernächte beschreiben die Anzahl der verbrachten Nächte durch die Gäste in einem bestimmten Hotel, Kurbetrieb bzw. Parahotellerie (Bundesamt für Statistik, 2023c). Diese wurden für die Jahre 2005 bis 2022 aus den oben vermerkten Berichten herausgenommen und tabellarisch sowie graphisch dargestellt. Die Tabelle dazu befindet sich im Anhang.

Zum Beitrag des Tourismus an die regionale Wertschöpfung zählt auch die Entwicklung der Beschäftigten. Dies wurde anhand der Betrachtung von Vollzeitäquivalenten (VZÄ) vorgenommen, da sich diese Kennzahl besonders gut eignet, die absolute Entwicklung und die zu befriedigende Nachfrage darzustellen. Sie sind ein Mass für das Total der Arbeitspensen der Leistungserbringer. Sie werden in Stellenprozenten gemessen, wobei eine Vollzeitstelle einem Vollzeitäquivalent entspricht. Dazu wurden im oben erwähnten Statistikportal nach den Beschäftigten nach Branche für die Jahre 2011 bis 2021 gesucht und tabellarisch und graphisch dargestellt. Für frühere Jahre und das Jahr 2022 stehen keine Daten zu Verfügung. Auch diese Tabelle befindet sich im Anhang. Es wird dabei der gesamte Kanton Uri betrachtet, da keine Auswertung zu den einzelnen Gemeinden zur Verfügung steht. Eine Unterteilung gemäss Regionen dank veröffentlichten Zahlen in den Controllingberichten (Kanton Uri, 2017; Kanton Uri, 2023c) konnte angenähert werden.

Letztlich wurden zur Untersuchung der Wertschöpfung durch den Windpark Gütsch Zahlen zur Führung „Energie & Umwelt“ der Andermatt-Urserntal Tourismuss AG in Zusammenarbeit mit dem EW Ursern untersucht.

4.2 Variablen und Messung

4.2.1 Untersuchte Variablen

Um die unter Kapitel 3 formulierten Hypothesen zu testen, wurden folgende abhängige und unabhängige Variablen festgestellt:

- **Teilhypothese 1a:**
 - *Abhängige Variable:* Die Wahrnehmung der Nähe des Windparks zu bestehender, touristischer Infrastruktur (als sinnvoll oder nicht sinnvoll betrachtet)
 - *Unabhängige Variable:* Nähe der Windparks zu bestehender touristischer Infrastruktur (z.B. Windpark Gütsch und potenzieller Windpark in Laax)

-
- **Teilhypothese 1b:**
 - *Abhängige Variable:* Die Wahrnehmung des Landschaftsbilds (als gestört oder bereichert empfunden)
 - *Unabhängige Variable:* Anwesenheit der Windkraftanlage (existierender Windpark Gütsch vs. potenzieller Windpark in Laax)
 - **Teilhypothese 1c:**
 - *Abhängige Variable:* Mögliche Beeinträchtigung der Tourismusaktivitäten durch einen Windpark (als gestört oder bereichert empfunden)
 - *Unabhängige Variable:* Jahreszeit (Sommer vs. Winter) und Aktivitäten (z.B. Wandern vs. Ski- und Snowboardfahren).
 - **Teilhypothese 1d:**
 - *Abhängige Variable:* Die Wahrnehmung eines Windparks als Zeichen der Nachhaltigkeit und Fortschrittlichkeit
 - *Unabhängige Variable:* Einstellung der TouristInnen zur Bedeutung von Nachhaltigkeit und Fortschrittlichkeit
 - **Teilhypothese 2a:**
 - *Abhängige Variable:* Die Beeinträchtigung der Tourismusaktivitäten durch einen Windpark (als gestört oder bereichert empfunden)
 - *Unabhängige Variable:* Teilnahme an Führungen oder spezifischen Veranstaltungen im Zusammenhang mit dem Windpark Gütsch (oder anderen Windparks)
 - **Teilhypothese 2b:**
 - *Abhängige Variable:* Die generierte regionale Wertschöpfung durch den Windpark in Andermatt (positiv oder negativ)
 - *Unabhängige Variable:* Vorhandensein des Windparks Gütsch und die Auswirkungen auf die regionale Wertschöpfung in Andermatt

Zudem wurden folgende weitere, beschreibende Variablen untersucht:

- Persönliche Haltung zur Energiewende generell, zum Klimaschutz in der Schweiz und die persönliche Wichtigkeit von Nachhaltigkeit und Fortschrittlichkeit eines gewählten Tourismusorts
- Persönliche Haltung zur Windenergie, das Interesse an Windenergie und deren Technologie
- Regelmässigkeit des Besuchs am jeweiligen Ort

-
- Gesuchte Tourismusaktivität (z.B. Skifahren, Wandern) als auch in Zusammenhang mit einem Windpark
 - Alter, Geschlecht, höchst abgeschlossene Ausbildung und Heimatland (bei SchweizerInnen auch der Kanton)

Die Prüfung dieser Variablen wird nun im anschliessenden Kapitel ausführlich erläutert.

4.2.2 Datenanalyse und Messung

Die drei gewählten empirischen Ansätze einer quantitativen Umfrage sowie qualitativen Interviews und die Analyse von Statistiken und Kennzahlen wurden alle, jedoch teilweise unterschiedlich, für die Prüfung der Hypothesen verwendet.

Um die Wahrnehmung der Nähe eines Windparks zu bestehender, touristischer Infrastruktur zu untersuchen (Teilhypothese 1a), wurde in beiden Umfragebögen zu Laax und Andermatt jeweils eine direkte Frage gestellt, wobei die Aussage „Ich finde einen Windpark in der Nähe von bestehender Infrastruktur zu platzieren sinnvoll“ anhand einer fünfstufigen Likert-Skala (Zustimmung) bewertet werden konnte. Im zweiten Block (B) zur Wahrnehmung wurde am konkreten Beispiel des Windpark Gütsch bzw. am konkreten Beispiel eines potenziellen Windparks in Laax erneut danach gefragt, ob der Standort als sinnvoll betrachtet wird, wobei dies anhand einer fünfstufigen, bipolaren Likert-Skala („überhaupt nicht sinnvoll“ bis „äusserst sinnvoll“) bewertet werden konnte.

Die Variable zur Wahrnehmung des Landschaftsbilds wurde ebenfalls anhand von Fragen in den beiden Umfragebögen sowie auch in den Interviews mit den beiden Tourismusdirektoren untersucht. In beiden Umfragebögen wurde am konkreten Beispiel des Windpark Gütsch bzw. am konkreten Beispiel eines potenziellen Windparks in Laax erfragt, inwiefern das Landschaftsbild durch den Windpark als gestört bzw. bereichert empfunden wird bzw. würde, was anhand einer fünfstufigen, bipolaren Likert-Skala bewertet werden konnte (von „stark gestört“ bis „sehr bereichert“). Durch den Vergleich der Antworten der beiden Umfragebögen auf diese Frage kann so auch die Variable untersucht werden, die den Unterschied zwischen einem bereits existierenden und einem sich vorzustellenden Windpark ausmacht (Teilhypothese 1b). Zudem wurden auch die beiden Tourismusdirektoren gefragt, wie sie den Einfluss des/eines Windparks auf das Landschaftsbild beurteilen.

Auch die Variable zu einer möglichen Beeinträchtigung einer Tourismusaktivität durch einen Windpark wurde vorwiegend durch die Umfrage, aber auch durch die qualitativen Interviews untersucht. Dazu wurde in beiden Umfragebögen, analog zur Frage zum Landschaftsbild, nach

dem Empfinden von „Störung“ bzw. „Bereicherung“ des/eines Windparks bei der jeweiligen Tourismusaktivität anhand einer fünfstufigen, bipolaren Likert-Skala gefragt (von „stark gestört“ bis „sehr bereichert“). Zudem wurden die befragten Personen eingangs der beiden Umfragen gefragt, für welche Tourismusaktivität sie in den jeweiligen Tourismusort gereist sind, was erlaubt, diese Variable auch auf den Unterschied zwischen Sommer- und Winteraktivitäten zu untersuchen (Teilhypothese 1c). Wie erwähnt wurden auch die beiden Tourismusdirektoren in den Interviews nach ihrer Wahrnehmung gefragt, wie der/ein Windpark Einfluss auf unterschiedliche Aktivitäten nimmt bzw. nehmen könnte (Sommer vs. Winter).

In Zusammenhang mit einer Frage unter Block C der beiden Umfragen, wobei gefragt wurde, ob die befragte Person bereits einmal an einer Führung durch den Windpark Gütsch bzw. irgendeines Windparks teilgenommen hat, wurde auch untersucht, ob sich die Person bei ihrer Tourismusaktivität durch den/einen Windpark eher gestört bzw. bereichert fühlt (Teilhypothese 2a).

Direkt wurde auch nach der Wahrnehmung des/eines Windparks als Zeichen der Nachhaltigkeit und Fortschrittlichkeit (Teilhypothese 1d) in beiden Umfragebögen sowie den beiden qualitativen Interviews mit den Tourismusdirektoren gefragt. Dazu konnte in den Umfragen der Aussage „Ich empfinde den Windpark Gütsch bzw. einen potenziellen Windpark in Laax als ein Zeichen der Nachhaltigkeit und Fortschrittlichkeit des jeweiligen Tourismusorts“ anhand einer fünfstufigen Likert-Skala zugestimmt werden. Diese Frage wurde auch in den Interviews, jedoch als offene Frage, an die beiden Tourismusdirektoren gerichtet. In diesem Zusammenhang wurde vorab auch die Frage unter Block A gestellt, ob die befragte Person generell darauf achtet, ob eine auszuwählende Reisedestination Wert auf Nachhaltigkeit und Fortschrittlichkeit legt. Diese Frage konnte ebenfalls anhand einer fünfstufigen Likert-Skala beantwortet werden.

Schliesslich wurde die Variable der regionalen Wertschöpfung in Andermatt anhand vorliegender Statistiken und Zahlen zu den Logiernächten, den Beschäftigten im Tourismus und der monetären Wertschöpfung durch die Führung „Energie und Umwelt“ untersucht. Zu den Logiernächten (gesamt als auch für In- und Ausland) wurden die Daten für jedes Jahr für die Gemeinde Andermatt zusammengestellt und tabellarisch dargestellt. Diese wurden auch um die Zahlen für die gesamte Schweiz ergänzt. Von beiden Spalten wurde jeweils die prozentuale Veränderung im Vergleich zum Vorjahr für einen besseren Vergleich und zur Eruierung von besonderen Geschehnissen zwischen den Jahren errechnet. Besonderes Augenmerk wurde anschliessend auf Schlüsselereignisse in der Entwicklung des Tourismusresorts der Andermatt Swiss Alps AG und die Etappen der Errichtung des Windparks sowie gesamtschweizerische und globale, wirtschaftliche Entwicklungen gelegt. Damit wurde ausgewertet, wie sich der Tourismus in der Zeit vor und nach der Errichtung des Windparks Gütsch entwickelt hat und welche Wertschöpfung auch explizit dank des Windparks Gütsch (mit Führungen) generiert werden konnte.

5 Resultate

5.1 Quantitative Umfrage

Insgesamt wurden bei der Umfrage für Andermatt 68 Rückmeldungen und bei der Umfrage für Laax 79 Rückmeldungen erfasst. Von den Antworten beim Umfragebogen für Andermatt konnten 54 Antworten (79.41%) verwertet, beim Umfragebogen für Laax konnten 65 Antworten (82.28%) berücksichtigt werden, was einem Total von 119 ausgewerteten Antworten entspricht. Es handelt sich hierbei um eine zielgerichtete Stichprobe.

5.1.1 Deskriptive Statistik

5.1.1.1 Haltung zu Energiewende, Klimaschutz, Nachhaltigkeit und Umweltfreundlichkeit einer Tourismusdestination

Insgesamt gaben 70.59% der Befragten an, dass sie die Geschwindigkeit der Energiewende „als zu langsam“ beurteilen würden. 14.29% der Befragten sehen diese als „genau richtig“, 13.45% als „zu überstürzt“ und zwei Personen haben dazu keine Aussage erfasst. Werden diese Ergebnisse mit den Resultaten des „Kundenbarometers für Erneuerbare Energien“ (Côté et al., 2022) verglichen, zeigt sich, dass die Zusammensetzung der Antworten im Jahr 2022 eine ähnliche Gestalt zeigte und dies einem zunehmenden Trend entspricht, dass seit 2015 eine zunehmende Anzahl an Personen der Meinung sind, dass die Energiewende tendenziell zu langsam vorangeht.

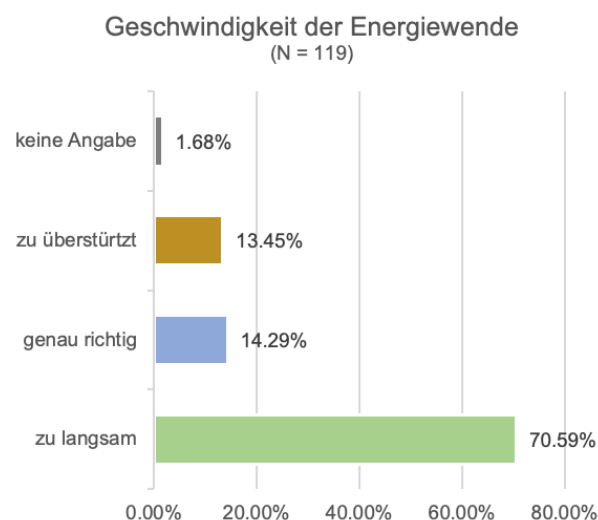


Abbildung 8: Umfrageresultate zur Geschwindigkeit der Energiewende



Abbildung 9: Resultate des „Kundenbarometers für Erneuerbare Energien“ (Côté et al., 2022)

Währenddessen entschieden sich 79.83% der Befragten für ein breitdiversifiziertes Portfolio von erneuerbaren Energien (Wasser, Sonne und Wind), wenn sie den Strommix der Zukunft für die Schweiz bestimmen könnten. 9.24% würde den Fokus auf Solarenergie und Wasserkraft legen und auf die Windenergie verzichten. 8.4% entschieden sich für verstärkte Förderung der Kernenergie. Keine der befragten Personen hat sich für den Bau von Gaskraftwerken entschieden, wobei jedoch 3 Personen keine Auswahl getroffen haben.

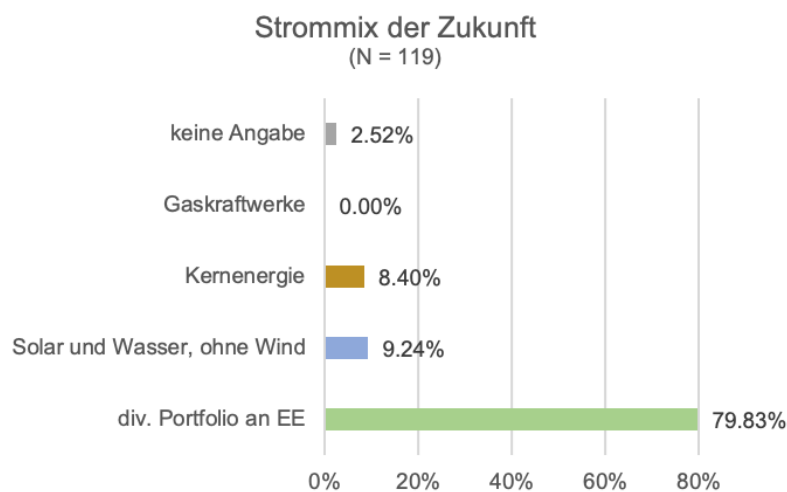


Abbildung 10: Umfrageresultate zum Strommix der Zukunft

Etwas mehr als die Hälfte der Befragten (57.14%) sagen aus, dass sie es für „sehr wichtig“ halten, dass die Schweiz im Klimaschutz vorankommt. Etwas mehr als ein Drittel (36.13%) empfindet dies als „wichtig“, 4.2% äussern sich dazu mit „weder noch“ und eine Person empfindet dies als „unwichtig“ sowie eine Person als „sehr unwichtig“. Zudem besitzen 39.5% der Befragten ein Generalabonnement des öffentlichen Verkehrs und ebenso 39.5% ein Halbtax, 7.56% besitzen ein Abonnement des Regionalverkehrs und 10.92% besitzen gar kein Abonnement.

Klimaschutz in der Schweiz
(N = 119)

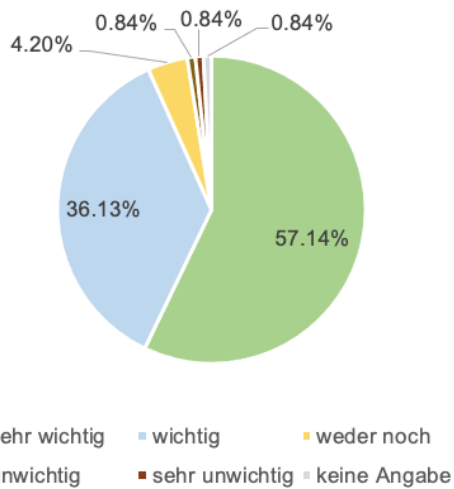


Abbildung 12: Umfrageresultate zum Klimaschutz in der Schweiz

ÖV-Abonnement
(N = 119)

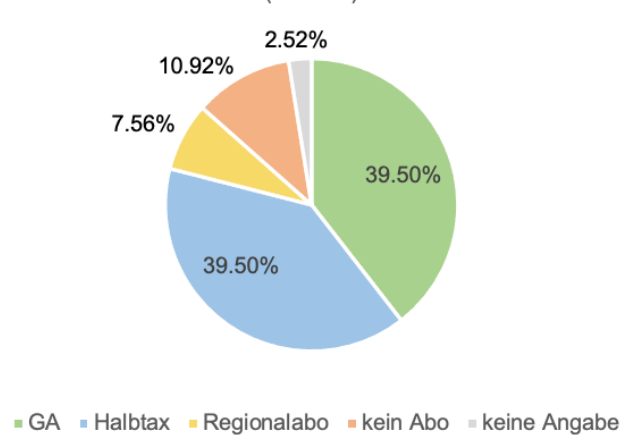


Abbildung 11: Umfrageresultate zum ÖV-Abonnement

Bei der Wahl ihrer Reisedestination achten sich 42.86% der befragten Personen „ab und zu“ darauf, ob diese Wert auf Nachhaltigkeit und Umweltfreundlichkeit legt. Nur 3 Personen (2.52%) haben angegeben, dass sie sich „stark darauf achten“ und 20.17% der Personen haben angegeben, dass sie sich „darauf achten“. Knapp ein Drittel der Befragten (28.57%) haben an-

Achtsamkeit Nachhaltigkeit und Umweltfreundlichkeit bei Reisedestination
(N = 119)

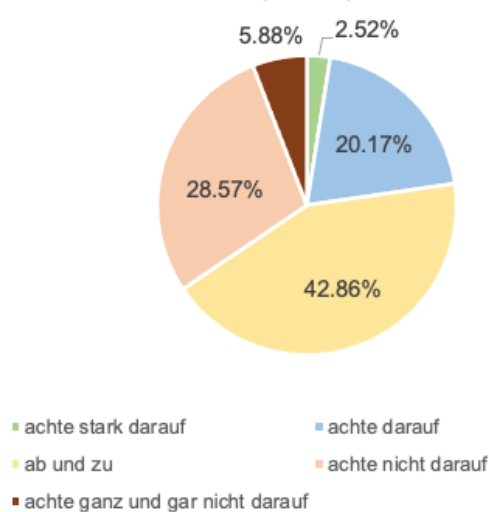


Abbildung 13: Umfrageresultate zu den Werten Nachhaltigkeit und Umweltfreundlichkeit einer Reisedestination

gegeben, dass sie sich nicht darauf achten und sieben Personen (5.88%) „achten sich ganz und gar nicht“ darauf.

5.1.1.2 Haltung zu Windenergie

Ungefähr die Hälfte (48.15% für Andermatt bzw. 49.23% für Laax) der befragten Personen stimmen der Aussage zu, dass der Windpark Gütsch in Andermatt respektive ein potentieller Windpark in Laax ein Zeichen der Nachhaltigkeit und Fortschrittlichkeit für den jeweiligen Tourismusort darstellt. Nahezu ein Drittel stimmt dieser Aussage sogar „voll und ganz zu“ (27.78% bzw. 32.31%). Neun Personen haben sich mit „weder noch“ dazu geäußert, jeweils 5 Personen stimmen dieser Wahrnehmung nicht zu, und 3 Personen in Andermatt bzw. 2 Personen in Laax stimmen dieser Aussage „ganz und gar nicht zu“.

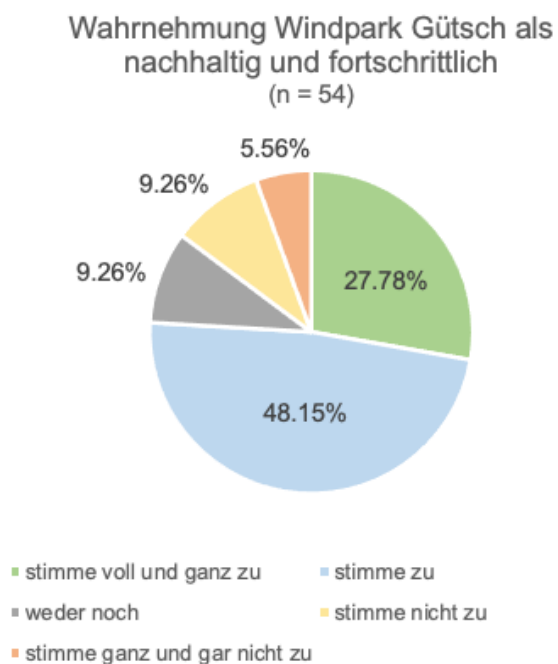


Abbildung 14: Umfrageresultate zur Wahrnehmung des Windparks Gütsch als nachhaltig und fortschrittlich

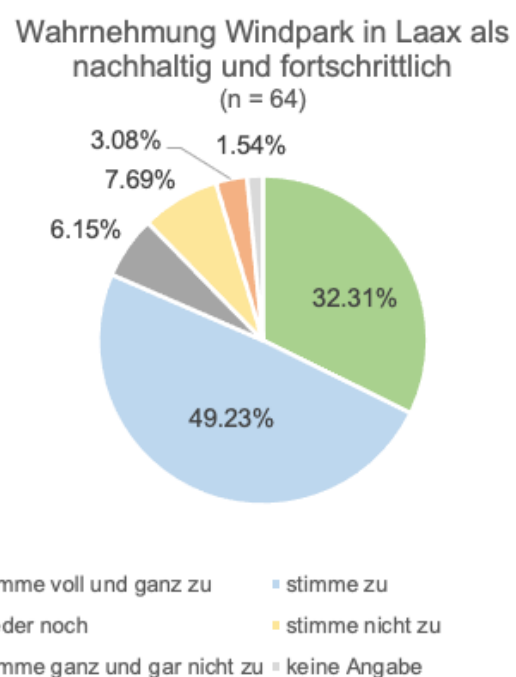


Abbildung 15: Umfrageresultate zur Wahrnehmung eines Windparks in Laax als nachhaltig und fortschrittlich

Von den befragten Personen stimmten 42.02% der Aussage zu, dass sie sich für Windenergie und ihre Technologie interessieren und 12 Personen (10.08%) stimmten dieser Aussage sogar „voll und ganz zu“. Etwas mehr als ein Drittel (34.45%) beantwortete die Frage mit „weder noch“, 9.24% stimmten dieser Aussage nicht zu und 3.36% stimmten der Aussage „ganz und gar nicht zu“. Mehr als die Hälfte der Befragten (61.34%) haben geantwortet, dass sich die nächstgelegene Windkraftanlage entweder ausserhalb ihres gewöhnlichen Bewegungsradius befindet oder sie nicht wissen, wo sich die nächstgelegene Windkraftanlage befindet.

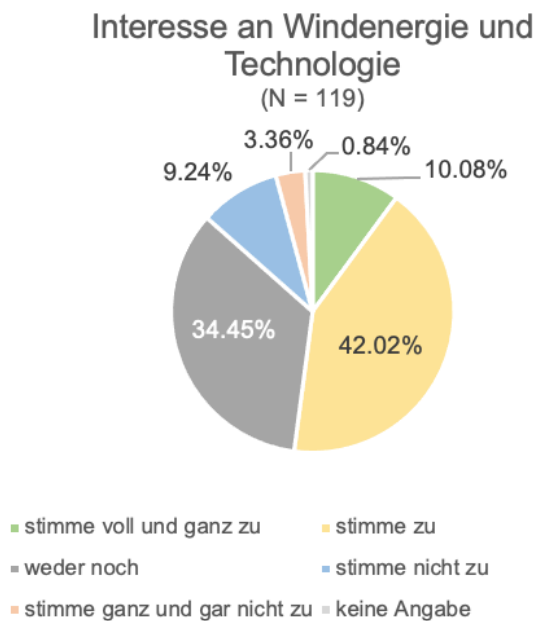


Abbildung 16: Umfrageresultate zum Interesse an Windenergie

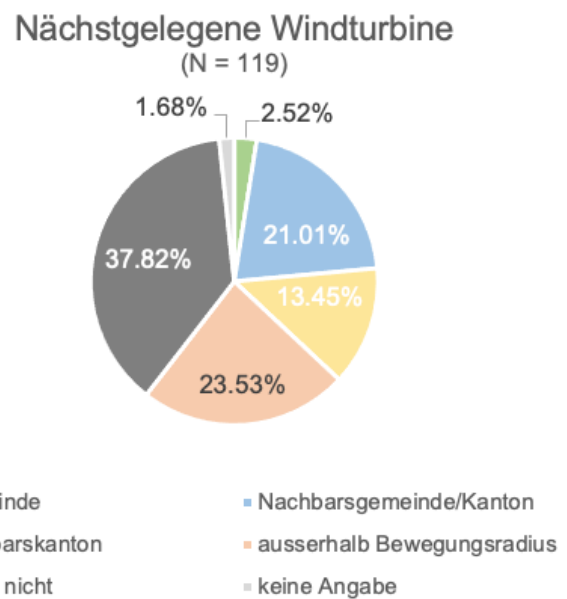


Abbildung 17: Umfrageresultate zur nächstgelegenen Windturbine

Etwa ein Fünftel der Befragten (21.01%) hat angegeben, dass sich die nächste Windturbine in einer Nachbarsgemeinde oder innerhalb ihres Wohnkantons befindet, drei Personen (2.52%) haben angegeben, dass sich diese in ihrer Gemeinde befindet. Zudem haben 13.45% angegeben, dass sich eine solche in einem Nachbarskanton befindet.

5.1.1.3 Regelmässigkeit des Besuchs und Tourismusaktivitäten

Ungefähr ein Viertel der befragten Personen (26.89%) besuchen den jeweiligen Tourismusort Andermatt oder Laax in einer gewissen Regelmässigkeit und ungefähr ebenso viele (25.21%)

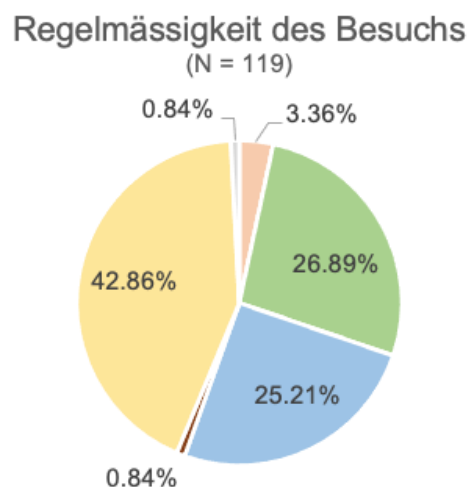


Abbildung 18: Umfrageresultate zur Regelmässigkeit von Besuchen

besuchen die beiden Orte zwar nicht regelmässig, aber auch nicht zum ersten Mal. Lediglich vier Personen (3.36%) besuchten die jeweiligen Orte zum ersten Mal und 42.86% der Befragten waren mindestens bereits einmal am jeweiligen Ort, haben den Fragebogen jedoch nicht zum Zeitpunkt, an dem sie anwesend waren, vor Ort ausgefüllt.

Beim Fragebogen für Andermatt wurden als Tourismusaktivität am häufigsten Skifahren, Snowboarden und andere Wintersportarten genannt (34.72% der Aussagen). Ungefähr ein Fünftel (19.44%) der Aussagen haben Wandern als Aktivität genannt und ebenso viele Aussagen wurden zu nicht auswählbaren Aktivitäten gemacht (Antwortoption „Andere“). Diese beinhalteten oftmals den Besuch von Familie oder Bekannten, das kulinarische Angebot oder etwa die konkrete Nennung einer Attraktion (z.B. Teufelsbrücke). Eine Aussage nannte den Windpark Gütsch, zwei Aussagen nannten eine Tour (2.78%) und sechs Aussagen kulturelles Angebot oder ein Event (8.33%), neun weitere Aussagen bezogen sich auf einen Wellnessaufenthalt (12.5%). Total wurden 71 Angaben von allen 53 Personen abgegeben. Es konnten mehrere Gründe ausgewählt werden.

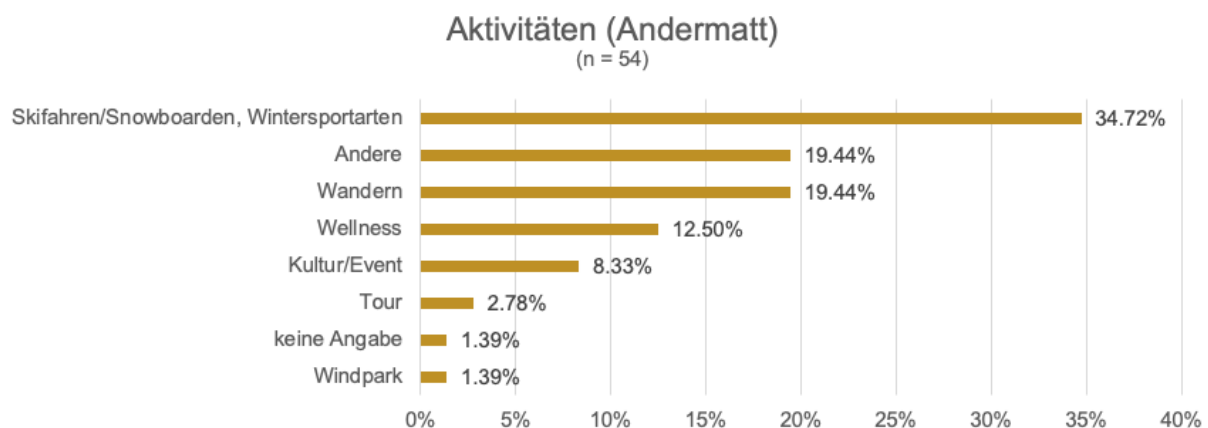


Abbildung 19: Umfrageresultate zu den gesuchten Tourismusaktivitäten in Andermatt

Die Aussagen zum Fragebogen für Laax verteilen sich ähnlich. 58.51% der Aussagen haben Wintersportarten, Ski- und Snowboardfahren genannt, 16 Personen (17.02%) haben Wandern angegeben und acht Aussagen wurden für Biken abgegeben (8.51%). Weitere acht Aussagen bezogen sich auf das kulturelle Angebot oder ein Event (8.51%), vier Aussagen auf einen Wellnnessaufenthalt (4.26%) und drei Aussagen nannten nicht gelistete, andere Gründe (meistens zum Besuch der Familie; 3.19%). Total wurden 76 Aussagen von allen 65 Personen erfasst. Auch hier konnten mehrere Gründe angegeben werden.

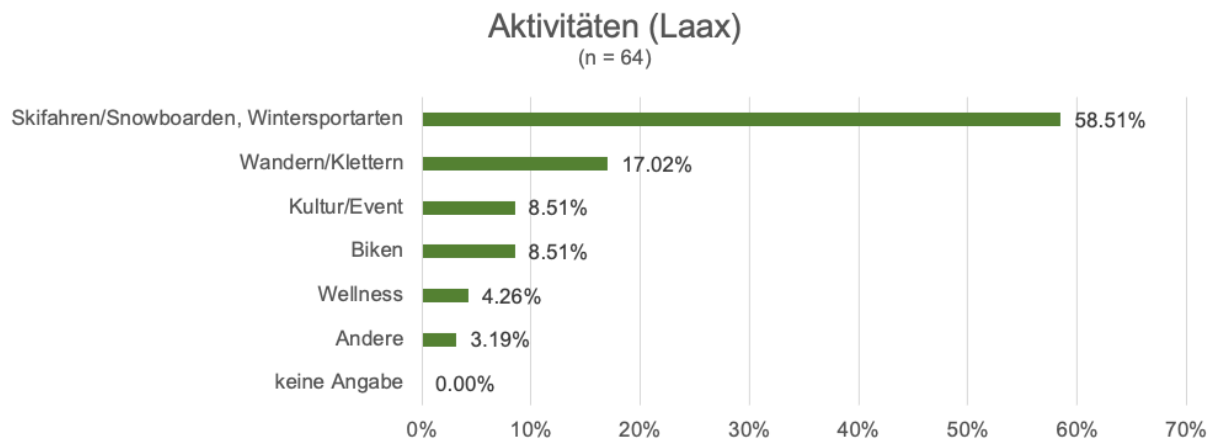


Abbildung 20: Umfrageresultate zu den gesuchten Tourismusaktivitäten in Laax

5.1.1.4 Demografie

Von den 119 Umfrageteilnehmenden waren 46.15% zwischen 25 und 34, 16.92% zwischen 18 und 24, 10.77% zwischen 35 und 44, 13.85% zwischen 55 und 64 Jahre alt und 4.62% zwischen 65 und 74 Jahre alt. Von diesen haben 60% einen Universitäts- oder Fachhochschulabschluss, 24.62% einen eidgenössischen Lehrabschluss, 12.31% einen Maturaabschluss und eine Person

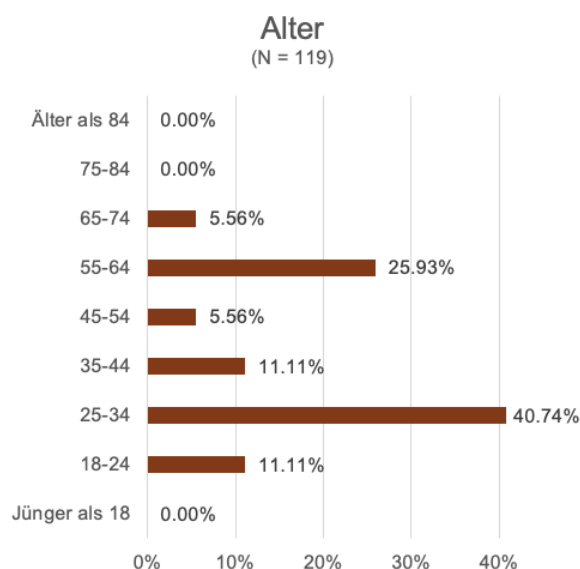


Abbildung 22: Umfrageresultate zum Alter

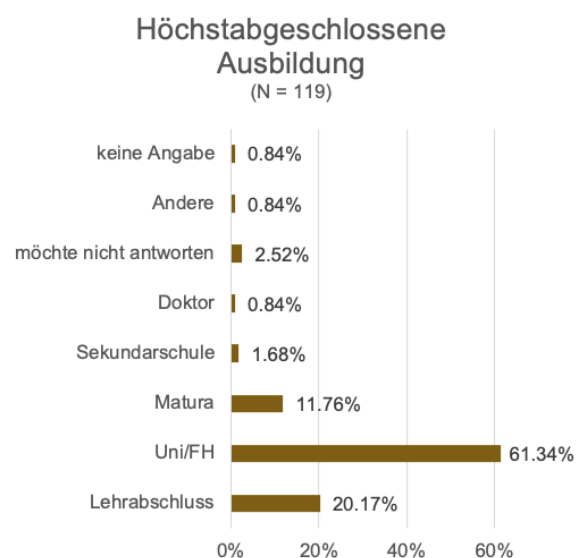


Abbildung 21: Umfrageresultate zur Ausbildung

einen Dokortitel. Zudem sind 89.08% der befragten Personen in der Schweiz wohnhaft, 6.72% leben in Deutschland und jeweils eine Person in Österreich und in Italien. Von den Teilnehmenden aus der Schweiz sind 65.09% wohnhaft im Kanton Zürich, die übrigen in den Kantonen Aargau (7; 6.6%), Basel-Stadt (1; 0.94%), Bern (2; 1.89%), Graubünden (6; 5.66%), Luzern (6; 5.66%), Nidwalden (1; 0.94%), Obwalden (2; 1.89%), Schwyz (1; 0.94%), St. Gallen (3; 2.83%), Thurgau (1; 0.94%) und Zug (6; 5.66%). Etwas mehr als die Hälfte der Teilnehmenden sind Männer (51.26%), 45.38% sind Frauen und 2.52% wollten keine Aussage zu ihrem Geschlecht abgeben.

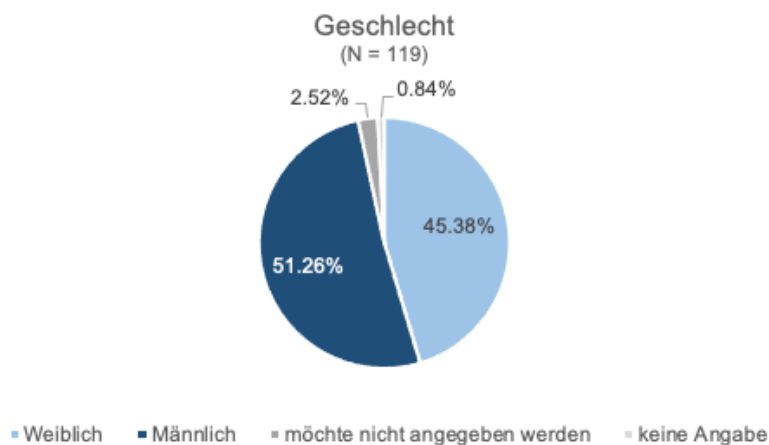


Abbildung 23: Umfrageresultate zum Geschlecht

5.1.2 Experimentelle Resultate

5.1.2.1 Wahrnehmung des Standorts

Von den 119 befragten Personen stimmen knapp die Hälfte (47.06%) der Aussage zu, dass sie es für sinnvoll finden, einen Windpark in der Nähe von bestehender Infrastruktur zu platzieren. Weitere 14.29% stimmen dieser Aussage sogar „voll und ganz zu“ und 26.89% bewerteten die Aussage mit „weder noch“. Zudem haben 6.72% angegeben, dass sie dieser Aussage nicht zustimmen und 5.04%, dass sie dieser „ganz und gar nicht“ zustimmen.

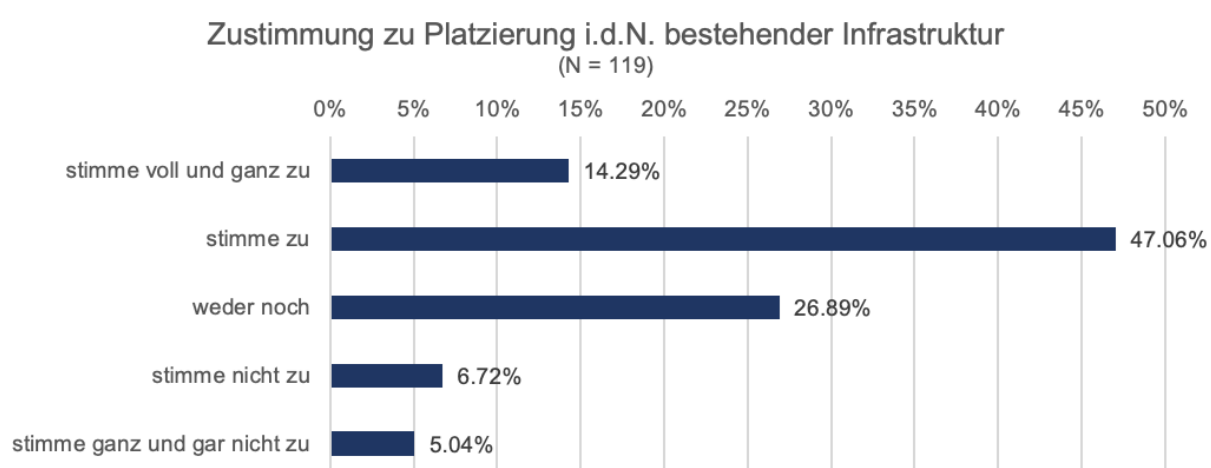


Abbildung 24: Umfrageresultate zur Wahrnehmung des Standorts in der Nähe von bestehender Infrastruktur

Anhand eines T-Tests wurden die Antworten in ihrem Mittelwert mit dem Mittelwert der Antwortmöglichkeiten auf Signifikanz geprüft, um herauszufinden, ob die unter Hypothese 1a formulierte Aussage bestätigt oder verworfen werden kann und auf statistische Signifikanz geprüft werden konnte. Dabei wurde die Skala so aufgestellt, dass 1 „stimme ganz und gar nicht zu“ darstellt und „stimme voll und ganz zu“ durch die Zahl 5 dargestellt wird. Dabei wurde ein Mittelwert der Antworten von 3.59, eine Standardabweichung von 0.99 und 118 Freiheitsgrade errechnet. Der Mittelwert der Antwortmöglichkeiten stellt 3 dar, wonach im Vergleich mit dem Mittelwert der Antworten von 3.59 ersichtlich ist, dass die Aussage, dass es die TouristInnen beider Orte für sinnvoll halten, einen Windpark in Nähe bestehender Infrastruktur zu platzieren, bestätigt ist. Der p-Wert (0.00000000195982) liegt deutlich unter der Signifikanzschwelle von 0.05, was bedeutet, dass die Aussage durch statistische Signifikanz bestätigt ist.

Anhand eines einseitigen T-Tests wurden die Antworten zu dieser Frage für Andermatt und Laax verglichen, wobei die Teilnehmenden aus Andermatt (Mittelwert = 3.61, Standardabweichung = 1.23) dieser Aussage etwas mehr zustimmen als Teilnehmende aus Laax (Mittelwert = 3.57, Standardabweichung = 0.86), da der Mittelwert bei Andermatt etwas höher ausfällt.

Zur Frage, ob die in Andermatt befragten Personen (N = 54) den Standort des Windparks Gütsch als sinnvoll erachten, haben sich 38.89% dazu ausgesprochen, dass der Standort als „sinnvoll“ erachtet wird, 14.81% als „sehr sinnvoll“ und 12.96% sogar als „äusserst sinnvoll“. Weitere 16.67% haben angegeben, dass sie nicht wissen, wie sie den Standort bewerten sollen, 7.41% finden den Standort „nicht sinnvoll“ und weitere 9.26% finden den Standort „überhaupt nicht sinnvoll“.

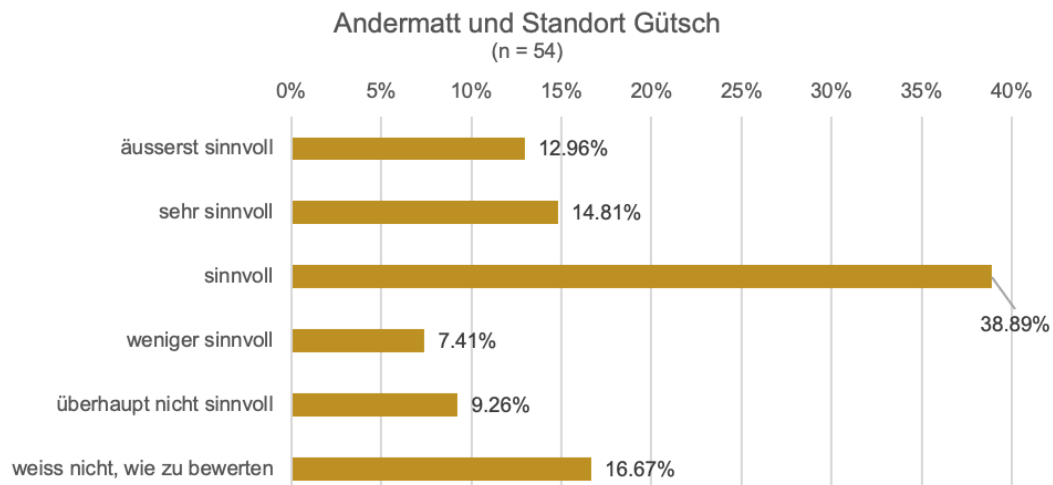


Abbildung 25: Umfrageresultate zur Wahrnehmung des Standorts Gütsch

Von den in Laax befragten Personen (N = 65) haben 35.38% dieselbe Frage für den Standort Crap Sogn Gion für einen potenziellen Windpark als „sinnvoll“ bewertet, 12.31% als „sehr sinnvoll“ und 20% als „äusserst sinnvoll“. Dabei haben den Standort 12.31% der Befragten als „weniger sinnvoll“ und 4.62% als „überhaupt nicht sinnvoll“ bezeichnet. Zudem haben 15.38% angegeben, dass sie nicht wissen, wie sie den Standort bewerten sollten.

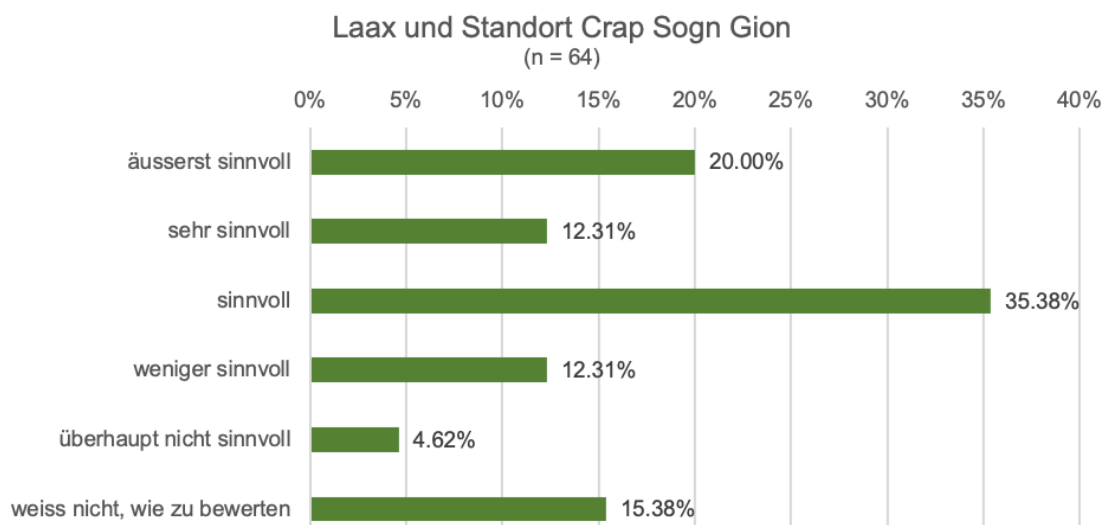


Abbildung 26: Umfrageresultate zur Wahrnehmung des Standorts Crap Sogn Gion

Betrachtet man die beiden Antwortgruppen zu Andermatt und Laax im Vergleich, lässt sich sagen, dass die Besuchenden in Andermatt (Mittelwert = 3.35, Standardabweichung = 2.53) den Standort Gütsch als sinnvoller empfinden, als Besuchende in Laax (Mittelwert = 3.15, Standardabweichung = 2.63) den Crap Sogn Gion für einen potentiellen Windpark, da hierbei der Mittelwert bei Andermatt höher liegt als der bei Laax. Die beiden Mittelwerte liegen jedoch so nahe beieinander, dass die T-Statistik auf keinen statistisch signifikanten Unterschied der Antworten der beiden Gruppen hinweist.

Die nachfolgende Graphik zeigt die Mittelwerte zu den beiden Fragen für die Antworten aus Andermatt und Laax im Vergleich.

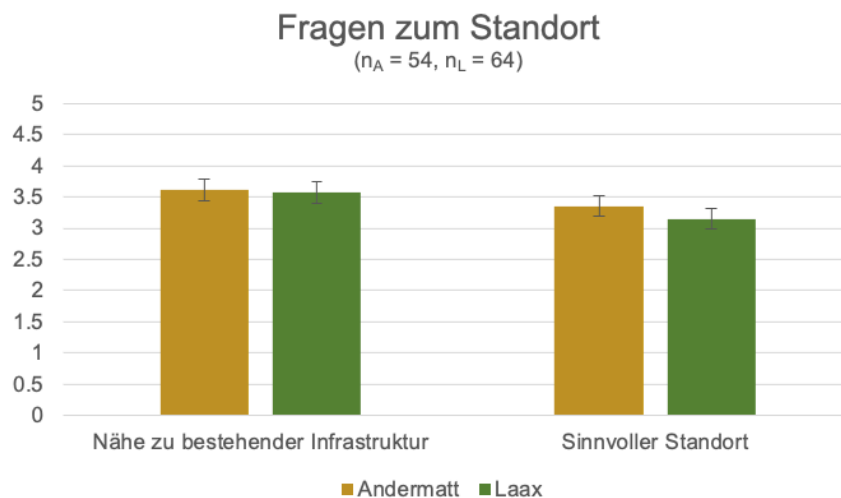


Abbildung 27: Umfrageresultate zur Wahrnehmung der beiden Standorte (im Vergleich)

Falls eine Person angegeben hat, dass sie den Standort als „nicht sinnvoll“, „überhaupt nicht sinnvoll“ bewertet oder nicht wusste, wie sie den Standort bewerten sollte, wurde sie zu zwei

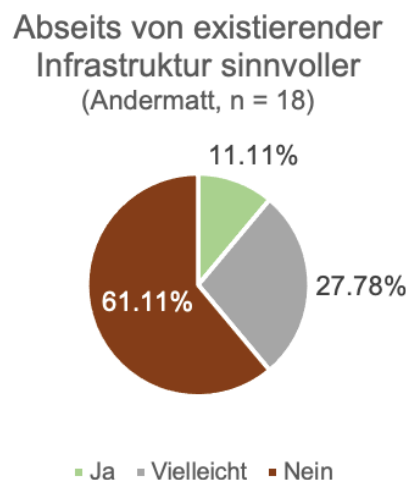


Abbildung 28: Umfrageresultate zur Wahrnehmung eines Standorts abseits existierender Infrastruktur (Andermatt)

weiteren Fragen weitergeleitet. Dies waren für Andermatt 18 Personen (33% aller Befragten in Andermatt) und für Laax 21 Personen (32% aller befragten Personen in Laax). Zur ersten Frage, ob sie einen Standort abseits von bereits existierender Infrastruktur, also in der „unberührten Landschaft“ sinnvoller fänden, haben in Andermatt zwei Personen mit „Ja“ (11.11%) geantwortet, fünf Personen mit „Vielleicht“ (27.78%) und elf Personen mit „Nein“ (61.11%).

Darauffolgend wurden von den 18 Personen insgesamt 21 Antworten dazu gesammelt, wo sie einen besseren Standort empfänden (jede Person hat mindestens eine Antwort abgegeben). Dazu wurde zehnmal „Industriezone“ angegeben (40% der Antworten), viermal das Mittelland (16% der Antworten), zweimal das Ausland (8% der Antworten), dreimal „Nirgends, ich bin

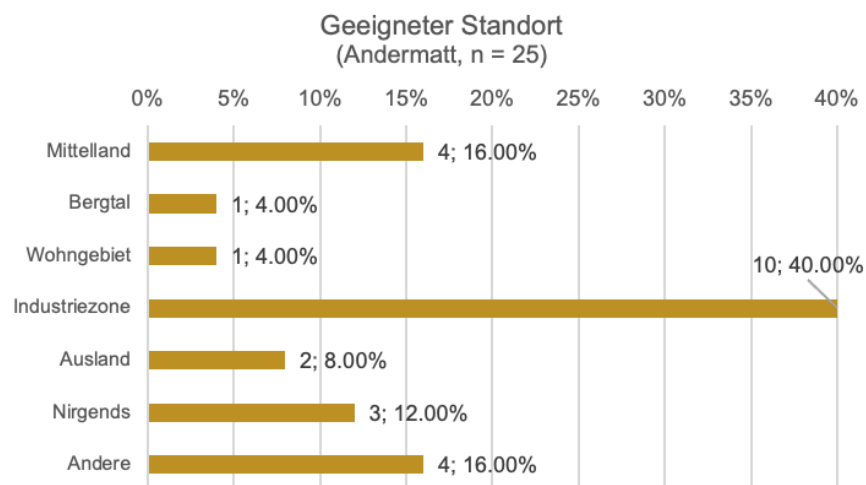


Abbildung 29: Umfrageresultate zur Wahrnehmung eines geeigneten Standorts (Andermatt)

grundsätzlich gegen die Nutzung von Windenergie“ (12% der Antworten) und je einmal wurde „Bergtal“ und „Wohngebiet“ genannt (je 4% der Antworten). Drei Personen haben zudem noch unter dem freien Eingabefeld „Andere“ einen Eintrag gemacht. Darin wurde beispielsweise erwähnt, dass Windkraftwerke auf das „offene Meer oder in die Tundra“ gehören oder generelle Bedenken gegenüber der Effizienz von Windkraftanlagen geteilt.

Bei den 21 befragten Personen in Laax wurde die Frage, ob sie einen Standort abseits von existierender Infrastruktur, also in der „unberührten Landschaft“, sinnvoller fänden, zweimal mit „Ja“ (9.52%), achtmal mit „Vielleicht“ (38.1%) und elfmal mit „Nein“ (52.38%) beantwortet.

**Abseits von existierender
Infrastruktur sinnvoller**
(Laax, n = 21)

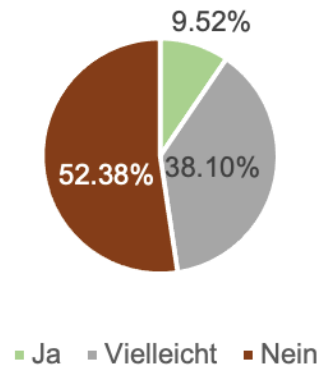


Abbildung 30: Umfrageresultate zur Wahrnehmung eines Standorts abseits existierender Infrastruktur (Laax)

Anschliessend wurden von diesen 21 Personen 31 Antworten dazu gesammelt, wo sie einen geeigneten Standort für einen Windpark empfänden (jede Person hat mindestens eine Antwort abgegeben). Dazu wurde ebenso am meisten „Industriezone“ (41.94% der Antworten) angegeben. Des Weiteren wurde sechsmal „Mittelland“ (19.35% der Antworten), fünfmal „Bergtal“ (16.23% der Antworten), dreimal „Ausland“ (9.68% der Antworten), einmal „Wohngebiet“ (3.23% der Antworten) und schliesslich noch dreimal „Nirgends, ich bin grundsätzlich gegen“ (9.68% der Antworten).

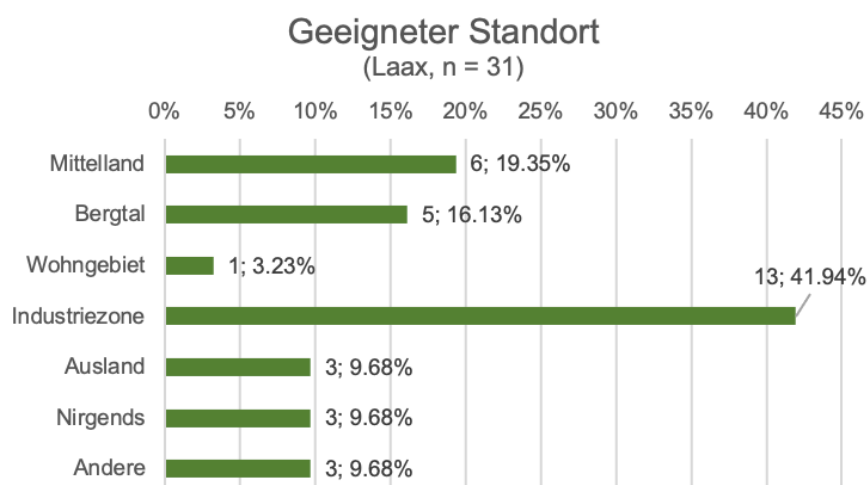


Abbildung 31: Umfrageresultate zur Wahrnehmung eines geeigneten Standorts (Laax)

die Nutzung von Windenergie“ (9.68% der Antworten) genannt. Zudem wurden von drei Personen unter dem offenen Feld „Andere“ Anmerkungen hinterlassen, die sich vor allem darauf bezogen haben, dass „da, wo es windtechnisch am meisten Sinn macht“ oder „dort, wo es optisch nicht stört“ als am sinnvollsten betrachtet wird.

5.1.2.2 Wahrnehmung des Landschaftsbilds

Von den 54 befragten Personen in Andermatt hat ein Grossteil (46.3%) angegeben, dass sie „indifferent“ demgegenüber sind, ob sie das Landschaftsbild durch den Windpark Gütsch als bereichert oder gestört empfinden. Acht Personen (14.81%) haben angegeben, dass sie das Landschaftsbild durch den Windpark als „bereichert“ empfinden und 3 Personen (5.56%) sehen dieses als „sehr bereichert“. 14 Personen (25.93%) empfinden das Landschaftsbild dadurch als „gestört“ und vier weitere als „sehr gestört“ (7.41%).

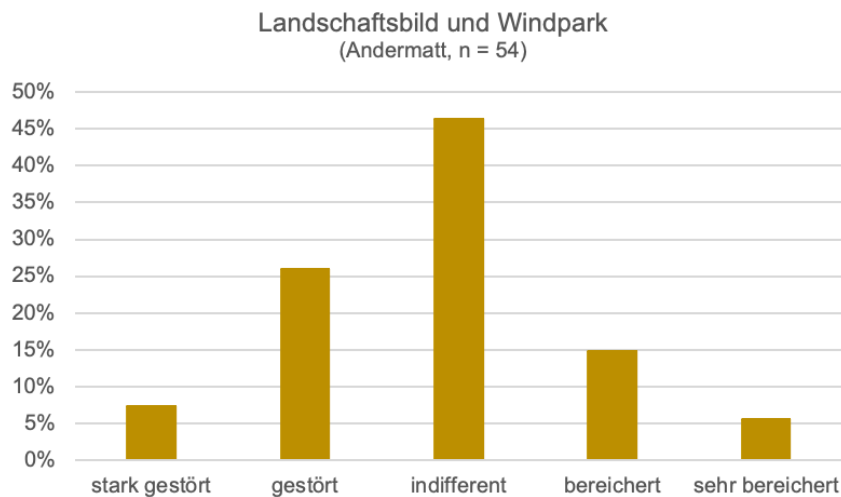


Abbildung 32: Umfrageresultate zur Wahrnehmung des Landschaftsbilds und den Windpark Gütsch (Andermatt)

Bei den Befragten in Laax hat ebenso ein Grossteil (46.15%) geantwortet, dass sie „indifferent“ sind, ob ein potenzieller Windpark in Laax das Landschaftsbild bereichert oder stört. 14 Personen (21.54%) haben angegeben, dass sie dieses durch einen potenziellen Windpark als „bereichert“ empfinden und vier weitere Personen (6.15%) empfinden dieses als „sehr bereichert“. Zudem haben 14 Personen (21.54%) geantwortet, dass sie das Landschaftsbild durch einen Windpark als „gestört“ empfinden und 3 weitere Personen (4.62%) empfinden es als „sehr gestört“.

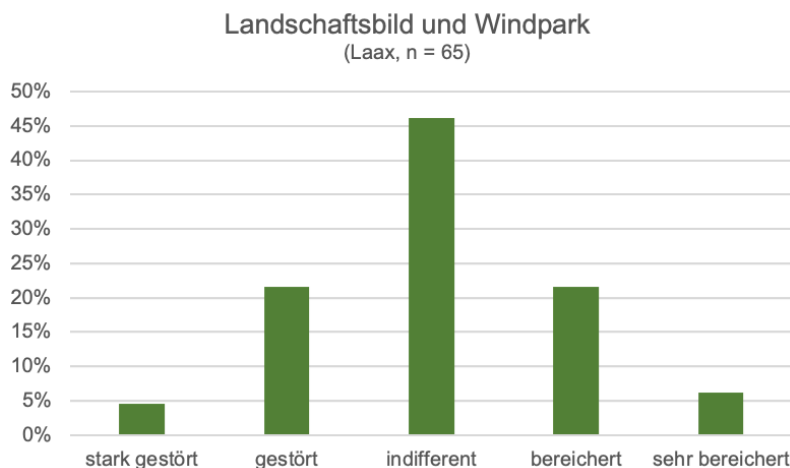


Abbildung 33: Umfrageresultate zur Wahrnehmung des Landschaftsbilds und einen potenziellen Windpark auf dem Crap Sogn Gion (Laax)

Keine Person hat sich ihrer Aussage enthalten. Die Antworten beider Orte sind normalverteilt und die beiden Gruppen wurden anhand eines einseitigen T-Tests statistisch verglichen, um die Teilhypothese 1b zu testen. Durch den leicht höheren Mittelwert der Antworten in Laax (Mittelwert = 3.03, Standardabweichung = 0.87) gegenüber dem Mittelwert von Andermatt (Mittelwert = 2.85, Standardabweichung = 0.92) wird ausgedrückt, dass sich die befragten Personen das Landschaftsbild in Andermatt durch den Windpark Gütsch als leicht „gestörter“ bzw. weniger „bereichert“ empfinden als die befragten Personen in Laax einen potenziellen Windpark empfinden (t -Wert = -1.02). Jedoch liegen beide Mittelwerte sehr nah am Mittelwert der Antwortmöglichkeiten, der durch die Zahl 3 beschrieben wird und für „Indifferenz“ steht. Auch der deutlich über dem Signifikanzniveau liegenden p -Wert (0.41) und der im Vergleich zum t -Wert höhere kritische t -Wert (1.66) weisen darauf hin, dass kein statistisch signifikanter Unterschied zwischen den beiden Gruppen vorliegt.

5.1.2.3 Wahrnehmung in Bezug auf die Tourismusaktivität

In Bezug auf die Frage, ob sich die befragte Person bei ihrer Tourismusaktivität durch den Windpark Gütsch in Andermatt bereichert bzw. gestört fühlt, hat die Mehrheit (70.37%) geant-

wortet, dass sie „indifferent“ sind. Acht Personen (14.81%) fühlen sich dabei durch den Windpark Gütsch „bereichert“ und zwei weitere „sehr bereichert“ (3.7%). Des Weiteren haben vier Personen (7.41%) angegeben, dass sie sich dabei „gestört“ fühlen und zwei weitere „sehr gestört“ (3.7%).

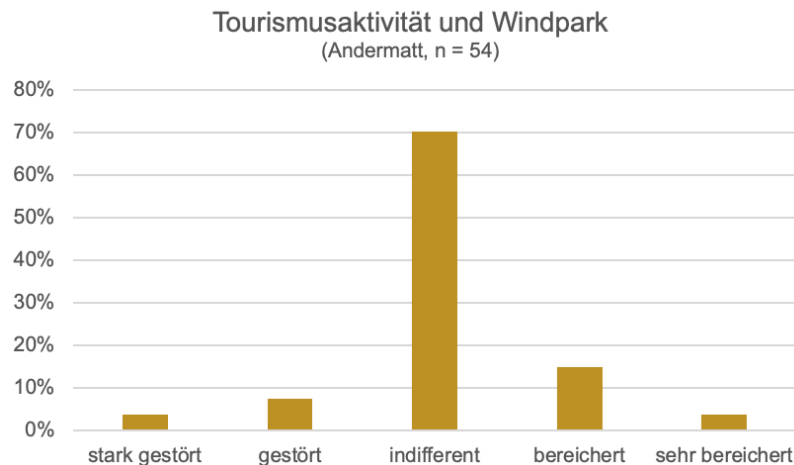


Abbildung 34: Umfrageresultate zur wahrgenommenen Auswirkung des Windparks Gütsch auf die Tourismusaktivität (Andermatt)

Bei den Befragten in Laax haben sich ebenfalls die Mehrheit (67.69%) als „indifferent“ dazu geäußert, ob sie sich durch einen potenziellen Windpark auf dem Crap Sogn Gion in Laax bei ihrer Tourismusaktivität bereichert bzw. gestört fühlen. Acht Personen (12.31%) empfinden einen Windpark als „bereichernd“ und vier weitere als „sehr bereichernd“. Zudem haben fünf Personen (7.69%) angegeben, dass sie sich bei ihrer Tourismusaktivität durch einen solchen „gestört“ fühlen würden und 4 Personen (6.15%) als sehr gestört.

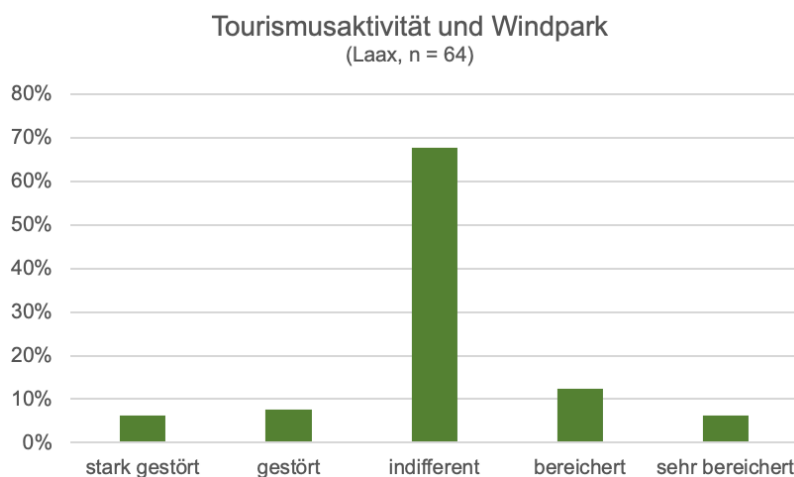


Abbildung 35: Umfrageresultate zur wahrgenommenen Auswirkung eines potenziellen Windparks auf dem Crap Sogn Gion auf die Tourismusaktivität (Laax)

Keine Person hat sich ihrer Aussage enthalten. Die Antworten beider Orte sind normalverteilt und auch hierzu wurden die beiden Gruppen anhand eines einseitigen T-Tests statistisch verglichen. Die beiden Mittelwerte für die Antworten aus Andermatt (Mittelwert = 3.07, Standardabweichung = 0.52) und für die Antworten aus Laax (Mittelwert = 3.05, Standardabweichung = 0.7) sind nahezu identisch und liegen beide sehr nah am Mittelwert der Antwortmöglichkeiten, der durch die Zahl 3 beschrieben wird und für „Indifferenz“ steht. Der t-Wert (0.19) gibt an, dass der Mittelwert von Andermatt leicht höher ist als der von Laax, was darauf hindeuten würde, dass sich Personen in Andermatt bei ihrer Tourismusaktivität weniger gestört bzw. eher bereichert fühlen durch den Windpark Gütsch als Personen in Laax durch einen potenziellen Windpark. Jedoch sprechen der höhere, kritische t-Wert (1.66) und der p-Wert, der über dem Signifikanzwert von 0.05 liegt, ebenso dafür, dass kein statistischer Unterschied zwischen den beiden Gruppen zu erkennen ist.

Um die Teilhypothese 1c zu untersuchen, wurden die Antworten aller Umfrageteilnehmenden zur Tourismusaktivität gemäss ihrer Angabe, für welche Aktivität sie zum jeweiligen Ort gereist sind, nach Sommer- und Winteraktivitäten kategorisiert. Dazu wurden die Antwortmöglichkeiten in „Sommeraktivität“ (= 1), „Winteraktivität“ (= 3) und „Beides“ (= 2) aufgeteilt und die Antworten entsprechend übersetzt. Den Sommeraktivitäten wurden Wandern, Klettern und Biken zugeordnet. Ski- und Snowboardfahren, andere Wintersportarten und Wellness (ausser es wurde gleichzeitig eine Sommeraktivität mit angekreuzt) wurden entsprechend den Winteraktivitäten zugeordnet. Die Angabemöglichkeiten Events, Kultur und Touren wurden zur Kategorie „Beides“ zugeordnet. Wenn eine Person beispielsweise Skifahren und Wandern angegeben hat, wurde dieser Datenpunkt ebenso in die Kategorie „Beides“ eingestuft. Bei den Antworten zu Andermatt wurden bedauerlicherweise von vier Personen keine Angabe gemacht, weshalb sich die untersuchte Datenmenge für die Regressionsanalyse auf $N = 115$ beläuft. Darunter befinden sich 17 Personen, die der Kategorie „Sommeraktivitäten“, 68 Personen, die der Kategorie „Winteraktivitäten“ und 30 Personen, die der Kategorie „Beides“ zugeteilt wurden. Zur Prüfung der Hypothese wurde eine einfaktorielle Varianzanalyse (ANOVA) durchgeführt, um die drei Gruppen zu vergleichen. Der Mittelwert der SommertouristInnen (Mittelwert = 3.24, Standardabweichung = 0.57) liegt über dem der WintertouristInnen (Mittelwert = 3.02, Standardabweichung = 0.55) und auch über der Gruppe, denen „beides“ zugeordnet wurde (Mittelwert = 3.07, Standardabweichung = 0.89). Dies deutet darauf hin, dass sich die SommertouristInnen bei ihrer Aktivität weniger durch den Windpark Gütsch respektive durch einen potenziellen Windpark in Laax gestört bzw. sich eher bereichert fühlen als WintertouristInnen. Der F-Wert (0.52) liegt deutlich unter dem kritischen F-Wert (3.08) und auch der p-Wert (0.6) liegt über dem Signifikanzniveau von 0.05, was darauf hindeutet, dass kein statistisch signifikanter Unterschied zwischen den Mittelwerten gegeben ist und damit kein deutlicher Effekt durch die unterscheidende Aktivität gegeben ist.

Zur Veranschaulichung wurde zudem eine Regressionsanalyse vorgenommen, die ebenso gezeigt hat, dass die Wahrnehmung, ob sich eine Person durch einen Windpark bei ihrer Tourismusaktivität gestört fühlt, nicht davon abhängt, ob diese Person eine Sommer- oder Winteraktivität verfolgt. Das Bestimmtheitsmass (R^2) von 0.82% zeigt deutlich, dass diese Wahrnehmung durch diese Variable nicht gut erklärt werden kann und wohl eher von anderen Faktoren abhängt.

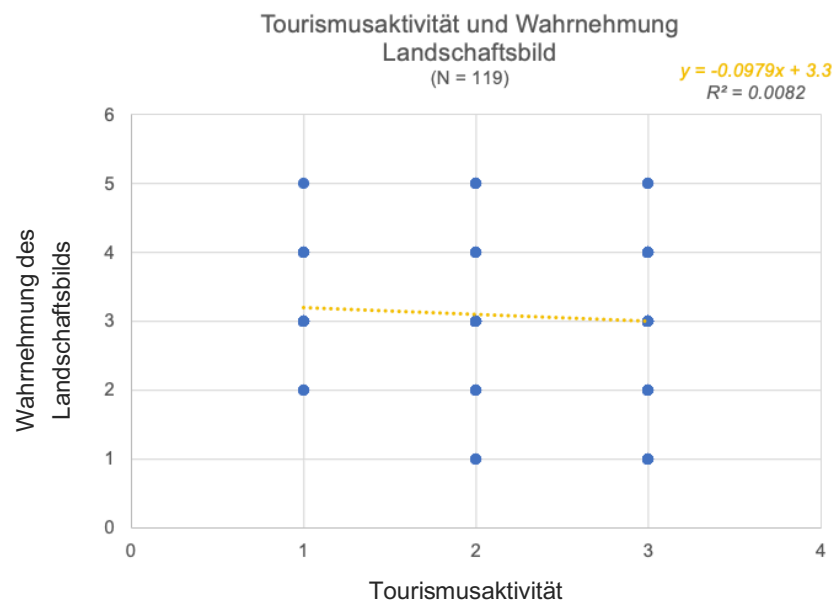


Abbildung 36: Regressionsanalyse zum Zusammenhang der Wahrnehmung des Landschaftsbilds und der Auswirkung eines Windparks auf die Tourismusaktivität

5.1.2.4 Windenergie als Zeichen der Nachhaltigkeit und Fortschrittlichkeit

Um die Hypothese 1d zu testen, wurden die Antworten der befragten Personen dazu, ob Nachhaltigkeit und Fortschrittlichkeit für sie Kriterien darstellen bei der Wahl ihrer Tourismusdestination den Antworten dazu gegenübergestellt, wie sie das Landschaftsbild in Anwesenheit des Windpark Gütschs bzw. eines potenziellen Windparks in Laax wahrnehmen. Um die Abhängigkeit dieser beiden Variablen zu prüfen, wurde ein Chi-Quadrat-Test durchgeführt. Der daraus resultierende p-Wert (0.99), der über dem Signifikanzniveau von 0.05 liegt, weist darauf hin, dass die beiden Variablen statistisch voneinander unabhängig sind. In einer Regressionsanalyse konnte gezeigt werden, dass zwar eine leichte Tendenz vorhanden ist, die darauf hindeuten würde, dass Personen, die sich bei der Wahl ihrer Reisedestination auf Werte wie Nachhaltigkeit und Fortschrittlichkeit achten, das Landschaftsbild weniger als durch einen Windpark „gestört“ oder eher „bereichert“ empfinden. Jedoch hat auch hier das tiefe Bestimmtheitsmass von 2.57% gezeigt, dass die Aussagekraft nicht gegeben ist und somit kein direkter Zusammenhang gezeigt werden kann.

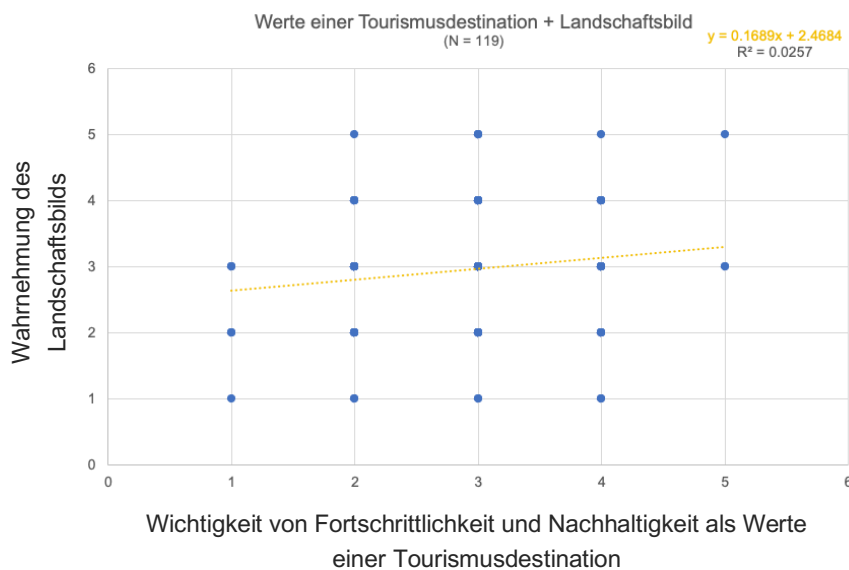


Abbildung 37: Regressionsanalyse zum Zusammenhand der Wahrnehmung des Landschaftsbilds und der Wichtigkeit der Werte Fortschrittlichkeit und Nachhaltigkeit einer Tourismusdestination

5.1.2.5 Windtourismus

Unter den befragten Personen haben 116 Personen angegeben, dass sie noch nie an einer Führung durch einen Windpark teilgenommen haben und lediglich drei Personen haben bereits einmal an einer Führung teilgenommen. Diese drei Personen haben alle am Umfragebogen zu Laax teilgenommen. Jeweils eine Person hat angegeben, dass sie im Rahmen einer beruflichen Veranstaltung, im Rahmen einer schulischen Veranstaltung oder aus (eigenem) Interesse an der Technologie teilgenommen hat.

Um die Teilhypothese 2a zu untersuchen, wurden die Antworten der drei Personen mit ihren Antworten zu den Fragen diesbezüglich betrachtet, ob sie die Landschaft durch einen Windpark als „gestört“ oder „bereichert“ betrachten und ob sie sich bei ihrer Tourismusaktivität durch einen Windpark „gestört“ oder „bereichert“ fühlen. Diese wurden anschliessend entsprechend mit den Datensätzen der Personen verglichen, die noch nie an einer Führung teilgenommen haben. Die nachfolgende Tabelle zeigt die Antworten der drei Personen, die bereits einmal an einer Führung teilgenommen haben.

	Wahrnehmung eines Windparks in Bezug auf das Landschaftsbild	Wahrnehmung eines Windparks in Bezug auf die Tourismusaktivität
Person 1	Sehr bereichert	Indifferent
Person 2	Indifferent	Indifferent
Person 3	Stark gestört	Stark gestört

Alle drei Personen haben unterschiedliche Wahrnehmungen von Windparks in Bezug auf das Landschaftsbild bzw. die Tourismusaktivität. Diese Gegebenheiten liefern keine Grundlage, eine statistisch signifikante Aussage zu formulieren.

5.1.2.6 Potential zu regionaler Wertschöpfung

Zur Frage, welche Aktivitäten die Umfrageteilnehmenden in Verbindung mit einem Windpark überzeugen würden, diesen zu besuchen, wurden insgesamt 276 Angaben von 118 Personen gemacht. Am meisten wurde dafür eine künstlerische Darstellung, wie beispielsweise eine Lichtshow auf Windrädern (44 Antworten), angegeben. Die Antwortmöglichkeit, dass eine Führung durch die Windkraftanlagen, um dabei auch mehr über die Technologie zu erfahren, bereits für einen Besuch überzeugen würde, wurde fast gleich so viel mal angegeben (43 Antworten). Auf dem dritten Platz mit 40 Antworten steht die Aussichtsplattform. Für ein Kombi-

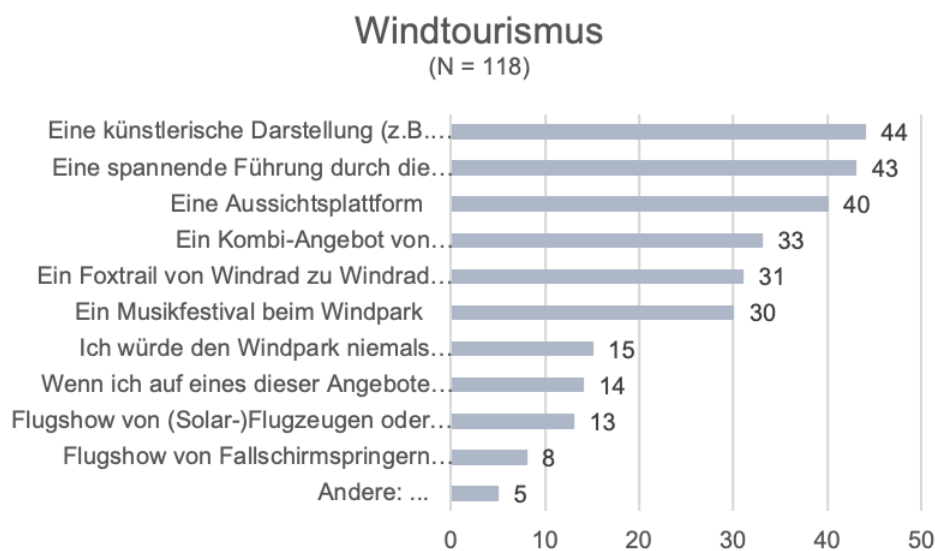


Abbildung 38: Umfrageresultate zu verschiedenen Tourismusaktivitäten im Windtourismus

Angebot von Transportmitteln (Bergbahn und SBB) und einer Führung durch den Windpark wurden 33 Antworten abgegeben und einen Foxtrail von Windrad zu Windrad oder andere spielerische Ansätze, um die Windkraft näher kennenzulernen, wurden 31 Antworten abgegeben. Des Weiteren wurde ein Musikfestival beim Windpark von 30 Personen gewählt. Weniger Stimmen haben die Antwortmöglichkeiten Flugshow von (Solar-)Flugzeugen oder Drohnen um die Windräder (13 Antworten), die Flugshow von Fallschirmspringern oder Wind-Drachen (8 Antworten) sowie die Antwortmöglichkeit, wenn man auf eines dieser Angebote auf Social Media aufmerksam gemacht würde und ein solches gewinnen könnte (14 Antworten). Letztlich haben 15 Personen angegeben, dass sie einen Windpark niemals besuchen würden. Eine Person hat entgegnet, dass sie den Windpark nur besuchen würde, sofern sie sich zufällig in der Region aufhalten würde und nicht spezifisch dafür anreisen würde. Eine weitere Person hat angegeben,

dass sie durch das nicht vorhandene Interesse an der Windenergie keine der Antwortmöglichkeiten wählen würde. Zudem hat eine Person im Umfragebogen zu Laax angegeben, dass sie es begrüßen würde, wenn ein Windpark in den Snowpark integriert würde.

5.2 Qualitative Interviews

Herr André Gisler⁴ bezeichnet sich als Befürworter von erneuerbaren Energien. Er betont dabei deren effiziente und effektive Nutzung, wobei die Konformität der jeweiligen Zonen beziehungsweise deren Nutzung unter Berücksichtigung der Energieeffizienz zentral sind. Einem potenziellen Windpark in Laax steht er insofern positiv gegenüber, als dass das Gebiet bereits touristisch genutzt wird. Somit sind Emissionen in Bezug auf Lärm und Optik dadurch bereits gegeben und so sollte dieser auch keine AnwohnerInnen stören. Ausserdem steht er einem Windpark positiv gegenüber, sofern die Effektivität durch das Windvorkommen am Standort gegeben wäre.

Herr Thomas Christen⁵ erachtet den Windpark Gütsch und seinen Standort als der Zeit angemessen und sinnvoll. Er unterstützt das Bestreben, die Benützung von fossilen Brennstoffen zu reduzieren und Alternativen der Energieproduktion zu fördern. Dies sollte aus seiner Sicht jedoch „nicht um jeden Preis“, sondern pragmatisch und sinnvoll geschehen, so dass auch Windturbinen oder Alternativen unter vernünftigen Bedingungen und mit natürlichen Stoffen hergestellt werden könnten. Dazu erläutert er zum Beispiel deren Produktion mit Kohlestrom aus China oder seltenen oder gar giftigen Erden und Rohstoffen.

Herr Gisler sieht die Natur und somit das Landschaftsbild als das höchstpriorisierte Bedürfnis der TouristInnen, die die Ferienregion Laax besuchen. Im „Triple Bottom Ansatz“ des Unternehmens ist Ökologie eine wichtige Säule, weshalb ihm der nachhaltige Ansatz in ihrer Strategie und im täglichen Umgang sehr wichtig ist. Mit ihrer Kampagne „Keep It Calm and Beautiful“ sowie anhand von Aktivitäten und Massnahmen gemeinsam mit den Gemeinden und privaten Firmen (insbesondere der Weissen Arena AG und ihrem Greenstyle Konzept) stehen sie als Unternehmen für Nachhaltigkeit ein. Somit machen die Themen „Nachhaltigkeit und Fortschrittlichkeit“ einen bedeutenden Teil ihrer Tätigkeit aus.

Den Windpark Gütsch empfindet Herr Christen nicht primär als Zeichen der Nachhaltigkeit und Fortschrittlichkeit. Für ihn steht der Nutzen im Vordergrund, vor allem im Winter Strom zu erzeugen, antizyklisch zur Stromerzeugung durch die Wasserkraft in der Region. Für ihn ist somit der Sinn bzw. die Sinnhaftigkeit in dieser Hinsicht „das oberste Gebot“ bei der Umsetzung des Windparkprojekts.

⁴ Herr André Gisler ist seit 2015 Tourismusdirektor bei Flims Laax Falera Management.

⁵ Herr Thomas Christen ist seit 2019 Tourismusdirektor bei Andermatt-Uri-Untertal Tourismus GmbH.

Einen direkten Einfluss eines Windparks in Laax auf gewisse Tourismusaktivitäten sieht Herr Gisler nicht, ausser, dass dieser den nachhaltigen Umgang mit Energie und die Glaubwürdigkeit ihres Handelns stärken würde. So hat auch Herr Reto Fry keinerlei Bedenken und sähe einen Windpark sogar eher als Vor- als als Nachteil für den Tourismus. Möglichkeiten, wie beispielsweise eine Aussichtsplattform an einer Windturbine anzubringen, sieht er positiv. Aus seiner Sicht ist es den TouristInnen ebenso gleichgültig, ob eine Seilbahn in der Landschaft steht. Diese wird genauso wie die Stromproduktion dafür gebraucht, dass sie ihre Urlaubsaktivitäten ausführen können. Zudem erklärt Herr Reto Fry, dass es auch zu „einem guten Gewissen“ der TouristInnen beiträgt.

Herr Christen erklärt, dass der Windpark Gütsch sowie die Windkraftanlagen auf dem Gotthard „ohne Ende“ von TouristInnen fotografiert werden und er bisher keinen „Angriff“ in Bezug auf einen negativen Einfluss auf das Landschaftsbild vernommen hat. Er stützt seine Aussage auch darauf ab, dass das Elektrizitätswerk Ursern, das den Windpark Gütsch betreibt, den Projektprozess „sauber durchgezogen“ hat und alle Beteiligten informiert hat. Die verschiedenen Tourismusaktivitäten, die in der Ferienregion Andermatt angeboten werden, sieht Herr Christen positiv durch den Windpark Gütsch beeinflusst. Die BesucherInnen sind sehr neugierig und interessiert, dies zeigt sich auch anhand der Führungen, die mit dem Elektrizitätswerk Ursern angeboten werden. Der Windpark Gütsch wird laut Herrn Christen als „ergänzendes Modul“ angesehen, das als positiv für die Tourismusentwicklung betrachtet wird. Er stört nicht im „Sichtfeld der Gemeinde“ und macht auch keine Geräusche im Dorf. Sobald Bedenken aufkommen, werden diese vom Elektrizitätswerk Ursern ernstgenommen, weswegen sie auch Studien zu Zugvögeln und Fledermäusen und weiteres durchführen, um sich stetig zu verbessern.

Herr Gisler erwähnt, dass sie in der Ferienregion Laax zum Thema rund um die Nutzung von Wasser und die Energieerzeugung durch Wasserkraft ein Kulturprojekt betreiben, wobei diese auf verschiedene Wege den BesucherInnen der Region nähergebracht wird. „Sinfonia d’aua“ ist ein technisches Kulturprojekt der Flims Electric AG, das in enger Zusammenarbeit mit der Gemeinde Flims und der Weissen Arena AG und zusammen mit dem Ingenieurbüro Straus aus Chur entwickelt und umgesetzt wurde (*Sinfonia d’aua*, 2023). Einerseits wird anhand der Webseite erklärt, welche sechs Wasserkraftwerke durch die Flims Electric AG betrieben werden und wie diese mit der Natur und dem Menschen zusammenspielen. Zudem wird das Anergienetz und konkret die Produktion von Ökostrom durch Wasserkraft erklärt. Andererseits wird konkret auf die verschiedenen Wassersysteme, den Umgang mit Trinkwasser, eine ökologische und nachhaltige Beschneidung, Photovoltaikanlagen und Mehrzweckspeicher eingegangen. Die verschiedenen Bereiche sind unter ihrer Vision vereint, die erste 100% CO₂-neutrale Alpendestination zu werden. Das Projekt beinhaltet nebst den Informationen auf der Webseite auch Informationstafeln vor Ort sowie einen Informations- und Seminarraum im Besucherzentrum Punt Gronda, wo Führungen und mögliche anschliessende Wanderungen angeboten werden.

Herr Gisler meint, dass es diese Möglichkeiten auch in Zusammenhang mit Windenergie umzusetzen gäbe, und sich so gewisse B2B-Interessen beziehungsweise Führungen ergeben könnten.

Auf die Frage zu Attraktionen und/oder Aktivitäten zusammen mit dem Windpark, die für eine zusätzliche Wertschöpfung zu nutzen wären, erwähnt Herr Christen einzig die Führungen, die heute im Sommer durchgeführt werden und dass diese infolge Eiswurfs von den Flügeln im Winter nicht stattfinden können.

5.3 Statistische Kennzahlen

5.3.1 Logiernächte

Die Logiernächte in Andermatt haben seit dem Jahr 2005 insgesamt zugenommen, wobei ein zwischenzeitlicher Rückgang in den Jahren 2009 bis 2012 zu verzeichnen ist (Abbildung 8). Dieser Zeitraum beinhaltet auch die Zeitperiode, in welcher der Windpark Gütsch grösstenteils errichtet und fertiggestellt wurde (2010-2012) und das Tourismusresort durch den Unternehmer und Investor Samih Sawiris geplant wurde (2009-2011).

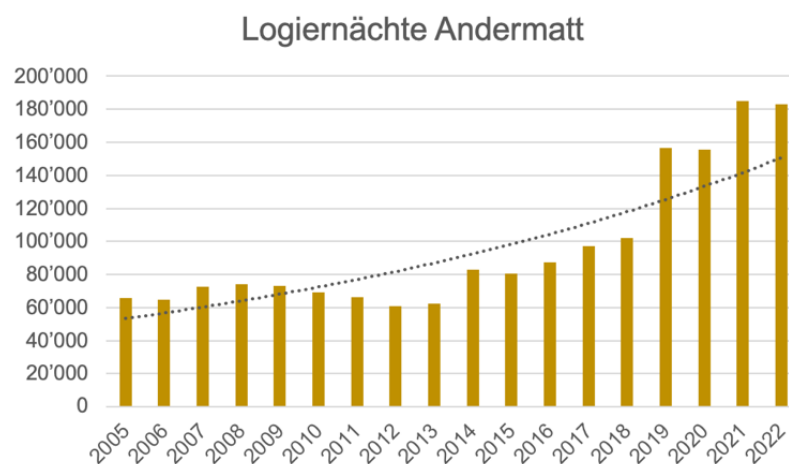


Abbildung 39: Darstellung der Logiernächte in Andermatt (2005 bis 2022)

Alleinige Rückschlüsse auf diese beiden Ereignisse zu ziehen, wäre jedoch nicht realistisch. Der Schweizer Tourismus verzeichnete in den Jahren 2009 bis 2012 landesweit einen Rückgang, besonders beeinflusst durch die Weltwirtschaftskrise und den relativ tiefen Wechselkurses des Euro gegenüber dem Schweizerfranken (Bundesamt für Statistik, 2021b). Die nachfolgende Abbildung zeigt demnach die prozentuale Veränderung der Logiernächte im Vergleich zum vorgängigen Jahr für die Schweiz und Andermatt seit 2005.

In den Jahren 2013 und 2014 nehmen die Logiernächte gesamtschweizerisch und in Andermatt im Vergleich zum jeweiligen Vorjahr wieder zu. Andermatt verzeichnet eine Zunahme der Logiernächte im Jahr 2014 im Vergleich zum Vorjahr von 33.54%, was sich hauptsächlich durch die Eröffnung des ersten Luxushotels, dem „The Chedi Andermatt“, im Jahr 2013 im Rahmen des Tourismusresorts der Andermatt Swiss Alps AG von Samih Sawiris erklären lässt. Nach einem kurzweiligen, erneuten Rückgang im Vergleich zum Vorjahr im Jahr 2015 von gerade einmal 2.79% in Andermatt, was mit der Aufhebung des „Fixen Wechselkurses“ der SNB einhergeht, steigt die Anzahl der Logiernächte in Andermatt wieder an.

Ab dem Jahr 2016 wurde das Skigebiet „SkiArena“ durch die Andermatt Swiss Alps AG weitergehend ausgebaut, wobei besonders auch die Gondelbahn von Andermatt auf den Nätschen und weiter auf den Gütsch erfolgte und somit die erste Etappe der Skigebietserweiterung darstellte. Diese erschliesst damit direkt auch den Windpark Gütsch durch einen direkten Zugang von Andermatt aus. Die Logiernächte steigen demnach in den Jahren 2016 um 8.39% im Vergleich zum Vorjahr und um weitere 10.91% im Vergleich zum Vorjahr im Jahr 2017.

Im Jahr 2018 wurde das zweite grosse Hotel, das „Radisson Blu Hotel Reussen“ durch die Andermatt Swiss Alps AG errichtet, was den starken Anstieg der Logiernächte gegenüber dem Vorjahr im Jahr 2019 von 53.73% erklären dürfte.

Im Jahr 2020 wurden die negativen Zahlen zu den Logiernächten schweizweit stark durch die Covid-19 Pandemie beeinflusst, wobei Andermatt jedoch nur einen Rückgang zum Vorjahr von 0.51% verzeichnet, während gesamtschweizerisch ein Rückgang von 40.02% vorliegt. Die Logiernächte Ausland sind in Andermatt in diesem Jahr im Vergleich zum Vorjahr um 31.62% gesunken, während die Logiernächte Inland im Vergleich zum Vorjahr um 18.99% gestiegen sind. Ab 2021 steigen die Logiernächte sowohl gesamtschweizerisch (+24.56% im Vgl. zum Vorjahr) als auch in Andermatt (+18.82% im Vgl. zum Vorjahr) wieder. Dabei sind die Logiernächte Ausland im Vergleich zum Vorjahr um 19.27% und die Logiernächte Inland um weitere 18.66% im Vergleich zum Vorjahr gestiegen. Im Jahr 2022 liegt ein minimaler Rückgang der Logiernächte in Andermatt im Vergleich zum Vorjahr von 1.03% vor, wobei die Logiernächte

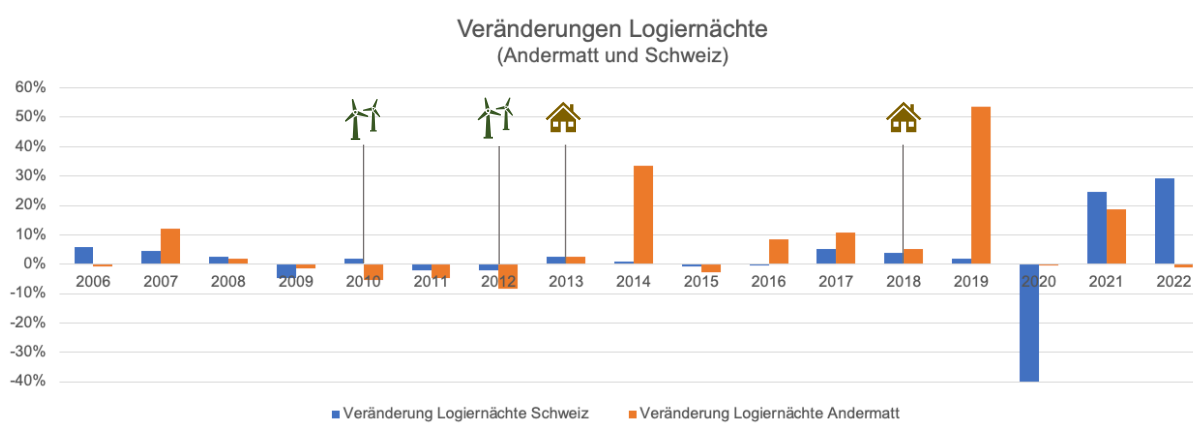


Abbildung 40: Darstellung der Veränderungen der Logiernächte im Vergleich zum Vorjahr in Andermatt und der Schweiz (2006 bis 2022)

Ausland im Vergleich zum Vorjahr um 46.32% gestiegen sind und die Logiernächte Inland im Vergleich zum Vorjahr um 18.17% gesunken sind.

5.3.2 Beschäftigte im Tourismus

Die Vollzeitäquivalente der Beschäftigten im Tourismus (dabei wurden die Bereiche Gastronomie und Beherbergung berücksichtigt) im Kanton Uri sind seit 2011 stetig angestiegen, ausgenommen eines Rückgangs von 6.7% im Vgl. zum Vorjahr im Jahr 2012 und eines minimalen Rückgangs im Vergleich zum Vorjahr im Jahr 2017 von 0.09%. Besonders angestiegen im Vergleich zum Vorjahr sind die VZÄ der Beschäftigten im Jahr 2013, und zwar um 18.86%. Zwei weitere deutliche Anstiege im Vergleich zum jeweiligen Vorjahr wurden im Jahr 2018 (+8.17%) und im Jahr 2020 (+9.72%) festgestellt.

Da der *Anteil der Beschäftigten im Tourismus im Urserntal*, welches die Gemeinden Andermatt, Hospental und Realp einschliesst und somit die relevante Tourismusregion um Andermatt darstellt, leider nur für die beiden Jahre 2015 und 2019 bekannt sind und die Entwicklung der Beschäftigten im Kanton annähernd linear ist, wurden die übrigen Anteile für die Jahre zwischen 2011 und 2021 ebenfalls statistisch linear angenähert. Demnach zeigen sich in dieser statistisch angenäherten Untersuchung ähnliche Schwankungen. Es wurde ein minimaler Rückgang von 0.68% im Vergleich zum Vorjahr im Jahr 2012 errechnet, ansonsten haben die VZÄ der Beschäftigten jedoch ausschliesslich zugenommen. Der Rückgang im Jahr 2012 im Vergleich zum Vorjahr könnte sich durch die rückgängigen Logiernächte seit 2009 erklären. Ein besonderer Anstieg (+26.06%) im Vgl. zum Vorjahr der VZÄ der Beschäftigten im Tourismus im Urserntal wurde im Jahr 2013 errechnet, sowie in den Jahren 2018 (+13.2%) und 2020 (+14.39%). Diese beiden zeitlichen Abschnitte dürften sich durch die Eröffnung der Hotels „The Chedi Andermatt“ und „The Radisson Blu Hotel Reussen“ erklären lassen.

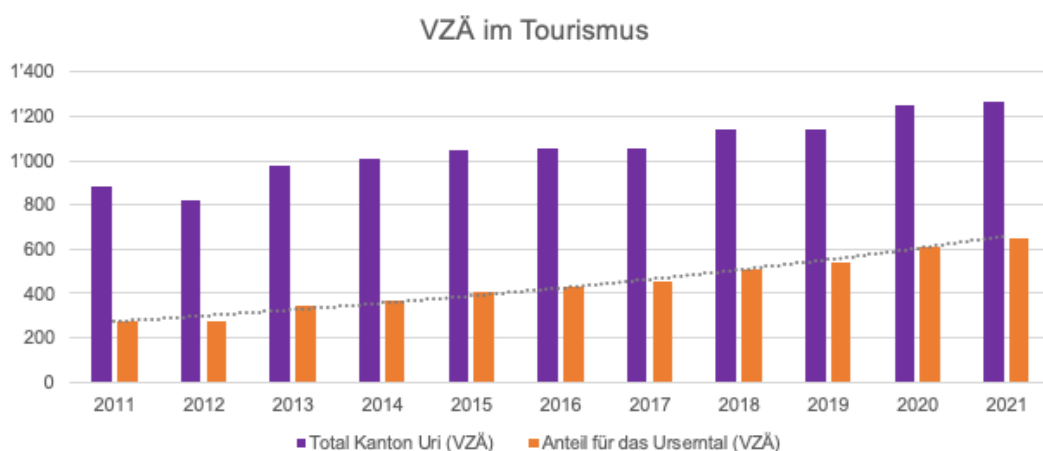


Abbildung 41: Darstellung der Beschäftigten (in Vollzeitäquivalenten) im Kanton Uri und anteilmässig im Urserental (2011 bis 2021)

5.3.3 Wertschöpfung durch die Führung „Energie und Umwelt“

Die Führung „Energie & Umwelt“ wurde im Winter 2020/21 lanciert und ab Sommer 2021 offiziell angeboten. Im ersten Jahr gab es insgesamt 4 Buchungen mit total ca. 30 Personen. Im Jahr 2022 waren es bereits 13 Buchungen mit insgesamt 197 Personen. Seit Beginn des Jahres 2023 wurde die Führung bereits fünfmal gebucht und insgesamt haben 140 Personen teilgenommen.

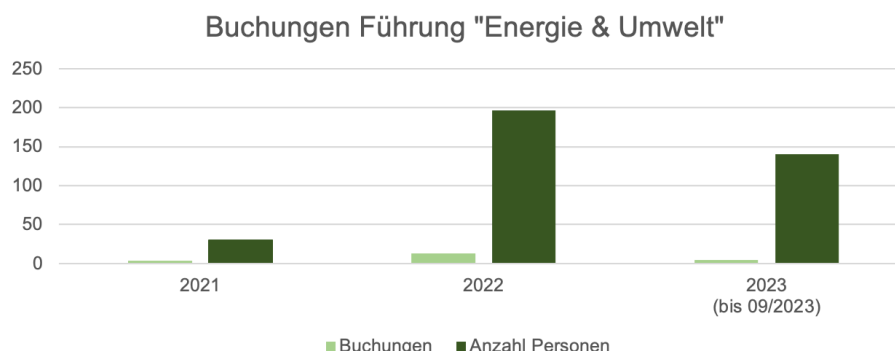


Abbildung 42: Darstellung der Anzahl Buchungen und Personen bei Führungen ("Energie & Umwelt") in Andermatt (2021 bis 2023)

Eine Führung kostet normalerweise CHF 150, kann jedoch auch zwischen CHF 200 und 250 kosten, sollte sie länger dauern. Demnach wurden im Jahr 2021 Einnahmen in der Höhe von CHF 745 verzeichnet, während im Jahr 2022 CHF 2112.50 verzeichnet wurden. Im Jahr 2023 wurden bisher CHF 1050 verzeichnet. Die Ausgaben werden vor allem durch die Löhne der vier Guides berechnet, welche zu einem Stundenlohn von CHF 60 angestellt sind. Diese belaufen sich auf ca. CHF 473.50 für das Jahr 2021, CHF 1440 für das Jahr 2022 und CHF 630 für das Jahr 2023. Dies führt zu einem Gewinn von ca. CHF 271.46 für das Jahr 2021, CHF 672.50 für das Jahr 2022, und CHF 420 für das Jahr 2023. Entsprechend ergeben sich mittlere Einnahmen pro Führung von CHF 186.25 und mittlere Ausgaben von CHF 118.38 pro Führung, was einem mittleren Gewinn von CHF 67.87 pro Führung entspricht.

Jahr	Buchungen	Anzahl Personen	Einnahmen	Ausgaben	Gewinn	Mittlere Einnahmen p.F.	Mittlere Ausgaben p.F.	Mittlerer Gewinn p.F.
2021	4	30	745*	473.54*	271.46*	-	-	-
2022	13	197	2112.50	1440	672.50	162.50	110.77	51.73
2023	5	140	1050	630	420	210	126	84
						186.25**	118.38**	67.87**

*) wurden errechnet anhand der Mittelwerte, da keine Daten zu Verfügung stehen

**) stellen die Mittelwerte über die Jahre 2022 und 2023 Jahre dar

Des Weiteren, und nicht in dieser Rechnung miteinbezogen, wird die Führung durch eine Mitarbeiterin, die als Produktmanagerin tätig ist, administrativ und organisatorisch verarbeitet und die Tourbuchung wird von den Mitarbeiterinnen der Touristeninformation erfasst. Eine Aussage darüber zu machen, wie vielen Vollzeitäquivalenten dies entspricht, ist leider durch die schwankende Nachfrage nicht explizit zu beziffern.

6 Diskussion

6.1 Diskussion der Ergebnisse

6.1.1 Deskriptive Statistik

Ein Grossteil der untersuchten Stichprobengruppe zeigte sich in Energiefragen positiv gestimmt darüber, dass die Energiewende in der Schweiz zu langsam vorangeht und entschied sich für ein breitdiversifiziertes Portfolio an erneuerbaren Energien, wenn sie den Strommix der Schweiz bestimmen könnten. Zudem anerkennt auch ein Grossteil die Wichtigkeit des Klimaschutzes in der Schweiz und verfügt über ein Abonnement des öffentlichen Verkehrs. Dennoch achtet sich nur knapp ein Viertel darauf, ob die von ihnen gewählte Reisedestination Wert auf Fortschrittlichkeit und Nachhaltigkeit legt. Eine deutliche Mehrheit sieht den Windpark Gütsch beziehungsweise einen potenziellen Windpark in Laax als ein solches Zeichen für Nachhaltigkeit und Fortschrittlichkeit des jeweiligen Tourismusorts. Etwas mehr als die Hälfte der untersuchten Stichprobengruppe zeigt sich zudem interessiert an der Windenergie und deren Technologie, allerdings weiss nur ungefähr ein Drittel der Studienteilnehmenden, wo sich die nächstgelegene Windturbine befindet. Es würde sich demnach anbieten, Massnahmen zu ergreifen, die die Sichtbarkeit von bestehenden Windkraftwerken und ihre Zugänglichkeit entsprechend erhöhen.

Nahezu alle Umfrageteilnehmenden haben die jeweilige Tourismusdestination nicht zum ersten Mal, sondern in einer gewissen Regelmässigkeit besucht. Sowohl in Andermatt als auch in Laax sind Wintersportarten die meisterwähnten Aktivitäten, die am jeweiligen Ort nachgefragt werden. Die zweitmeist gewählte Aktivität bezog sich auf Wandern oder andere Aktivitäten in der freien Natur.

Hervorzuheben ist zudem, dass ein Grossteil der Umfrageteilnehmenden zwischen 25 bis 24 Jahre als ist und ein Viertel zwischen 55 und 64 Jahre als ist. Zudem haben über die Hälfte der Teilnehmenden einen Universitäts- oder Fachhochschulabschluss.

Die betrachtete zielgerichtete Stichprobe stellt in sich eine diversifizierte Gruppe dar, jedoch zeigt sich eine starke Tendenz dazu, dass die Personen ein gewisses Bewusstsein zu den Themen erneuerbare Energien, Klima, Nachhaltigkeit und Fortschrittlichkeit mitbringen und sie besonders einer beziehungsweise zweier Altersgruppen entspringen.

6.1.2 Experimentelle Resultate

Besonders die Auswirkungen einer Windkraftanlage auf das Landschaftsbild hat sich als ein wichtiger Faktor erwiesen, für welchen die Akzeptanz von ebendiesen positiv oder negativ abhängig sein kann. Dies kann insbesondere davon abhängen, ob eine Person einen Standort für eine Windkraftanlage als sinnvoll betrachtet. Gemäss Hunziker et al. (2014) wird der Errichtung einer Windkraftanlage in der Nähe von bestehender Infrastruktur mit höherer Akzeptanz

begegnet. Auch die Resultate der vorliegenden Studie haben gezeigt, dass die untersuchte Stichprobengruppe es als sinnvoll betrachtet, eine Anlage in der Nähe von bestehender Infrastruktur zu errichten, womit die Teilhypothese 1a bestätigt werden konnte. Auch Herr André Gisler, Tourismusdirektor in Laax, sieht den Standort als sinnvoll, da die Landschaft bereits durch die touristische Infrastruktur vorbelastet ist. Wie auch einzelne Wortmeldungen von den Umfrageteilnehmenden erwähnt haben, ist die Berücksichtigung der Effizienz einer Anlage und der Zonnennutzung für Herrn Gisler sehr wichtig. Dies unterstreicht auch Herr Thomas Christen, Tourismusdirektor in Andermatt, der zudem noch auf die ökologische Produktion der Windkraftanlage hinweist.

Des Weiteren wurde unter anderem von Bidwell (2023) und Seitz (2011) erklärt, dass die Akzeptanz von Windenergieanlagen in Bezug auf das Landschaftsbild höher ist, wenn eine bestehende Anlage betrachtet wird, als bei einer potenziellen Anlage. Dies konnte durch den Vergleich der beiden Umfragegruppen Andermatt und Laax untersucht werden. Durch eine leicht höhere gemittelte Aussage, dass der Standort Gütsch als sinnvoller betrachtet wird als ein potenzieller Windpark in Laax, lässt sich die zweite Teilhypothese 1b in Hinsicht der Sinnhaftigkeit zwar bestätigen, ist statistisch gesehen jedoch eher unzureichend. Dabei zu berücksichtigen ist, dass die Werte für die beiden Gruppen sehr nahe beieinander liegen und deshalb keinen statistisch signifikanten Unterschied zu verzeichnen haben. Die Aussagen beider Gruppen können jedoch so interpretiert werden, dass beide Standorte insgesamt als „sinnvoll“ beurteilt wurden. Personen, die die Standorte als weniger sinnvoll bezeichnet haben oder denen die Beurteilung schwerfiel, haben besonders Industriezonen und das Mittelland erwähnt, wo sich ein Windpark eher zu befinden habe. Eine Industriezone könnte als ein durch bestehende Infrastruktur vorbelastetes Gebiet beschrieben werden, wonach sich die Frage ergeben würde, ob ein Tourismusgebiet möglicherweise als zu wenig „vorbelastet“ angesehen wird durch das Vorhandensein von beispielsweise Mästen und Bergbahnen, oder diese durch den Betrachter eher toleriert werden. Besonders die Umfrageteilnehmenden zu Andermatt lehnen die Errichtung einer Windkraftanlage abseits von existierender Infrastruktur in der „unberührten Natur“ eher ab als Umfrageteilnehmende zu Laax. Dies lässt die Spekulation offen, ob BesucherInnen in Andermatt durch das bestehende Windparkprojekt Gütsch als Vorzeigebispiel von einem Windpark, der in touristische Infrastruktur integriert ist, überzeugt wurden.

Zudem wurden die Umfrageteilnehmenden konkret dazu gefragt, ob sie den Windpark Gütsch beziehungsweise einen potenziellen Windpark in Laax als für das Landschaftsbild bereichernd oder störend wahrnehmen. Hierzu wurden die beiden Gruppen ebenfalls verglichen, um einen Unterschied in der Wahrnehmung zwischen einem bereits bestehenden und einem sich vorzustellenden Windpark zu untersuchen. Auch bei dieser Untersuchung lagen die Ergebnisse beider Gruppen sehr nah beieinander. Wider der gestellten Teilhypothese 1b sehen die Umfrageteilnehmenden zu Andermatt das Landschaftsbild durch den Windpark eher als „gestört“ im Gegensatz zu Umfrageteilnehmenden zu Laax. Auch bei dieser Untersuchung lagen die Ergebnisse beider Gruppen sehr nah aneinander und eine besonders hervorzuhebende Erkenntnis ist

hierbei, dass die Aussagen beider Gruppen darauf hinweisen, dass das Landschaftsbild durch einen Windpark als weder bereichernd noch störend empfunden wird und es somit die meisten Teilnehmenden demgegenüber indifferent stehen. Bei dieser Betrachtung ist auch nicht zu vergessen, dass sich die beiden Untersuchungsgruppen in ihrer Grösse unterscheiden.

Ähnlich haben sich die Umfrageteilnehmenden beider Gruppen dazu geäußert, ob sie sich bei ihrer Tourismusaktivität durch den Windpark Gütsch bzw. einen potenziellen Windpark auf dem Crap Sogn Gion eher bereichert oder gestört fühlen. Hierbei wurde sogar noch stärker diesbezügliche Indifferenz ausgedrückt als beim Landschaftsbild. Dies dürfte darauf hinweisen, dass der wahrgenommene Einfluss auf das Landschaftsbild von höherer Relevanz ist. Nur wenige Personen beider Gruppen haben sich dazu geäußert, dass sie sich bei ihrer Tourismusaktivität durch den bzw. einen Windpark gestört oder bereichert fühlen, wobei sich Personen in der Tendenz eher durch diesen bereichert als gestört fühlen. In Studien von Frantál & Kunc (2011), Tverijonaite et al. (2022) oder Weiss (2015) wurde hergeleitet, dass sich SommertouristInnen eher durch eine Windkraftanlage gestört fühlen als WintertouristInnen. In der vorliegenden Studie konnte dies jedoch nicht ausreichend bestätigt werden. Obwohl die Untersuchungsgruppe an WintertouristInnen deutlich grösser war als die Gruppe der SommertouristInnen, haben sich die SommertouristInnen im Mittel eher dazu geäußert, dass sie sich weniger „gestört“ bzw. eher „bereichert“ fühlen durch den bzw. einen Windpark bei ihrer Tourismusaktivität. Unter der Berücksichtigung der bereits erwähnten vorherrschenden Indifferenz, lässt sich jedoch eher eine Aussage dazu machen, dass in der vorliegenden Studie kein signifikanter Unterschied zwischen Sommer- und WintertouristInnen gezeigt wird und der Einfluss auf die Tourismusaktivität womöglich durch andere Faktoren besser beschrieben werden könnte. Auch Herr Gisler erwähnte, dass er sich keinen negativen Einfluss einer Windkraftanlage auf eine Tourismusaktivität vorstellen könnte. Er empfinde eine solche eher noch als Unterstreichung der Glaubwürdigkeit im Handeln des Unternehmens, Nachhaltigkeit und Ökologie einen hohen Stellenwert einzuräumen. Herr Reto Fry, Umweltbeauftragter in Laax, sieht sogar Potential für den Windtourismus – beispielsweise anhand einer Aussichtsplattform – und erwähnt eher den positiven Effekt, der bei TouristInnen ausgelöst wird, wie dass beispielsweise „zu einem guten Gewissen“ beitragen wird. Die Erfahrungen von Herrn Thomas Christen zeigen auch, dass sie bisher keine negativen Rückmeldungen dazu erhalten haben, dass der Windpark Gütsch bei einer Tourismusaktivität „gestört“ hätte. Er sieht diese sogar positiv durch den Windpark beeinflusst. Diese Berichte unterstreichen die Ergebnisse der Umfrage, dass die Anwesenheit eines Windparks keinen spezifisch negativen Einfluss auf das Tourismuserlebnis hat.

Die Mehrheit der Umfrageteilnehmenden haben angegeben, dass sie den Windpark Gütsch bzw. einen potenziellen Windpark in Laax als ein Zeichen von „Nachhaltigkeit und Fortschrittlichkeit“ der jeweiligen Tourismusdestination sehen und so hat auch die Mehrheit angegeben, dass sie sich zumindest „ab und zu“ bei der Wahl einer Reisedestination auf diese Werte achtet.

Herr André Gisler erwähnt, dass ein Windpark „die Glaubwürdigkeit ihres Handelns“ unterstützen würde, da sie sich in der Unternehmensstrategie diesen Werten klar verpflichten. Herr Reto Fry erwähnt zudem, dass damit den TouristInnen auch vor Augen geführt würde, dass durch den Tourismus Strom verbraucht wird, der irgendwo anders erzeugt werden muss, womit die Stromproduktion und -konsumation direkt ersichtlich werden. Herr Thomas Christen weist hingegen eher darauf hin, dass „der Nutzen“ im Vordergrund steht. Dieser besteht darin, dass der Windpark Gütsch besonders im Winter Strom produziert.

Ob die Wahrnehmung des Landschaftsbilds durch die Wichtigkeit der Werte von „Nachhaltigkeit und Fortschrittlichkeit einer Tourismusdestination“ geprägt wird, wurde unter Teilhypothese 1d untersucht, wozu aber kein statistisch signifikanter Zusammenhang der beiden Variablen festgestellt werden konnte. Es liegt jedoch eine gewisse Tendenz vor, dass Personen, denen solche Werte wichtig sind, das Landschaftsbild weniger als „gestört“ bzw. als „bereichert“ wahrnehmen durch einen Windpark. Hinzuzufügen ist der Bericht von Herrn Fry, insofern als dass die BesucherInnen in Laax einen Windpark zu 80% befürworteten und die Destination Laax durch ihre Vision und Werte in ihrem Sinne ebenfalls BesucherInnen anziehen dürfte, denen solche Werte von höherer Bedeutung sind.

Einen Windpark touristisch zu nutzen (Windtourismus) birgt das Potential, die Akzeptanz diesem gegenüber zu erhöhen. Herr Thomas Christen spricht von vielen interessierten und neugierigen TouristInnen, die die Windkraftanlagen fotografieren und ihr Interesse zeigt sich auch anhand der Führungen mit dem EW Ursern, was die Tourismusentwicklung positiv beeinflusst. Um die Teilhypothese 2a zu bestätigen, ob ein positiver Effekt der Akzeptanzerhöhung bei TouristInnen durch eine Führung durch den Windpark Gütsch oder einen anderen Windpark vorliegt, konnte anhand der untersuchten Stichprobengruppe nicht gezeigt werden. Lediglich drei Personen der Umfrageteilnehmenden haben jemals an einer Führung durch einen Windpark teilgenommen und bei allen Personen ist die Wahrnehmung des Einflusses eines Windparks auf das Landschaftsbild und die Tourismusaktivität unterschiedlich ausgefallen. Da das Interesse an der Windenergie und ihrer Technologie bei den Umfrageteilnehmenden vorhanden ist, jedoch nur drei Personen jemals an einer Führung teilgenommen haben, besteht das Potential, diese interessierten Personen durch eine Führung abzuholen. Die Zahlen zur Führung „Energie & Umwelt“ in Andermatt und auch die Aussagen von Herrn Thomas Christen haben gezeigt, dass die Nachfrage und somit das Bedürfnis steigt, dass sich interessierte Personen durch Führungen zur Windenergie weiterbilden können. Zudem wurde im Fragebogen direkt abgefragt, welche Aktivitäten in Verbindung mit einer Windkraftanlage die Person überzeugen würden, eine solche zu besuchen, wobei die Antwortmöglichkeit „Führung“ mit 43 Stimmen die zweitmeist genannte Antwort war.

Herr Fry sieht Windkraftanlagen ebenso als Potenzial, weitere Tourismusaktivitäten damit zu verbinden, wie beispielsweise eben anhand einer Aussichtsplattform. Das Beispiel von *Sinfonia d'aua* (2023) zeigt zudem, dass bereits praktische Kenntnisse in der Tourismusregion Laax

vorhanden sind, künstlerische und spielerische Ansätze zu nutzen, um TouristInnen über die regionale Stromproduktion aufzuklären und zu unterrichten. Ebensolche Ansätze wurden auch in der Umfrage durch die Mehrheit der Umfrageteilnehmenden befürwortet, um einen erleichterten Zugang zum Thema Windenergie zu schaffen

Schliesslich wurde anhand der letzten Teilhypothese 2a untersucht, wie sich der Tourismussektor und seine regionale Wertschöpfung unter der Anwesenheit des Windparks Gütsch entwickelt haben. Dazu wurde unter der Berücksichtigung von schweizweiten Entwicklungen festgestellt, dass die Errichtung und stark positive Entwicklung des Tourismusresorts der Andermatt Swiss Alps AG unter der Anwesenheit des Windparks Gütsch erfolgreich entstanden und weiterentwickelt wurde. Trotz eines Rückgangs der Logiernächte in den Jahren der Errichtung des Windparks, was durch einen schweizweiten Rückgang begleitet wurde, stiegen die Logiernächte besonders in den Jahren der Eröffnung zweier grosser Hotels deutlich an und es folgten Jahre, in denen die Logiernächte weiterhin zunahmen. Dies dürfte auch mit dem Ausbau des Bergbahnnetzes zusammenhängen, der den Windpark Gütsch indes für TouristInnen noch zugänglicher gemacht hat. Entsprechend wurde auch die regionale Wertschöpfung anhand der Schaffung von Arbeitsplätzen betrachtet. Die positive Entwicklung des Arbeitsmarktes im Tourismussektor weist zusätzlich darauf hin, dass der Windpark Gütsch keinen negativen Einfluss auf die Tourismusentwicklung in der Region genommen hat. Zudem wurde gezeigt, dass durch die Führung zusätzliche Gewinne erzielt werden können und in diesem Rahmen ebenfalls weitere Arbeitsplätze geschaffen werden. Somit konnte die letzte Teilhypothese 2b bestätigt werden, dass unter der Anwesenheit des Windparks Gütsch positive regionale Wertschöpfung generiert wird und durch diesen sogar zusätzliche positive Wertschöpfung erzeugt wird.

Um das zukünftige Potential der Wertschöpfungsgenerierung durch den Windpark Gütsch oder einen potenziellen Windpark in Laax zu untermauern, wird hierbei zudem auf die Antworten hingewiesen zur Frage, welche Aktivität die befragte Person davon überzeugen würde, den bzw. einen Windpark zu besuchen. Lediglich 15 Personen haben angegeben, dass sie den bzw. einen Windpark niemals besuchen würden, und besonders künstlerische Darstellungen, spannende Führungen und eine Aussichtsplattform sind bei vielen der befragten Personen auf Interesse gestossen. Des Weiteren wurde auch ein Kombi-Angebot von Transportleistungen zum Windpark, ein Foxtrail von Windrad zu Windrad oder ein Musikfestival von bis zu einem Drittel der befragten Personen als Attraktion genannt.

6.2 Implikationen

Die untersuchte Stichprobengruppe hat gezeigt, dass das Interesse an einer in Bezug auf die Energiewende progressiven Entwicklung vorhanden und dass auch ein grundlegendes Interesse an der Windenergie und ihrer Technologie gegeben ist. Es wird gewünscht, dass diese gefördert

wird und sie wird zudem als Zeichen für „Nachhaltigkeit und Fortschrittlichkeit“ einer Tourismusdestination betrachtet. Ein ressourcenschonender Umgang im Tourismussektor gewinnt zunehmend an Wichtigkeit.

Es hat sich auch gezeigt, dass die Umfrageteilnehmenden nicht darüber informiert sind, wo sich die nächstgelegene Windturbine ihres Wohnorts befindet, jedoch das Interesse für die Technologie vorhanden ist. Die Windenergie ist den TouristInnen in der Schweiz nicht bekannt und somit ist das Potential nach wie vor vorhanden, weitere positive Beispiele zu erschaffen, um die Windenergie näher an die Bevölkerung zu bringen. Unter anderem hat dazu auch das Beispiel der Windkraftanlage in Entlebuch, Luzern, untersucht durch Seitz (2011), die Wichtigkeit von Vorzeigebispielen gezeigt.

Zu beachten dabei ist besonders die Wahl eines geeigneten Standorts, der auch von den Personen, die sich in dieser Region aufhalten, als sinnvoll erachtet wird. Die Nähe zu bestehender Infrastruktur scheint hierbei ein wichtiger, leitender Entscheidungsfaktor zu sein.

Wenn sich die Pläne für die Errichtung eines Windparks in Laax weitergehend entwickeln würden, würde dieser Anlage aus touristischer Sicht wohl keine Opposition entgegenstehen, da der Standort von TouristInnen als sinnvoll eingestuft wurde. Besonders zu betonen ist auch die ausgedrückte Indifferenz dazu, ob ein Windpark das Landschaftsbild oder eine Tourismusaktivität spezifisch beeinflusst. Vielen Personen scheint es auch „egal“ zu sein, ob ein Windpark beim Wandern oder Skifahren gesichtet werden kann. Diese Ergebnisse und die positive Entwicklung der Tourismusregion Andermatt dürfte ein gutes Beispiel dafür sein, dass ein alpiner Windpark in der Schweiz keine Abschreckung darstellt.

Im Gegenteil dazu sollte das Potential nicht vergessen gehen, Windkraftanlagen für zusätzliche Wertschöpfung zu nutzen und damit die Akzeptanz gegenüber der Windenergie generell anzuheben. Das vorhandene Angebot sowohl in Andermatt als auch in Laax dürfte zeigen, dass die Nutzung von Marketingwerkzeugen und spielerischen Ansätzen, die Umgebung näher kennenzulernen und etwas über die Region zu lernen, bereits bekannt und vorhanden sind. So wie in Andermatt der Berghüttenpass als „Sammelspiel“ eingesetzt wird und mit den Sozialen Medien verknüpft wird oder Audio-Touren und digital unterstützte Schnitzeljagden angeboten werden, könnte dies auch auf den Windpark erweitert werden. Die steigende Nachfrage nach der Führung „Energie & Umwelt“ in Andermatt deutet bereits daraufhin, dass ein Wertschöpfungspotential durch zusätzliche Einnahmen künftig interessant sein könnte. Auch in Laax besteht mit den Themenwanderungen, die durch ExpertInnen begleitet werden und interaktiv mit einer App erlebt werden können, ein Angebot, dass sich auf eine Windkraftanlage übertragen liesse. Zudem zeigt auch das Projekt *Sinfonia d’aua* (2023) das Potenzial eines künstlerischen Ansatzes, den Besuchenden die Energiewelt in der Region näher zu bringen und es könnte durch eine Windkraftanlage ergänzt werden. Des Weiteren § könnte bei der Planung einer Windkraftanlage

in Laax daher besonders darauf geachtet werden, diese in Events miteinzubauen und Attraktionen, wie beispielsweise eine künstlerische Installation, ein Musikfestival oder den Snowpark damit zu verbinden. Das bereits bestehende Angebot beider Regionen bietet zahlreiche Beispiele, die auch von anderen Tourismusregionen als Inspiration genutzt werden können. Dabei zu beachten ist laut Bidwell (2013), Spiess et al. (2014) und Spiess et al. (2015) besonders die Einbindung der und die transparente Kommunikation mit der lokalen Bevölkerung und sämtlichen weiteren Stakeholdern, um die Sinnhaftigkeit und damit die Akzeptanz eines Projekts möglichst zu erhöhen. Dazu gehört, den entstehenden, lokalen Nutzen und Mehrwert einer geplanten Windkraftanlage klar zu kommunizieren. Dies wurde auch durch die Aussagen von Herrn Thomas Christen unterstrichen, dass die integrative und rücksichtsvolle Vorgehensweise des Elektrizitätswerk Ursern ein entscheidender Faktor für den Erfolg des Windparkprojekts Gütsch darstellt.

Schliesslich dürften auch Marketingwerkzeuge von gewisser Wichtigkeit sein. Einen Windpark aktiv zu bewerben und als integrativen Teil des Landschaftsbildes zu etablieren, könnte sich entsprechend auf verschiedene Tourismusfaktoren erfolgreich auswirken. Darunter beispielsweise die Steigerung des Interesses an der Windkraft und den damit verbundenen Angeboten oder die Aufwertung des „Ortimages“.

6.3 Limitationen

Die vorliegende Studie hat sich auf Personen fokussiert, die in Andermatt bzw. Laax touristisch unterwegs sind, weshalb sich die Teilnahme an der Studie aus unterschiedlichen Gründen nicht als sehr einfach erwies. Personen vor Ort waren häufig gedanklich darauf fokussiert, wofür sie in die jeweilige Reisedestination gekommen sind und haben den QR-Code entgegengenommen, um die Umfrage später auszufüllen. Dass die Personen die Umfrage schliesslich ausgefüllt haben, konnte so nicht garantiert werden. Personen, die darum gebeten worden sind, die Umfrage auf einem iPad vor Ort auszufüllen, waren dieser Anfrage teilweise eher abweisend eingestellt. Diese Umstände haben möglicherweise dazu geführt, dass die Personen die Umfrage unter gewissen Voreinnahmen oder gar nicht ausgefüllt haben. Aufgrund dieser Bedingungen wurde die Umfrage im privaten Umfeld der Autorin gestreut, um weitere Umfrageteilnehmende zu gewinnen. Um die Aussagekraft der Hypothesentests zu stärken, wäre die Betrachtung einer grösseren Stichprobengruppe von Vorteil. Zukünftige Forschung könnte sich dabei insbesondere darauf fokussieren, mehr Zeit und personelle Ressourcen für die Gewinnung repräsentativer Umfrageteilnehmenden einzuräumen.

Besonders die nicht gefundenen Zusammenhänge unter Teilhypothese 1d und 2a weisen darauf hin, dass die individuelle Wahrnehmung von Windparks im Zusammenspiel mit dem Tourismus nicht von einzelnen, klar zu bestimmenden Faktoren abhängt, sondern für jede Person mit mehreren Perspektiven und Einflussfaktoren beschrieben werden kann. Besonders der Einfluss

von demografischen Merkmalen, der Regelmässigkeit des Besuchs und die grundsätzliche Haltung gegenüber erneuerbaren Energien wurden zwar untersucht, jedoch nicht anhand einer Hypothese genauer experimentell geprüft. Es wäre daher sinnvoll, diesen Faktoren höhere Aufmerksamkeit zu schenken. Zukünftige Forschung könnte sich demnach darauf fokussieren, ein umfassenderes Abbild der verschiedenen Einflussfaktoren zu generieren – unter anderem auch anhand einer höher diversifizierten Stichprobengruppe.

Eine weitere, nicht inkludierte Variable stellt der Markt der erneuerbaren Energien in der Schweiz dar. Nebst den politischen Hürden im Planungs- und Bauverfahren wurde in beiden Beispielen – Andermatt und Laax – darüber gesprochen, dass die Finanzierung und die Attraktivität für Stromproduzenten und Windkraftanlagenbauer in der Schweiz weitere Hürden sind. Zukünftige Forschung könnte ergänzend eine Marktanalyse der Windenergie miteinbeziehen und auf ungenutztes Potential schliessen.

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Anhang

A. Umfragebogen

Fragebogen Andermatt

Block A: Einstiegsfragen

1. Bitte geben Sie die Regelmässigkeit für ihren Besuch in Andermatt an
 - a. Ich bin zum ersten Mal in Andermatt
 - b. Ich besuche Andermatt in gewisser Regelmässigkeit (bspw. jedes Jahr, alle zwei Jahre, etc.)
 - c. Ich besuche Andermatt nicht zum ersten Mal, jedoch nicht regelmässig
 - d. Ich war noch nie in Andermatt und bin auch zum jetzigen Zeitpunkt, zu dem ich die Umfrage ausfülle, nicht vor Ort
 - e. Ich war bereits mind. Einmal in Andermatt, bin aber zum jetzigen Zeitpunkt, zu dem ich die Umfrage ausfülle, gerade nicht da
2. Bitte geben Sie den Grund/die Gründe (Mehrauswahl möglich) für Ihren Besuch in Andermatt an
 - a. Ich besuche Andermatt zum Wandern
 - b. Ich besuche Andermatt, um den Windpark zu besichtigen
 - c. Ich besuche Andermatt (hauptsächlich) zum Skifahren/Snowboarden und für andere Wintersportarten
 - d. Ich besuche Andermatt für ein kulturelles Angebot oder Event (Museum, Theater, o.ä.)
 - e. Ich nehme an einer Tour teil (z.B. Dorftour, «Hexenjagd im Urserntal», Audio-Tour, o.ä.)
 - f. Ich bin für einen Wellness-Aufenthalt in Andermatt
 - g. Andere Gründe: ...
3. Wie beurteilen Sie die Geschwindigkeit der Energiewende?
 - a. Die Energiewende geht zu langsam voran
 - b. Das Tempo der Energiewende ist genau richtig
 - c. Bei der Energiewende wird überstürzt gehandelt
4. Für wie wichtig halten Sie es, dass die Schweiz beim Klimaschutz vorankommt?
 - a. Sehr wichtig
 - b. Wichtig
 - c. Weder noch

-
- d. Nicht wichtig
 - e. Sehr unwichtig
5. Wenn Sie den Strommix der Zukunft für die Schweiz bestimmen könnten, welche der folgenden Optionen würden Sie wählen:
- a. breit diversifiziertes Portfolio von erneuerbaren Energien (Wasser, Sonne und Wind)
 - b. Fokus auf Solarenergie und Wasserkraft, Verzicht auf Windenergie
 - c. verstärkte Förderung der Kernenergie
 - d. Bau von Gaskraftwerken
6. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie sich bei der Wahl Ihrer Reisedestination darauf achten, dass diese Wert auf Nachhaltigkeit und Umweltfreundlichkeit legt:
- a. Ich achte mich sehr stark darauf
 - b. Ich achte mich darauf
 - c. Ich achte mich ab und zu darauf
 - d. Ich achte mich nicht darauf
 - e. Ich achte mich ganz und gar nicht darauf
7. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie der folgenden Aussage zustimmen:
- Ich interessiere mich für Windkraft und ihre Technologie.
- a. Ich stimme voll und ganz zu
 - b. Ich stimme zu
 - c. Ich stimme weder zu noch nicht zu
 - d. Ich stimme nicht zu
 - e. Ich stimme überhaupt nicht zu
8. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie der folgenden Aussage zustimmen:
- Ich finde einen Windpark in Nähe von bestehender Infrastruktur zu platzieren sinnvoll.
- a. Ich stimme voll und ganz zu
 - b. Ich stimme zu
 - c. Ich stimme weder zu noch nicht zu
 - d. Ich stimme nicht zu
 - e. Ich stimme überhaupt nicht zu
9. In welcher Entfernung Ihres Wohnorts befindet sich die nächste Windturbine?

- a. In meiner Gemeinde
- b. In einer Nachbargemeinde oder innerhalb meines Kantons
- c. In einem Nachbarkanton
- d. Ausserhalb meines gewöhnlichen Bewegungsradius
- e. Ich weiss nicht, wo sich die nächstgelegene Windturbine befindet

Block B: Fragen zur Wahrnehmung

Nachfolgend sehen Sie zwei reale Bilder des Windparks Gütsch in Andermatt. Bitte betrachten Sie diese als Gedankenstütze und beantworten Sie die folgenden Fragen möglichst basierend auf Ihrer eigenen Wahrnehmung.



Quelle: [SkiArena Andermatt-Sedrun](#) (Facebook)



Quelle: [Tages Anzeiger](#), Keystone

10. Wie empfinden Sie die Anwesenheit des Windparks Gütsch **in Bezug auf das Landschaftsbild**?

Ich empfinde das Landschaftsbild durch den Windpark ...

- a. stark gestört
- b. gestört
- c. indifferent
- d. bereichert
- e. sehr bereichert

11. Wie empfinden Sie die Anwesenheit des Windparks Gütsch **in Bezug auf die Tourismusaktivität**, wofür Sie nach Andermatt gekommen sind (z.B. Skifahren, Wandern, etc.)?

Ich fühle mich in meinem Vorhaben durch den Windpark ...

- a. stark gestört
- b. gestört

-
- c. indifferent
 - d. bereichert
 - e. sehr bereichert
12. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie der folgenden Aussage zustimmen:
- Ich empfinde den Windpark Gütsch als ein Zeichen der **Nachhaltigkeit und Fortschrittlichkeit** des Tourismusorts Andermatt.
- a. Ich stimme voll und ganz zu
 - b. Ich stimme zu
 - c. Ich stimme weder zu noch nicht zu
 - d. Ich stimme nicht zu
 - e. Ich stimme überhaupt nicht zu
13. Bitte geben Sie nachfolgend an, wie Sie dem Windpark Gütsch gegenüberstehen (Mehrfachauswahl möglich)
- a. Ich bin besorgt um den Lärm von Windturbinen
 - b. Ich sehe einen Windpark als eine Bereicherung für den Tourismusort
 - c. Ich bin besorgt um negative Auswirkungen wie Eiswurf, Schattenwurf oder Vogelschlag
 - d. Ich denke, der Windpark fördert die Energieunabhängigkeit der Region
 - e. Ich bin skeptisch gegenüber der Wirtschaftlichkeit von Windparks in dieser Region
 - f. Ich bin besorgt um die Sicherheit des Windparks
 - g. Für mich trifft keine der Aussagen zu
14. Finden Sie die Erzeugung von Strom durch den Windpark Gütsch an diesem Standort sinnvoll?
- a. Ich empfinde den Standort als äusserst sinnvoll
 - b. Ich empfinde den Standort als sehr sinnvoll
 - c. Ich empfinde den Standort als sinnvoll
 - d. Ich empfinde den Standort weniger sinnvoll
 - e. Ich empfinde den Standort als überhaupt nicht sinnvoll
 - f. Ich weiss nicht, wie ich den Standort bewerten soll
15. Falls die befragte Person den Standort als weniger bis überhaupt nicht sinnvoll bewertet:

Fänden Sie einen Standort abseits von existierender Infrastruktur, also in der «unberührten» Landschaft sinnvoller?

- a. Ja
- b. Vielleicht
- c. Nein

16. Wo empfänden Sie einen geeigneten Standort für einen Windpark?

- a. Im Mittelland
- b. In einem Bergtal
- c. Im Wohngebiet
- d. In einer Industriezone
- e. Im Ausland
- f. Nirgends, ich bin grundsätzlich gegen die Nutzung von Windenergie
- g. Andere: ...

Block C: Fragen zum Windtourismus

17. Haben Sie bereits einmal an einer Führung durch den Windpark Gütsch teilgenommen?

- a. Ja
- b. Nein

18. *Falls ja:* Aus welchem Grund/welchen Gründen (Mehrauswahl möglich) haben Sie den Windpark besucht?

- a. Im Rahmen einer beruflichen Veranstaltung (z.B. Team-Event)
- b. Im Rahmen einer schulischen Veranstaltung (z.B. Schülerreise)
- c. Aus (eigenem) Interesse an der Technologie
- d. Für einen Ausflug mit Freunden oder Familie
- e. Andere: ...

19. Welche der folgenden Attraktionen oder Angebote (Mehrfachauswahl möglich) würde Sie in Zukunft ermuntern, den Windpark zu besuchen?

- a. Flugshow von (Solar-)Flugzeugen oder Drohnen um die Windräder
- b. Flugshow von Fallschirmspringern oder Wind-Drachen um die Windräder
- c. Eine Aussichtsplattform
- d. Ein Foxtrail von Windrad zu Windrad oder anderen spielerischen Ansätzen, um die Windkraft näher kennenzulernen

-
- e. Ein Kombi-Angebot von Transportmitteln (Bergbahn oder SBB) und einer Führung durch den Windpark
 - f. Eine künstlerische Darstellung (z.B. eine Lichtshow) auf den Windrädern
 - g. Eine spannende Führung durch die Windkraftanlagen und dabei mehr über die Technologie zu erfahren, würde mich bereits überzeugen
 - h. Ein Musikfestival beim Windpark
 - i. Wenn ich auf eines dieser Angebote auf Social Media aufmerksam gemacht würde oder eines dieser Angebote da gewinnen könnte
 - j. Ich würde den Windpark niemals besuchen
 - k. Andere: ...

Block D: Allgemeine Fragen

20. Wie alt sind Sie?

- a. Jünger als 18
- b. 18-24
- c. 25-34
- d. 35-44
- e. 45-54
- f. 55-64
- g. 65-74
- h. 75-84
- i. Älter als 84

21. Bitte geben Sie Ihre höchst abgeschlossene Ausbildung an

- a. Eidg. Lehrabschluss
- b. Universität / Hochschule (FH, Uni, ETH)
- c. Gymnasiale oder berufliche Matura
- d. Sekundarschule
- e. Doktorabschluss
- f. Ich möchte diese Frage nicht beantworten
- g. Andere: ...

22. Bitte geben Sie ihr Heimatland an

- a. Schweiz
- b. Deutschland

-
- c. Österreich
 - d. Italien
 - e. Niederlanden
 - f. England
 - g. Andere: ...

23. *Wenn Schweiz angewählt:* In welchem Kanton der Schweiz wohnen Sie?

- a. Aargau
- b. Appenzell Ausserrhoden
- c. Appenzell Innerrhoden
- d. Basel-Landschaft
- e. Basel-Stadt
- f. Bern
- g. Freiburg
- h. Genf (Genève)
- i. Glarus
- j. Graubünden (Grigioni/Grischun)
- k. Jura
- l. Luzern
- m. Neuenburg (Neuchâtel)
- n. Nidwalden
- o. Obwalden
- p. Schaffhausen
- q. Schwyz
- r. Solothurn
- s. St. Gallen
- t. Tessin (Ticino)
- u. Thurgau
- v. Uri
- w. Waadt (Vaud)
- x. Wallis (Valais)
- y. Zug
- z. Zürich

24. Bitte geben Sie Ihr Geschlecht an.

- a. Weiblich

-
- b. Männlich
 - c. Non-binär
 - d. Möchte ich nicht angeben
 - e. Anderes: ...

25. Bitte geben Sie an, ob Sie eines der folgenden Abonnements des öffentlichen Verkehrs besitzen.

- a. Ich besitze ein GA (General-Abonnement)
- b. Ich besitze ein Halbtax
- c. Ich besitze ein Abonnement meines Regionalverkehrsverbunds
- d. Ich besitze kein Abonnement

Herzlichen Dank für Ihre Teilnahme an dieser Umfrage!

Ihre Antworten werden hilfreich sein, um mehr über das Zusammenspiel von Windkraft und Tourismus zu erfahren und sind eine wertvolle Ergänzung zu meiner Masterarbeit. Dafür schätze ich Ihre Unterstützung sehr!

Falls Ihnen Limitationen dieser Umfrage aufgefallen sind oder Sie ein anderes Feedback zur Umfrage abgeben möchten, freuen ich mich über Ihre Rückmeldung: ...

Fragebogen Laax

Block A: Einstiegsfragen

2. Bitte geben Sie die Regelmässigkeit für ihren Besuch in Laax an
 - a. Ich bin zum ersten Mal in Laax
 - b. Ich besuche Laax in gewisser Regelmässigkeit (bspw. jedes Jahr, alle zwei Jahre, etc.)
 - c. Ich besuche Laax nicht zum ersten Mal, jedoch nicht regelmässig
 - e. Ich war noch nie in Laax und bin auch zum jetzigen Zeitpunkt, zu dem ich die Umfrage ausfülle, nicht vor Ort
 - f. Ich war bereits mind. Einmal in Laax, bin aber zum jetzigen Zeitpunkt, zu dem ich die Umfrage ausfülle, gerade nicht da
3. Bitte geben Sie den Grund/die Gründe (Mehrauswahl möglich) für Ihren Besuch in Laax an
 - a. Ich besuche Laax zum Wandern oder Klettern
 - b. Ich besuche Laax zum Biken
 - c. Ich besuche Laax (hauptsächlich) zum Skifahren/Snowboarden und anderen Wintersportarten
 - d. Ich besuche Laax für ein kulturelles Angebot oder Event (Snowboard/Freestyle Contest, Museum, Kunst, o.ä.)
 - e. Ich bin für einen Wellness-Aufenthalt in Laax
 - f. Andere Gründe: ...
4. Wie beurteilen Sie die Geschwindigkeit der Energiewende?
 - a. Die Energiewende geht zu langsam voran
 - b. Das Tempo der Energiewende ist genau richtig
 - c. Bei der Energiewende wird überstürzt gehandelt
5. Für wie wichtig halten Sie es, dass die Schweiz beim Klimaschutz vorankommt?
 - a. Sehr wichtig
 - b. Wichtig
 - c. Weder noch
 - d. Nicht wichtig
 - e. Sehr unwichtig
6. Wenn Sie den Strommix der Zukunft für die Schweiz bestimmen könnten, welche der folgenden Optionen würden Sie wählen:

-
- a. breit diversifiziertes Portfolio von erneuerbaren Energien (Wasser, Sonne und Wind)
 - b. Fokus auf Solarenergie und Wasserkraft, Verzicht auf Windenergie
 - c. verstärkte Förderung der Kernenergie
 - d. Bau von Gaskraftwerken
7. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie sich bei der Wahl Ihrer Reisedestination darauf achten, dass diese Wert auf Nachhaltigkeit und Umweltfreundlichkeit legt:
- a. Ich achte mich sehr stark darauf
 - b. Ich achte mich darauf
 - c. Ich achte mich ab und zu darauf
 - d. Ich achte mich nicht darauf
 - e. Ich achte mich ganz und gar nicht darauf
8. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie der folgenden Aussage zustimmen:
- Ich interessiere mich für Windkraft und ihre Technologie.
- a. Ich stimme voll und ganz zu
 - b. Ich stimme zu
 - c. Ich stimme weder zu noch nicht zu
 - d. Ich stimme nicht zu
 - e. Ich stimme überhaupt nicht zu
9. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie der folgenden Aussage zustimmen:
- Ich finde einen Windpark in Nähe von bestehender Infrastruktur zu platzieren sinnvoll.
- a. Ich stimme voll und ganz zu
 - b. Ich stimme zu
 - c. Ich stimme weder zu noch nicht zu
 - d. Ich stimme nicht zu
 - e. Ich stimme überhaupt nicht zu
10. In welcher Entfernung Ihres Wohnorts befindet sich die nächste Windturbine?
- a. In meiner Gemeinde
 - b. In einer Nachbargemeinde oder innerhalb meines Kantons
 - c. In einem Nachbarkanton
 - d. Ausserhalb meines gewöhnlichen Bewegungsradius

- e. Ich weiss nicht, wo sich die nächstgelegene Windturbine befindet

Block B: Fragen zur Wahrnehmung

Nachfolgend sehen Sie zwei Bilder, welche einen möglichen Windpark in Laax abbilden. Bitte betrachten Sie diese als Gedankenstütze und beantworten Sie die folgenden Fragen möglichst basierend auf Ihrer eigenen Wahrnehmung.



Quelle: mtb-news.de; eigens ergänzt durch Windturbinen



Quelle: skiresort.at; eigens ergänzt durch Windturbinen

11. Wie empfänden Sie die Anwesenheit eines Windparks in Laax (auf dem Crap Sogn Gion) **in Bezug auf das Landschaftsbild**?

Ich empfinde das Landschaftsbild durch den Windpark ...

- a. stark gestört
- b. gestört
- c. indifferent
- d. bereichert
- e. sehr bereichert

12. Wie empfänden Sie die Anwesenheit eines Windparks in Laax **in Bezug auf die Tourismusaktivität**, wofür Sie nach Laax gekommen sind (z.B. Skifahren, Wandern, etc.)?

Ich würde mich in meinem Vorhaben durch den Windpark ...

- a. stark gestört
- b. gestört
- c. indifferent
- d. bereichert
- e. sehr bereichert

13. Bitte bewerten Sie anhand folgender Skala, inwiefern Sie der folgenden Aussage zustimmen:

Ich empfinde einen Windpark als ein Zeichen der **Nachhaltigkeit und Fortschrittlichkeit** des Tourismusorts Laax.

- a. Ich stimme voll und ganz zu
- b. Ich stimme zu
- c. Ich stimme weder zu noch nicht zu
- d. Ich stimme nicht zu
- e. Ich stimme überhaupt nicht zu

14. Bitte geben Sie nachfolgend an, wie Sie einem Windpark in Laax (auf dem Crap Sogn Gion) gegenüberstehen (Mehrfachauswahl möglich)

- a. Ich bin besorgt um den Lärm von Windturbinen
- b. Ich sehe einen Windpark als eine Bereicherung für den Tourismusort
- c. Ich bin besorgt um negative Auswirkungen wie Eiswurf, Schattenwurf oder Vogelschlag
- d. Ich denke, ein Windpark könnte die Energieunabhängigkeit der Region fördern
- e. Ich bin skeptisch gegenüber der Wirtschaftlichkeit von Windparks in dieser Region
- f. Ich bin besorgt um die Sicherheit des Windparks
- g. Für mich trifft keine der Aussagen zu

15. Fänden Sie die Erzeugung von Strom durch einen Windpark in Laax an diesem Standort (auf dem Crap Sogn Gion) sinnvoll?

- a. Ich empfinde den Standort als äusserst sinnvoll
- b. Ich empfinde den Standort als sehr sinnvoll
- c. Ich empfinde den Standort als sinnvoll
- d. Ich empfinde den Standort weniger sinnvoll
- e. Ich empfinde den Standort als überhaupt nicht sinnvoll
- f. Ich weiss nicht, wie ich den Standort bewerten soll

16. *Falls die befragte Person den Standort als weniger bis überhaupt nicht sinnvoll bewertet:*

Fänden Sie einen Standort abseits von existierender Infrastruktur, also in der «unberührten» Landschaft sinnvoller?

- a. Ja
- b. Vielleicht
- c. Nein

17. Wo empfänden Sie einen geeigneten Standort für einen Windpark?

-
- a. Im Mittelland
 - b. In einem Bergtal
 - c. Im Wohngebiet
 - d. In einer Industriezone
 - e. Im Ausland
 - f. Nirgends, ich bin grundsätzlich gegen die Nutzung von Windenergie
 - g. Andere: ...

Block C: Fragen zum Windtourismus

18. Haben Sie bereits einmal an einer Führung durch einen Windpark teilgenommen?

- a. Ja
- b. Nein

19. *Falls ja:* Aus welchem Grund/welchen Gründen (Mehrauswahl möglich) haben Sie den Windpark besucht?

- a. Im Rahmen einer beruflichen Veranstaltung (z.B. Team-Event)
- b. Im Rahmen einer schulischen Veranstaltung (z.B. Schülerreise)
- c. Aus (eigenem) Interesse an der Technologie
- d. Für einen Ausflug mit Freunden oder Familie
- e. Andere: ...

20. Welche der folgenden Attraktionen oder Angebote (Mehrfachauswahl möglich) würde Sie in Zukunft ermuntern, einen potentiellen Windpark in Laax zu besuchen?

- a. Flugshow von (Solar-)Flugzeugen oder Drohnen um die Windräder
- b. Flugshow von Fallschirmspringern oder Wind-Drachen um die Windräder
- c. Eine Aussichtsplattform
- d. Ein Foxtrail von Windrad zu Windrad oder anderen spielerischen Ansätzen, um die Windkraft näher kennenzulernen
- e. Ein Kombi-Angebot von Transportmitteln (Bergbahn oder SBB) und einer Führung durch den Windpark
- f. Eine künstlerische Darstellung (z.B. eine Lichtshow) auf den Windrädern
- g. Eine spannende Führung durch die Windkraftanlagen und dabei mehr über die Technologie zu erfahren, würde mich bereits überzeugen
- h. Ein Musikfestival beim Windpark

-
- i. Wenn ich auf eines dieser Angebote auf Social Media aufmerksam gemacht würde oder eines dieser Angebote da gewinnen könnte
 - j. Ich würde den Windpark niemals besuchen.
 - k. Andere: ...

Block D: Allgemeine Fragen

21. Wie alt sind Sie?

- a. Jünger als 18
- b. 18-24
- c. 25-34
- d. 35-44
- e. 45-54
- f. 55-64
- g. 65-74
- h. 75-84
- i. Älter als 84

22. Bitte geben Sie Ihre höchst abgeschlossene Ausbildung an

- a. Eidg. Lehrabschluss
- b. Universität / Hochschule (FH, Uni, ETH)
- c. Gymnasiale oder berufliche Matura
- d. Sekundarschule
- e. Doktorabschluss
- f. Ich möchte diese Frage nicht beantworten

23. Bitte geben Sie ihr Heimatland an

- a. Schweiz
- b. Deutschland
- c. Oesterreich
- d. Italien
- e. Niederlanden
- f. England
- g. Andere: ...

24. *Wenn Schweiz angewählt:* In welchem Kanton der Schweiz wohnen Sie?

- a. Aargau

-
- b. Appenzell Ausserrhoden
 - c. Appenzell Innerrhoden
 - d. Basel-Landschaft
 - e. Basel-Stadt
 - f. Bern
 - g. Freiburg
 - h. Genf (Genève)
 - i. Glarus
 - j. Graubünden (Grigioni/Grischun)
 - k. Jura
 - l. Luzern
 - m. Neuenburg (Neuchâtel)
 - n. Nidwalden
 - o. Obwalden
 - p. Schaffhausen
 - q. Schwyz
 - r. Solothurn
 - s. St. Gallen
 - t. Tessin (Ticino)
 - u. Thurgau
 - v. Uri
 - w. Waadt (Vaud)
 - x. Wallis (Valais)
 - y. Zug
 - z. Zürich

25. Bitte geben Sie Ihr Geschlecht an.

- a. Weiblich
- b. Männlich
- c. Non-binär
- d. Möchte ich nicht sagen
- e. Anderes: ...

26. Bitte geben Sie an, ob Sie eines der folgenden Abonnements des öffentlichen Verkehrs besitzen.

- a. Ich besitze ein GA (General-Abonnement)

-
- b. Ich besitze ein Halbtax
 - c. Ich besitze ein Abonnement meines Regionalverkehrsverbunds
 - d. Ich besitze kein Abonnement

Herzlichen Dank für Ihre Teilnahme an dieser Umfrage!

Ihre Antworten werden hilfreich sein, um mehr über das Zusammenspiel von Windkraft und Tourismus zu erfahren und sind eine wertvolle Ergänzung zu meiner Masterarbeit. Dafür schätze ich Ihre Unterstützung sehr!

Falls Ihnen Limitationen dieser Umfrage aufgefallen sind oder Sie ein anderes Feedback zur Umfrage abgeben möchten, freuen ich mich über Ihre Rückmeldung: ...

B. Interviews mit Tourismusdirektoren

Interviewleitfaden für André Gisler und Thomas Christen

1. Sind Sie selbst ein Befürworter von Windenergie bzw. wie stehen Sie zum Windpark Gütsch?
2. Inwiefern empfinden Sie, dass der Windpark ein Zeichen für Klimafreundlichkeit und Fortschrittlichkeit für Andermatt steht?
3. Wie empfinden Sie aus Sicht des Tourismus, dass das Landschaftsbild durch den Windpark Gütsch beeinflusst wird (positiv und/oder negativ)?
4. Wie empfinden Sie, dass der Windpark gewisse Tourismusaktivitäten (Sommer vs. Winter) beeinflusst?
5. Welchen Einfluss hat der Windpark auf die Tourismusentwicklung generell?
6. Welche Attraktionen/Aktivitäten haben Sie sich überlegt, wie man den Windpark für zusätzliche Wertschöpfung nutzen könnte?

C. Tourismuskennzahlen

Statistiken zur Hotellerie in Andermatt

Jahr	Logiernächte Schweiz	Veränderungen im Vgl. zum Vorjahr [%] (Logiernächte Schweiz)	Logiernächte Andermatt	Veränderungen im Vgl. zum Vorjahr [%] (Logiernächte An- dermatt)	Logiernächte Inland (für Andermatt)	Logiernächte Ausland (für Andermatt)
2005	32'943'736	100.00%	65'623	100.00%	18 146	36 417
2006	34'848'426	5.78%	65'023	-0.91%	14 900	38 399
2007	36'364'800	4.35%	72'881	12.08%	16 937	42 086
2008	37'333'769	2.66%	74'174	1.77%	16 621	43 942
2009	35'588'893	-4.67%	73'146	-1.39%	17 321	42 459
2010	36'207'812	1.74%	69'280	-5.29%	18'222	40'097
2011	35'486'256	-1.99%	66'106	-4.58%	16'557	38'903
2012	34'766'273	-2.03%	60'678	-8.21%	15'552	33'223
2013	35'623'883	2.47%	62'174	2.47%	16'186	32'722
2014	35'933'512	0.87%	83'027	33.54%	23'405	43'032
2015	35'628'476	-0.85%	80'709	-2.79%	25'674	38'425
2016	35'532'576	-0.27%	87'478	8.39%	30'284	38'846
2017	37'392'740	5.24%	97'020	10.91%	35'448	42'000
2018	38'806'777	3.78%	101'877	5.01%	35'227	45'187
2019	39'562'039	1.95%	156'613	53.73%	57'130	60'350
2020	23'730'738	-40.02%	155'814	-0.51%	63'066	41'268
2021	29'558'849	24.56%	185'142	18.82%	72'466	49'219
2022	38'241'145	29.37%	183'243	-1.03%	58'948	72'017

Quelle: Bundesamt für Statistik (2023b)

Statistiken zu den Beschäftigten im Tourismus in Andermatt

Jahr	Total Kanton Uri (VZÄ)	Anteil für das Urserntal (VZÄ)	Veränderungen im Vgl. zum Vorjahr Kanton Uri [%]	Veränderungen im Vgl. zum Vorjahr Urserntal [%]
2011	881	273.11	100.00%	100.00%
2012	822	271.26	-0.68%	-6.70%
2013	977	341.95	26.06%	18.86%
2014	1'006	372.22	8.85%	2.97%
2015	1'044	407.16	9.39%	3.78%
2016	1'054	432.14	6.14%	0.96%
2017	1'053	452.79	4.78%	-0.09%
2018	1'139	512.55	13.20%	8.17%
2019	1'142	536.74	4.72%	0.26%
2020	1'253	613.97	14.39%	9.72%
2021	1'266	645.66	5.16%	1.04%

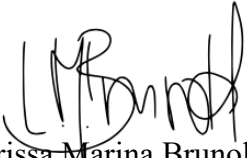
Quelle: Bundesamt für Statistik (2023c)

Eigenständigkeitserklärung

"Ich erkläre hiermit,

- dass ich die vorliegende Arbeit eigenständig verfasst habe,
- dass ich die Arbeit nur unter Verwendung der im Verzeichnis angegebenen Hilfsmittel verfasst habe;
- dass alle mit Hilfsmitteln erbrachten Teile der Arbeit präzise deklariert wurden;
- dass ich sämtliche verwendeten Quellen erwähnt und gemäss gängigen wissenschaftlichen Zitierregeln korrekt zitiert habe;
- dass ich sämtliche immateriellen Rechte an von mir allfällig verwendeten Materialien wie Bilder oder Grafiken erworben habe oder dass diese Materialien von mir selbst erstellt wurden;
- dass das Thema, die Arbeit oder Teile davon nicht bereits Gegenstand eines Leistungsnachweises einer anderen Veranstaltung oder Kurses waren, sofern dies nicht ausdrücklich mit der Referentin oder dem Referenten im Voraus vereinbart wurde und in der Arbeit ausgewiesen wird;
- dass ich mir über die rechtlichen Bestimmungen zur Publikation und Weitergabe von Teilen oder der ganzen Arbeit bewusst bin und ich diese entsprechend einhalte;
- dass ich mir bewusst bin, dass meine Arbeit elektronisch auf Plagiate und auf Drittautorschaft menschlichen oder technischen Ursprungs überprüft werden kann und ich hiermit der Universität St.Gallen laut Prüfungsordnung das Urheberrecht soweit einräume, wie es für die Verwaltungshandlungen notwendig ist;
- dass ich mir bewusst bin, dass die Universität einen Verstoß gegen diese Eigenständigkeitserklärung verfolgt und dass daraus disziplinarische wie auch strafrechtliche Folgen resultieren können, welche zum Ausschluss von der Universität resp. zur Titelaberkennung führen können.“

Mit Einreichung der schriftlichen Arbeit stimme ich mit konkludentem Handeln zu, die Eigenständigkeitserklärung abzugeben, diese gelesen sowie verstanden zu haben und, dass sie der Wahrheit entspricht.


Larissa Marina Brunold

St. Gallen, 6. November 2023